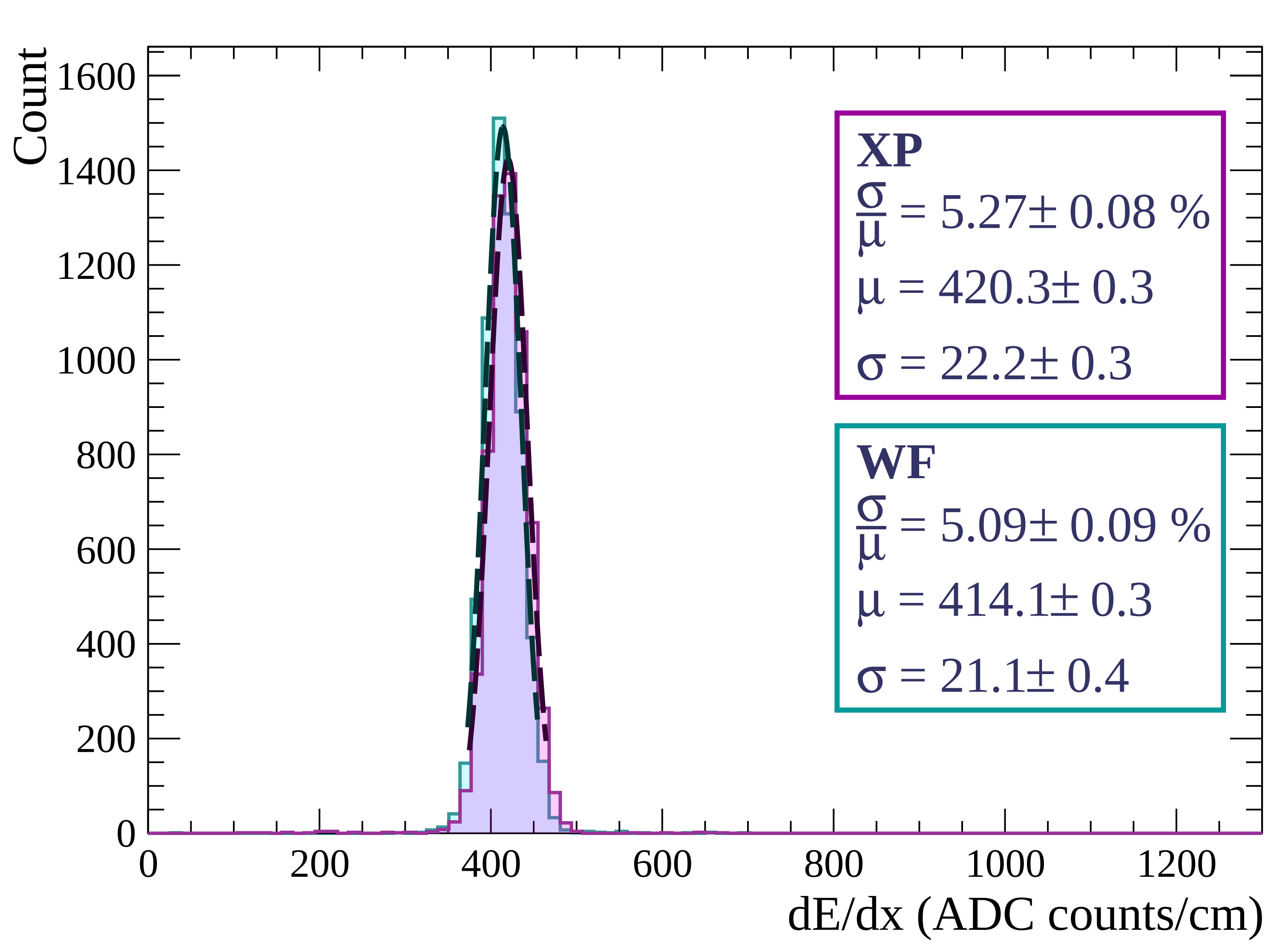
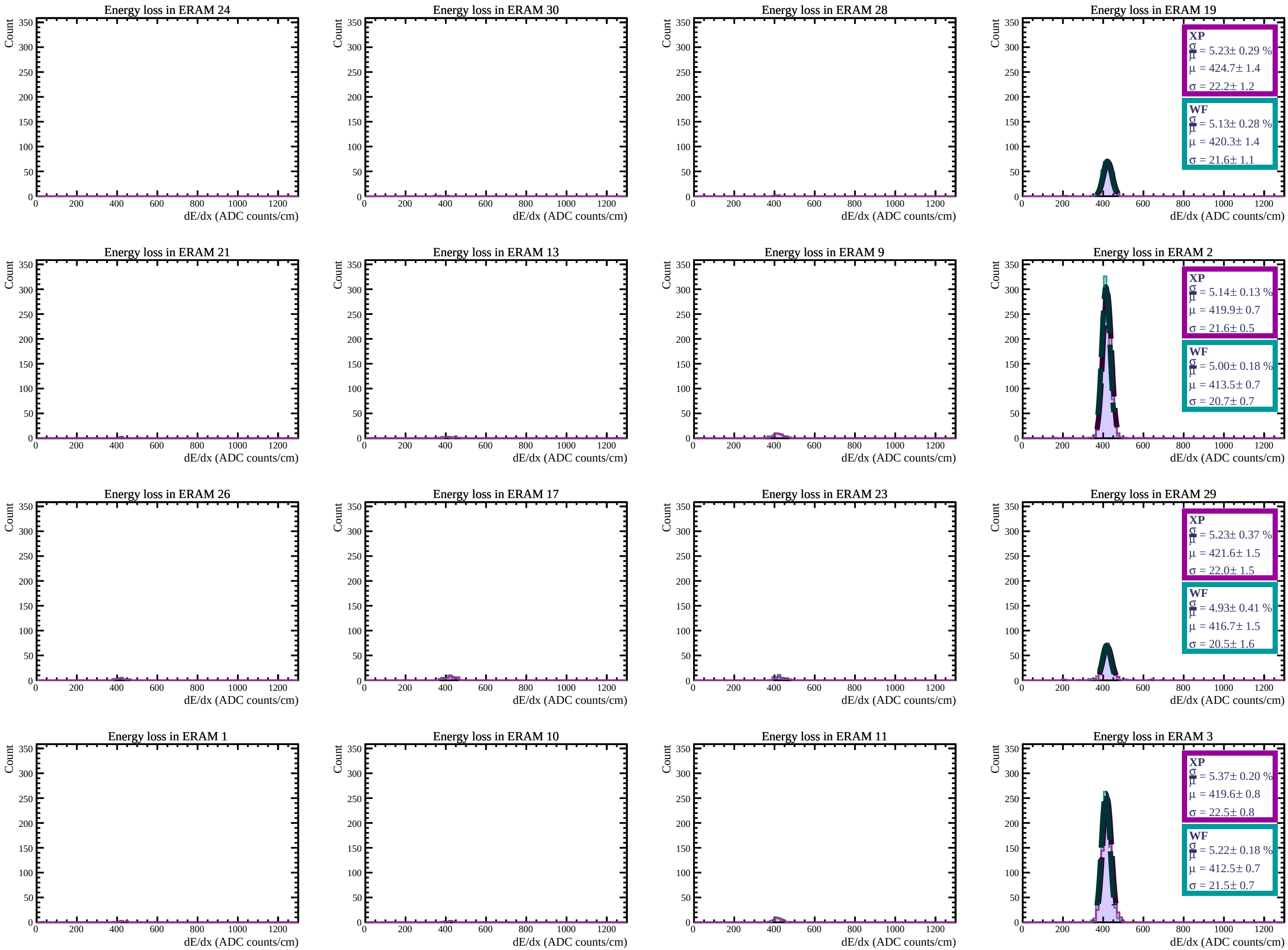
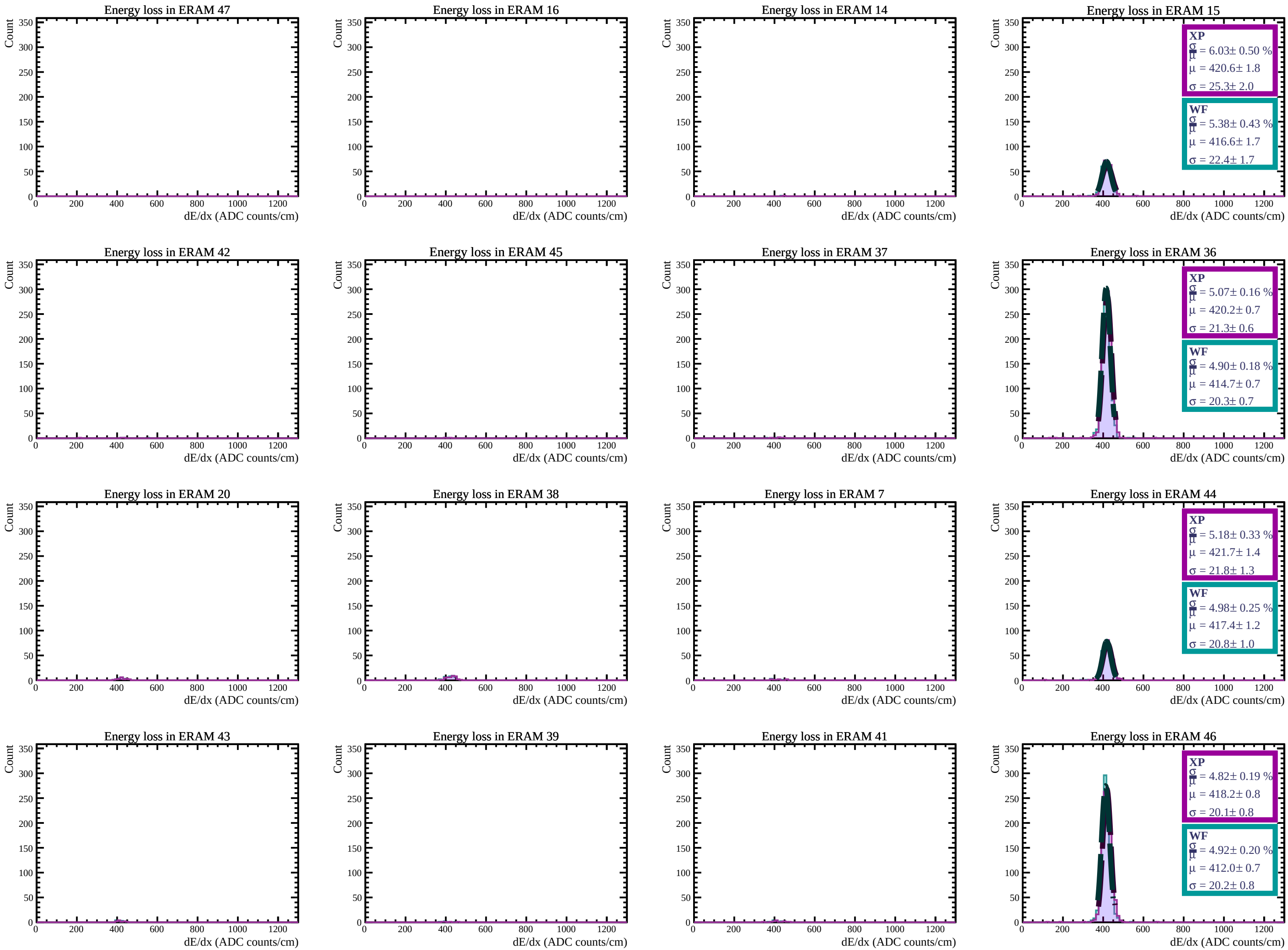
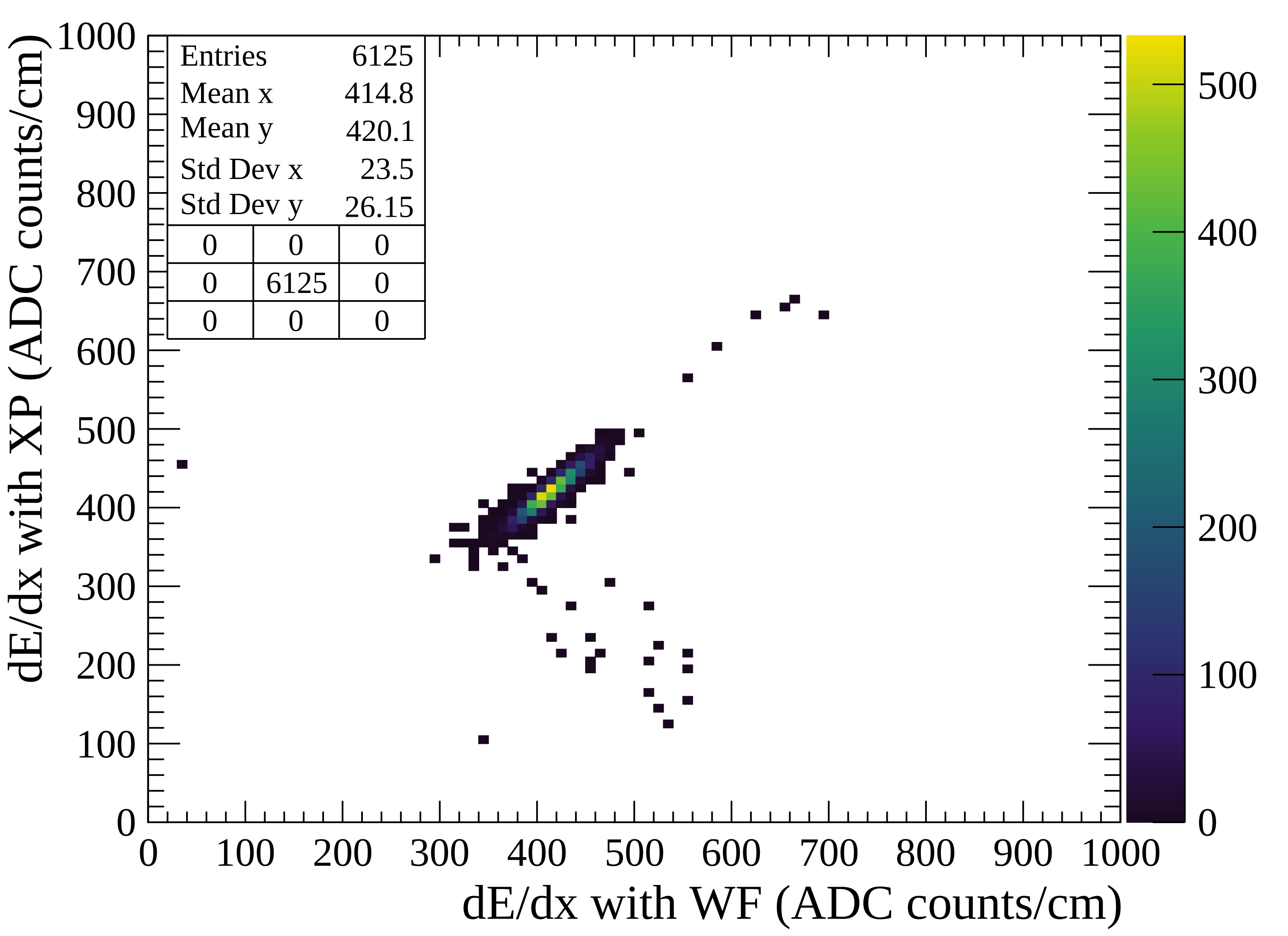

Count
1600
1400
1200
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 5.27± 0.08 %
μ = 420.3± 0.3
σ = 22.2± 0.3
WF
σ/μ = 5.09± 0.09 %
μ = 414.1± 0.3
σ = 21.1± 0.4

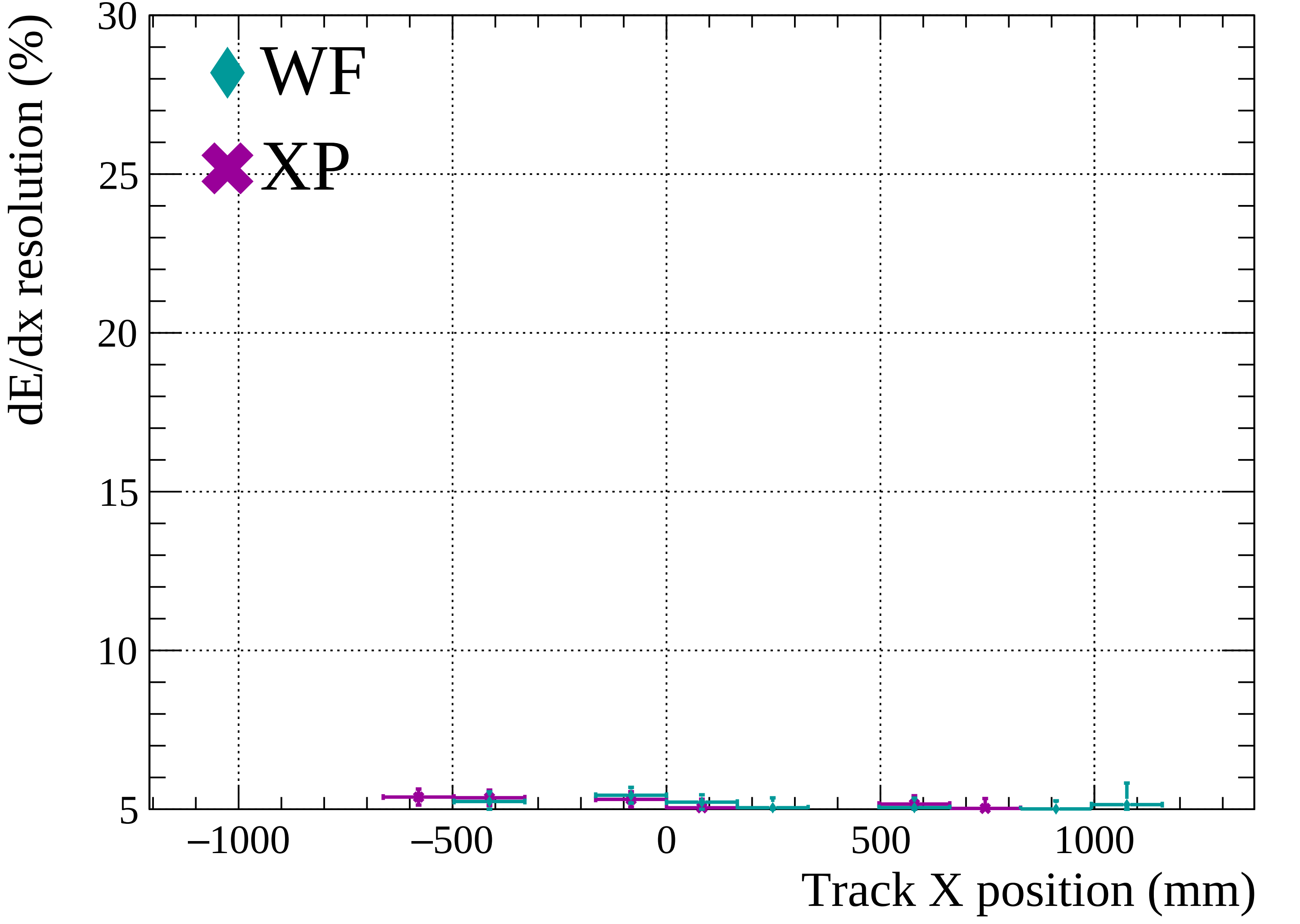


Energy loss in ERAM 24
Energy loss in ERAM 30
Energy loss in ERAM 28
Energy loss in ERAM 19
XP
σ/μ = 5.23± 0.29 %
μ = 424.7± 1.4
σ = 22.2± 1.2
WF
σ/μ = 5.13± 0.28 %
μ = 420.3± 1.4
σ = 21.6± 1.1
Energy loss in ERAM 21
Energy loss in ERAM 13
Energy loss in ERAM 9
Energy loss in ERAM 2
XP
σ/μ = 5.14± 0.13 %
μ = 419.9± 0.7
σ = 21.6± 0.5
WF
σ/μ = 5.00± 0.18 %
μ = 413.5± 0.7
σ = 20.7± 0.7
Energy loss in ERAM 26
Energy loss in ERAM 17
Energy loss in ERAM 23
Energy loss in ERAM 29
XP
σ/μ = 5.23± 0.37 %
μ = 421.6± 1.5
σ = 22.0± 1.5
WF
σ/μ = 4.93± 0.41 %
μ = 416.7± 1.5
σ = 20.5± 1.6
Energy loss in ERAM 1
Energy loss in ERAM 10
Energy loss in ERAM 11
Energy loss in ERAM 3
XP
σ/μ = 5.37± 0.20 %
μ = 419.6± 0.8
σ = 22.5± 0.8
WF
σ/μ = 5.22± 0.18 %
μ = 412.5± 0.7
σ = 21.5± 0.7
Count
dE/dx (ADC counts/cm)

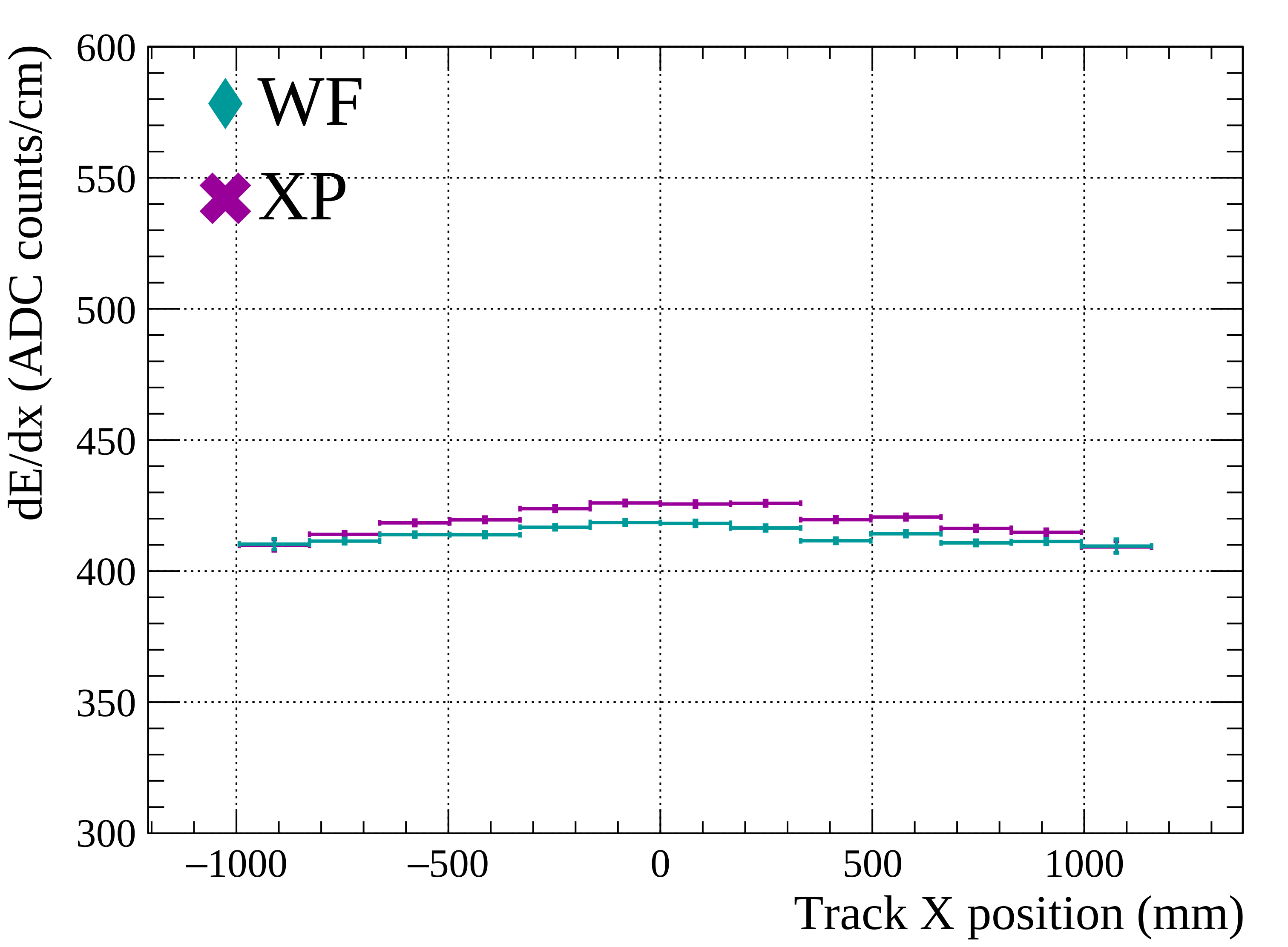


Energy loss in ERAM 47
Energy loss in ERAM 16
Energy loss in ERAM 14
Energy loss in ERAM 15
XP
σ/μ = 6.03± 0.50 %
μ = 420.6± 1.8
σ = 25.3± 2.0
WF
σ/μ = 5.38± 0.43 %
μ = 416.6± 1.7
σ = 22.4± 1.7
Energy loss in ERAM 42
Energy loss in ERAM 45
Energy loss in ERAM 37
Energy loss in ERAM 36
XP
σ/μ = 5.07± 0.16 %
μ = 420.2± 0.7
σ = 21.3± 0.6
WF
σ/μ = 4.90± 0.18 %
μ = 414.7± 0.7
σ = 20.3± 0.7
Energy loss in ERAM 20
Energy loss in ERAM 38
Energy loss in ERAM 7
Energy loss in ERAM 44
XP
σ/μ = 5.18± 0.33 %
μ = 421.7± 1.4
σ = 21.8± 1.3
WF
σ/μ = 4.98± 0.25 %
μ = 417.4± 1.2
σ = 20.8± 1.0
Energy loss in ERAM 43
Energy loss in ERAM 39
Energy loss in ERAM 41
Energy loss in ERAM 46
XP
σ/μ = 4.82± 0.19 %
μ = 418.2± 0.8
σ = 20.1± 0.8
WF
σ/μ = 4.92± 0.20 %
μ = 412.0± 0.7
σ = 20.2± 0.8
Count
dE/dx (ADC counts/cm)

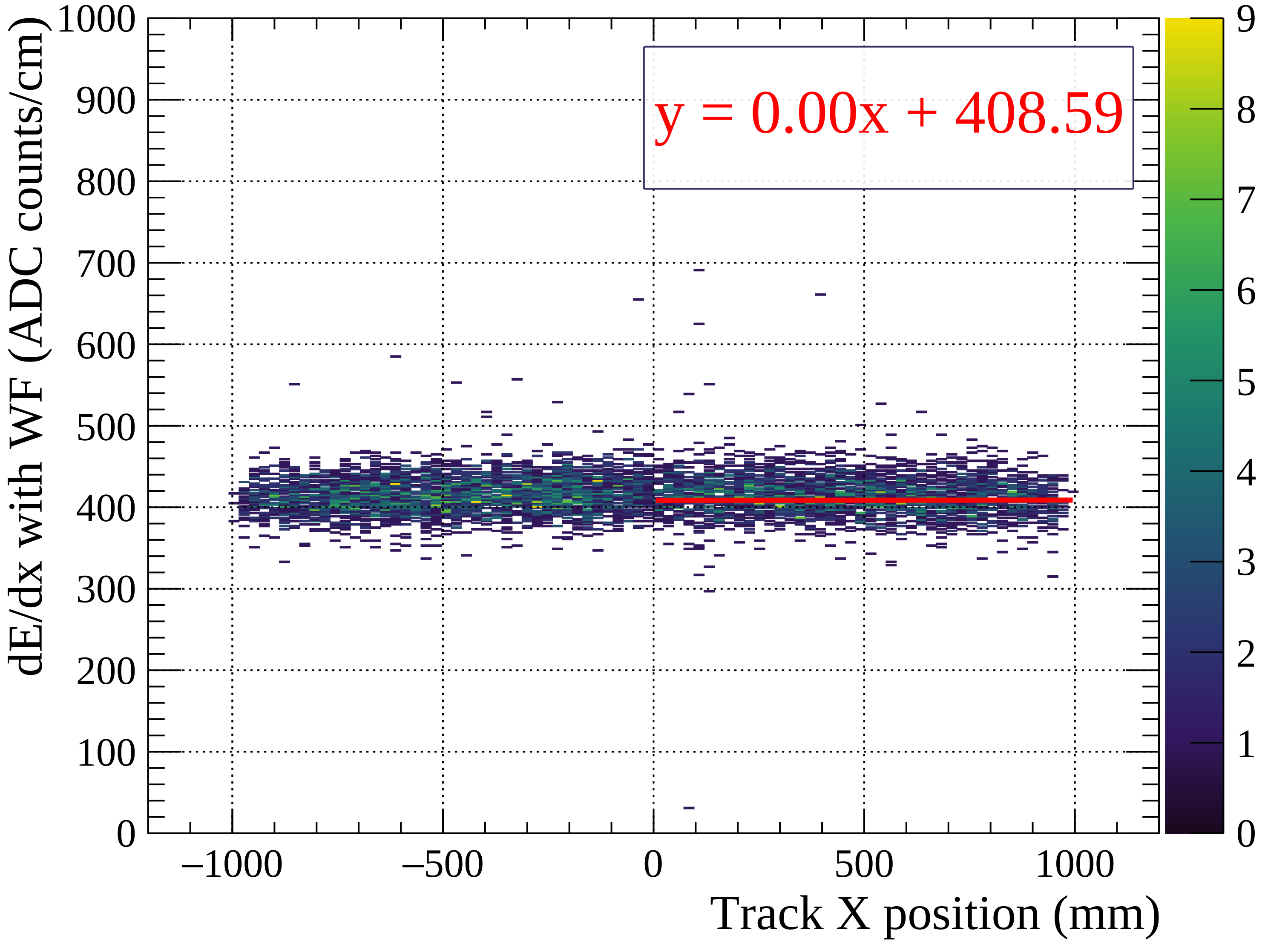

Entries 6125
Mean x 414.8
Mean y 420.1
Std Dev x 23.5
Std Dev y 26.15
0 0 0
0 6125 0
0 0 0
dE/dx with XP (ADC counts/cm)
dE/dx with WF (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
1000
500
400
300
200
100



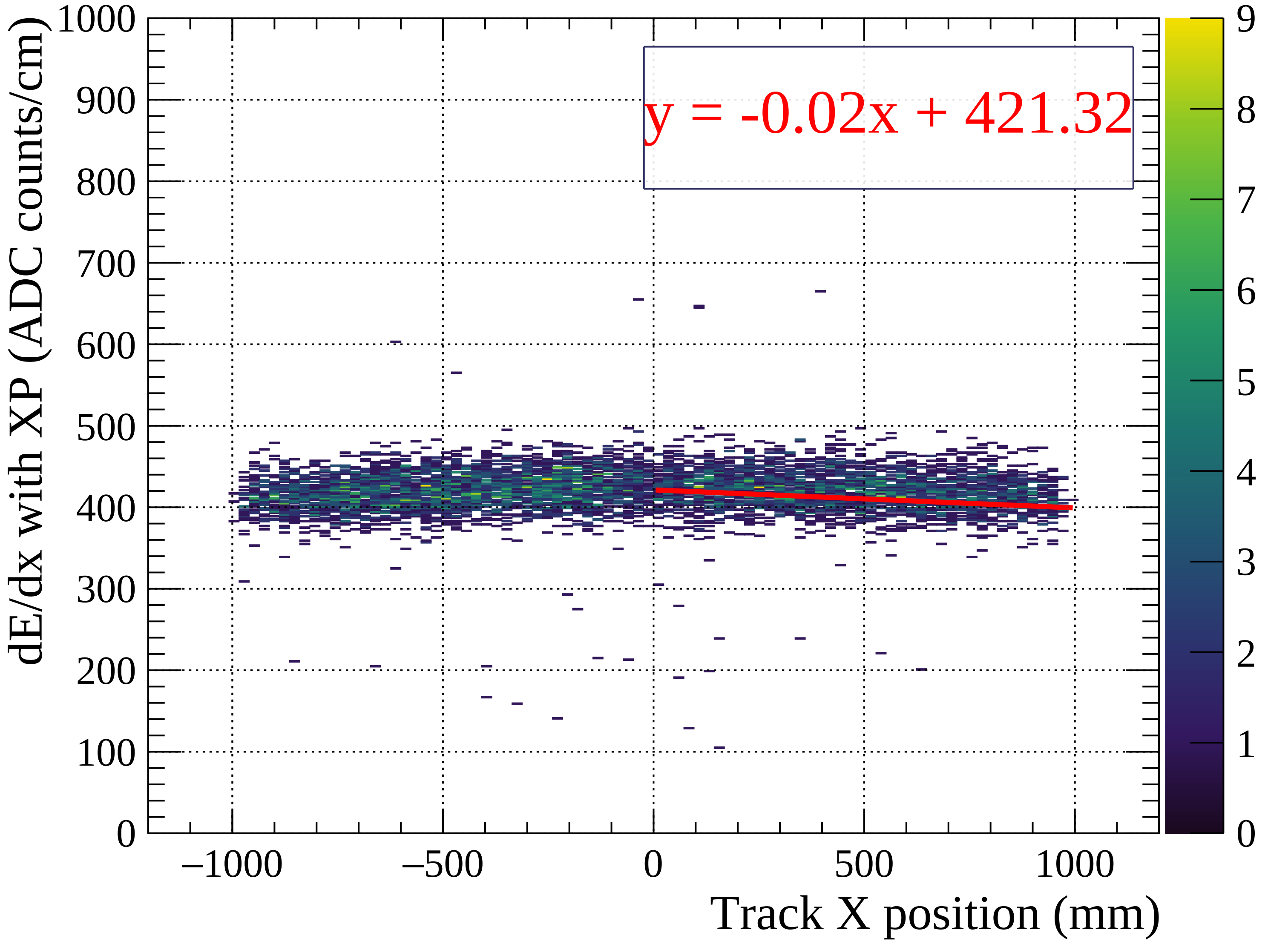
WF
XP
dE/dx resolution (%)
Track X position (mm)
30
25
20
15
10
5
−1000
−500
0
500
1000



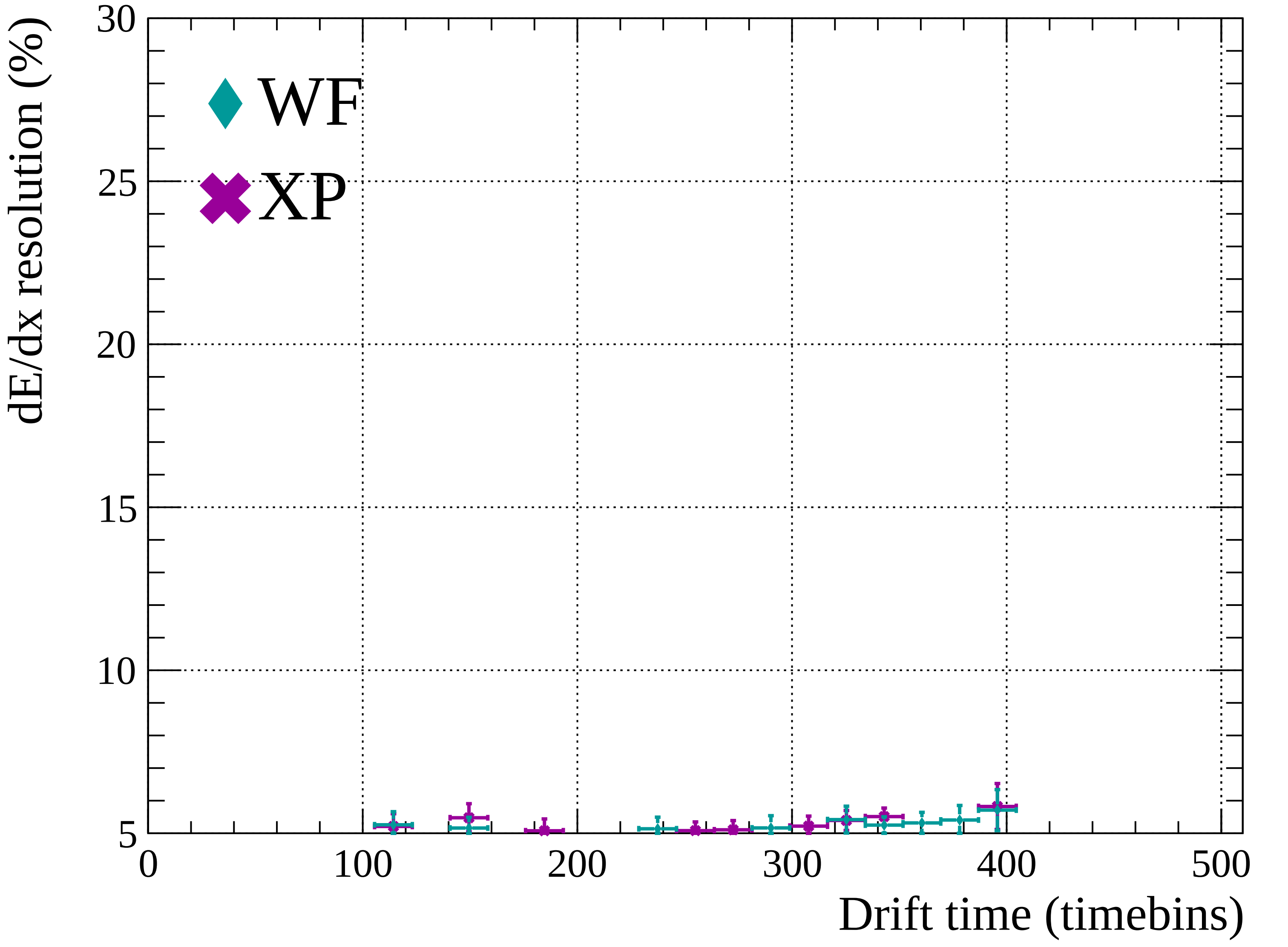
WF
XP
dE/dx (ADC counts/cm)
600
550
500
450
400
350
300
−1000
−500
0
500
1000
Track X position (mm)



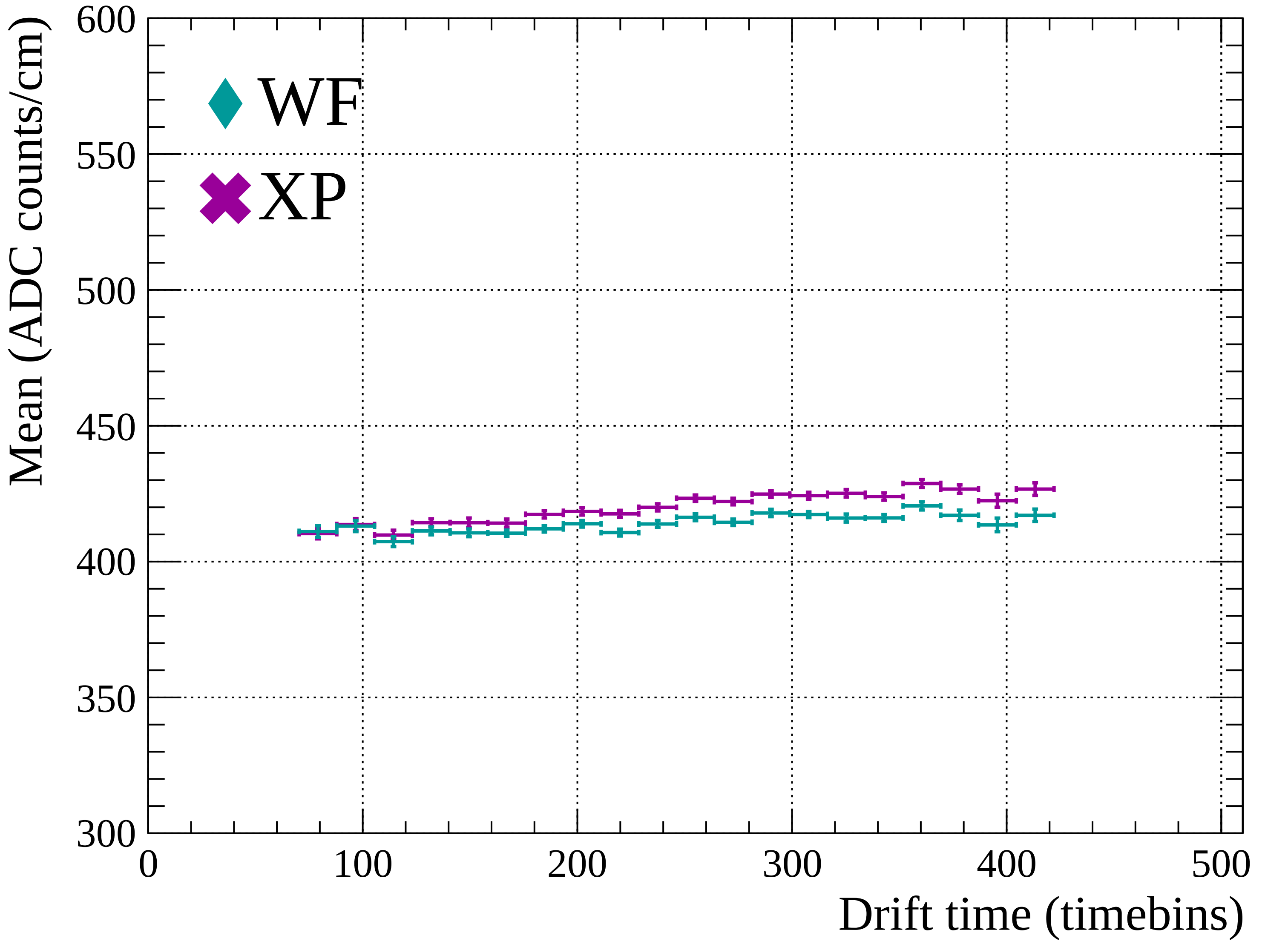
y = 0.00x + 408.59
dE/dx with WF (ADC counts/cm)
Track X position (mm)
1000
900
800
700
600
500
400
300
200
100
0
−1000
−500
0
500
1000
9
8
7
6
5
4
3
2
1
0



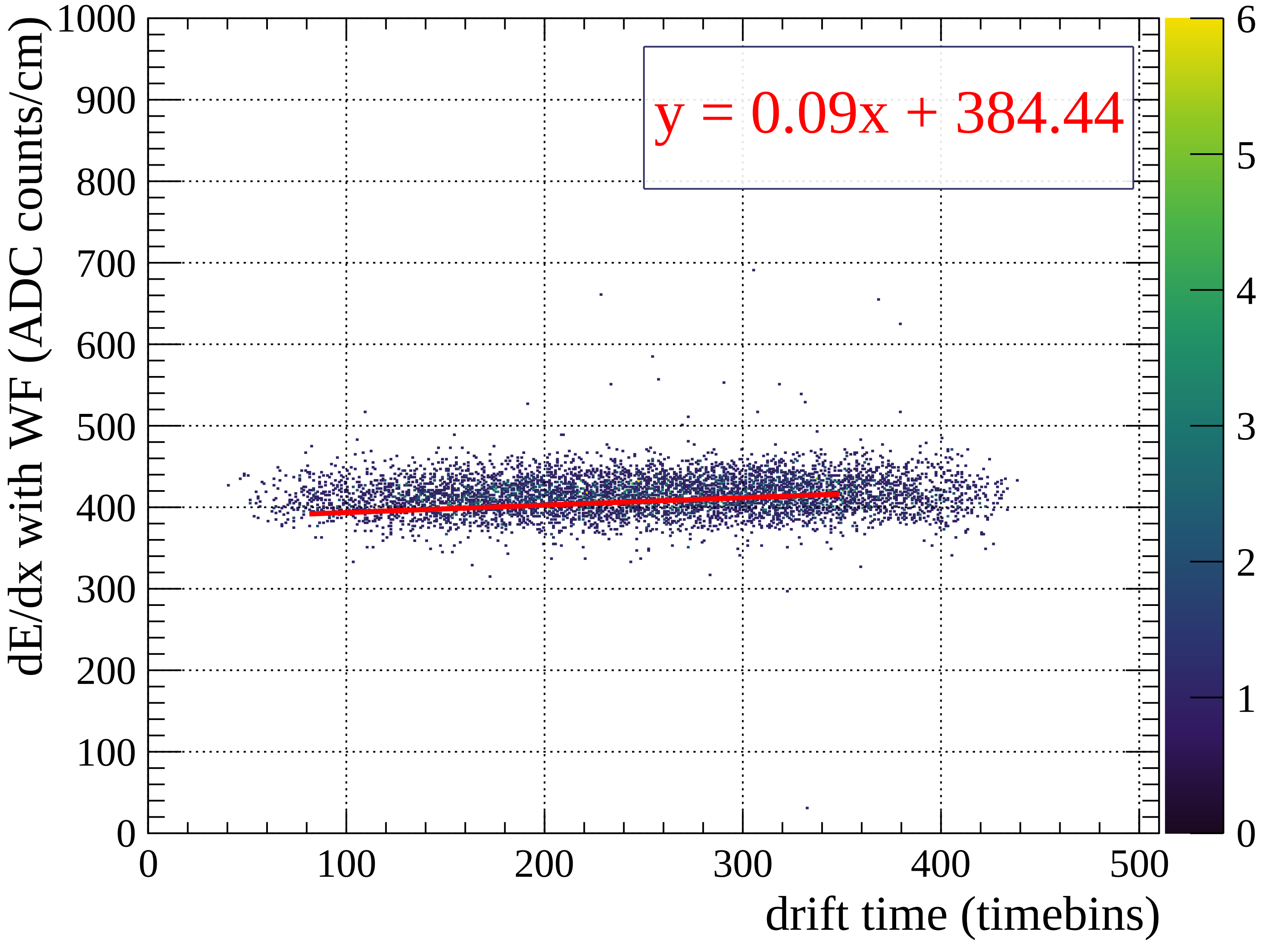
y = -0.02x + 421.32
dE/dx with XP (ADC counts/cm)
Track X position (mm)
0
100
200
300
400
500
600
700
800
900
1000
−1000
−500
0
500
1000
0
1
2
3
4
5
6
7
8
9



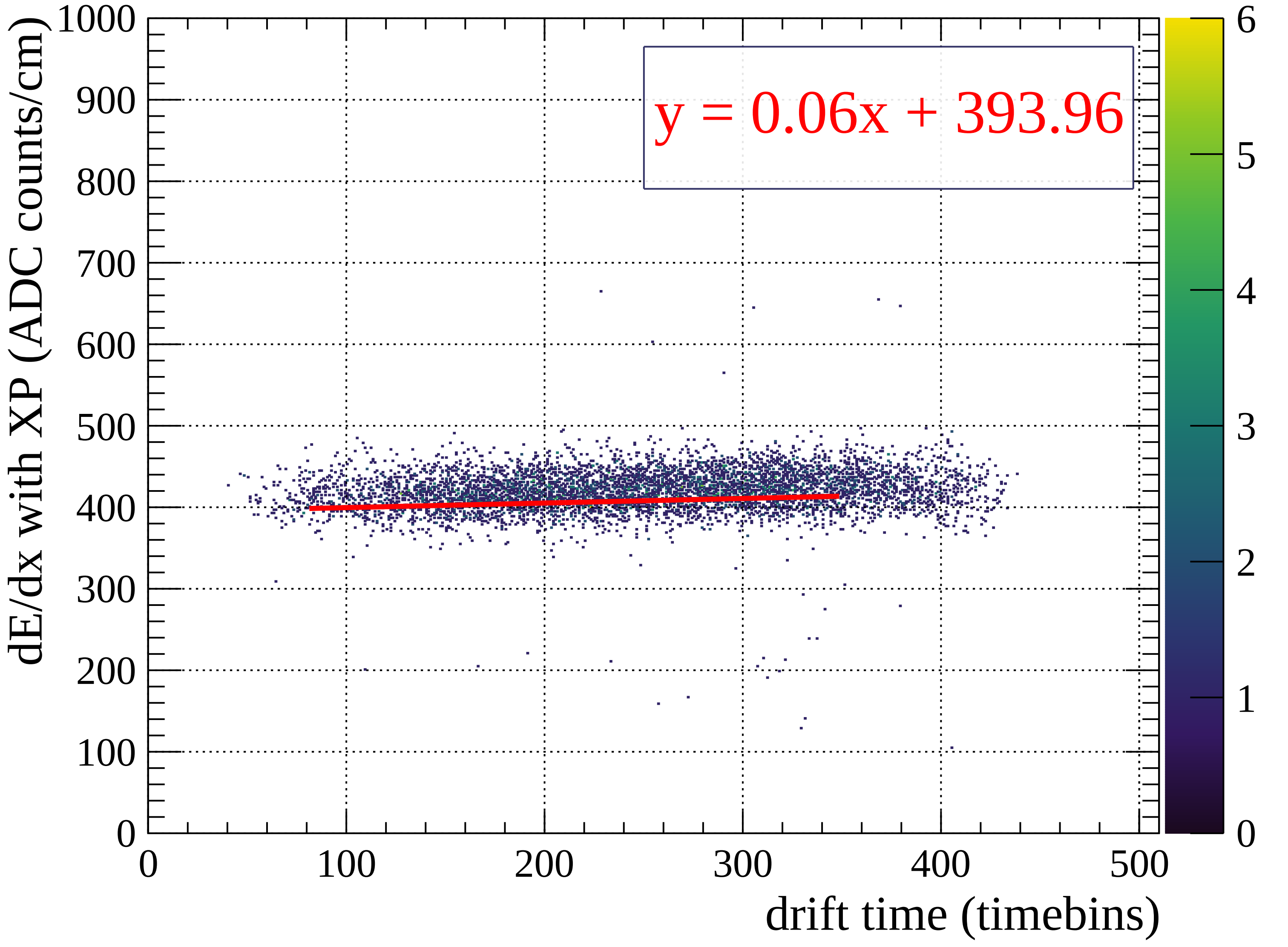
dE/dx resolution (%)
30
25
20
15
10
5
0
100
200
300
400
500
Drift time (timebins)
WF
XP



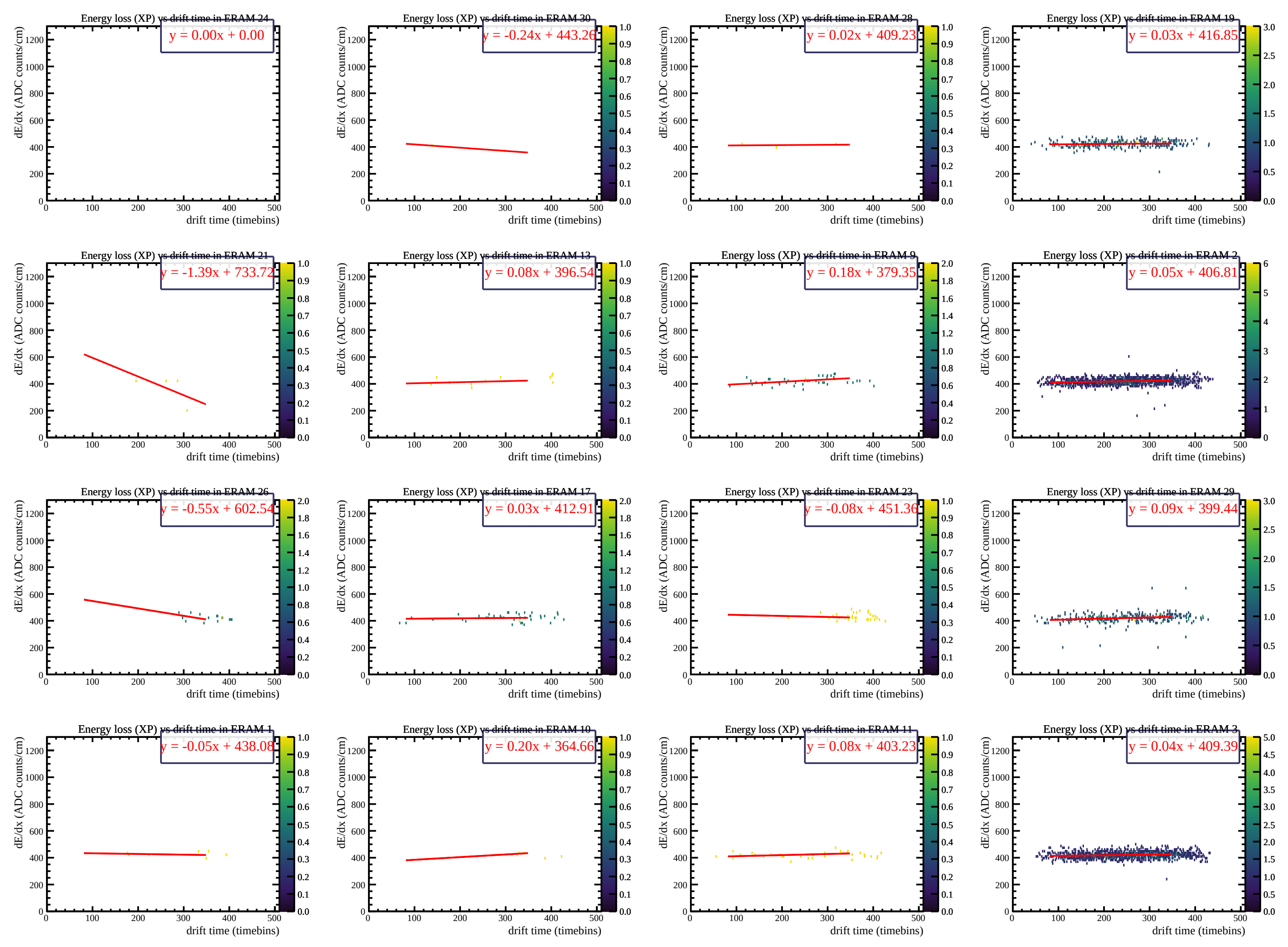
WF
XP
Mean (ADC counts/cm)
Drift time (timebins)
600
550
500
450
400
350
300
0
100
200
300
400
500

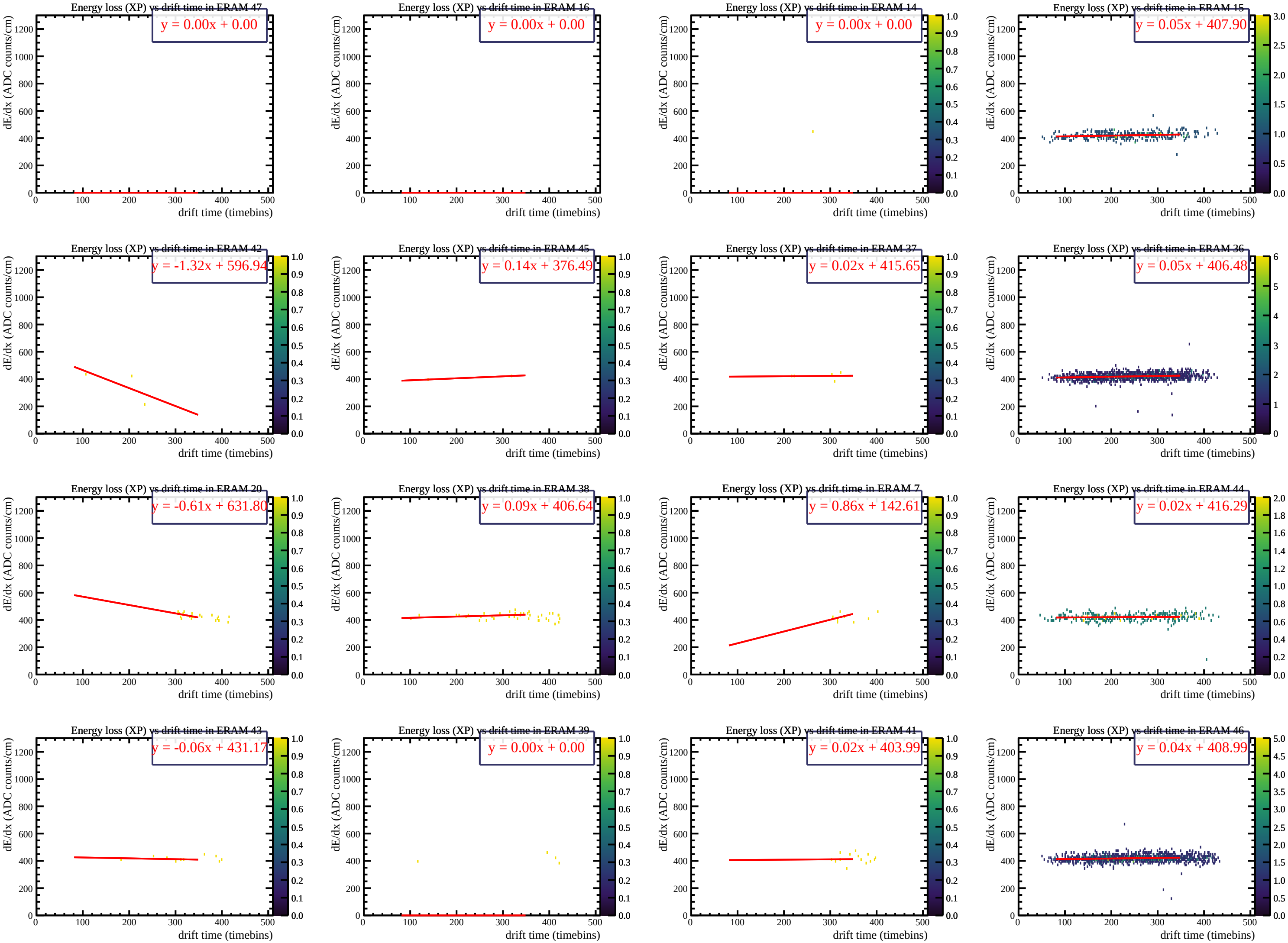


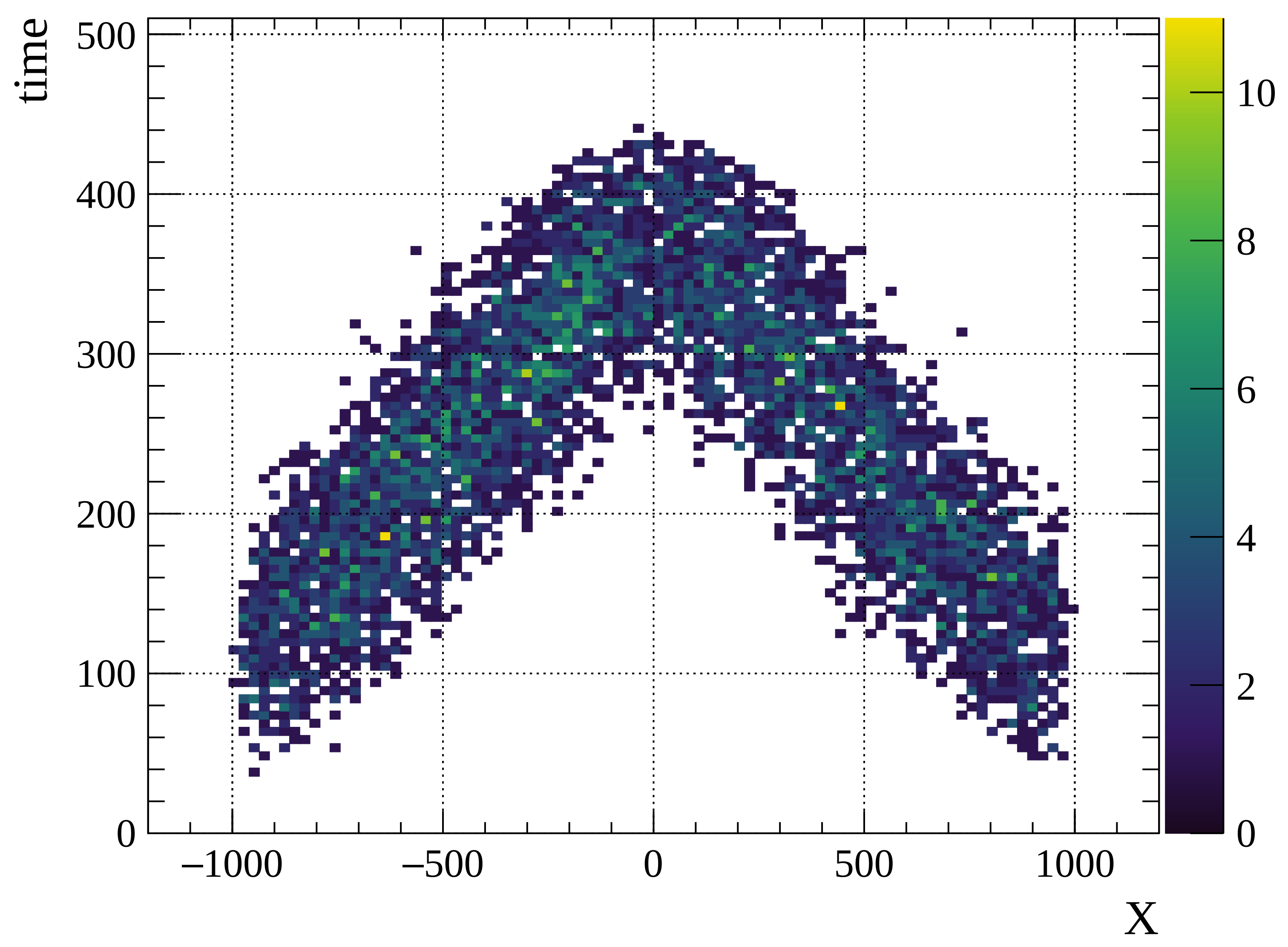
y = 0.09x + 384.44
dE/dx with WF (ADC counts/cm)
drift time (timebins)
0
100
200
300
400
500
600
700
800
900
1000
0
100
200
300
400
500
0
1
2
3
4
5
6

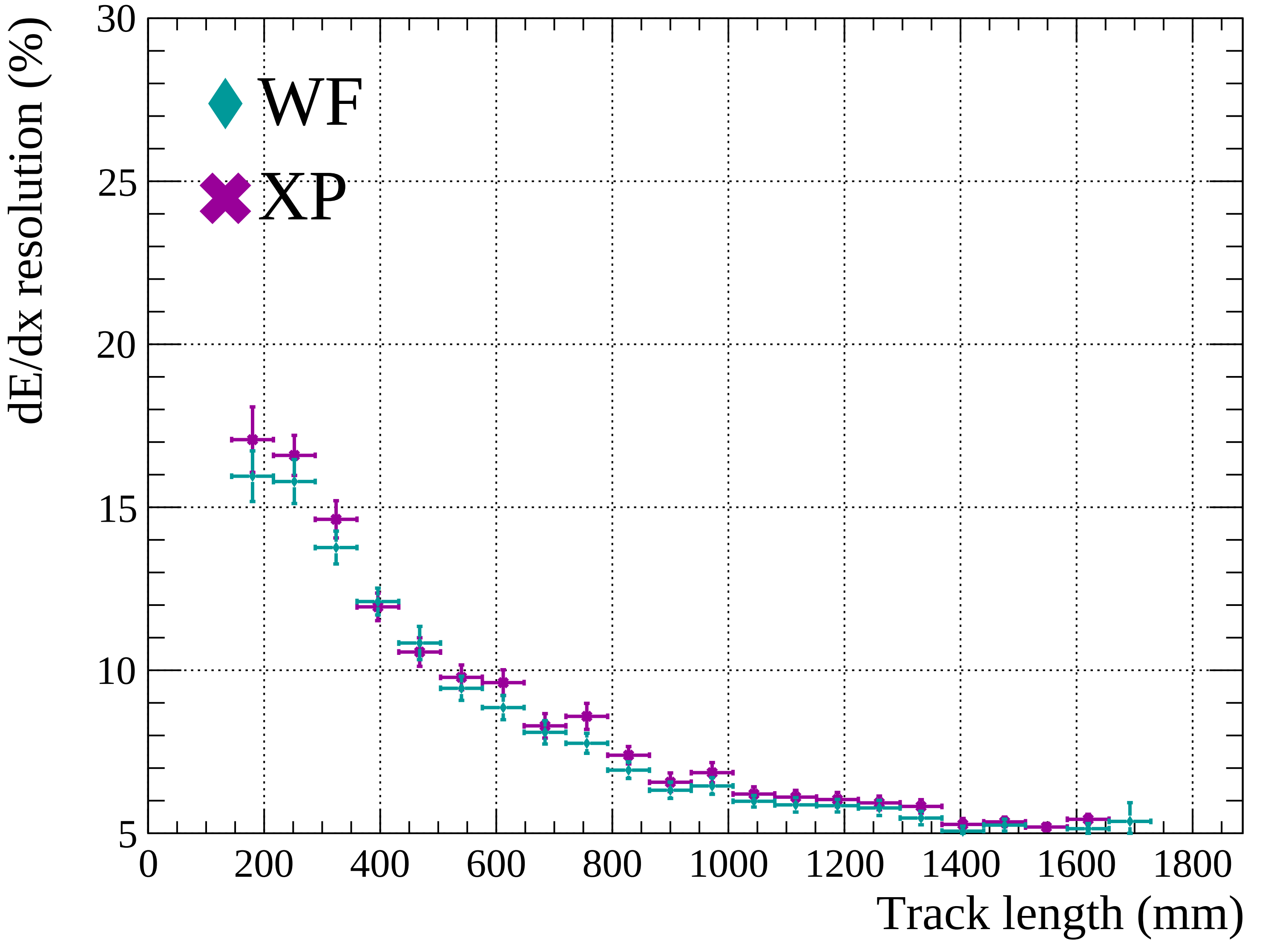


y = 0.06x + 393.96
dE/dx with XP (ADC counts/cm)
drift time (timebins)
0
100
200
300
400
500
600
700
800
900
1000
0
100
200
300
400
500
0
1
2
3
4
5
6

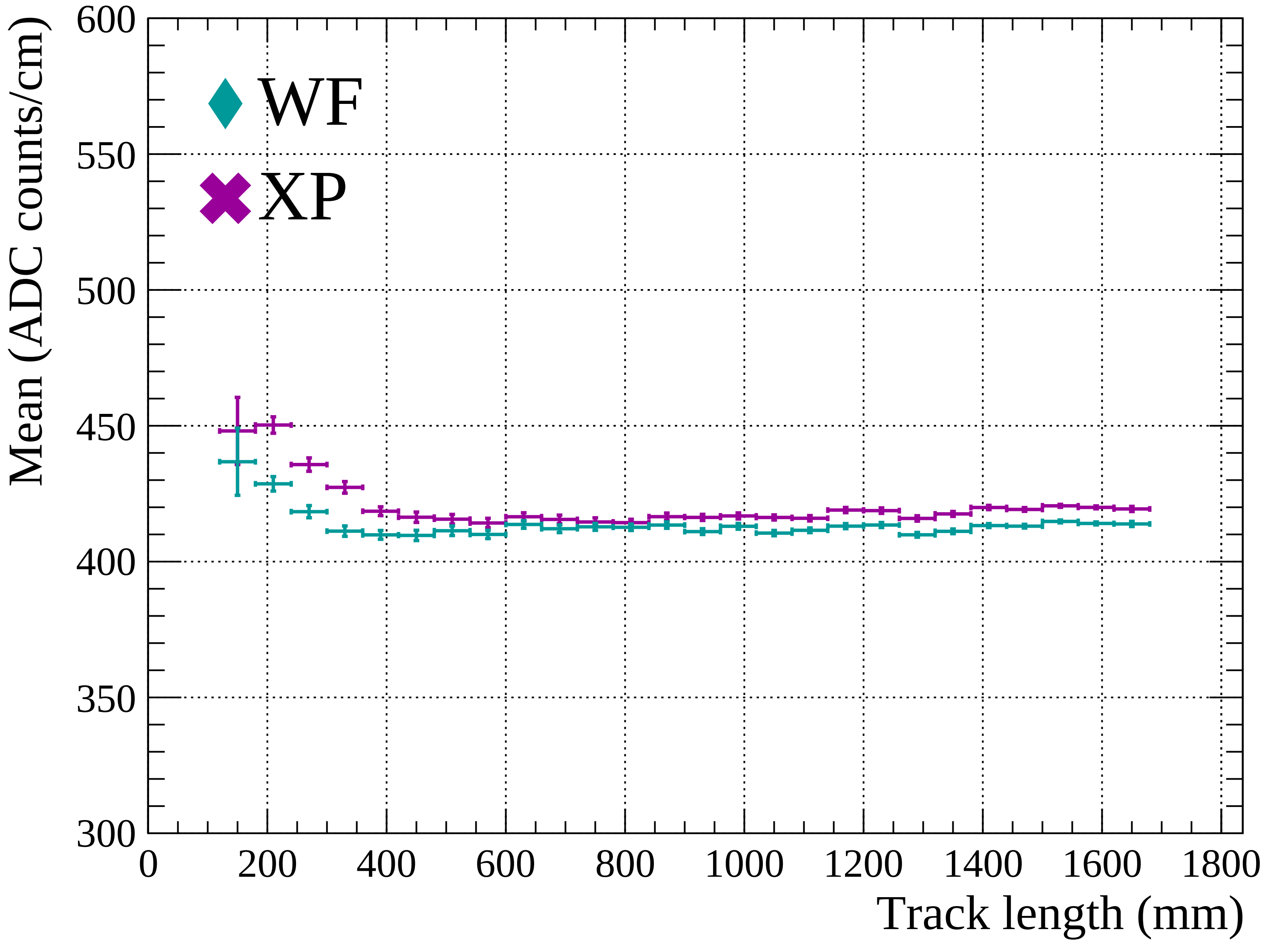

Energy loss (XP) vs drift time in ERAM 24
y = 0.00x + 0.00
Energy loss (XP) vs drift time in ERAM 30
y = -0.24x + 443.26
Energy loss (XP) vs drift time in ERAM 28
y = 0.02x + 409.23
Energy loss (XP) vs drift time in ERAM 19
y = 0.03x + 416.85
Energy loss (XP) vs drift time in ERAM 21
y = -1.39x + 733.72
Energy loss (XP) vs drift time in ERAM 13
y = 0.08x + 396.54
Energy loss (XP) vs drift time in ERAM 9
y = 0.18x + 379.35
Energy loss (XP) vs drift time in ERAM 2
y = 0.05x + 406.81
Energy loss (XP) vs drift time in ERAM 26
y = -0.55x + 602.54
Energy loss (XP) vs drift time in ERAM 17
y = 0.03x + 412.91
Energy loss (XP) vs drift time in ERAM 23
y = -0.08x + 451.36
Energy loss (XP) vs drift time in ERAM 29
y = 0.09x + 399.44
Energy loss (XP) vs drift time in ERAM 1
y = -0.05x + 438.08
Energy loss (XP) vs drift time in ERAM 10
y = 0.20x + 364.66
Energy loss (XP) vs drift time in ERAM 11
y = 0.08x + 403.23
Energy loss (XP) vs drift time in ERAM 3
y = 0.04x + 409.39
dE/dx (ADC counts/cm)
drift time (timebins)


Energy loss (XP) vs drift time in ERAM 47
y = 0.00x + 0.00
Energy loss (XP) vs drift time in ERAM 16
y = 0.00x + 0.00
Energy loss (XP) vs drift time in ERAM 14
y = 0.00x + 0.00
Energy loss (XP) vs drift time in ERAM 15
y = 0.05x + 407.90
Energy loss (XP) vs drift time in ERAM 42
y = -1.32x + 596.94
Energy loss (XP) vs drift time in ERAM 45
y = 0.14x + 376.49
Energy loss (XP) vs drift time in ERAM 37
y = 0.02x + 415.65
Energy loss (XP) vs drift time in ERAM 36
y = 0.05x + 406.48
Energy loss (XP) vs drift time in ERAM 20
y = -0.61x + 631.80
Energy loss (XP) vs drift time in ERAM 38
y = 0.09x + 406.64
Energy loss (XP) vs drift time in ERAM 7
y = 0.86x + 142.61
Energy loss (XP) vs drift time in ERAM 44
y = 0.02x + 416.29
Energy loss (XP) vs drift time in ERAM 43
y = -0.06x + 431.17
Energy loss (XP) vs drift time in ERAM 39
y = 0.00x + 0.00
Energy loss (XP) vs drift time in ERAM 41
y = 0.02x + 403.99
Energy loss (XP) vs drift time in ERAM 46
y = 0.04x + 408.99
dE/dx (ADC counts/cm)
drift time (timebins)

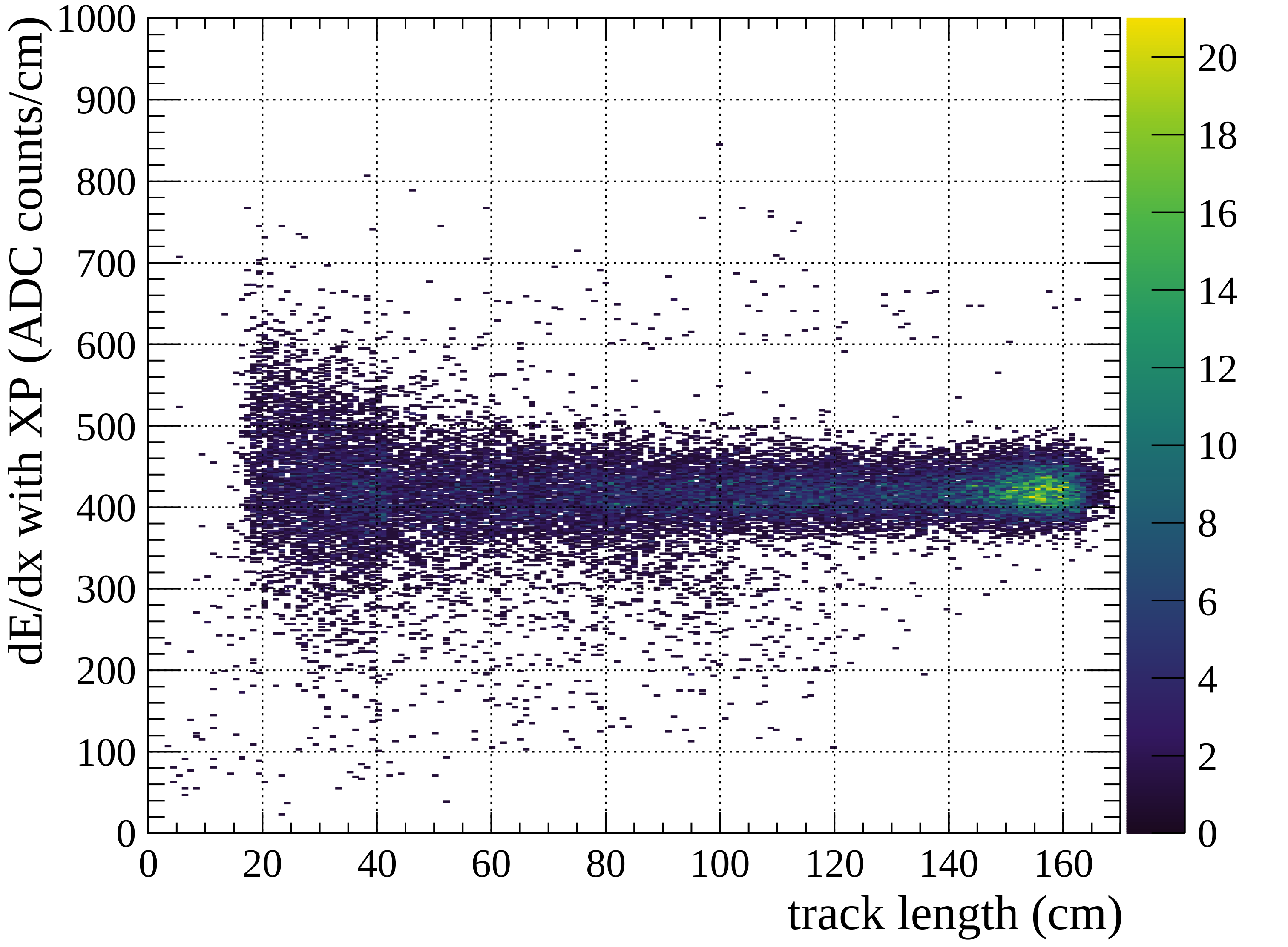

time
500
400
300
200
100
0
−1000
−500
0
500
1000
X
10
8
6
4
2
0

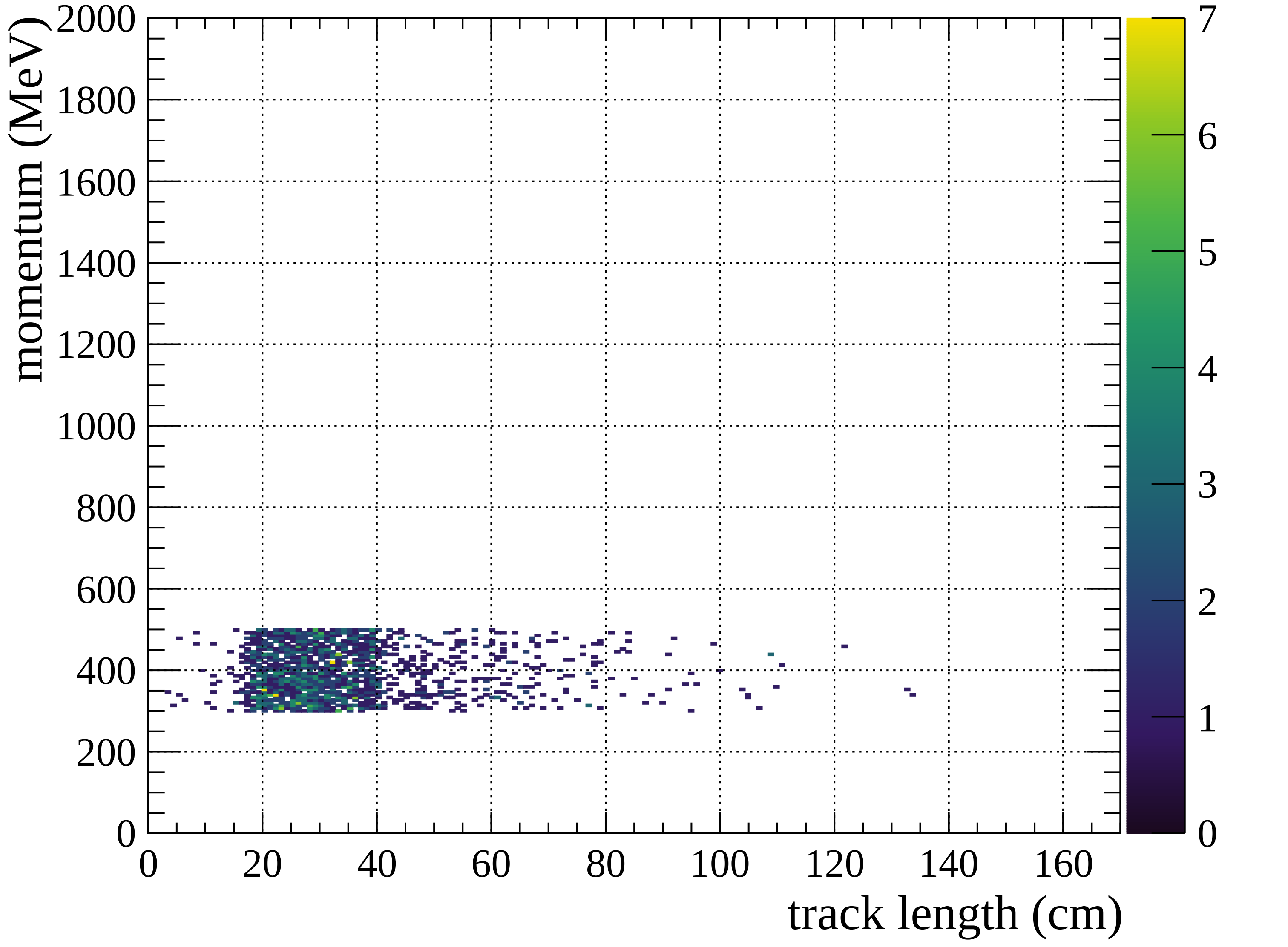


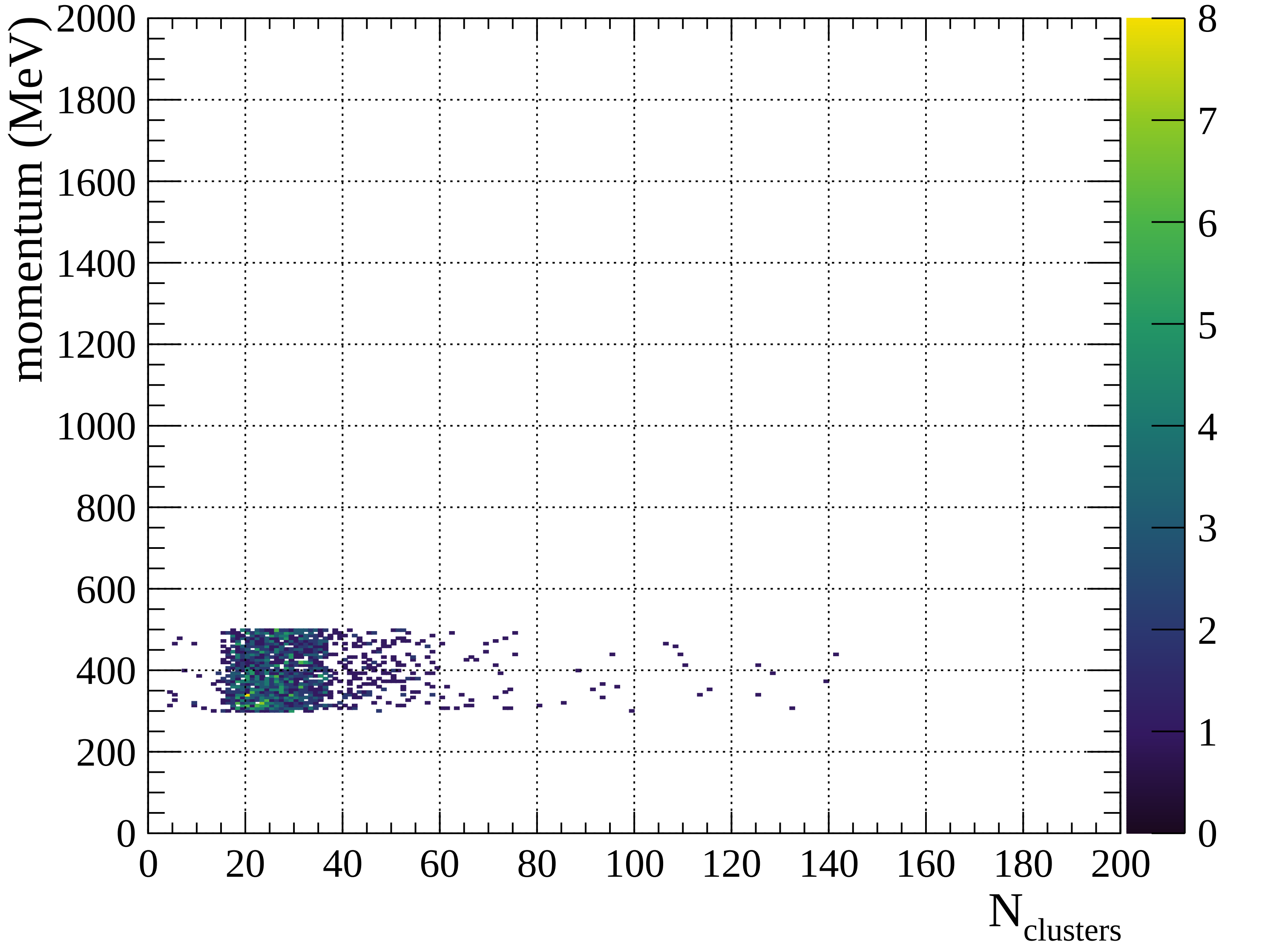
dE/dx resolution (%)
WF
XP
Track length (mm)
0
200
400
600
800
1000
1200
1400
1600
1800
5
10
15
20
25
30

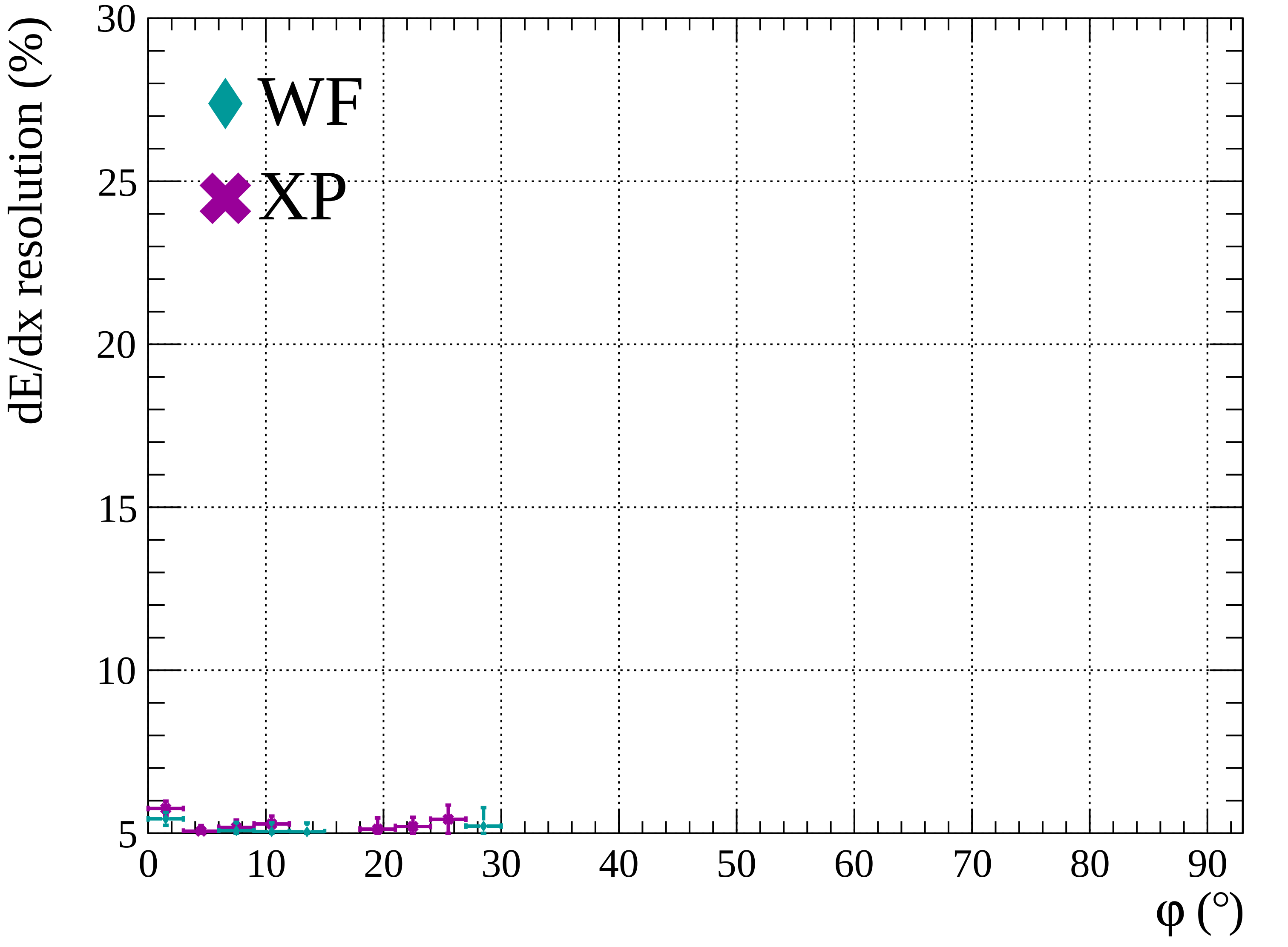


WF
XP
Mean (ADC counts/cm)
600
550
500
450
400
350
300
0
200
400
600
800
1000
1200
1400
1600
1800
Track length (mm)

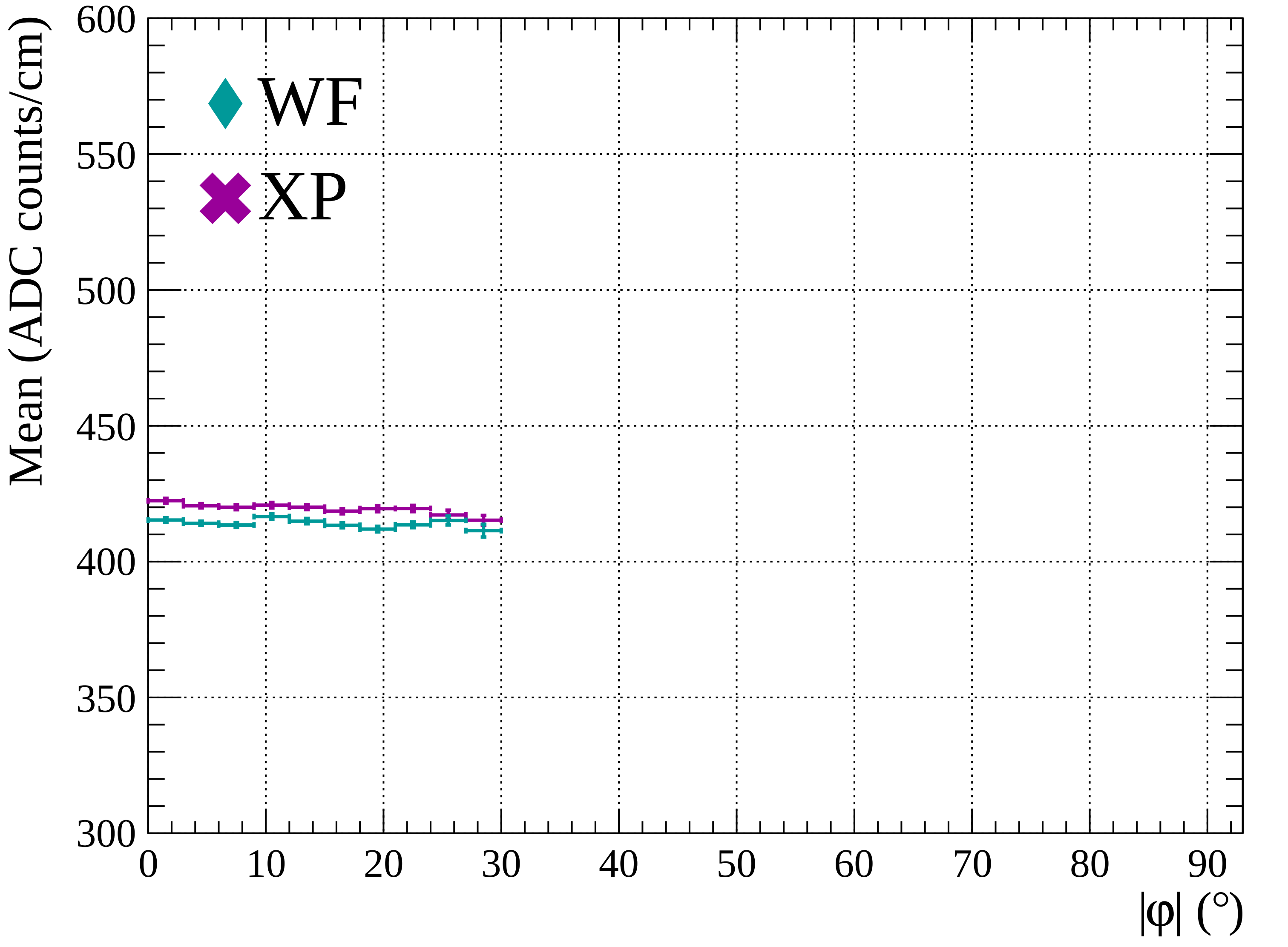

dE/dx with XP (ADC counts/cm)
track length (cm)
0
100
200
300
400
500
600
700
800
900
1000
20
40
60
80
100
120
140
160
2
4
6
8
10
12
14
16
18


momentum (MeV)
2000
1800
1600
1400
1200
1000
800
600
400
200
0
0
20
40
60
80
100
120
140
160
track length (cm)
7
6
5
4
3
2
1
0

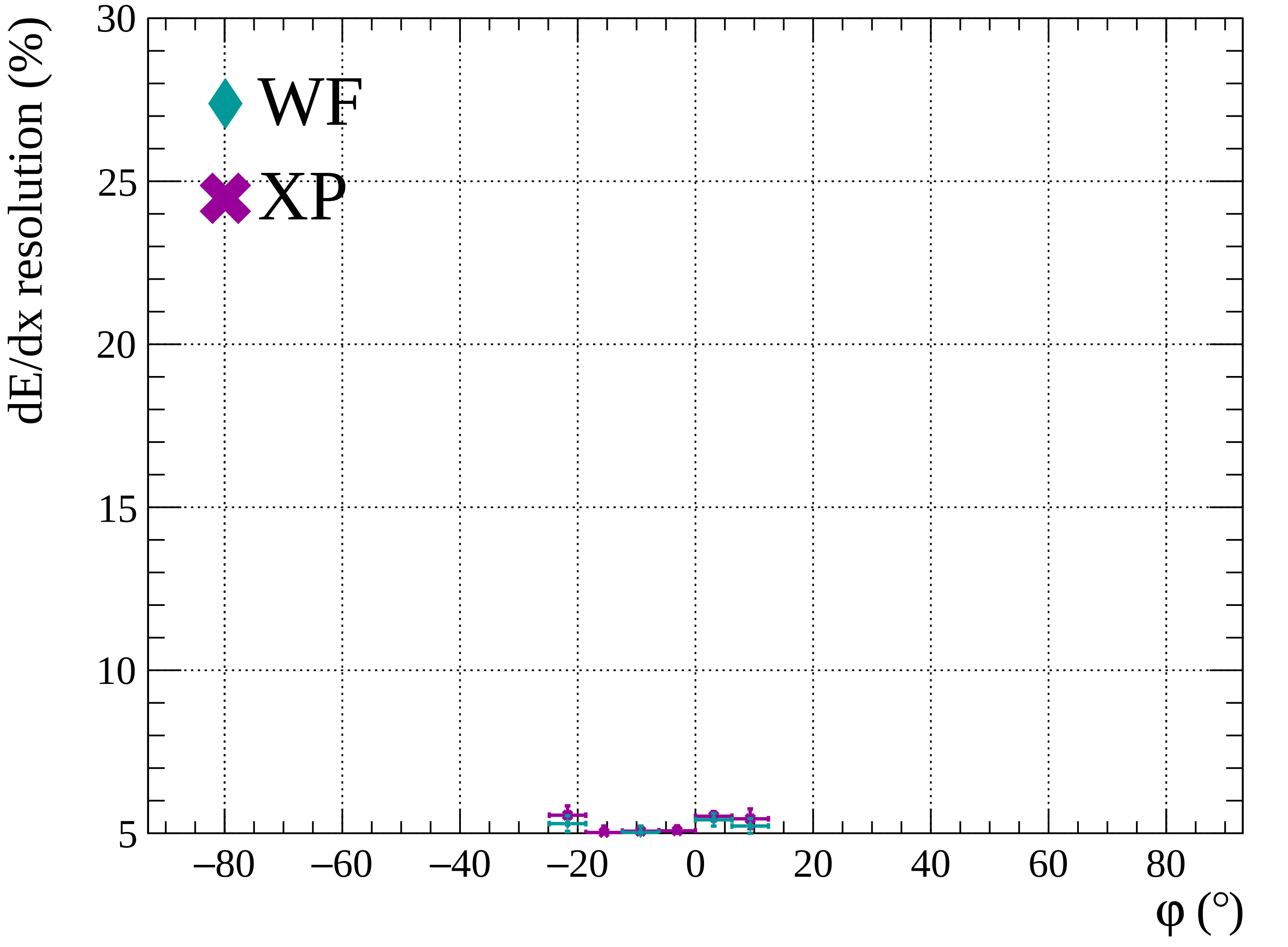

momentum (MeV)
2000
1800
1600
1400
1200
1000
800
600
400
200
0
0
20
40
60
80
100
120
140
160
180
200
N clusters
8
7
6
5
4
3
2
1
0



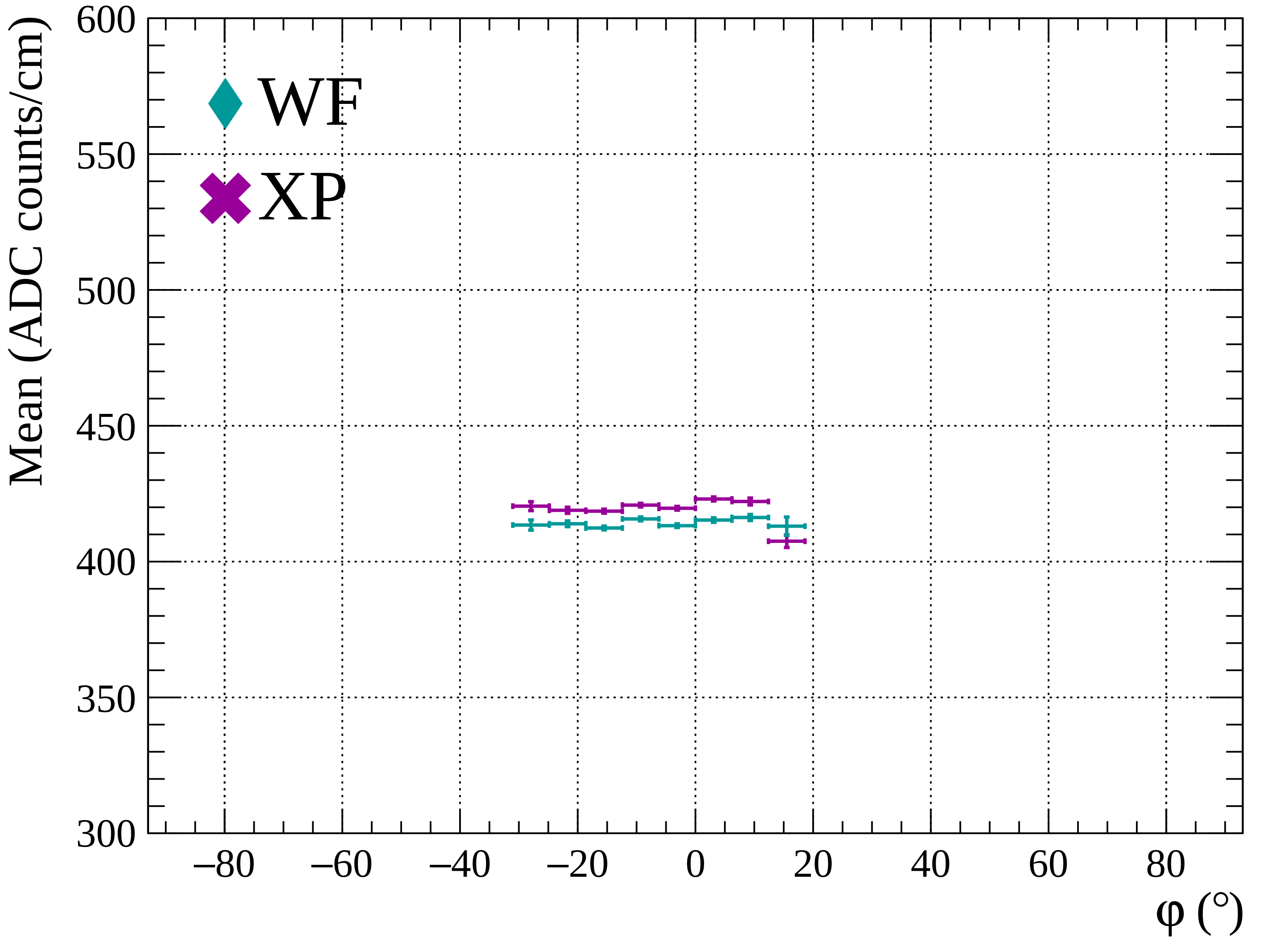
dE/dx resolution (%)
30
25
20
15
10
5
0
10
20
30
40
50
60
70
80
90
φ (°)
WF
XP

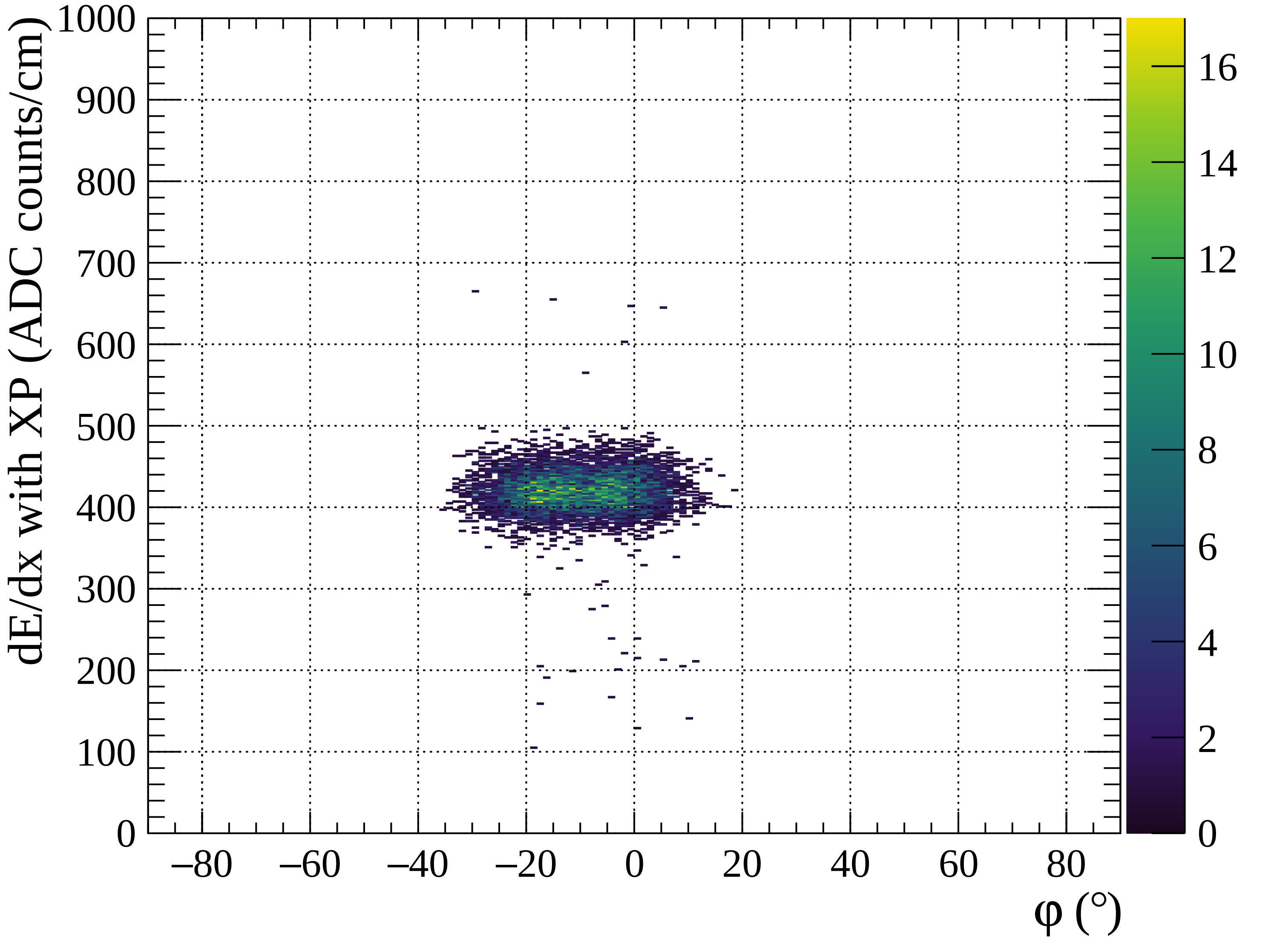


WF
XP
Mean (ADC counts/cm)
|φ| (°)
600
550
500
450
400
350
300
0
10
20
30
40
50
60
70
80
90

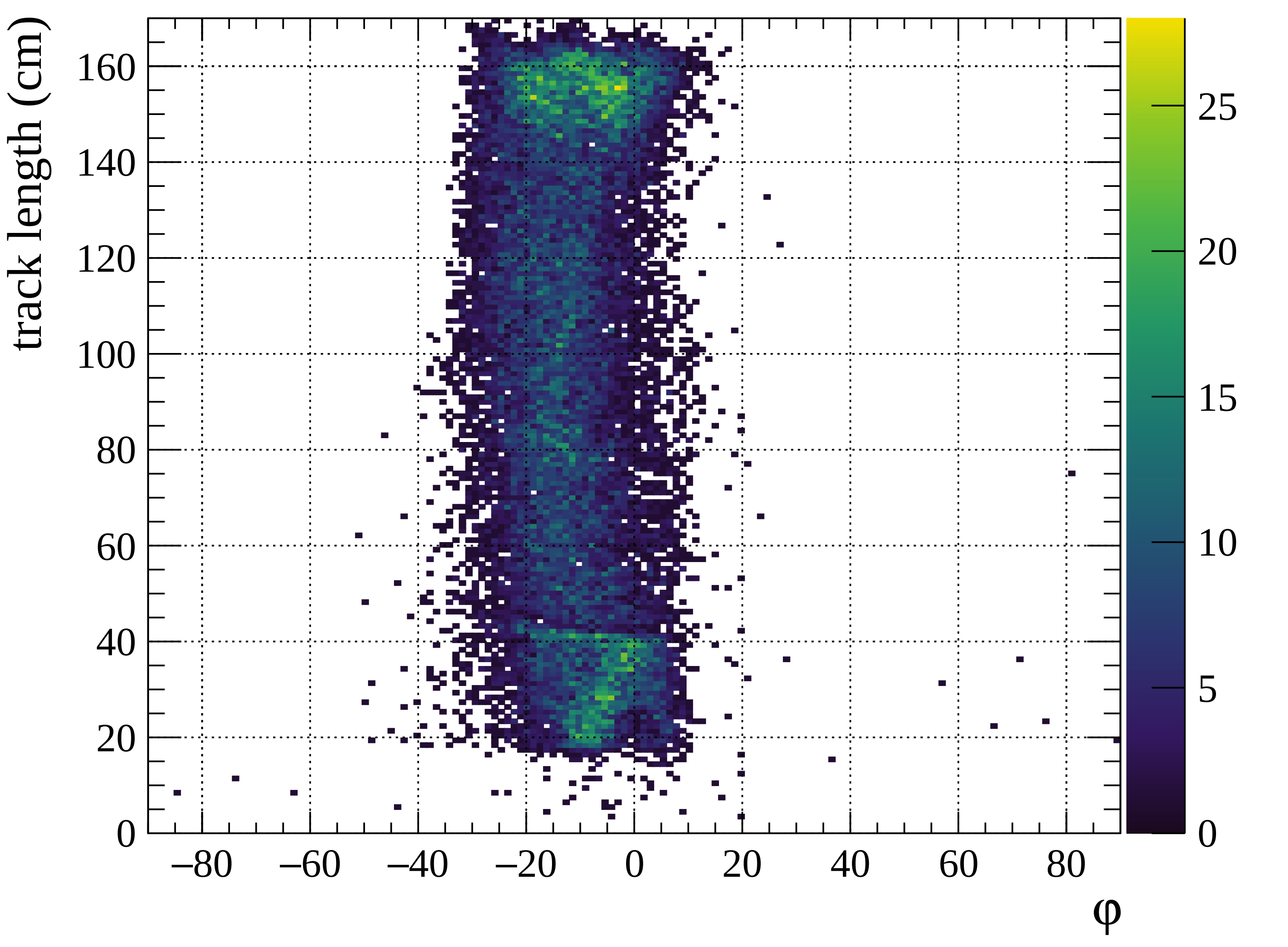


WF
XP
dE/dx resolution (%)
φ (°)
30
25
20
15
10
5
−80
−60
−40
−20
0
20
40
60
80

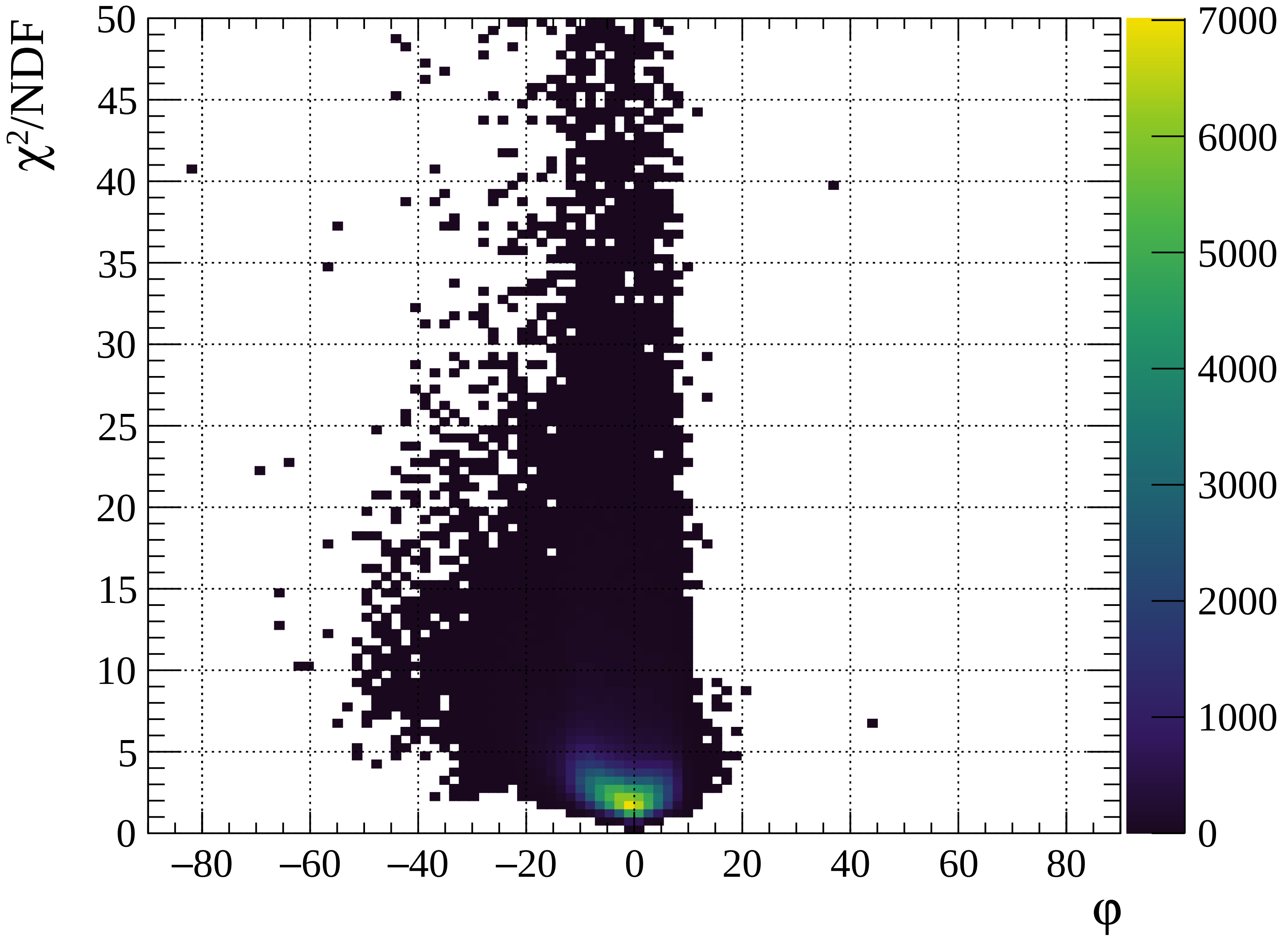

WF
XP
Mean (ADC counts/cm)
600
550
500
450
400
350
300
−80
−60
−40
−20
0
20
40
60
80
φ (°)



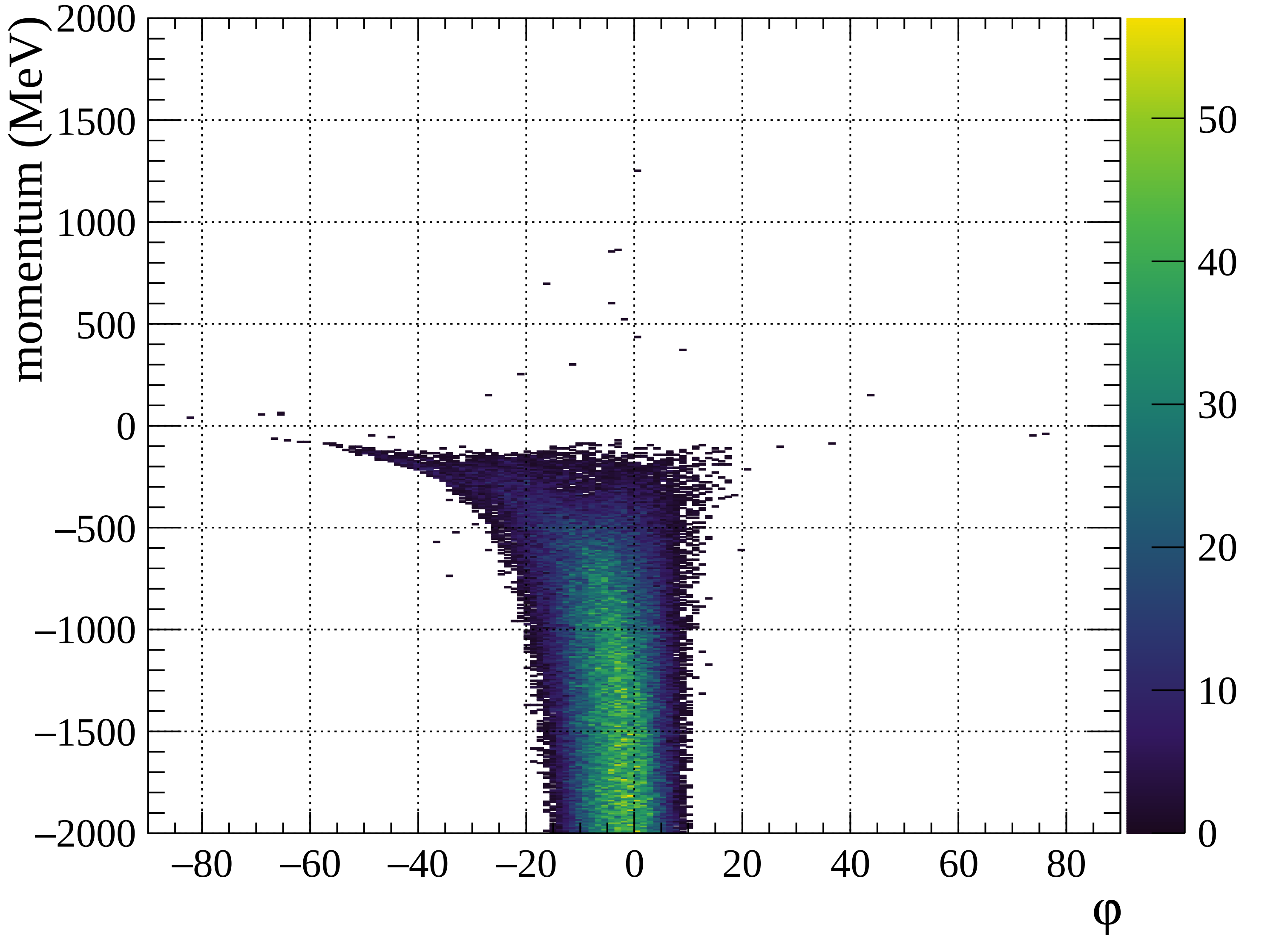
dE/dx with XP (ADC counts/cm)
φ (°)
1000
900
800
700
600
500
400
300
200
100
0
−80
−60
−40
−20
0
20
40
60
80
16
14
12
10
8
6
4
2
0

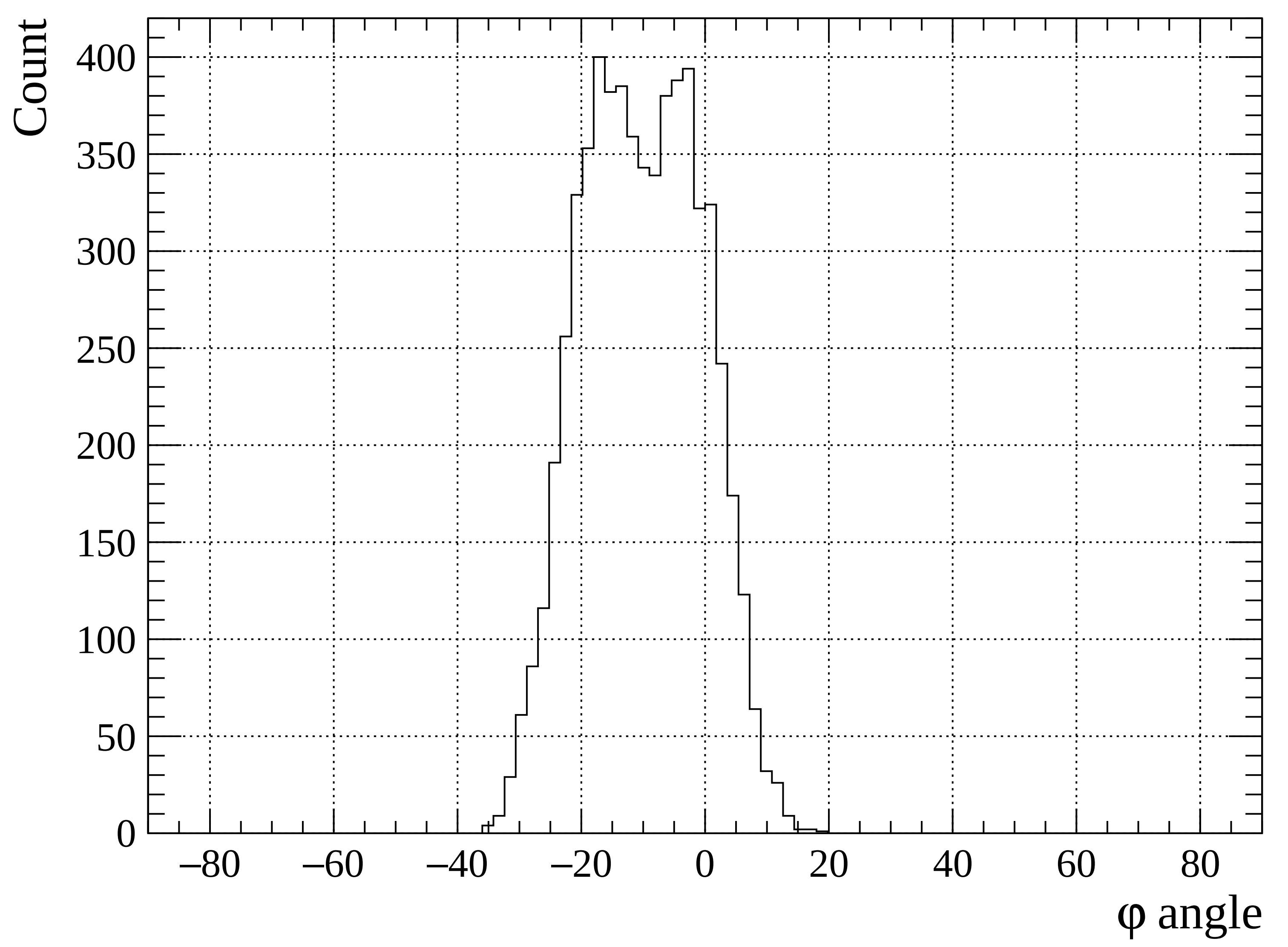


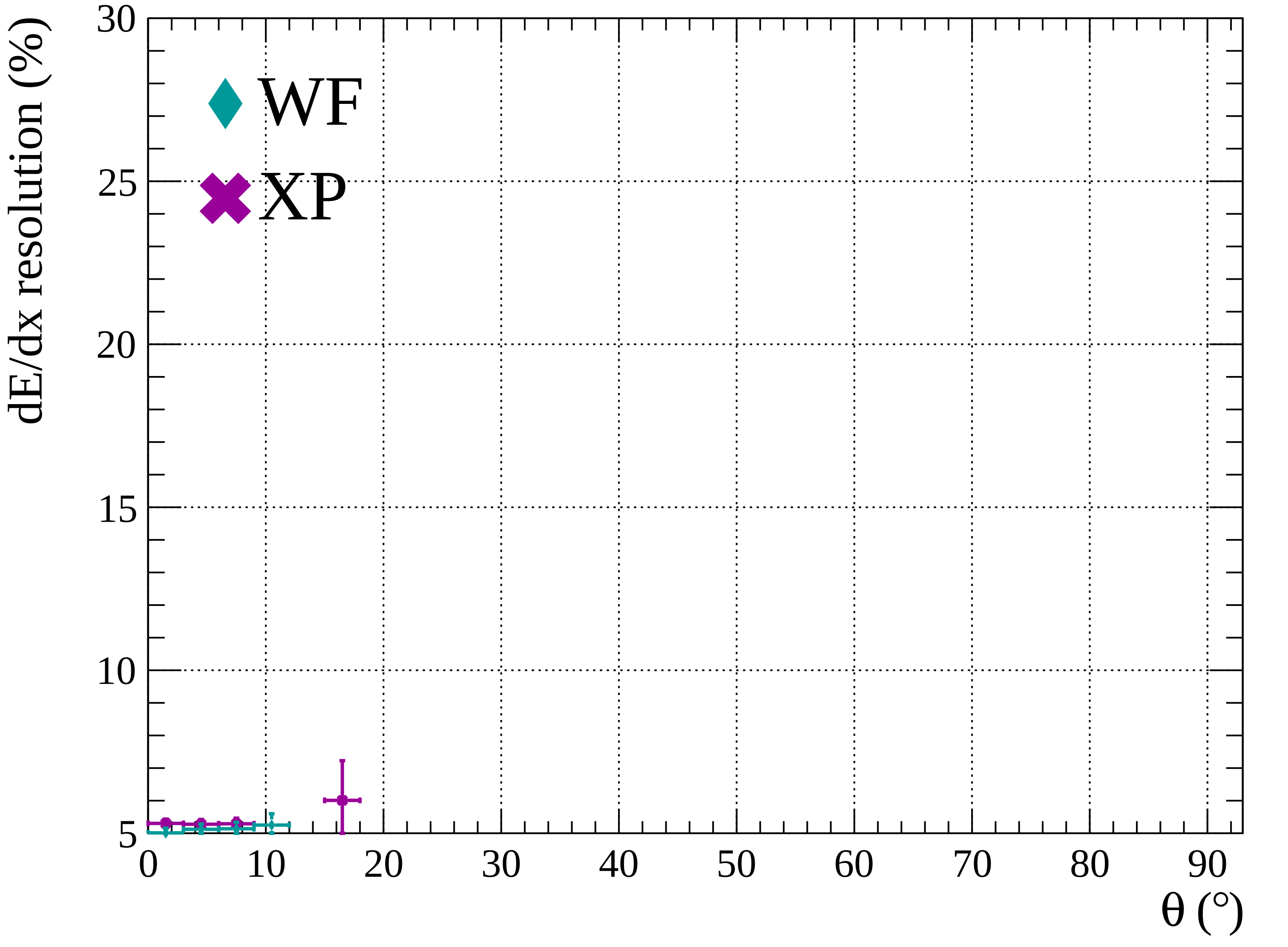
track length (cm)
160
140
120
100
80
60
40
20
0
−80
−60
−40
−20
0
20
40
60
80
φ
25
20
15
10
5
0



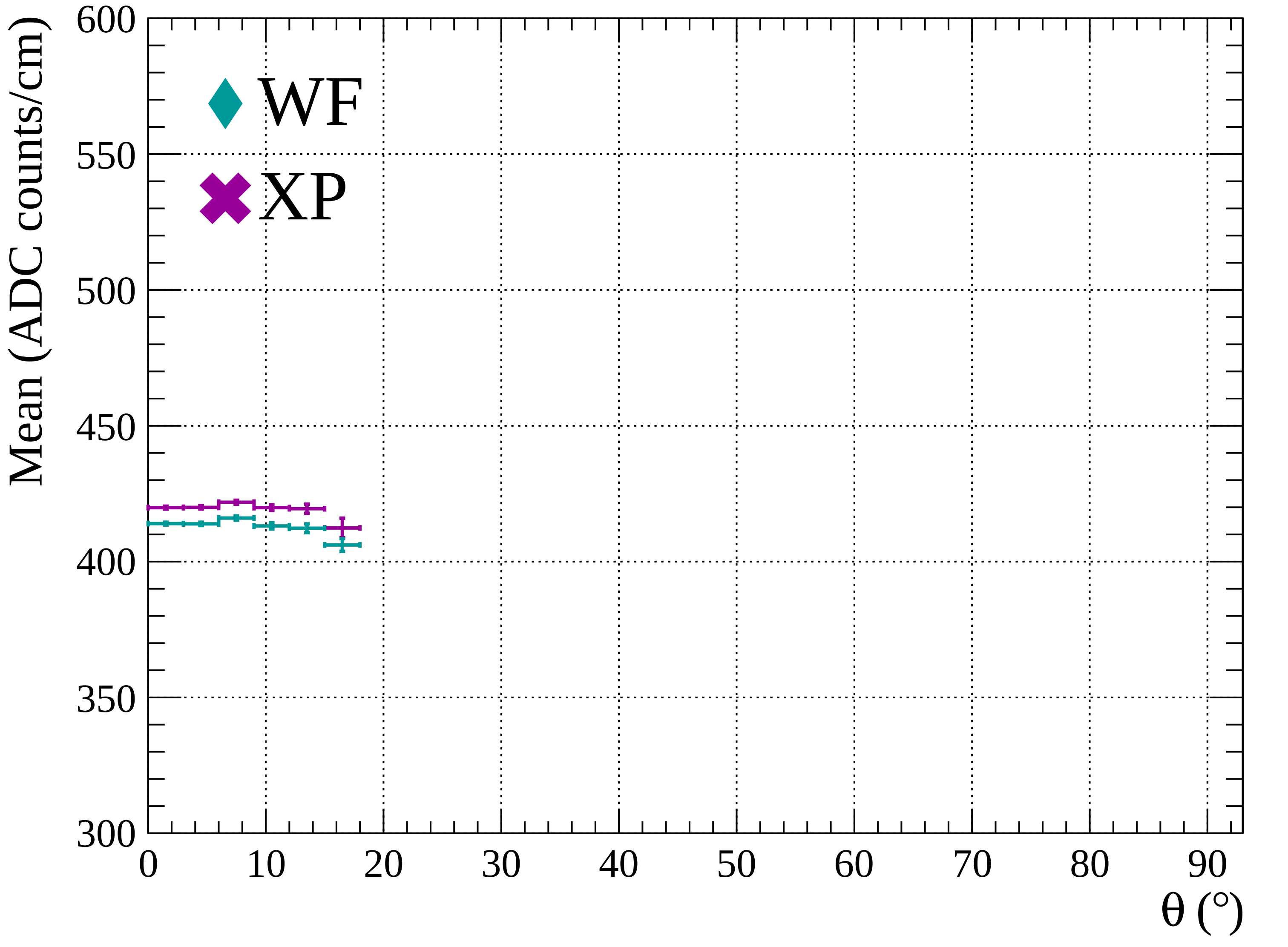
χ^2/NDF
50
45
40
35
30
25
20
15
10
5
0
−80
−60
−40
−20
0
20
40
60
80
φ
7000
6000
5000
4000
3000
2000
1000
0


momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−80
−60
−40
−20
0
20
40
60
80
φ
50
40
30
20
10
0

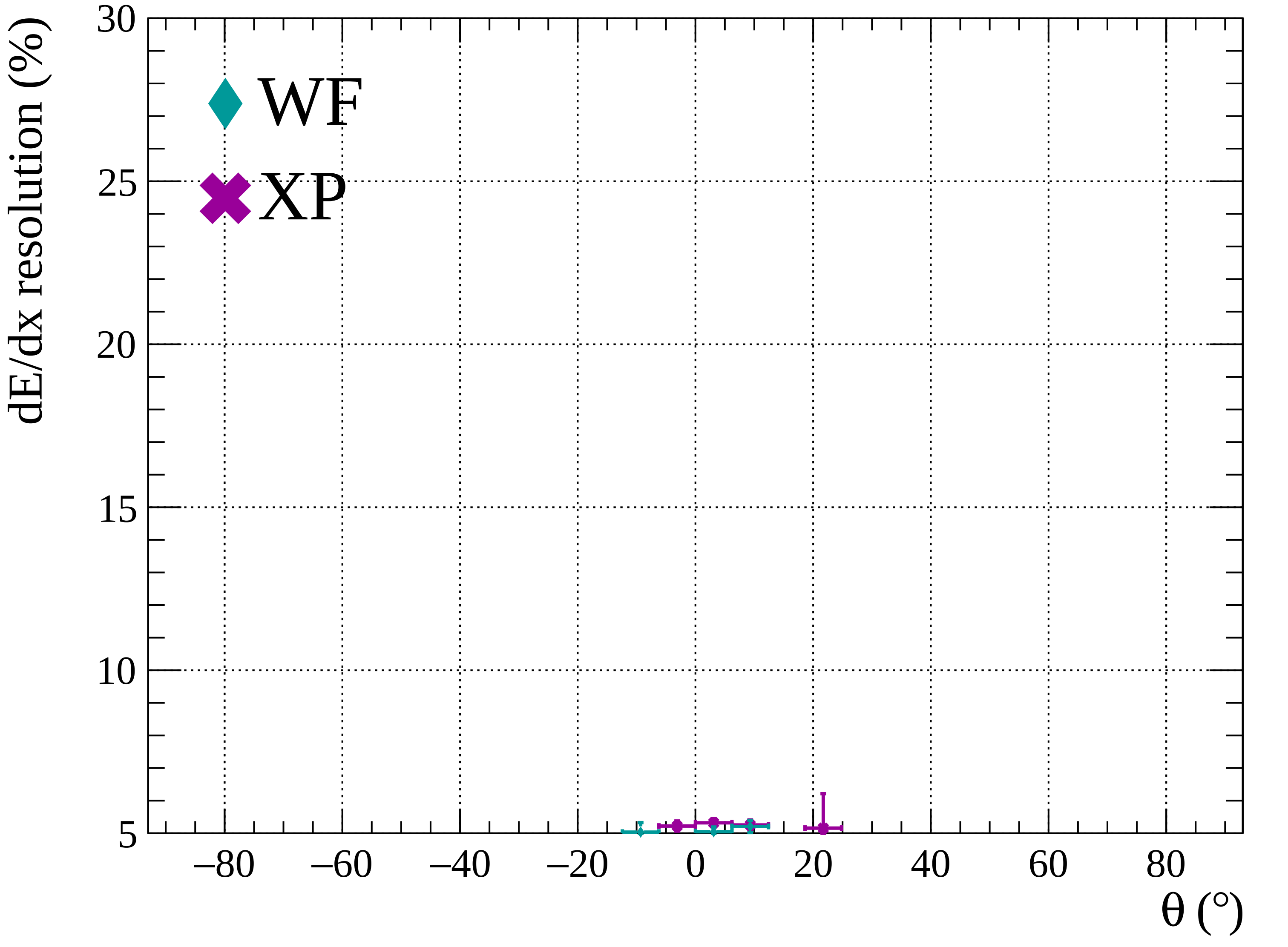

Count
400
350
300
250
200
150
100
50
0
−80
−60
−40
−20
0
20
40
60
80
φ angle

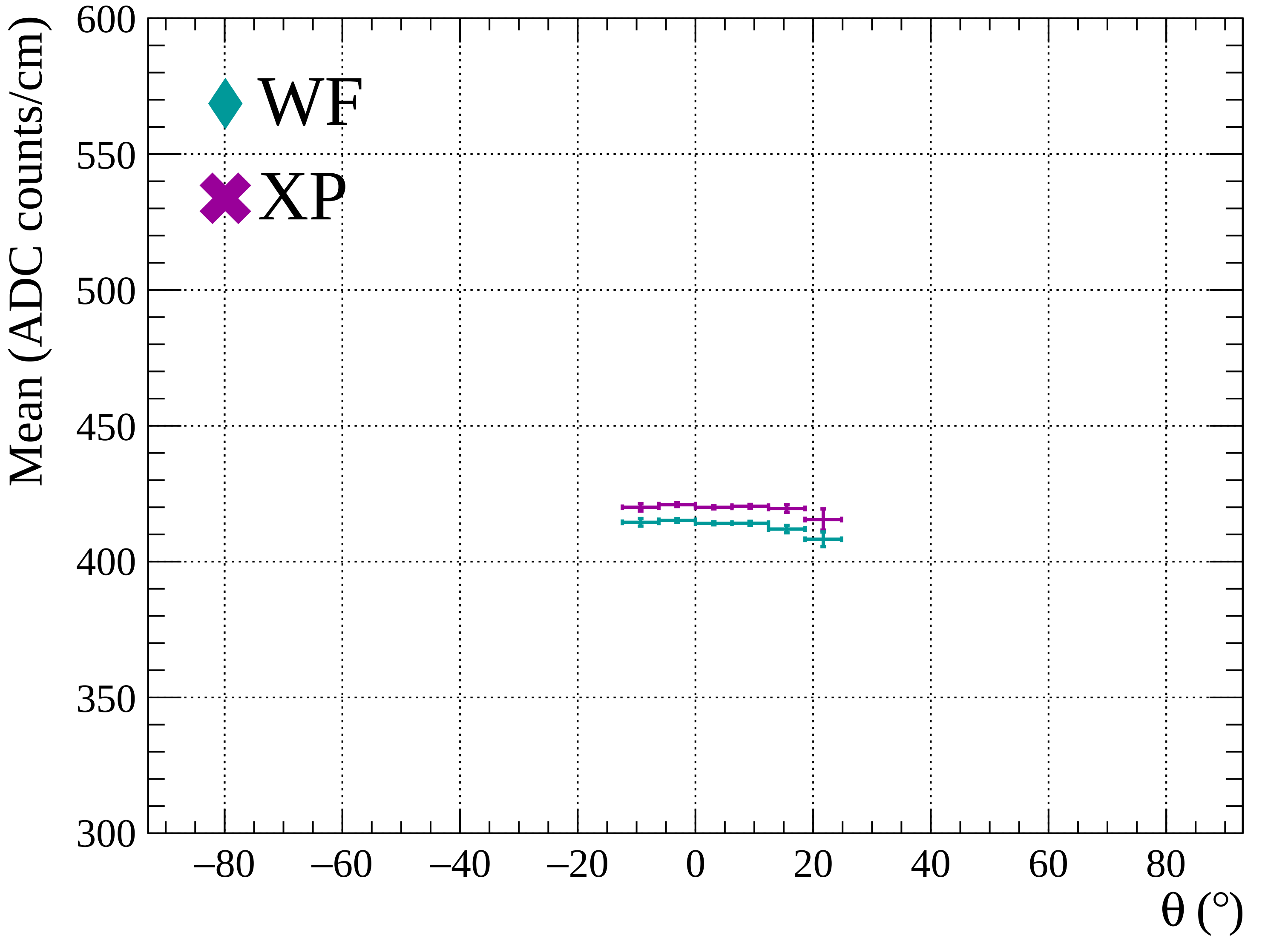


WF
XP
dE/dx resolution (%)
θ (°)
30
25
20
15
10
5
0
10
20
30
40
50
60
70
80
90

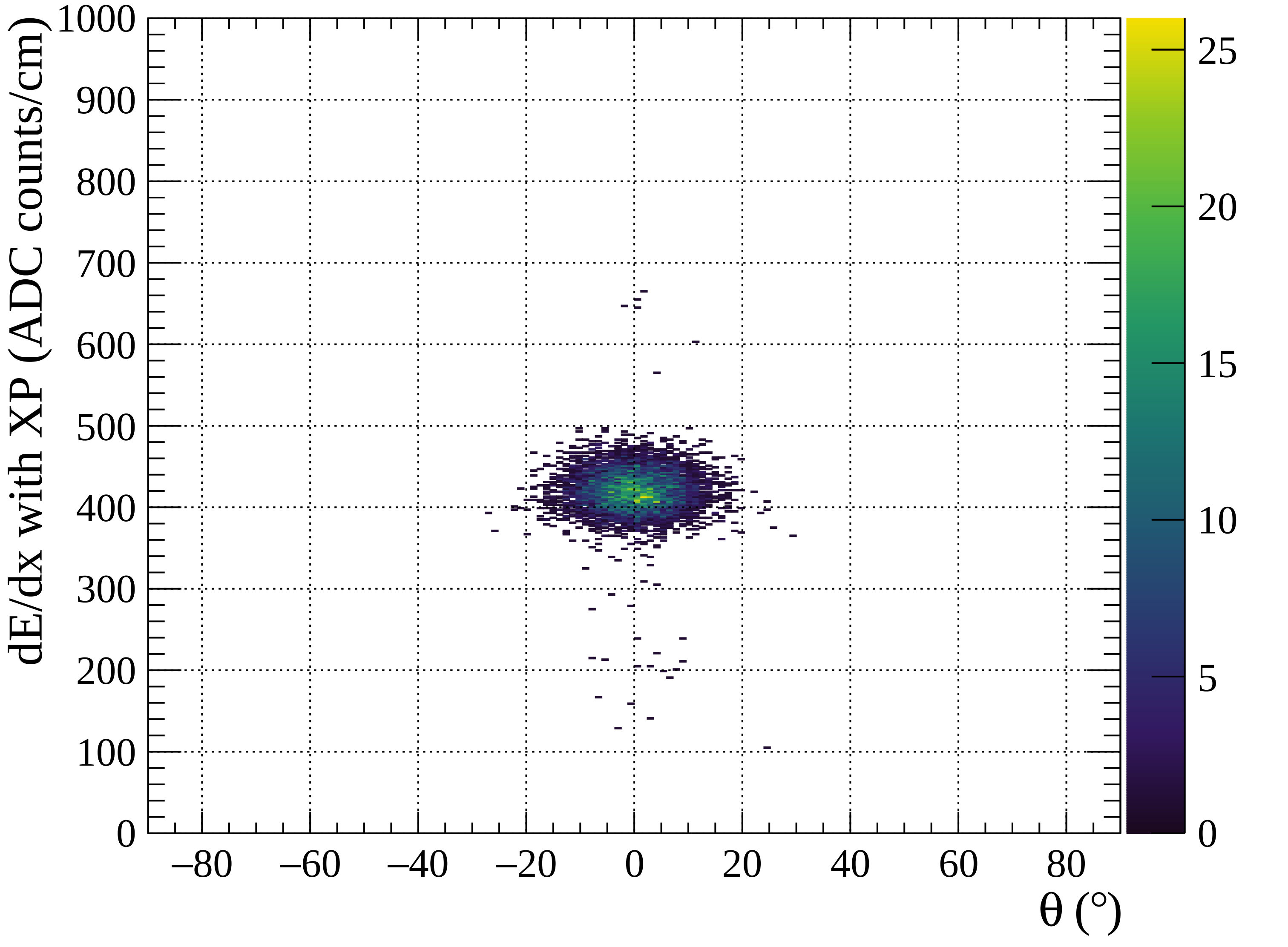


Mean (ADC counts/cm)
WF
XP
600
550
500
450
400
350
300
0
10
20
30
40
50
60
70
80
90
θ (°)

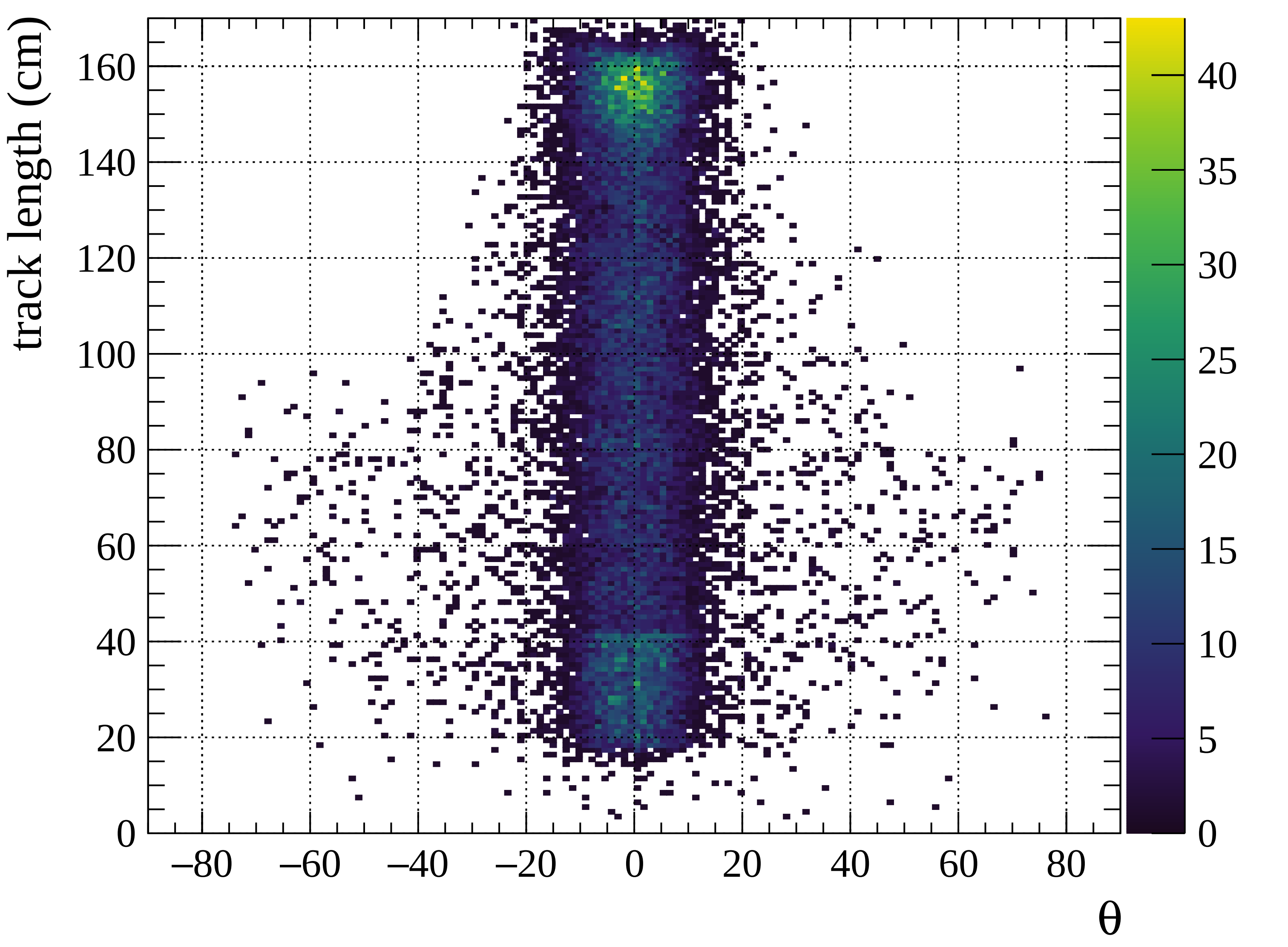

WF
XP
dE/dx resolution (%)
θ (°)
30
25
20
15
10
5
−80
−60
−40
−20
0
20
40
60
80



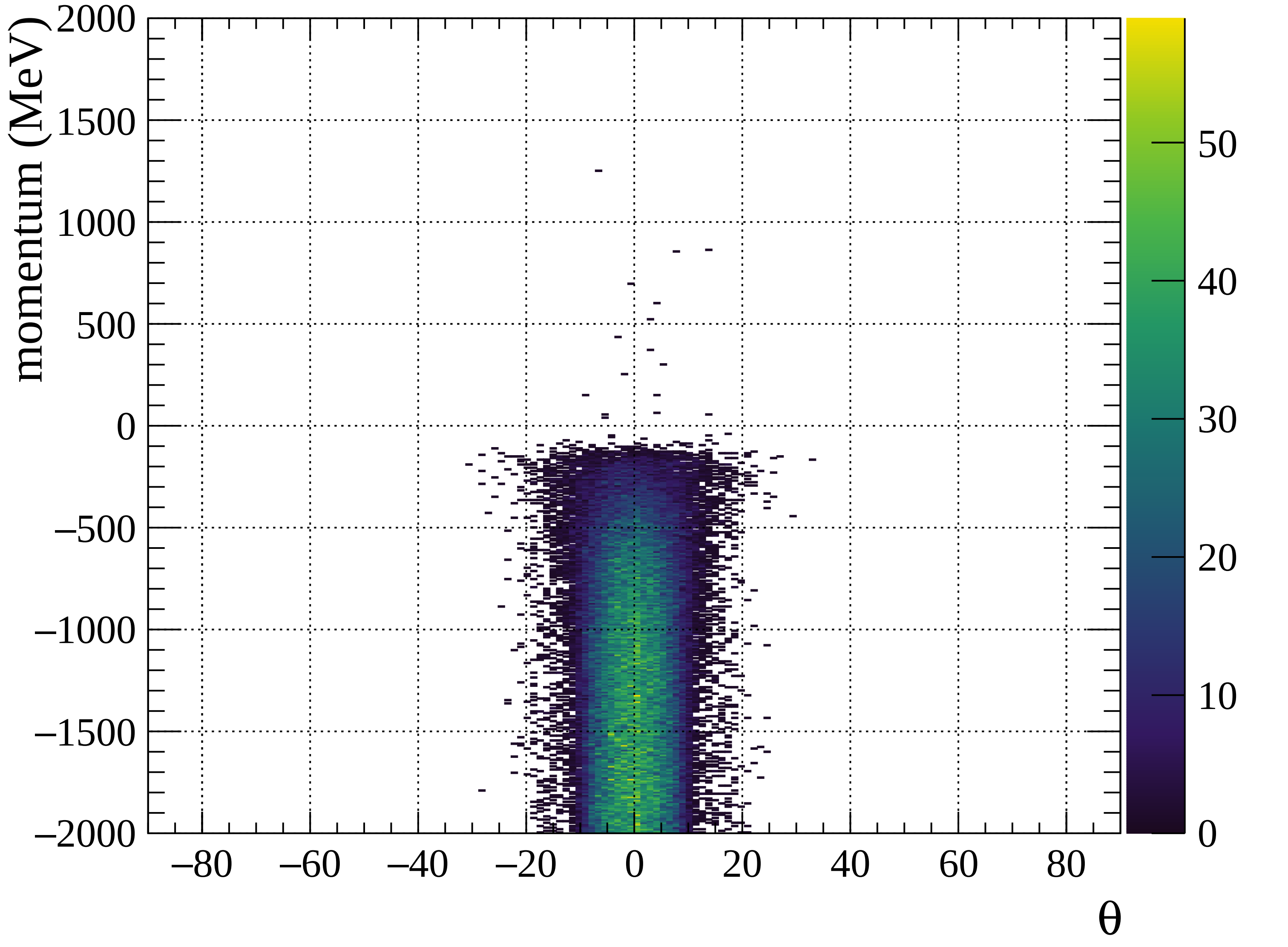
WF
XP
Mean (ADC counts/cm)
θ (°)
600
550
500
450
400
350
300
−80
−60
−40
−20
0
20
40
60
80



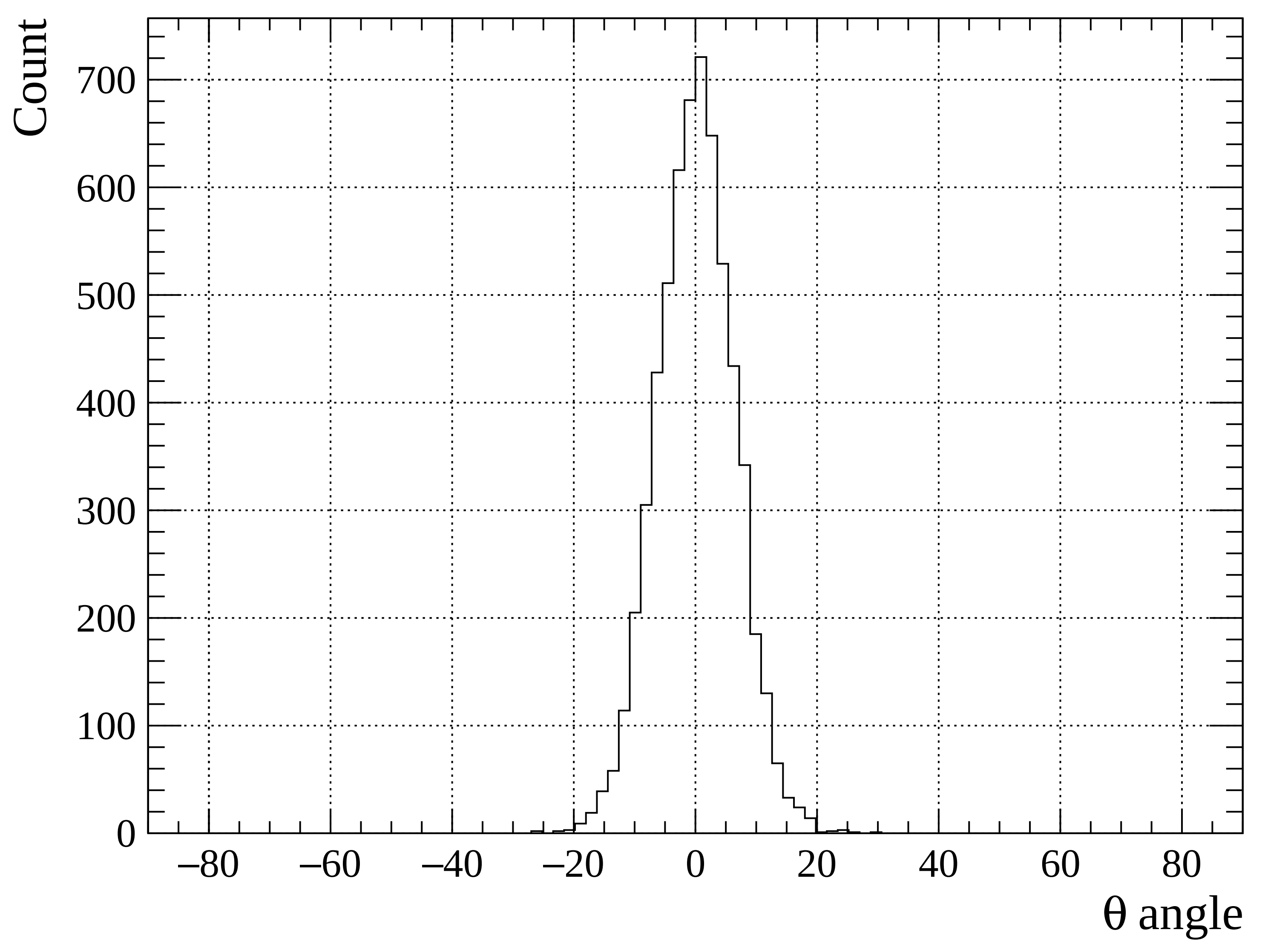
dE/dx with XP (ADC counts/cm)
θ (°)
1000
900
800
700
600
500
400
300
200
100
0
−80
−60
−40
−20
0
20
40
60
80
25
20
15
10
5
0



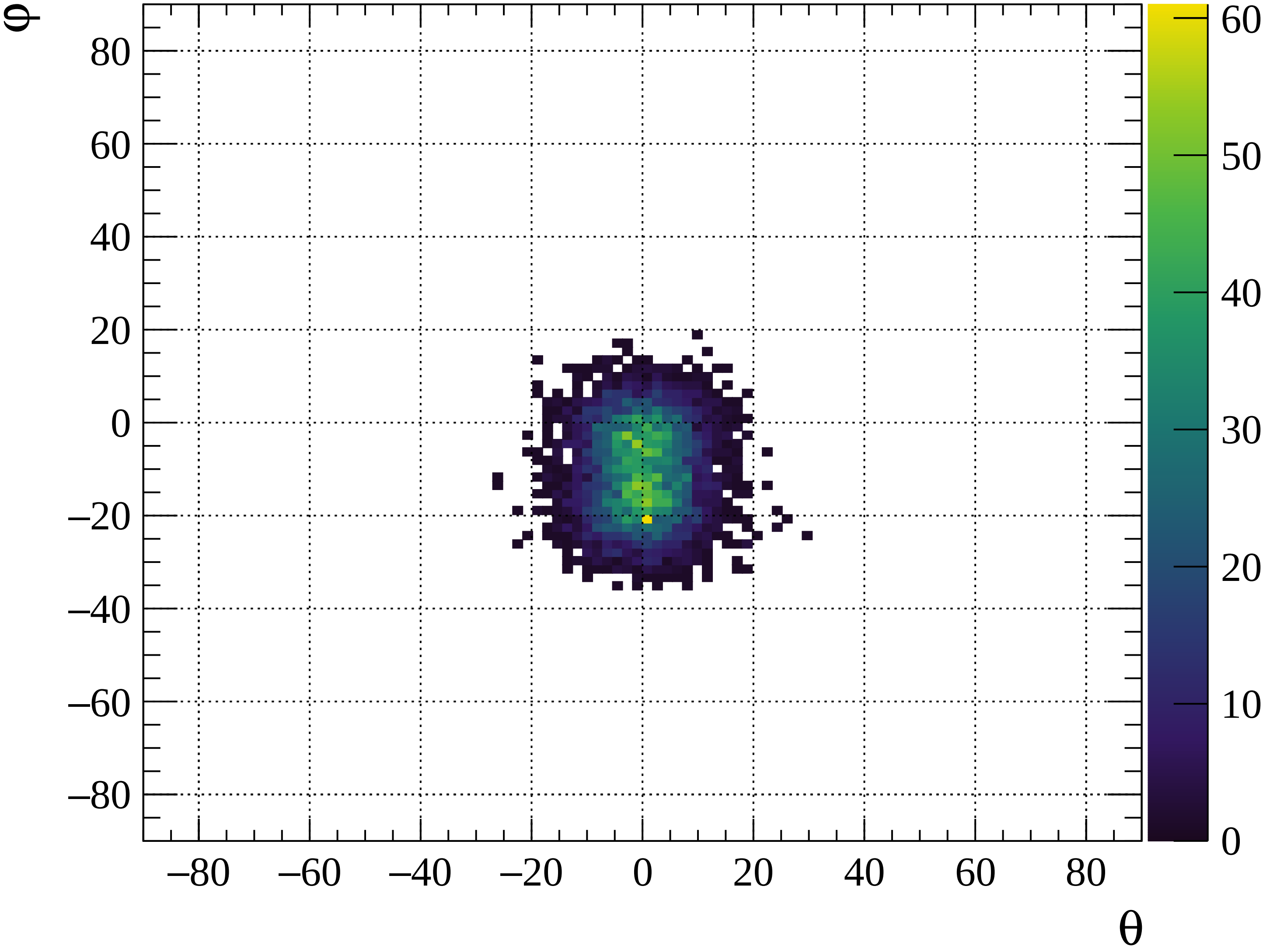
track length (cm)
θ
0
20
40
60
80
100
120
140
160
−80
−60
−40
−20
0
20
40
60
80
0
5
10
15
20
25
30
35
40



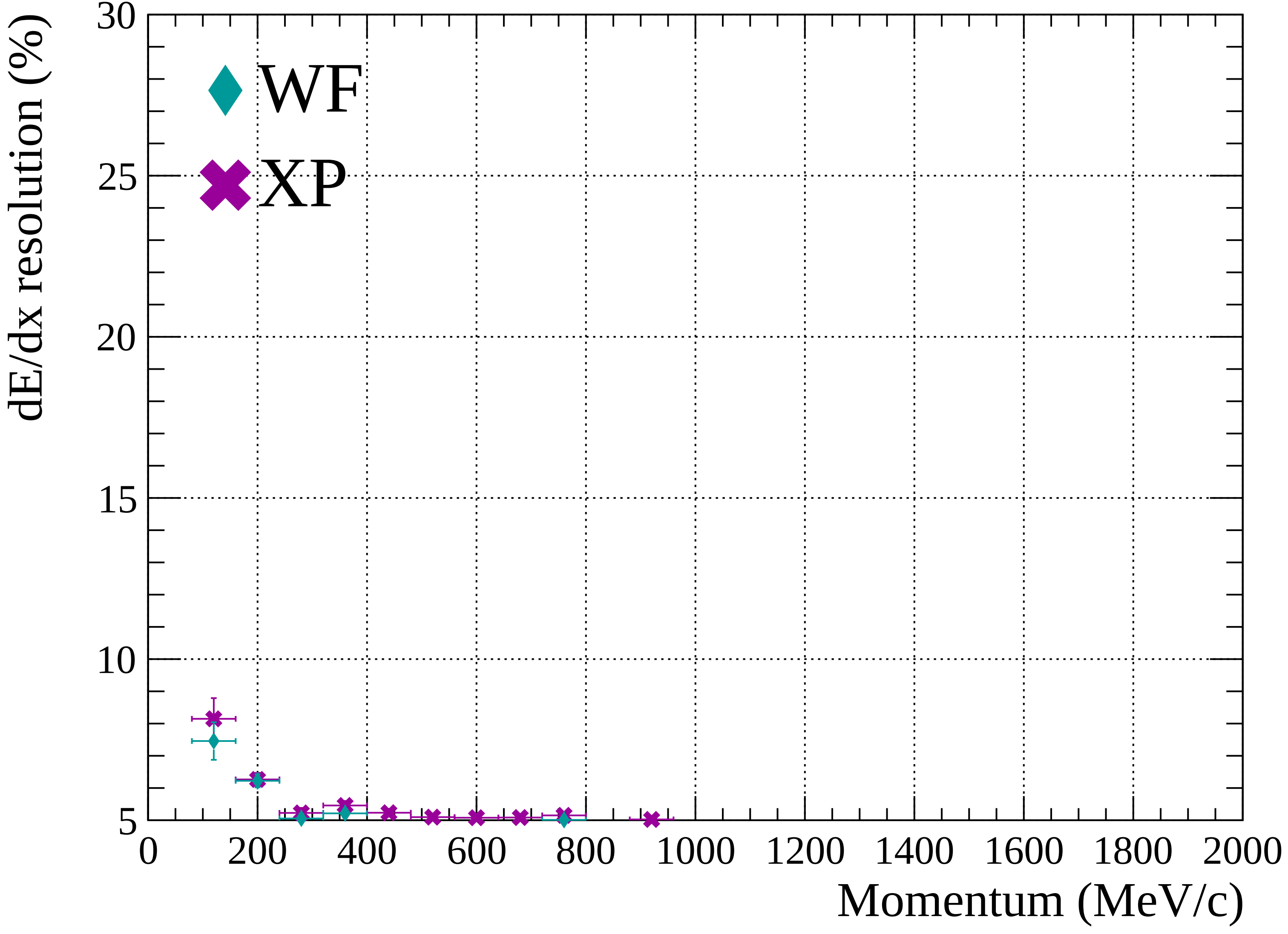
momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−80
−60
−40
−20
0
20
40
60
80
θ
50
40
30
20
10
0



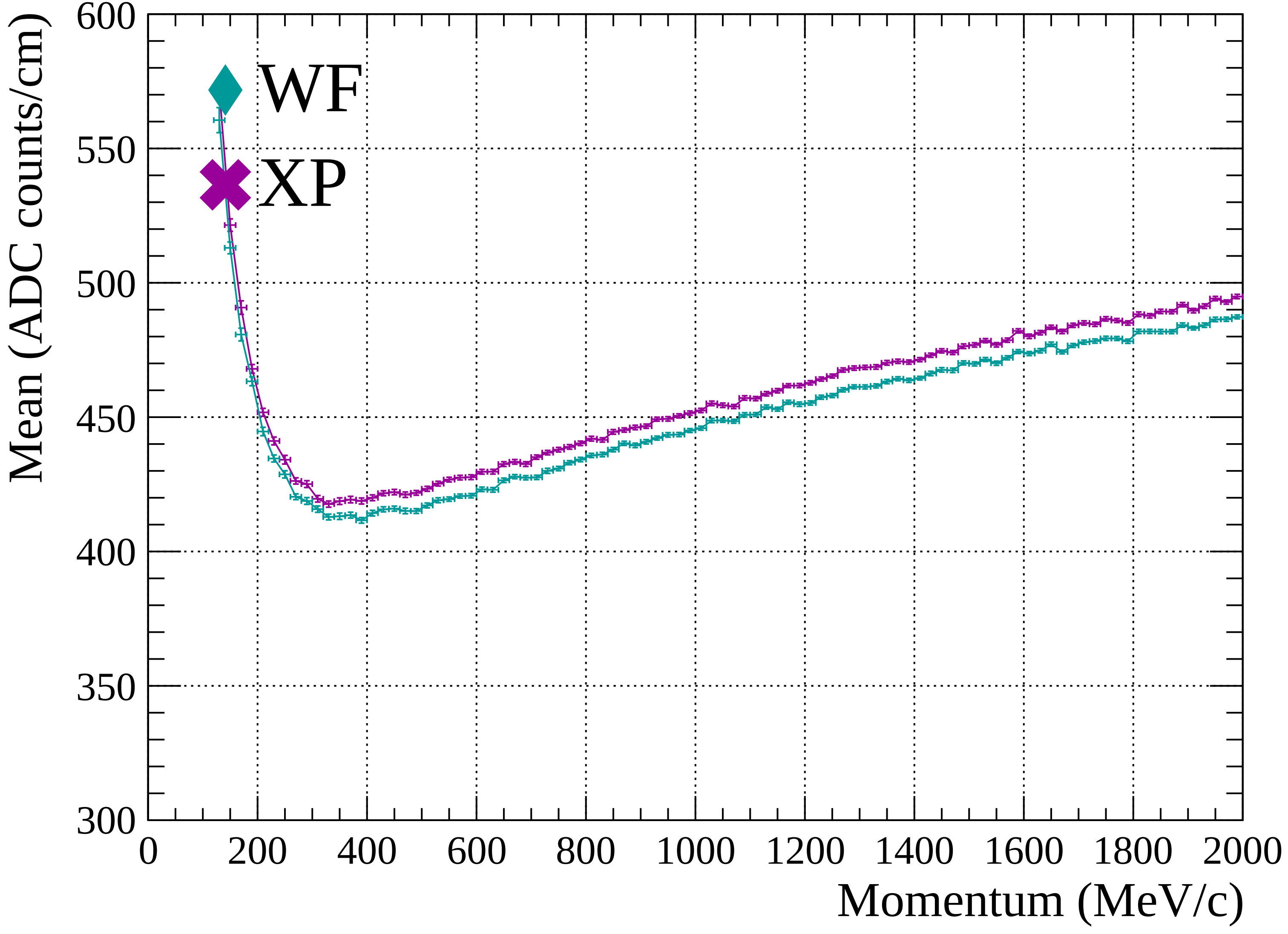
Count
700
600
500
400
300
200
100
0
−80
−60
−40
−20
0
20
40
60
80
θ angle



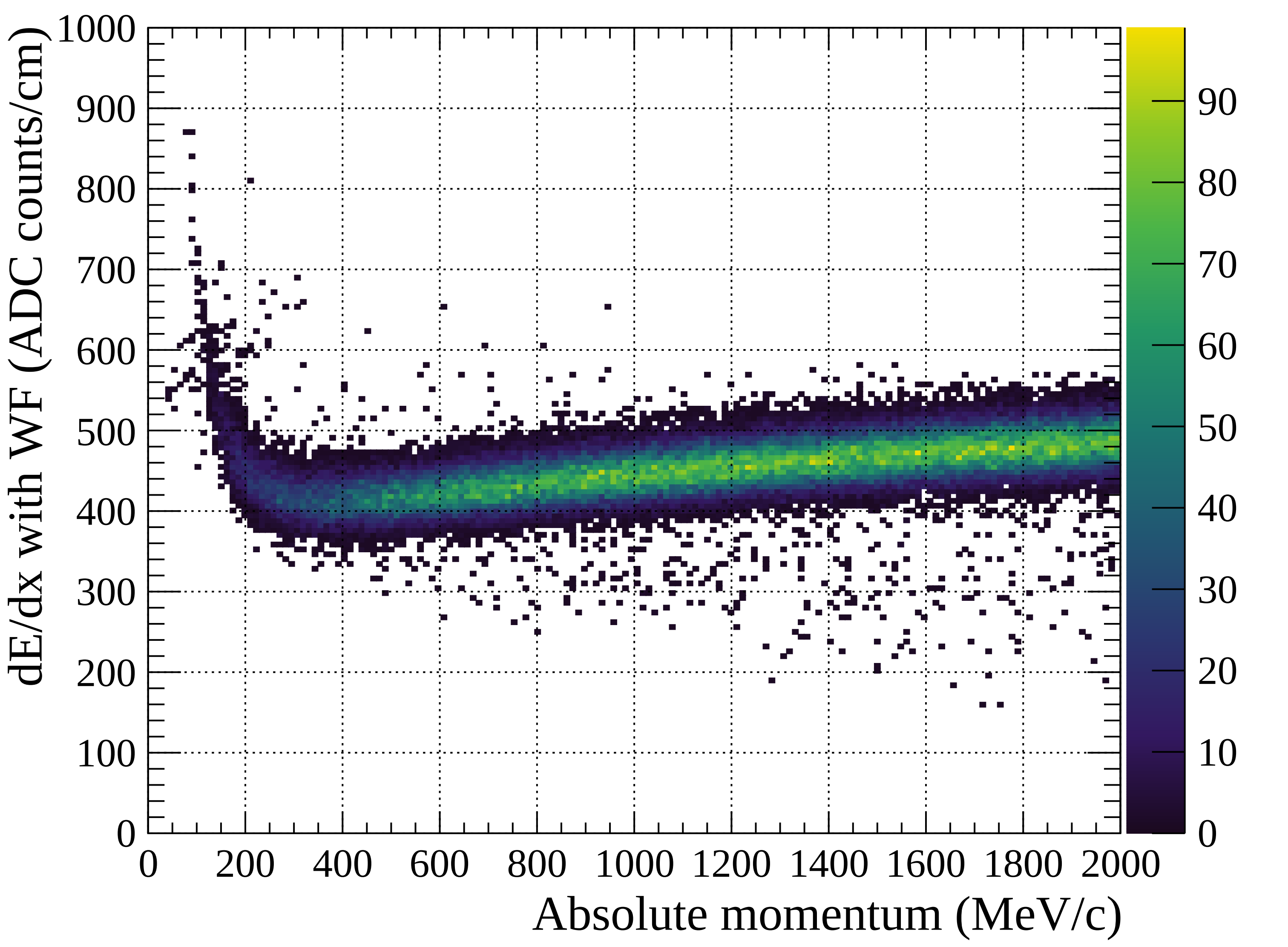
φ
80
60
40
20
0
−20
−40
−60
−80
−80
−60
−40
−20
0
20
40
60
80
θ
60
50
40
30
20
10
0

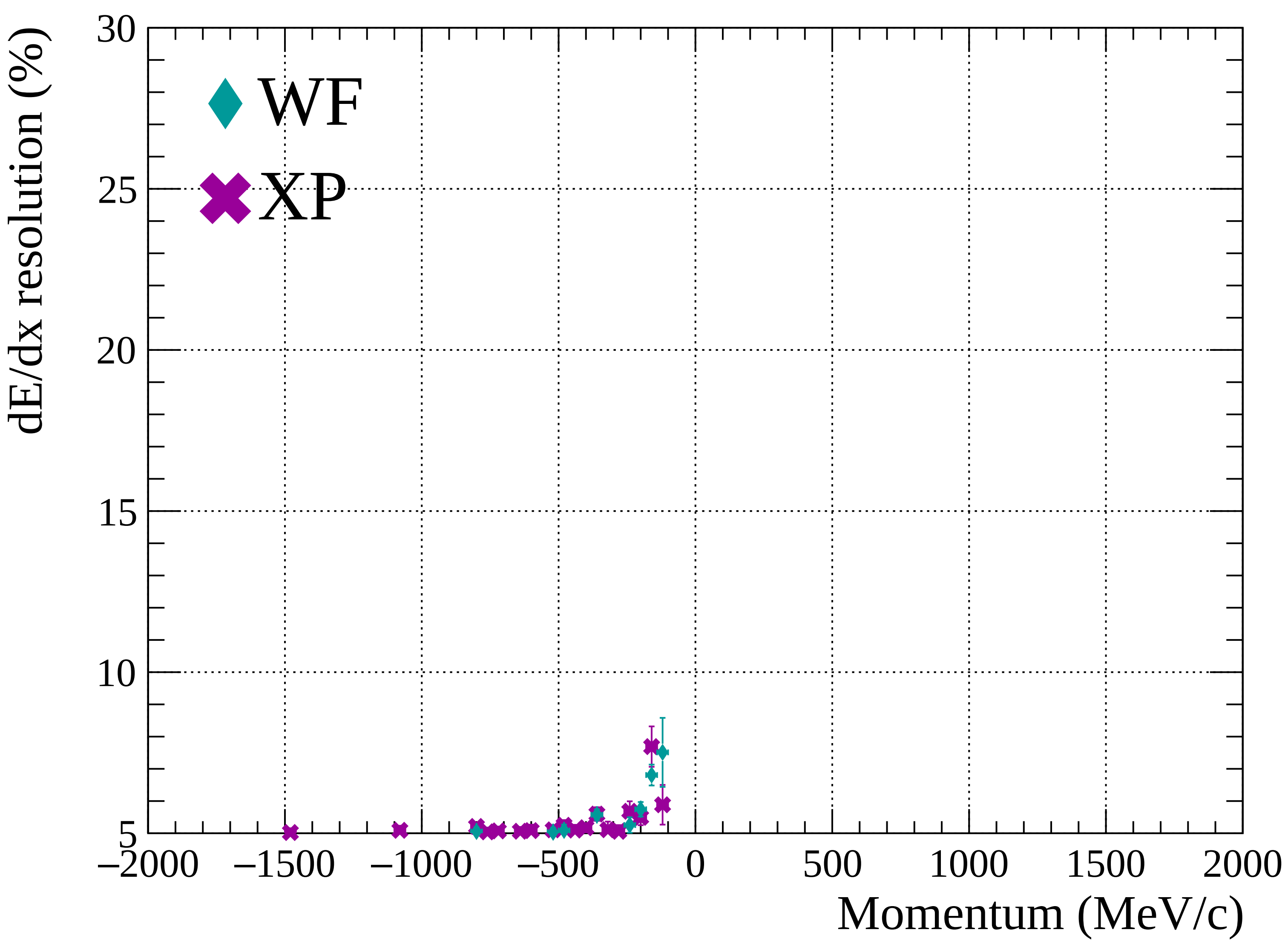


dE/dx resolution (%)
Momentum (MeV/c)
WF
XP
30
25
20
15
10
5
0
200
400
600
800
1000
1200
1400
1600
1800
2000

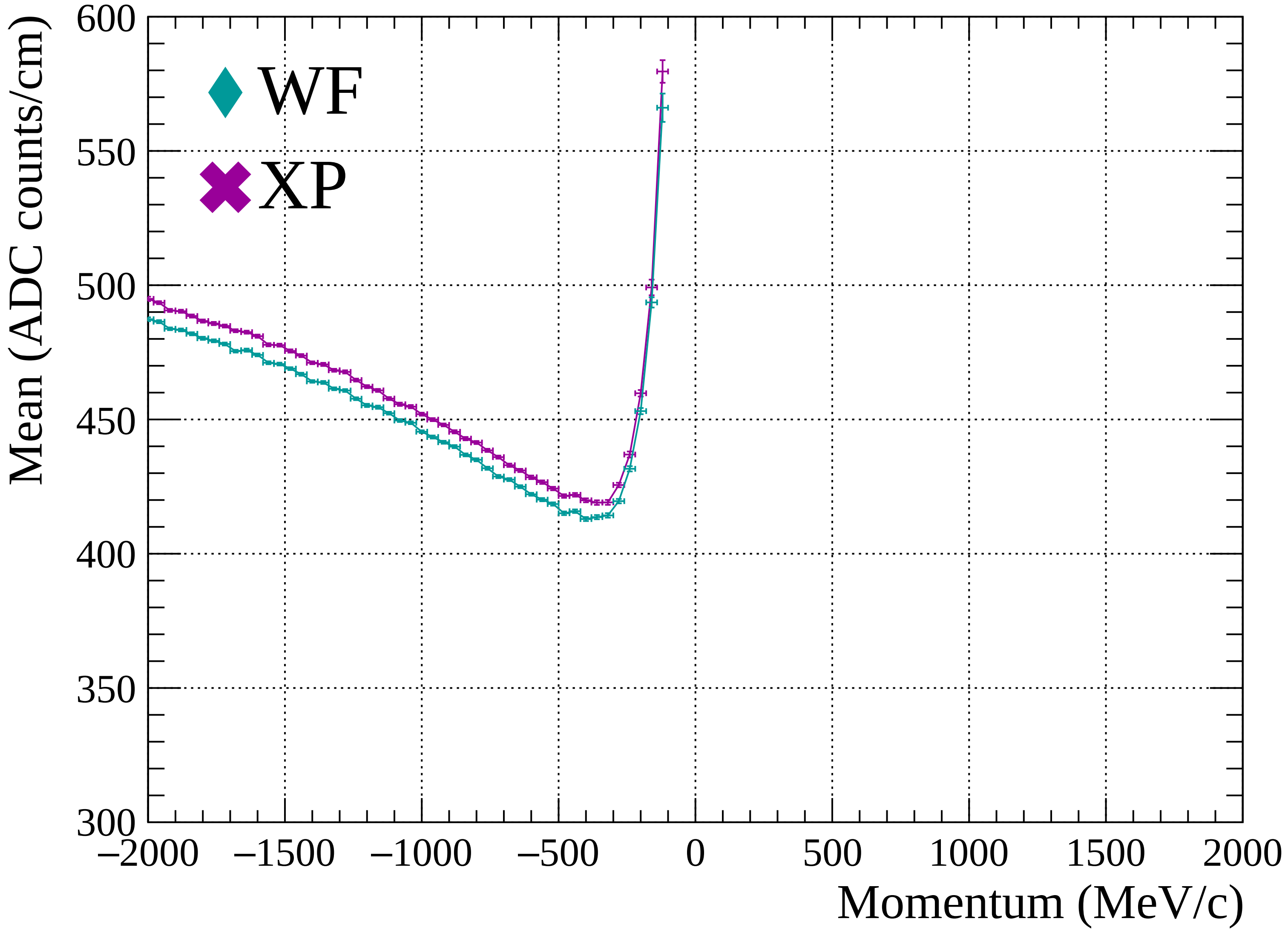


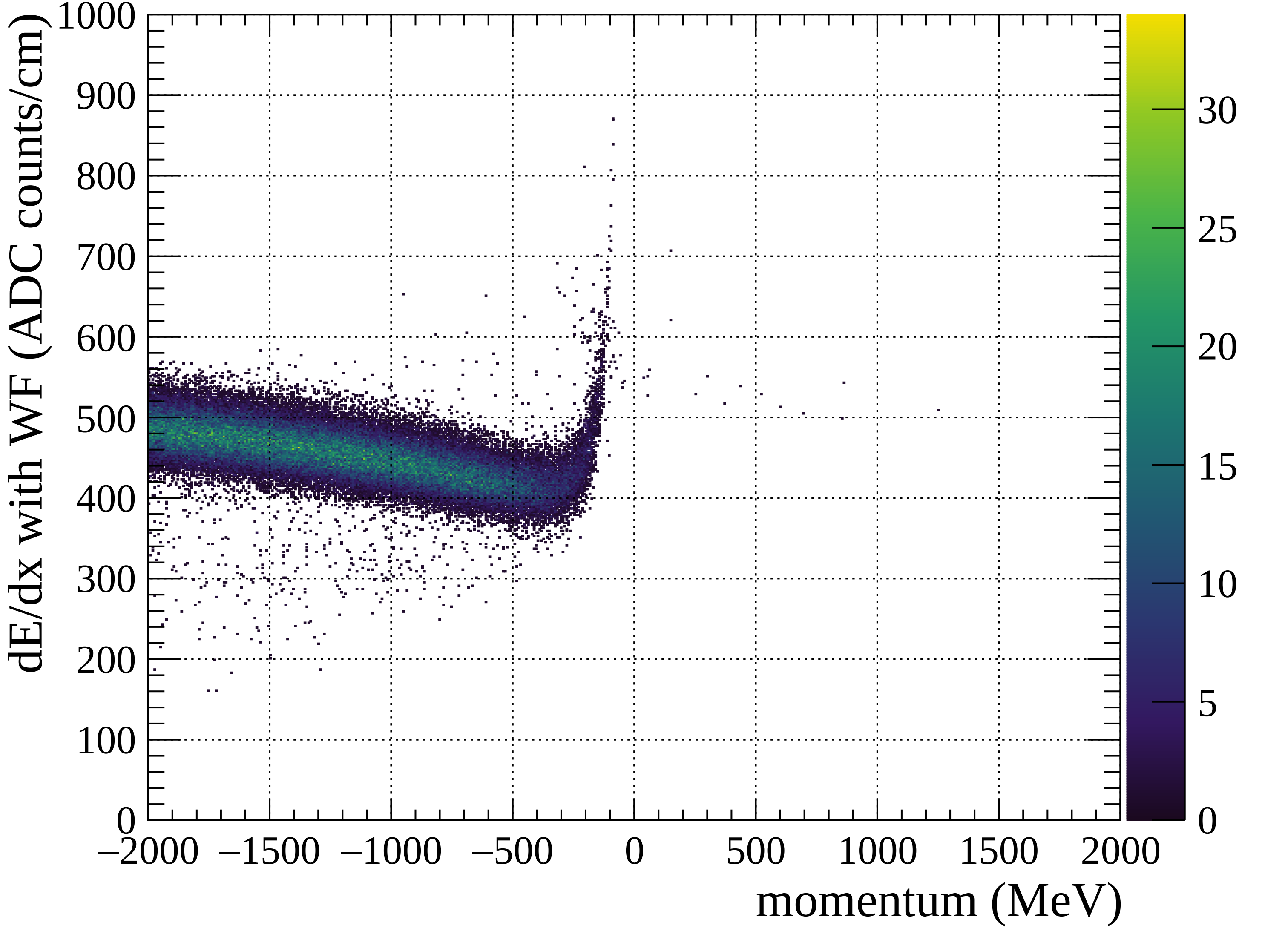
WF
XP
Mean (ADC counts/cm)
Momentum (MeV/c)
600
550
500
450
400
350
300
0
200
400
600
800
1000
1200
1400
1600
1800
2000


dE/dx with WF (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
1400
1600
1800
2000
Absolute momentum (MeV/c)
90
80
70
60
50
40
30
20
10
0

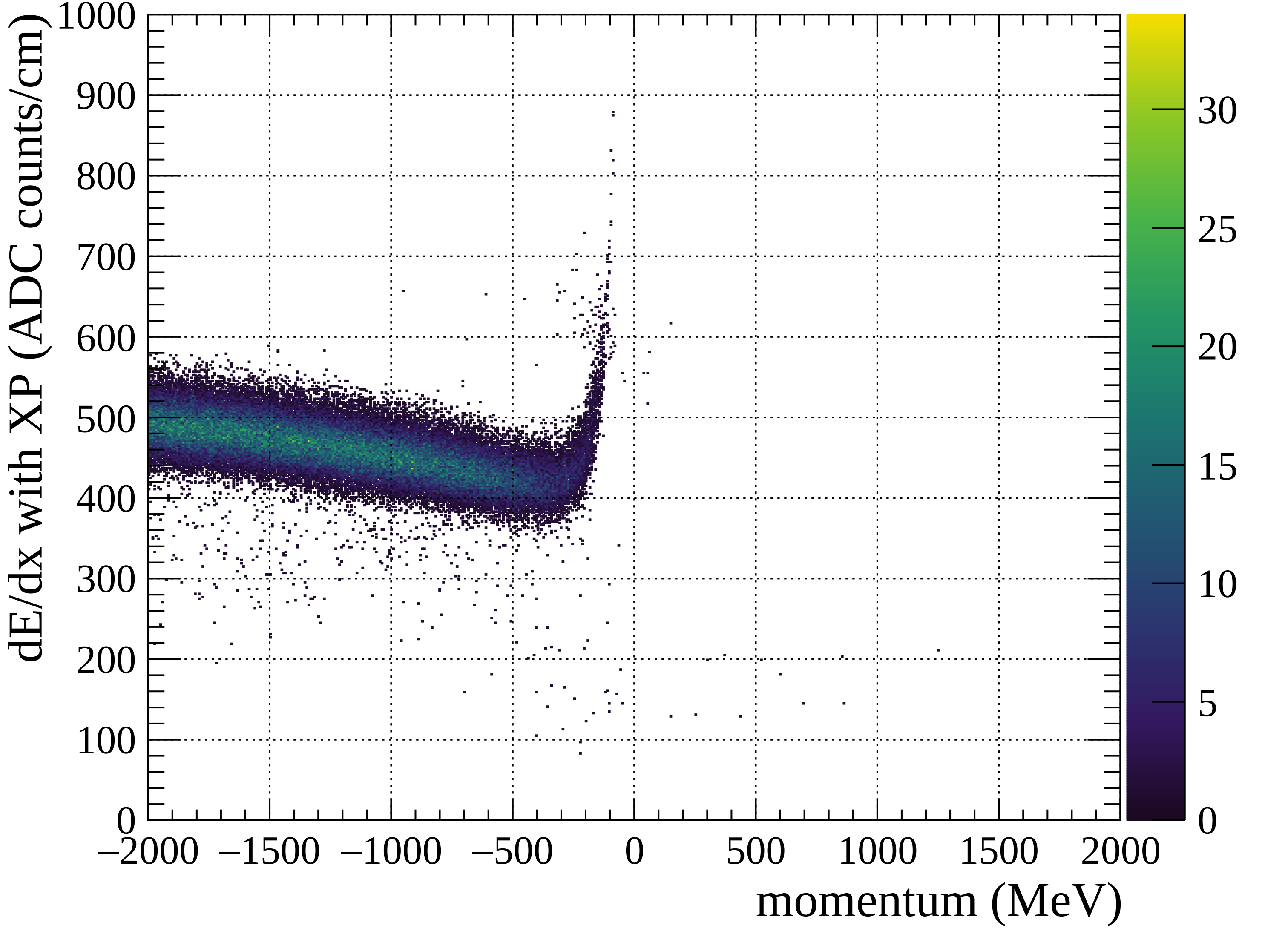


dE/dx resolution (%)
30
25
20
15
10
5
WF
XP
−2000
−1500
−1000
−500
0
500
1000
1500
2000
Momentum (MeV/c)

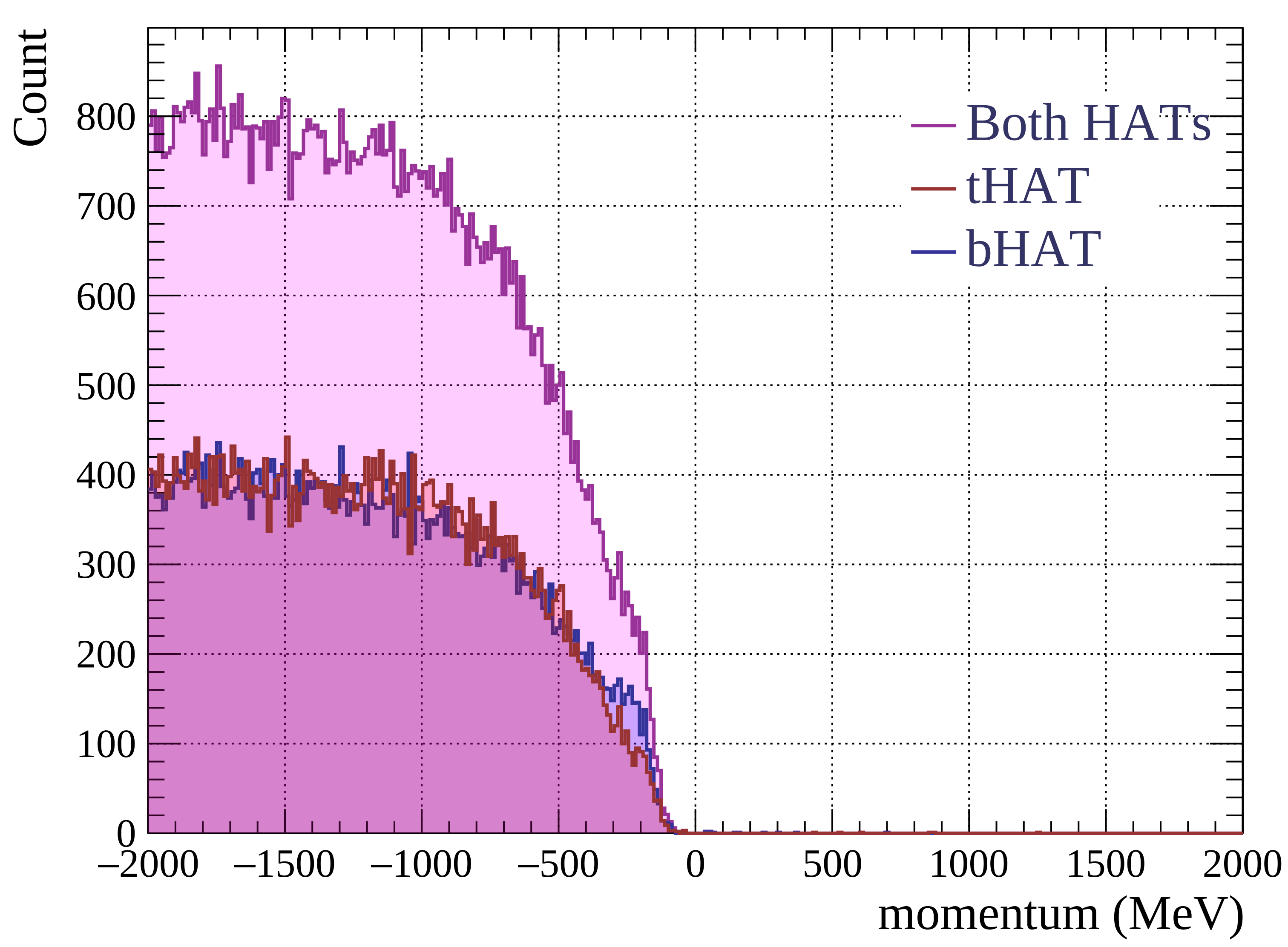

WF
XP
Mean (ADC counts/cm)
600
550
500
450
400
350
300
−2000
−1500
−1000
−500
0
500
1000
1500
2000
Momentum (MeV/c)



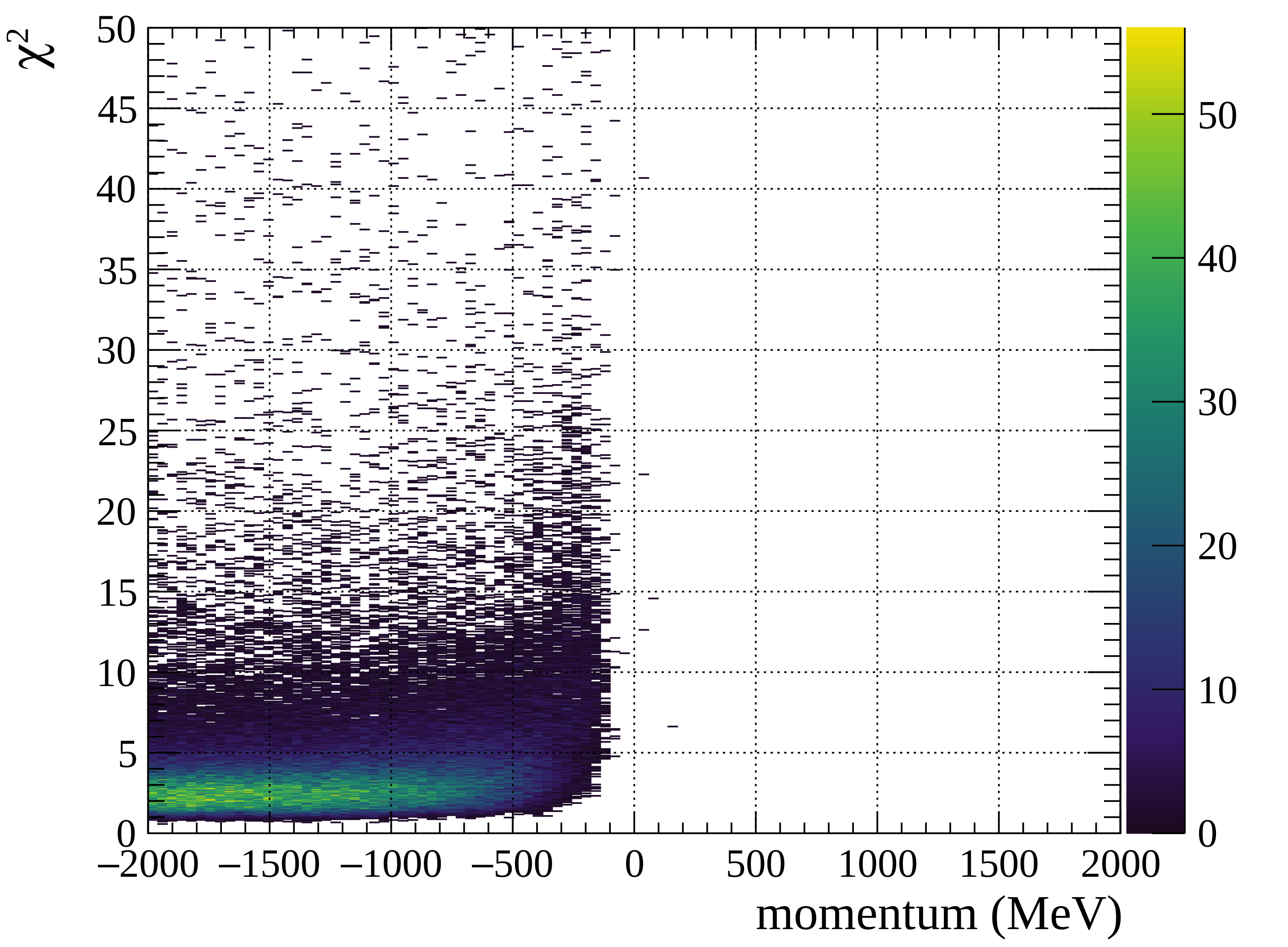
dE/dx with WF (ADC counts/cm)
momentum (MeV)
0
100
200
300
400
500
600
700
800
900
1000
−2000
−1500
−1000
−500
0
500
1000
1500
2000
0
5
10
15
20
25
30



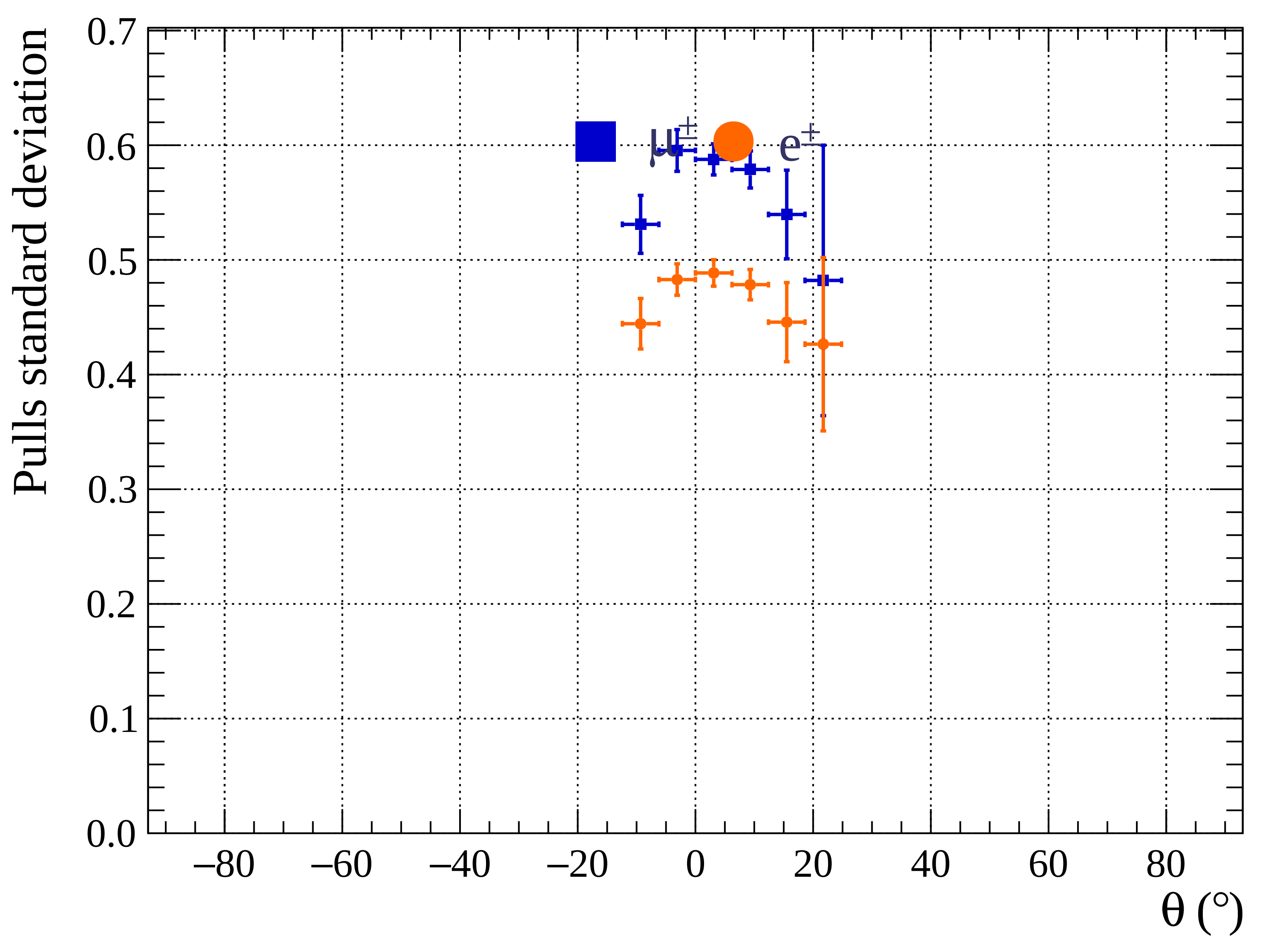
dE/dx with XP (ADC counts/cm)
momentum (MeV)
0
100
200
300
400
500
600
700
800
900
1000
−2000
−1500
−1000
−500
0
500
1000
1500
2000
0
5
10
15
20
25
30



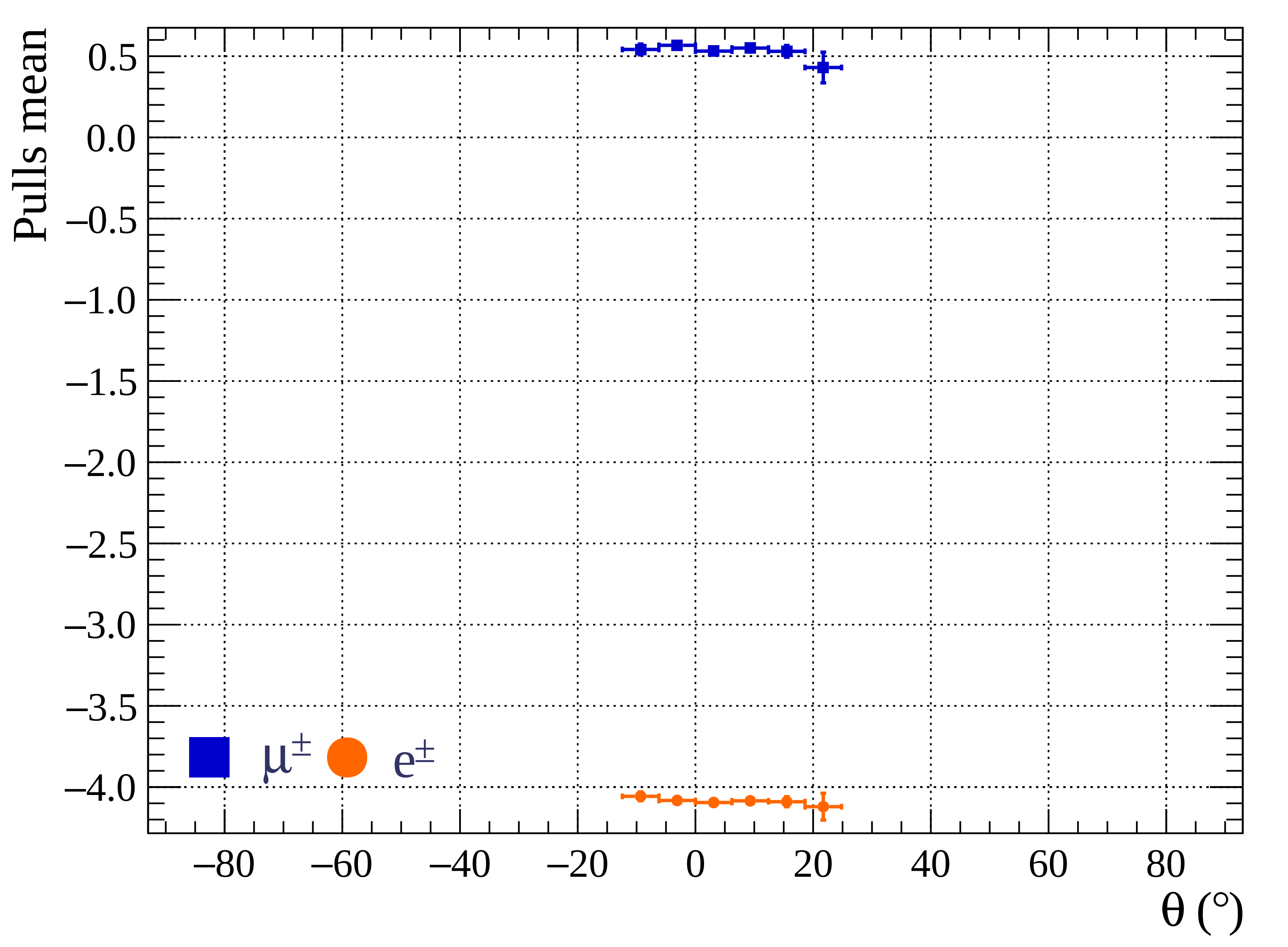
Count
800
700
600
500
400
300
200
100
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
momentum (MeV)
Both HATs
tHAT
bHAT



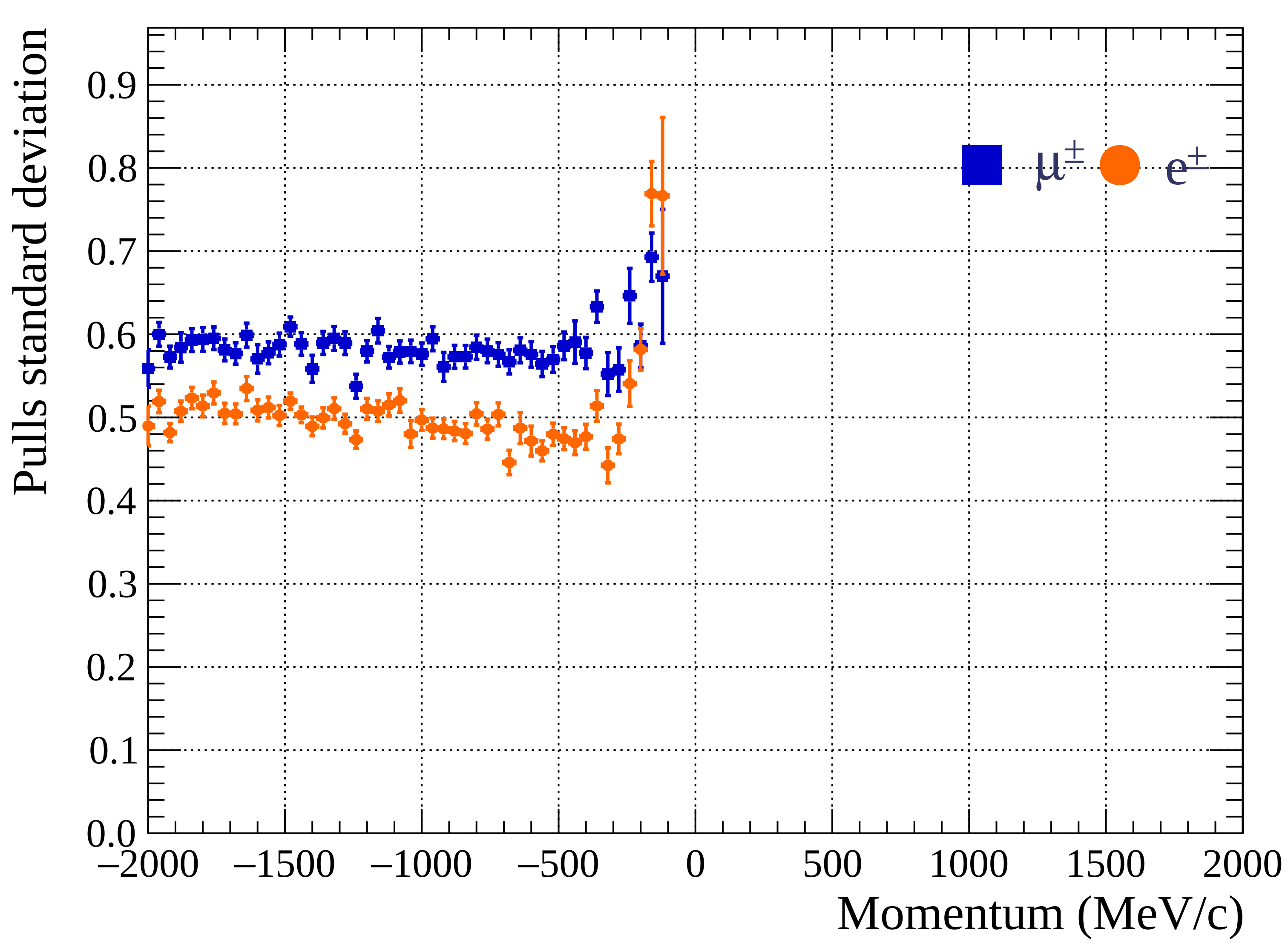
χ^2
50
45
40
35
30
25
20
15
10
5
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
momentum (MeV)
50
40
30
20
10
0



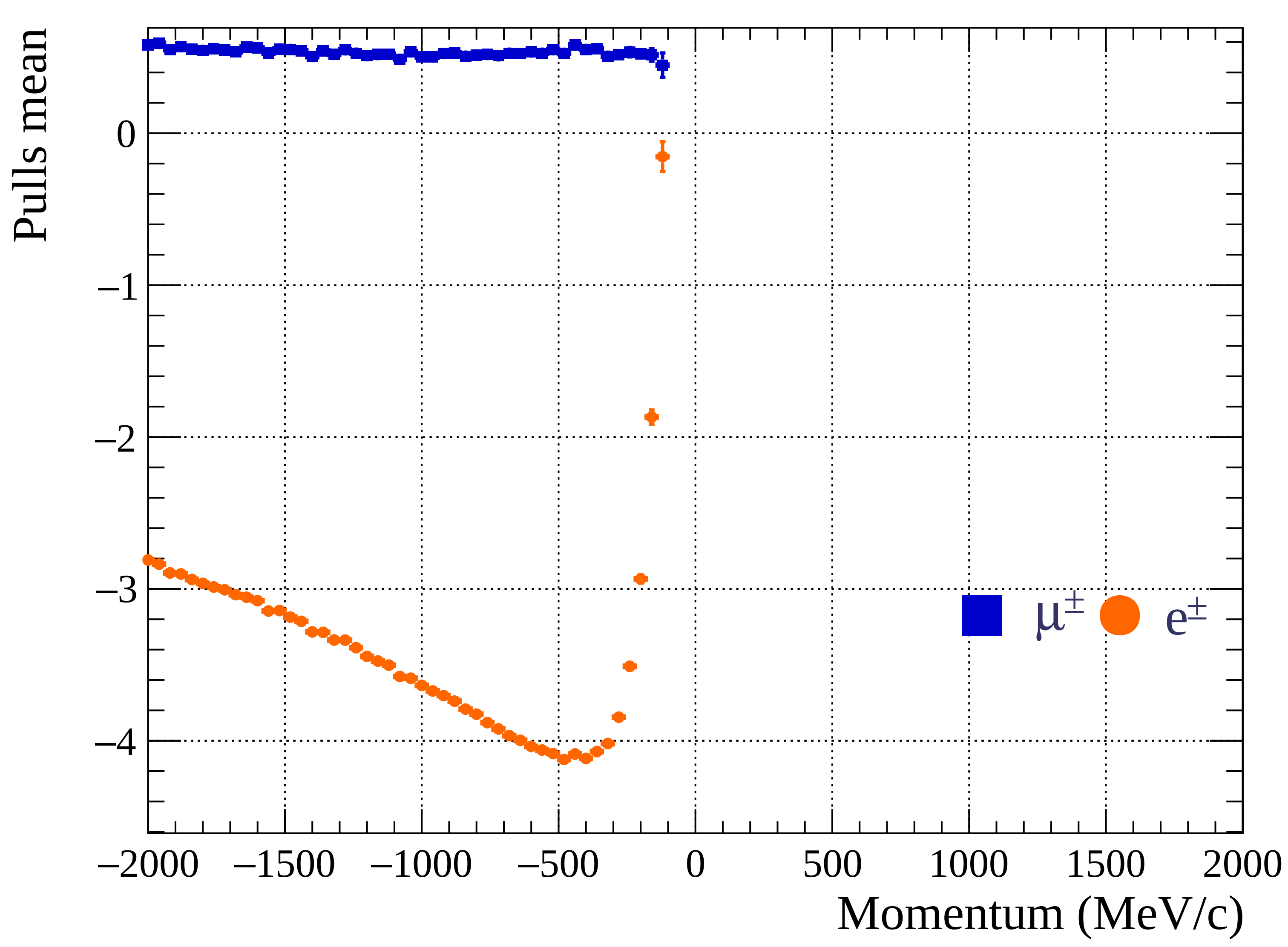
Pulls standard deviation
θ (°)
μ±
e±
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
−80
−60
−40
−20
0
20
40
60
80



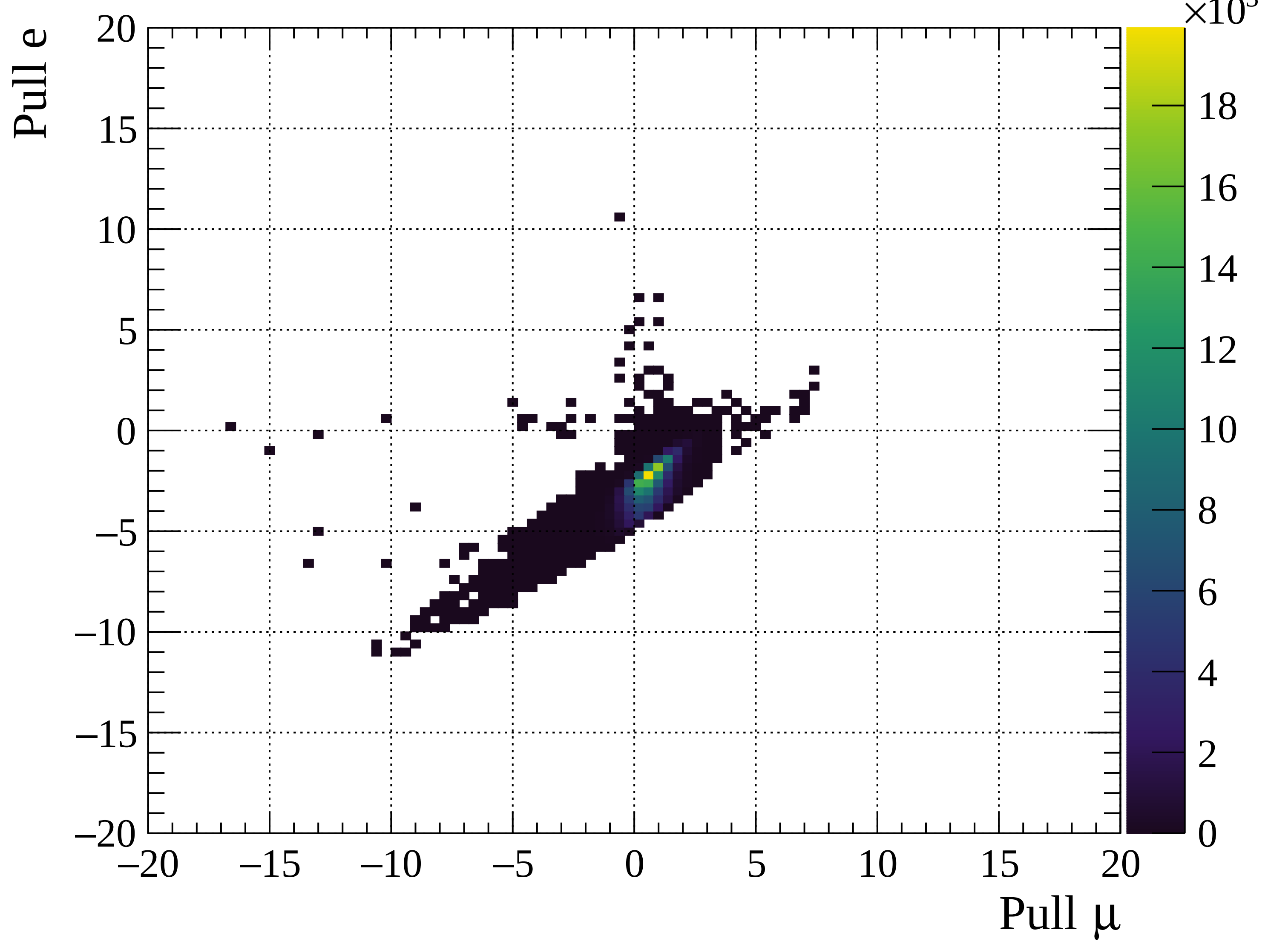
Pulls mean
0.5
0.0
−0.5
−1.0
−1.5
−2.0
−2.5
−3.0
−3.5
−4.0
μ±
e±
−80
−60
−40
−20
0
20
40
60
80
θ (°)



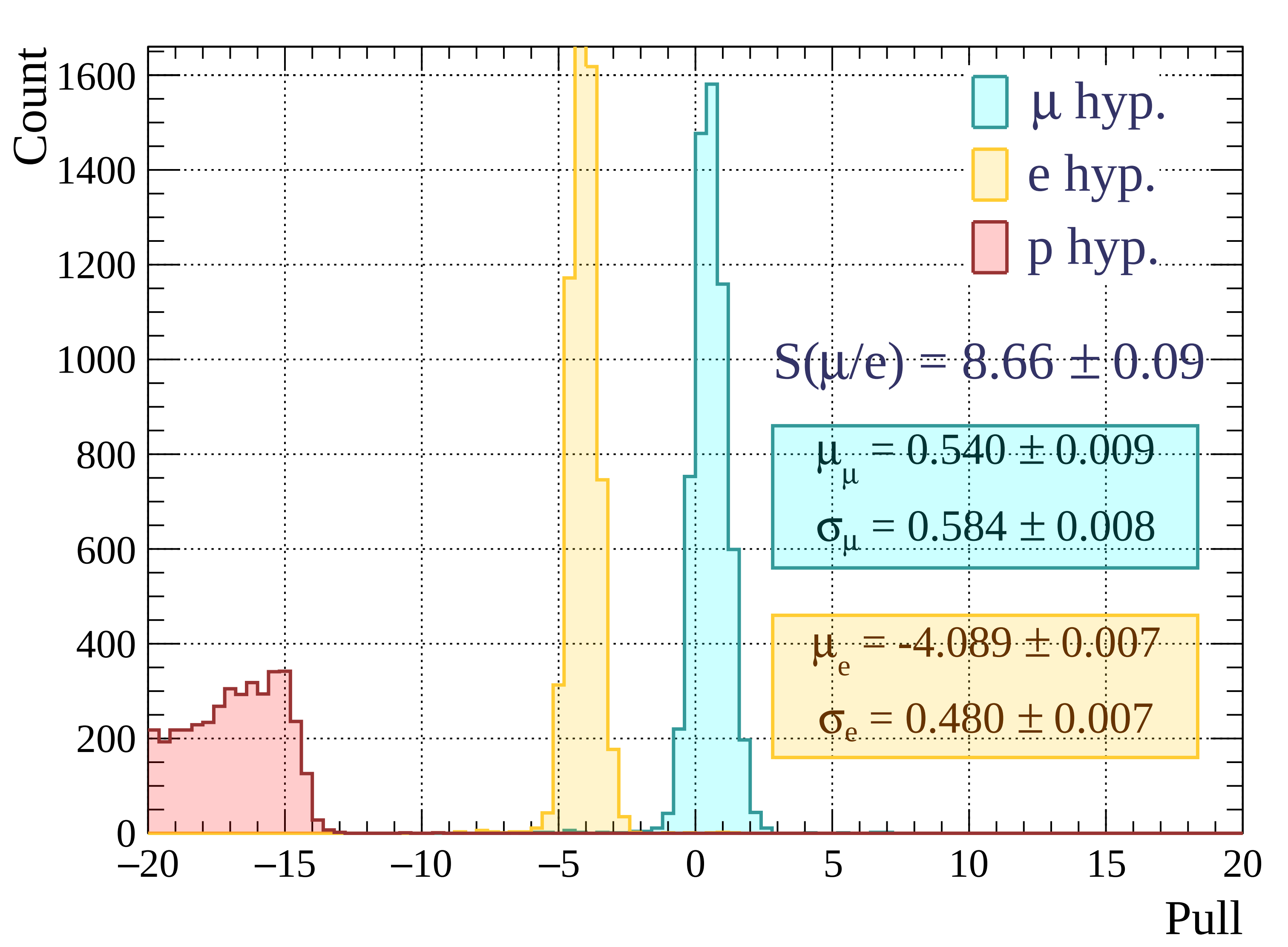
Pulls standard deviation
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
−2000
−1500
−1000
−500
0
500
1000
1500
2000
Momentum (MeV/c)
$\mu^{\pm}$
$e^{\pm}$



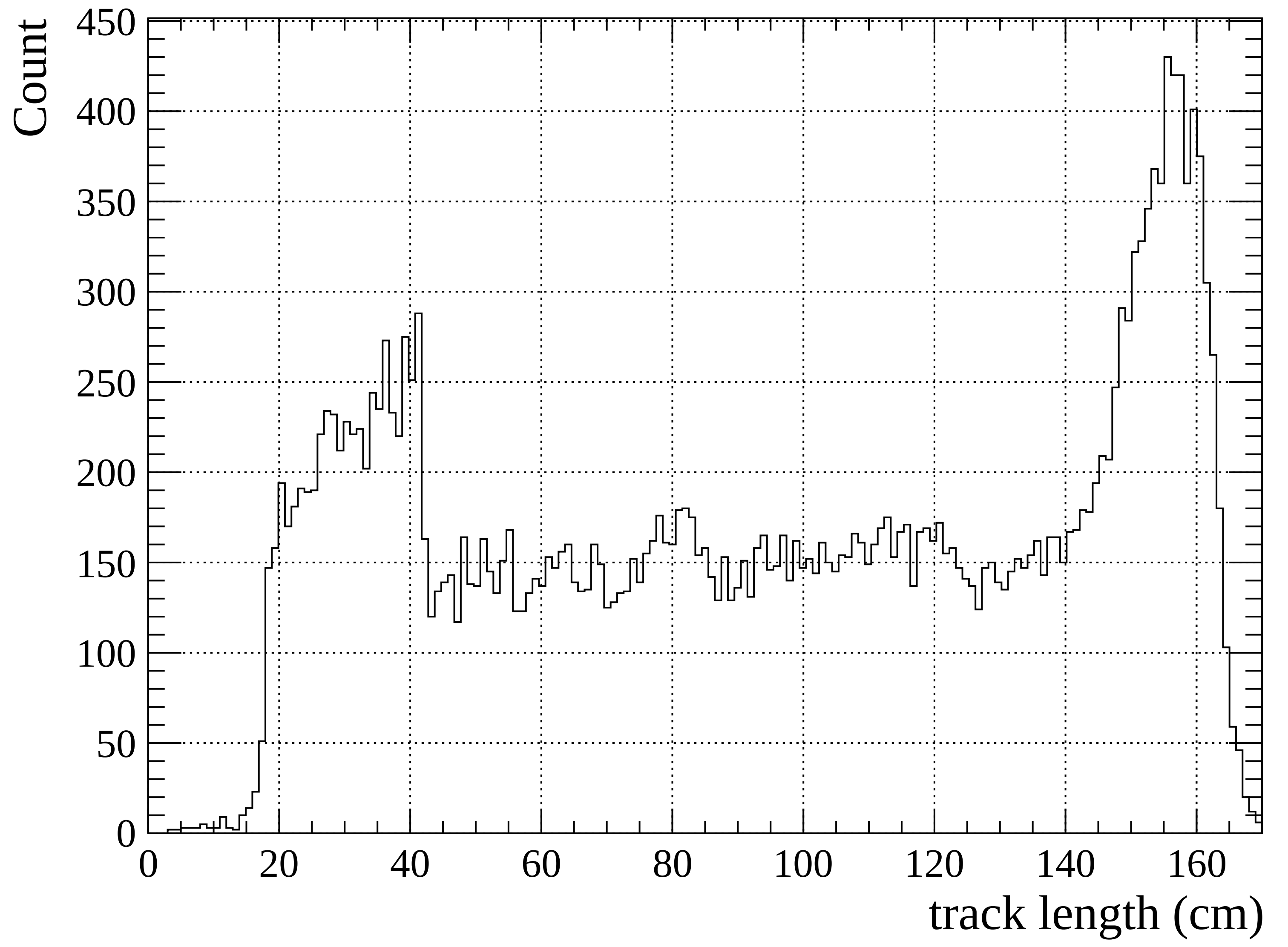
Pulls mean
0
−1
−2
−3
−4
−2000
−1500
−1000
−500
0
500
1000
1500
2000
Momentum (MeV/c)
$\mu^{\pm}$
$e^{\pm}$

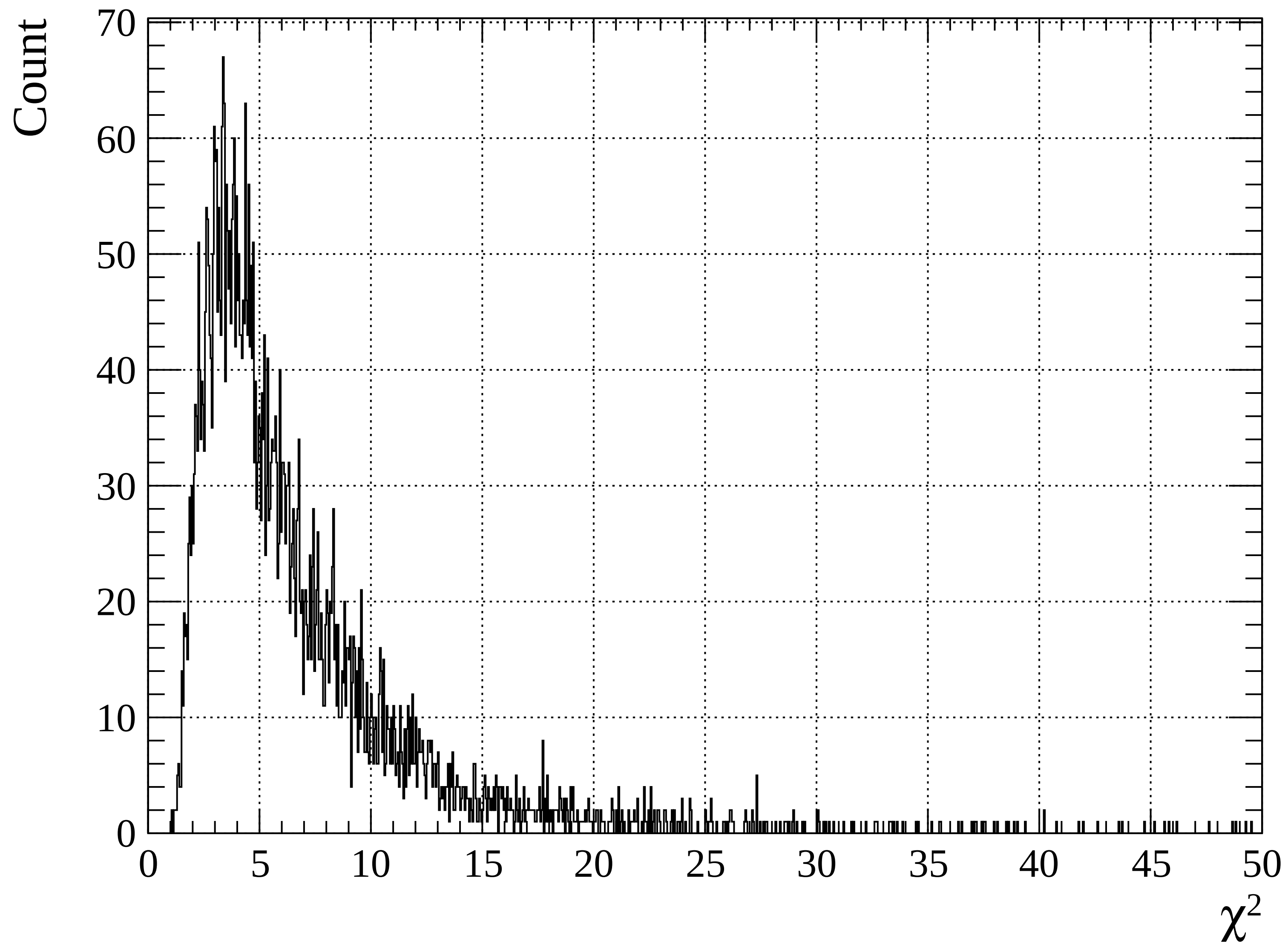


Pull e
Pull μ
×10³
20
15
10
5
0
−5
−10
−15
−20
18
16
14
12
10
8
6
4
2

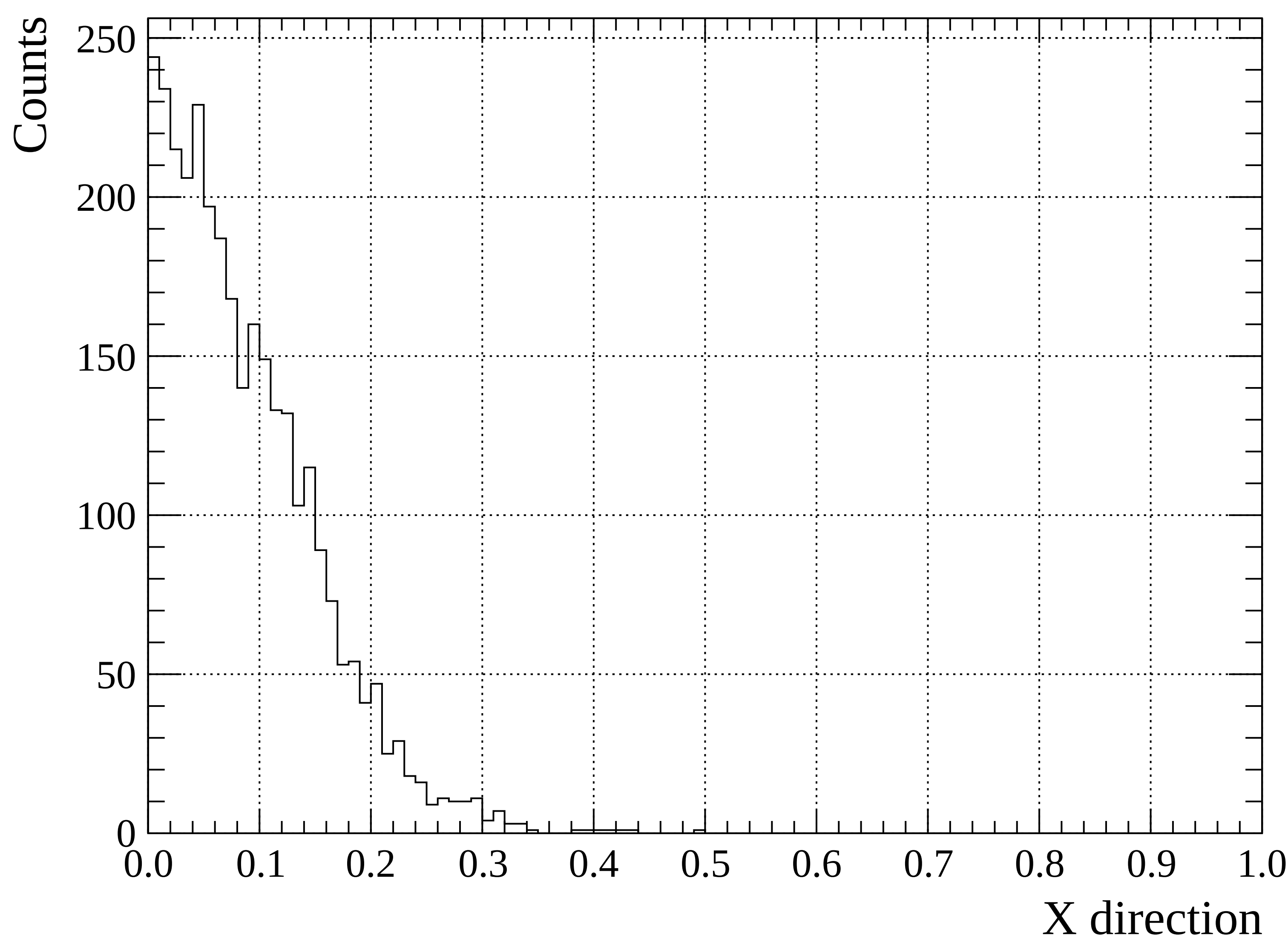


Count
Pull
μ hyp.
e hyp.
p hyp.
S(μ/e) = 8.66 ± 0.09
μ_μ = 0.540 ± 0.009
σ_μ = 0.584 ± 0.008
μ_e = -4.089 ± 0.007
σ_e = 0.480 ± 0.007

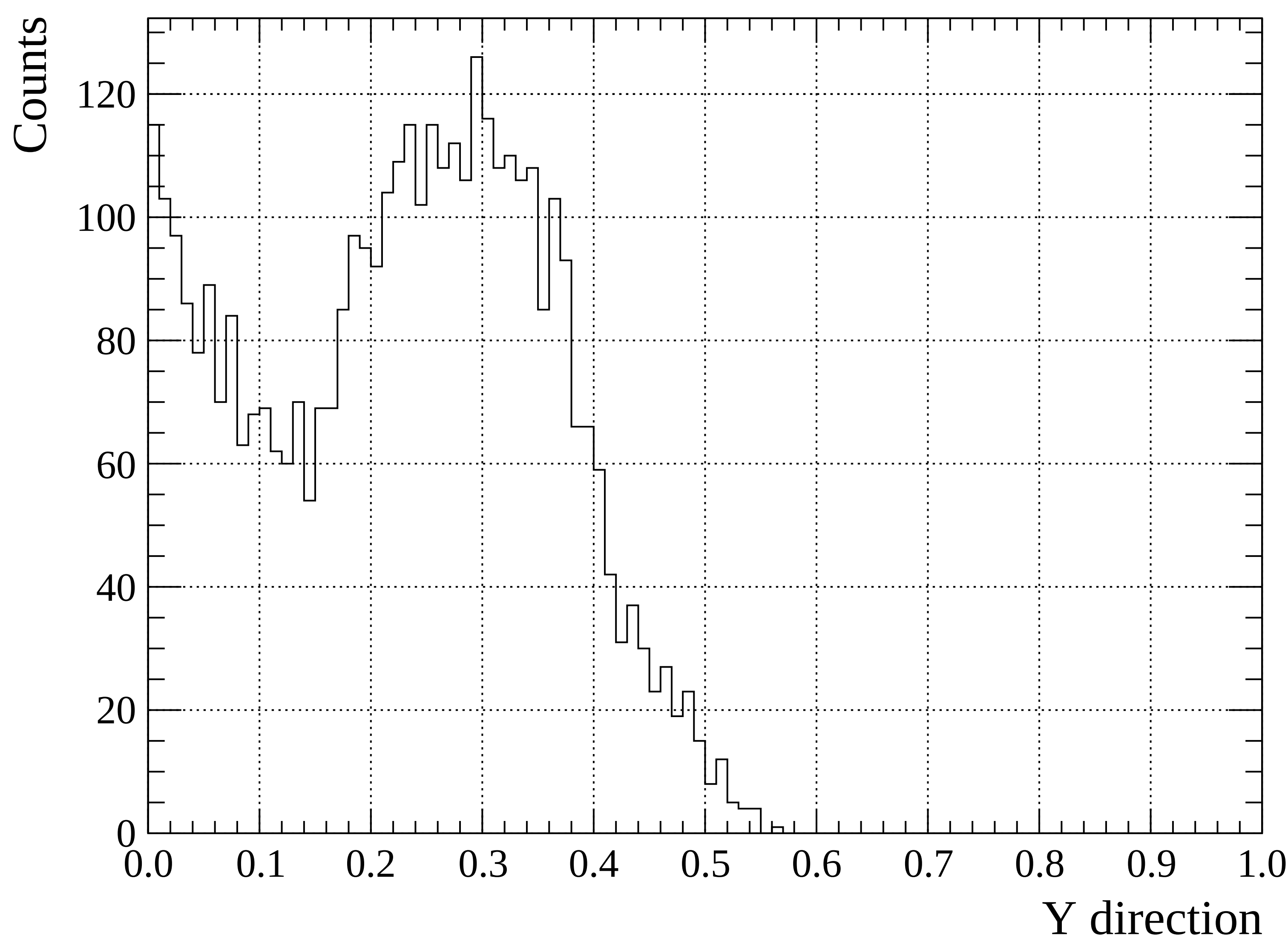

Count
450
400
350
300
250
200
150
100
50
0
0
20
40
60
80
100
120
140
160
track length (cm)

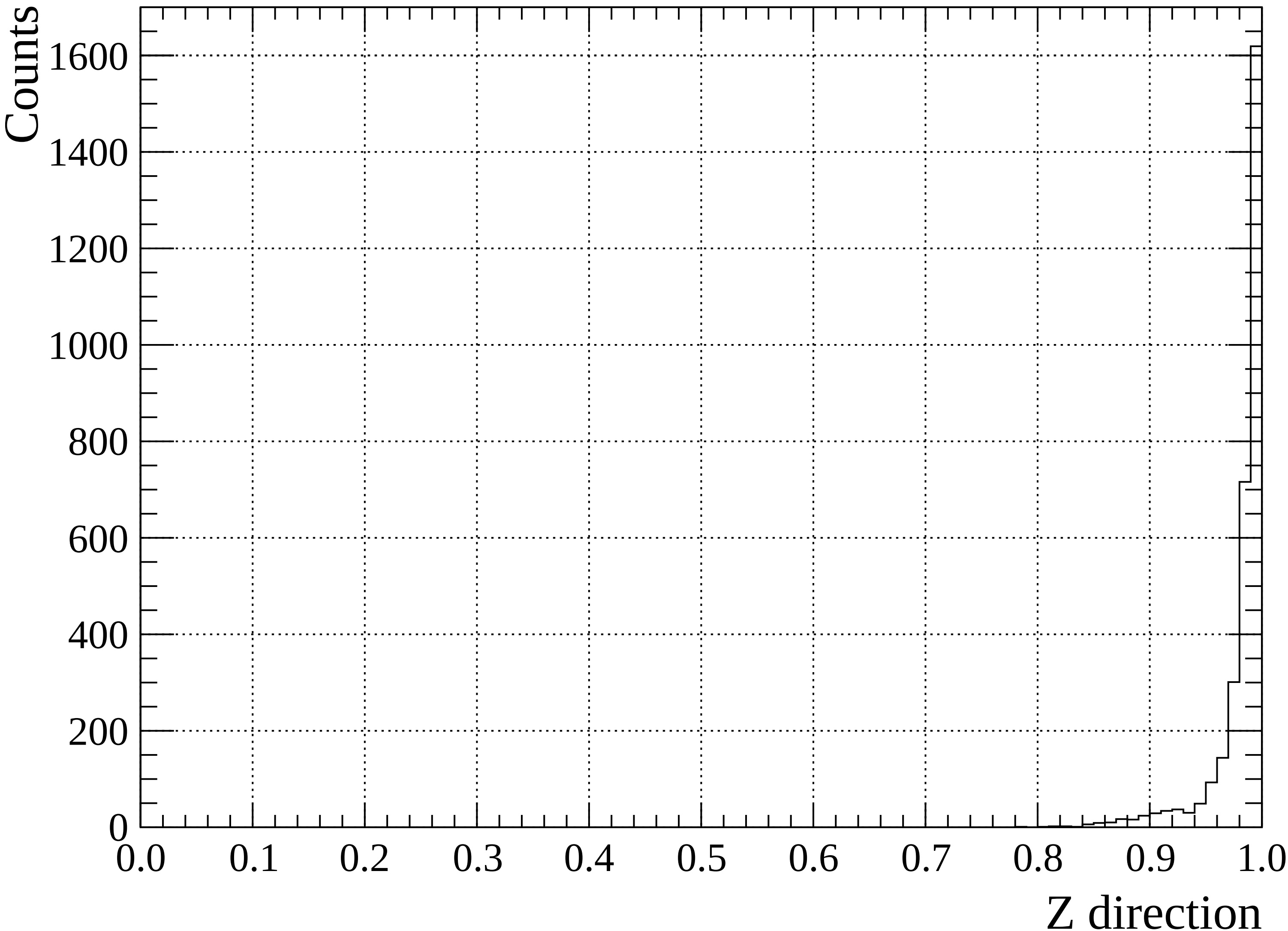


Count
0
10
20
30
40
50
60
70
0
5
10
15
20
25
30
35
40
45
50
χ^2

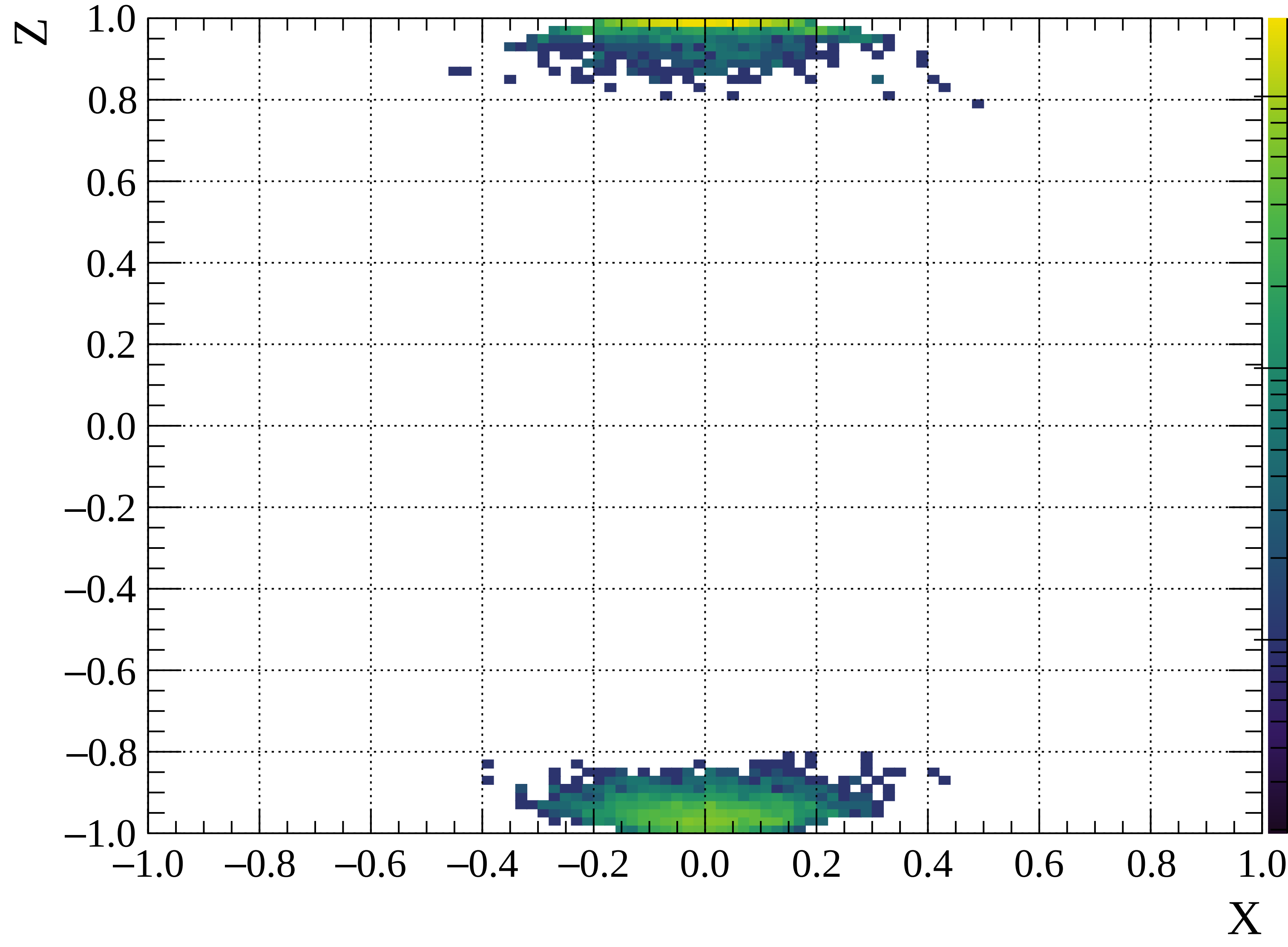


Counts
250
200
150
100
50
0
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
X direction

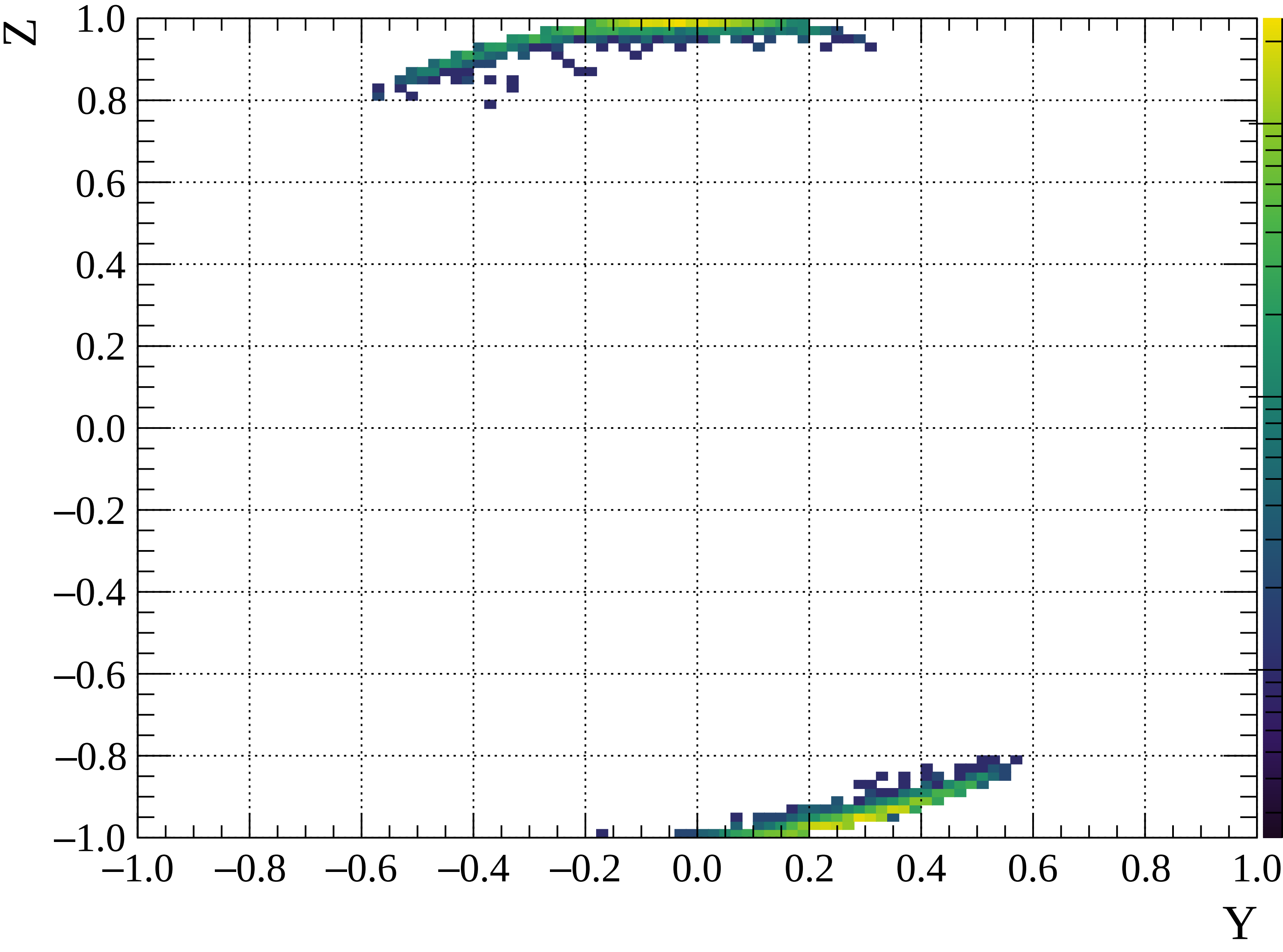

Counts
120
100
80
60
40
20
0
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
Y direction

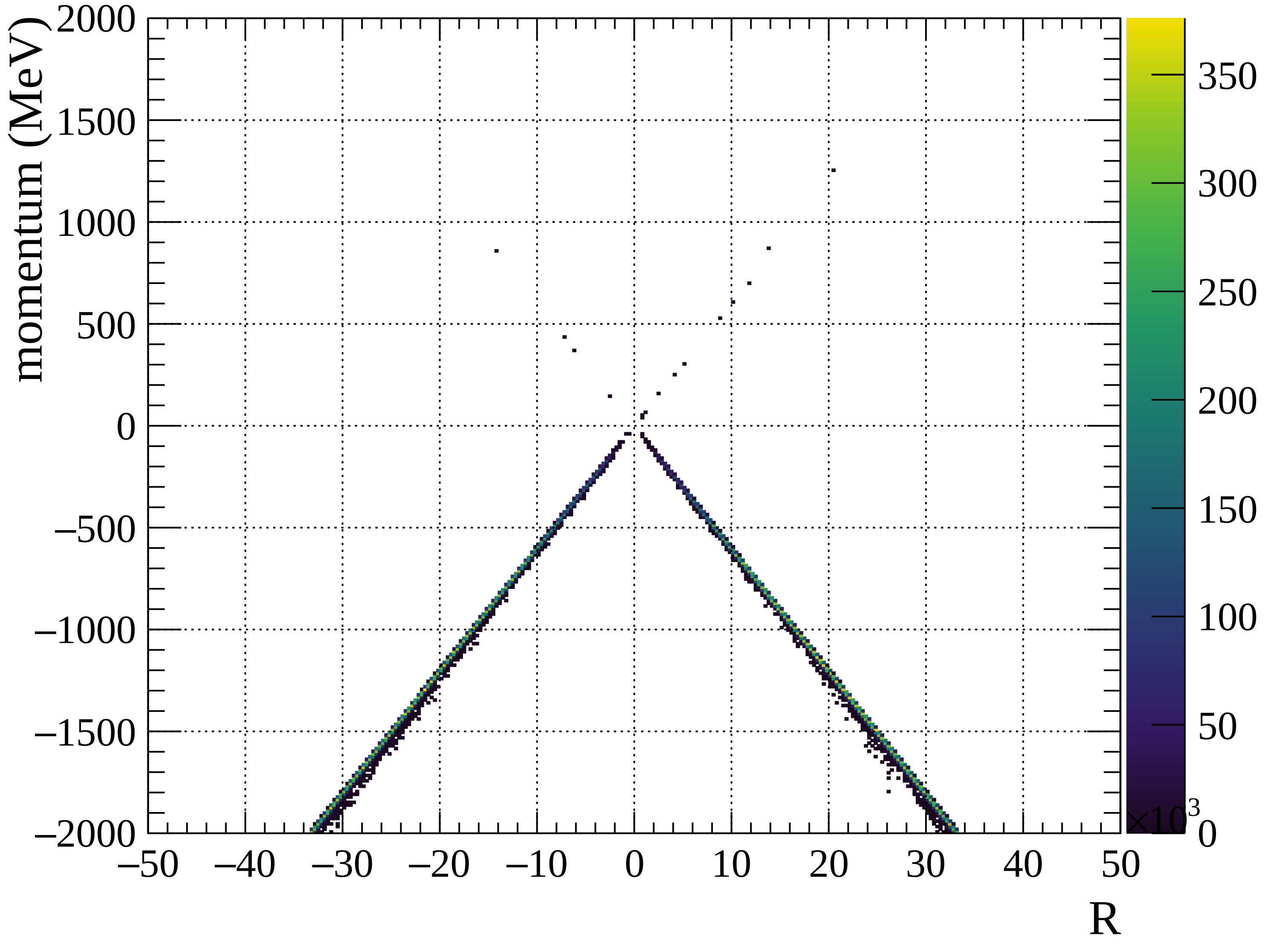


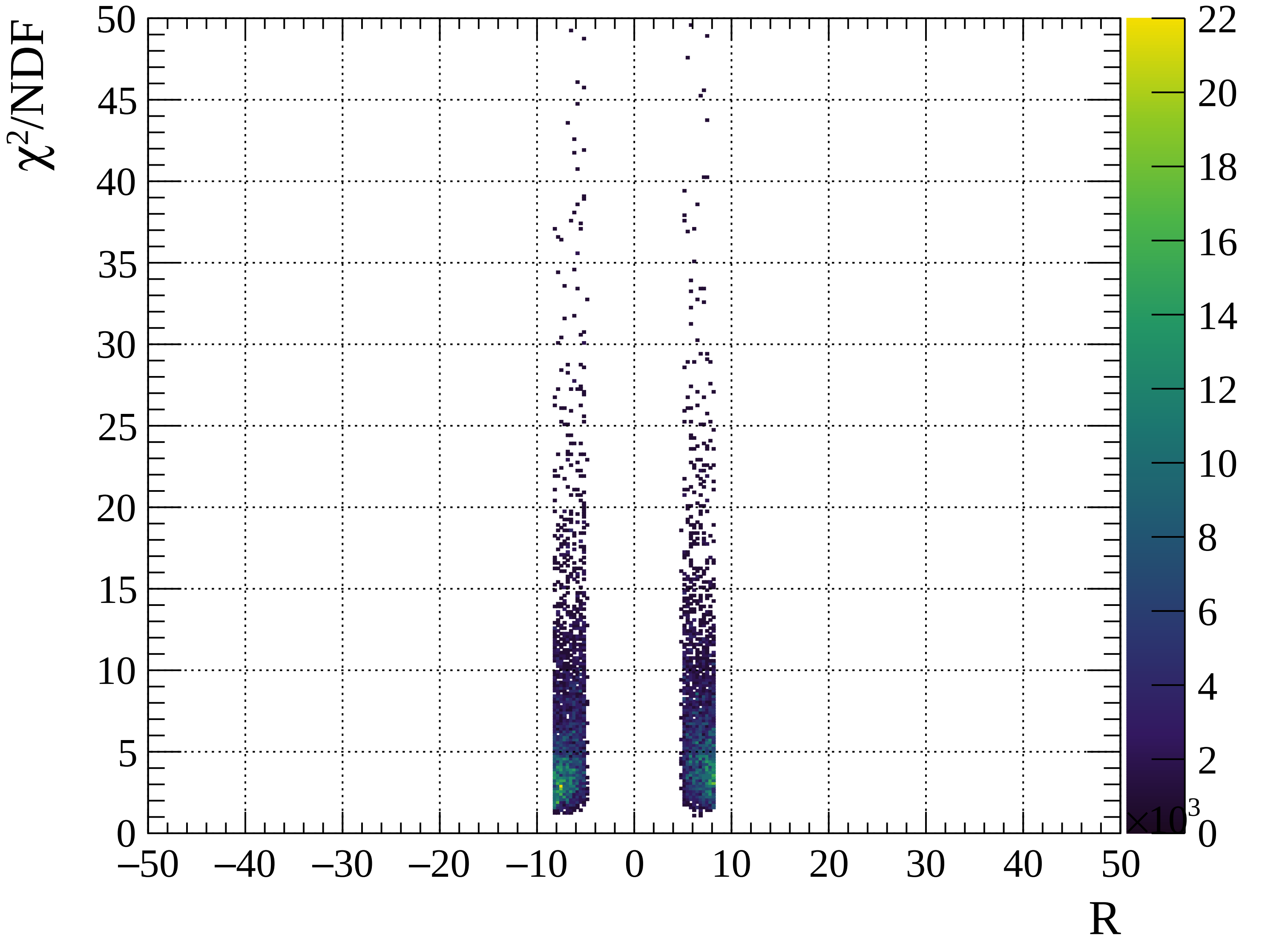
Counts
1600
1400
1200
1000
800
600
400
200
0
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
Z direction



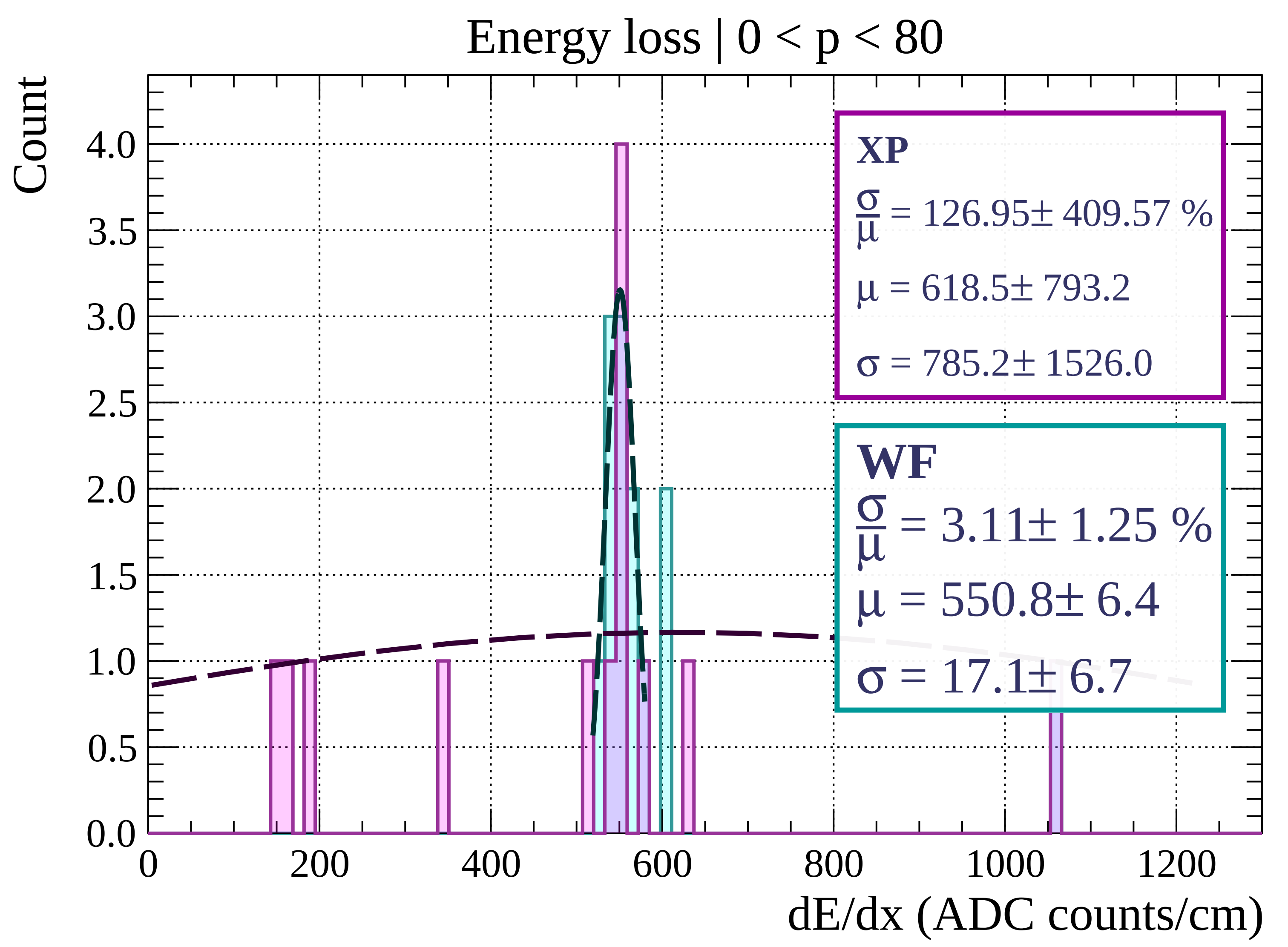
Z
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
X


Z
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
Y

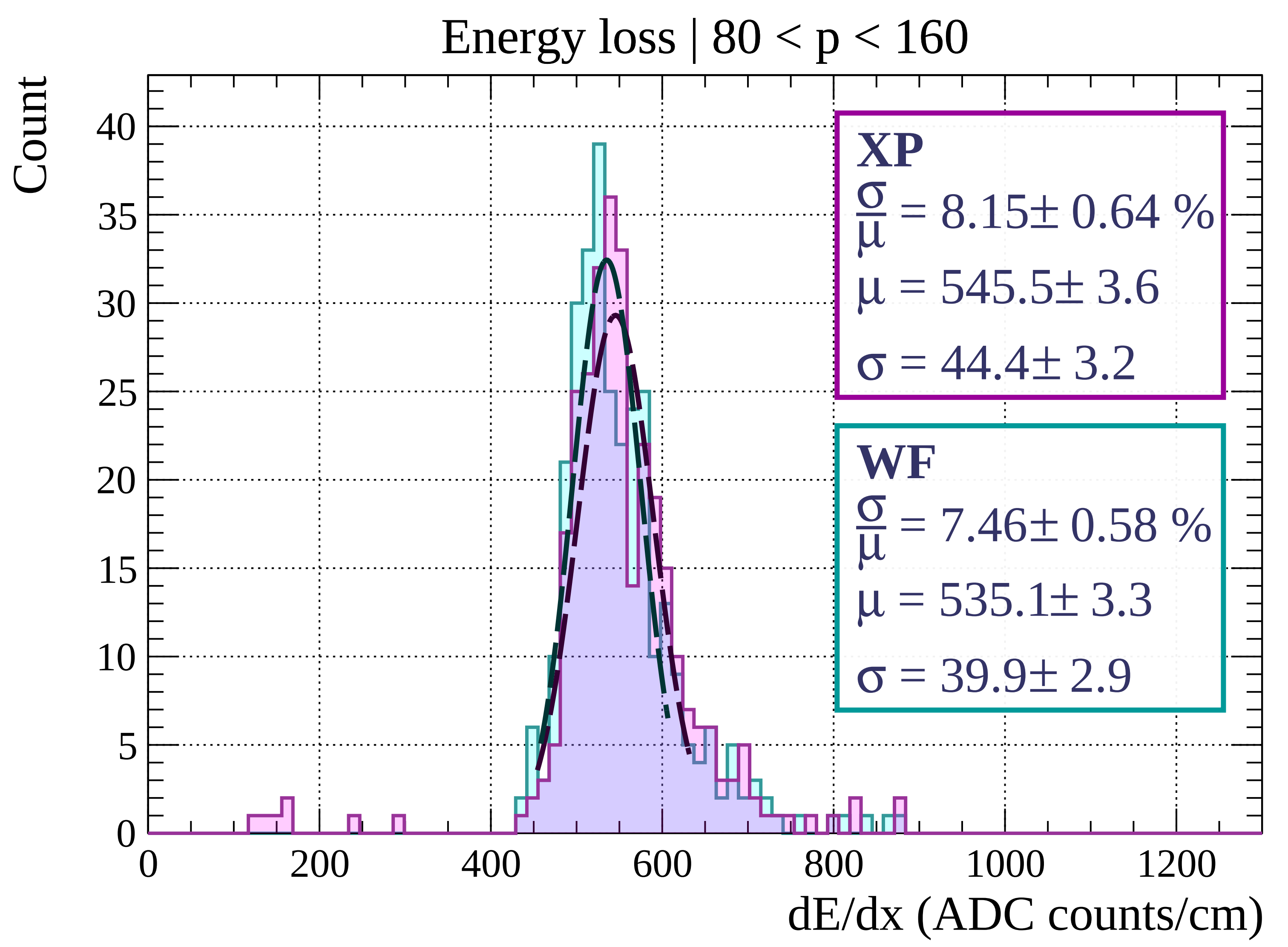

momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−50
−40
−30
−20
−10
0
10
20
30
40
50
R
350
300
250
200
150
100
50
0
×10^3



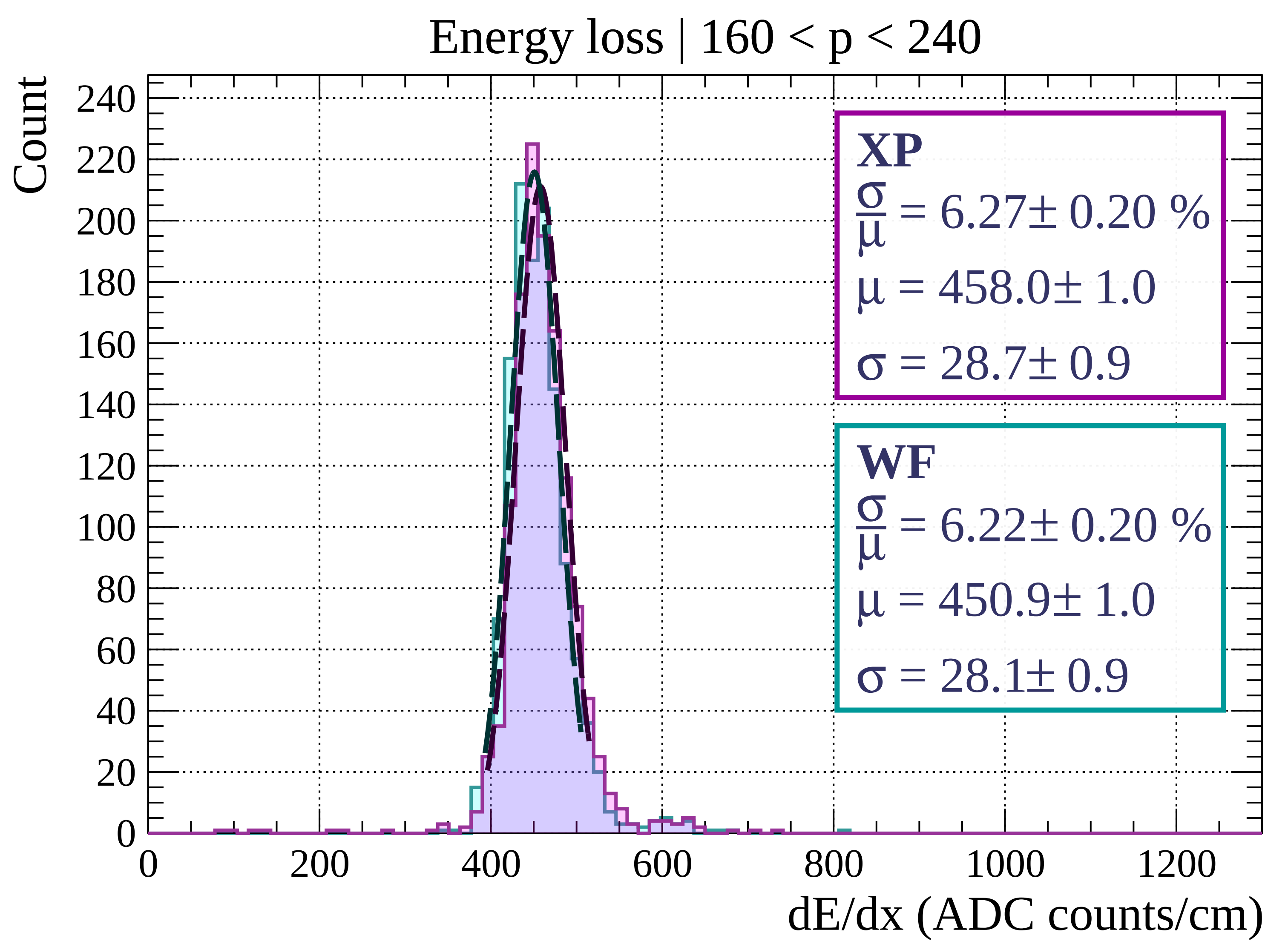
χ^2/NDF
R
$\times 10^3$
50
45
40
35
30
25
20
15
10
5
0
−50
−40
−30
−20
−10
0
10
20
30
40
50
22
20
18
16
14
12
10
8
6
4
2
0



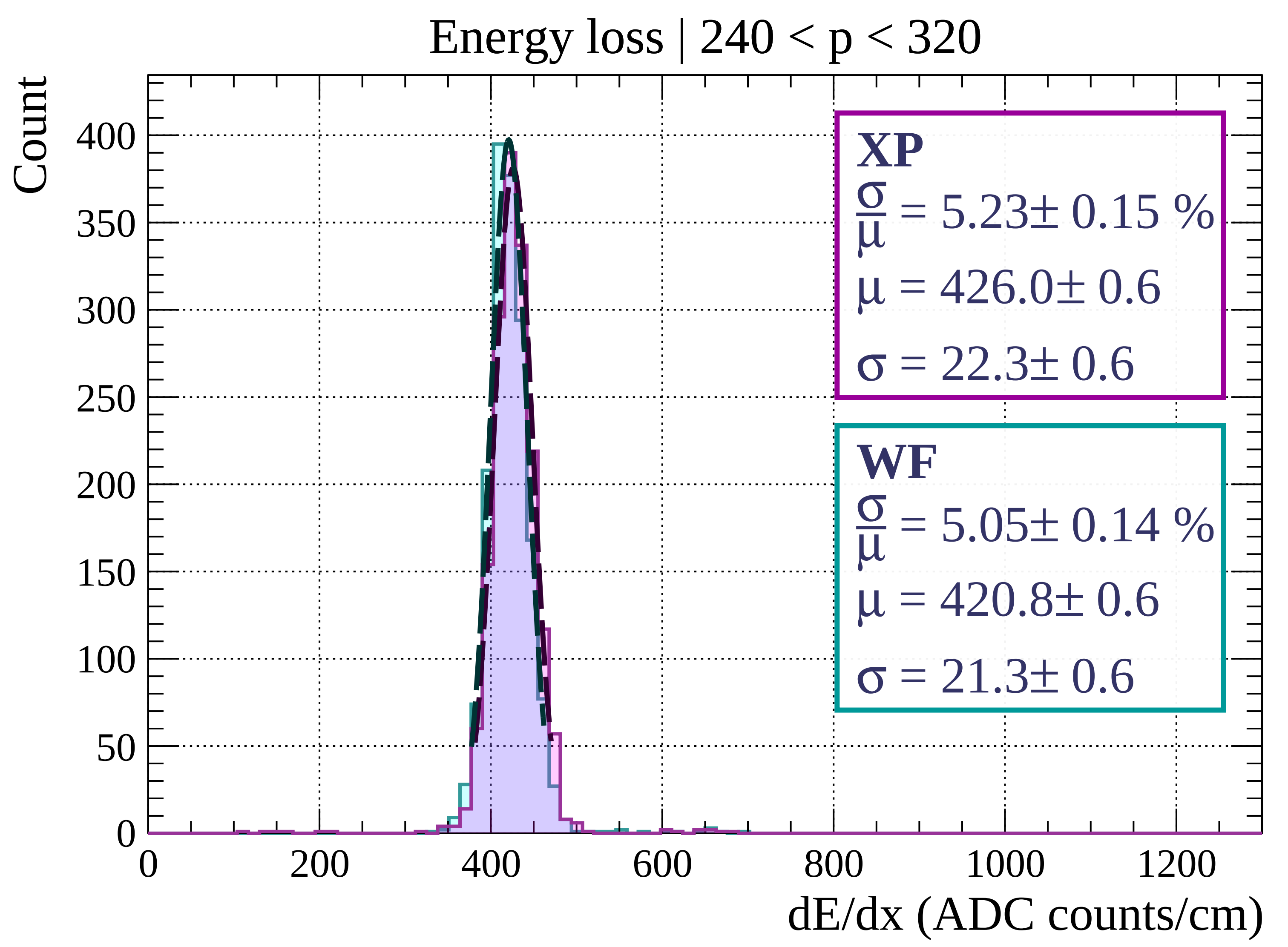
Energy loss | 0 < p < 80
Count
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 126.95± 409.57 %
μ = 618.5± 793.2
σ = 785.2± 1526.0
WF
σ/μ = 3.11± 1.25 %
μ = 550.8± 6.4
σ = 17.1± 6.7



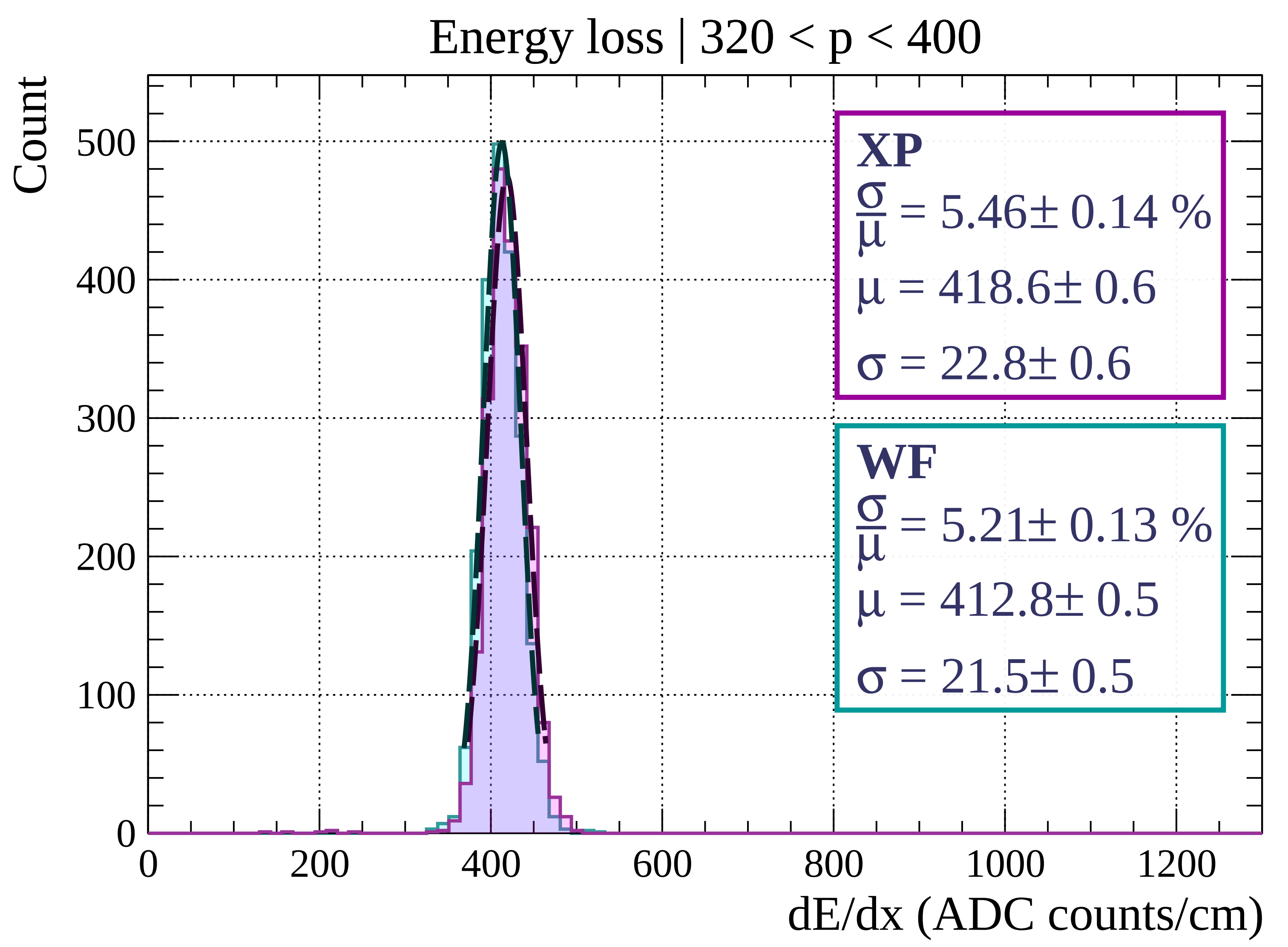
Energy loss | 80 < p < 160
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 8.15± 0.64 %
μ = 545.5± 3.6
σ = 44.4± 3.2
WF
σ/μ = 7.46± 0.58 %
μ = 535.1± 3.3
σ = 39.9± 2.9



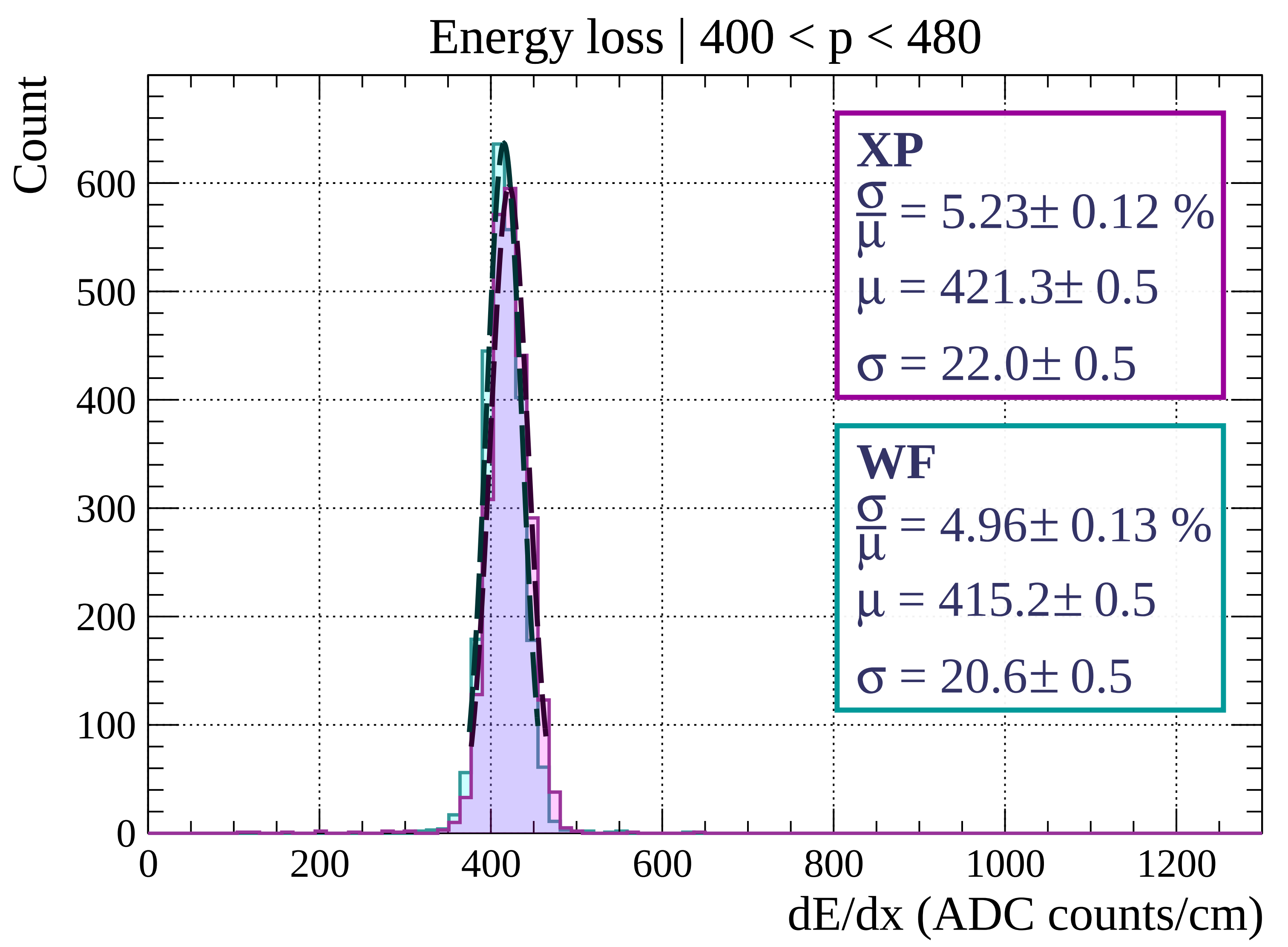
Energy loss | 160 < p < 240
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
220
240
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 6.27± 0.20 %
μ = 458.0± 1.0
σ = 28.7± 0.9
WF
$\frac{\sigma}{\mu}$ = 6.22± 0.20 %
μ = 450.9± 1.0
σ = 28.1± 0.9



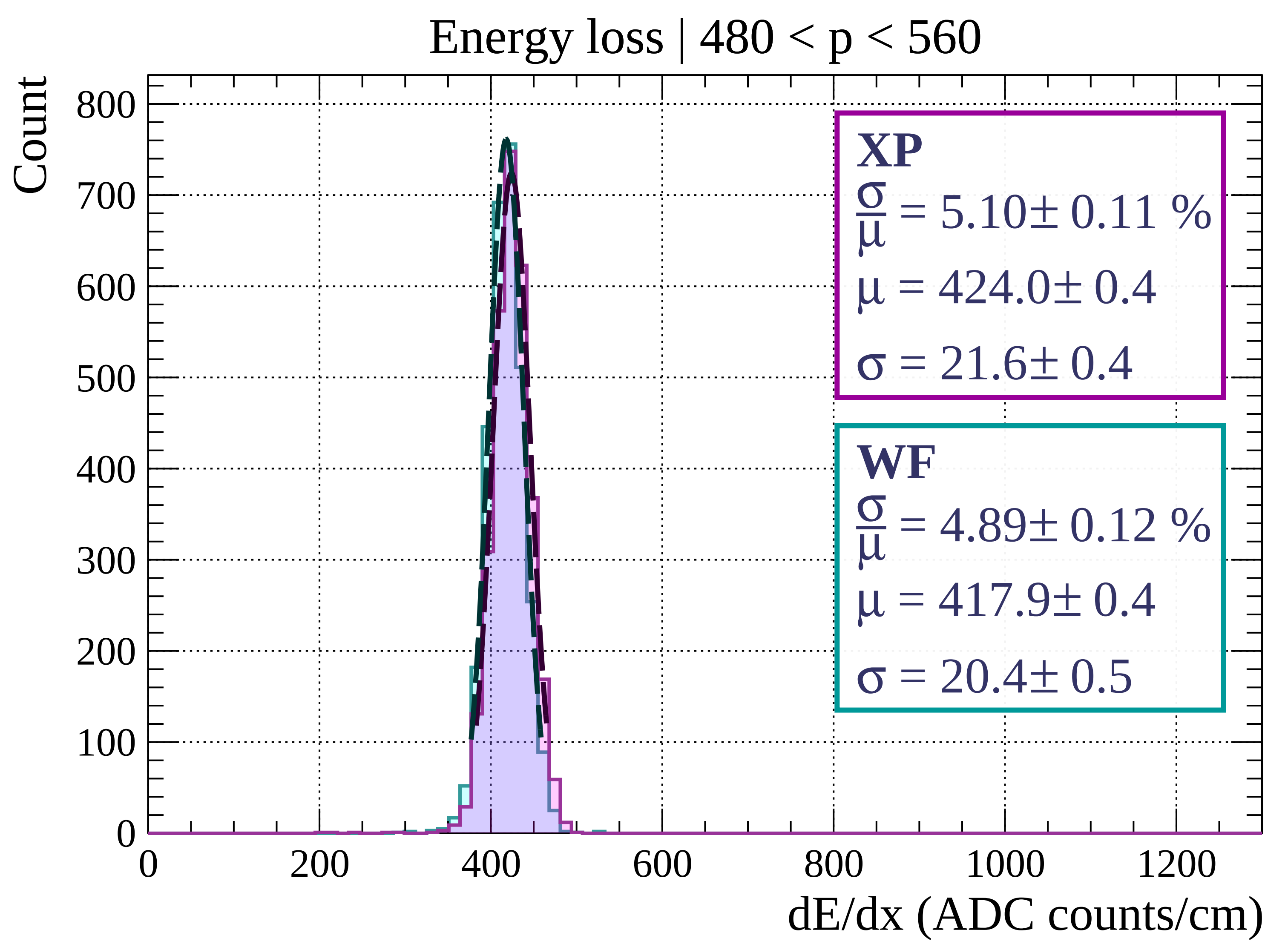
Energy loss | 240 < p < 320
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
50
100
150
200
250
300
350
400
XP
$\frac{\sigma}{\mu}$ = 5.23± 0.15 %
μ = 426.0± 0.6
σ = 22.3± 0.6
WF
$\frac{\sigma}{\mu}$ = 5.05± 0.14 %
μ = 420.8± 0.6
σ = 21.3± 0.6



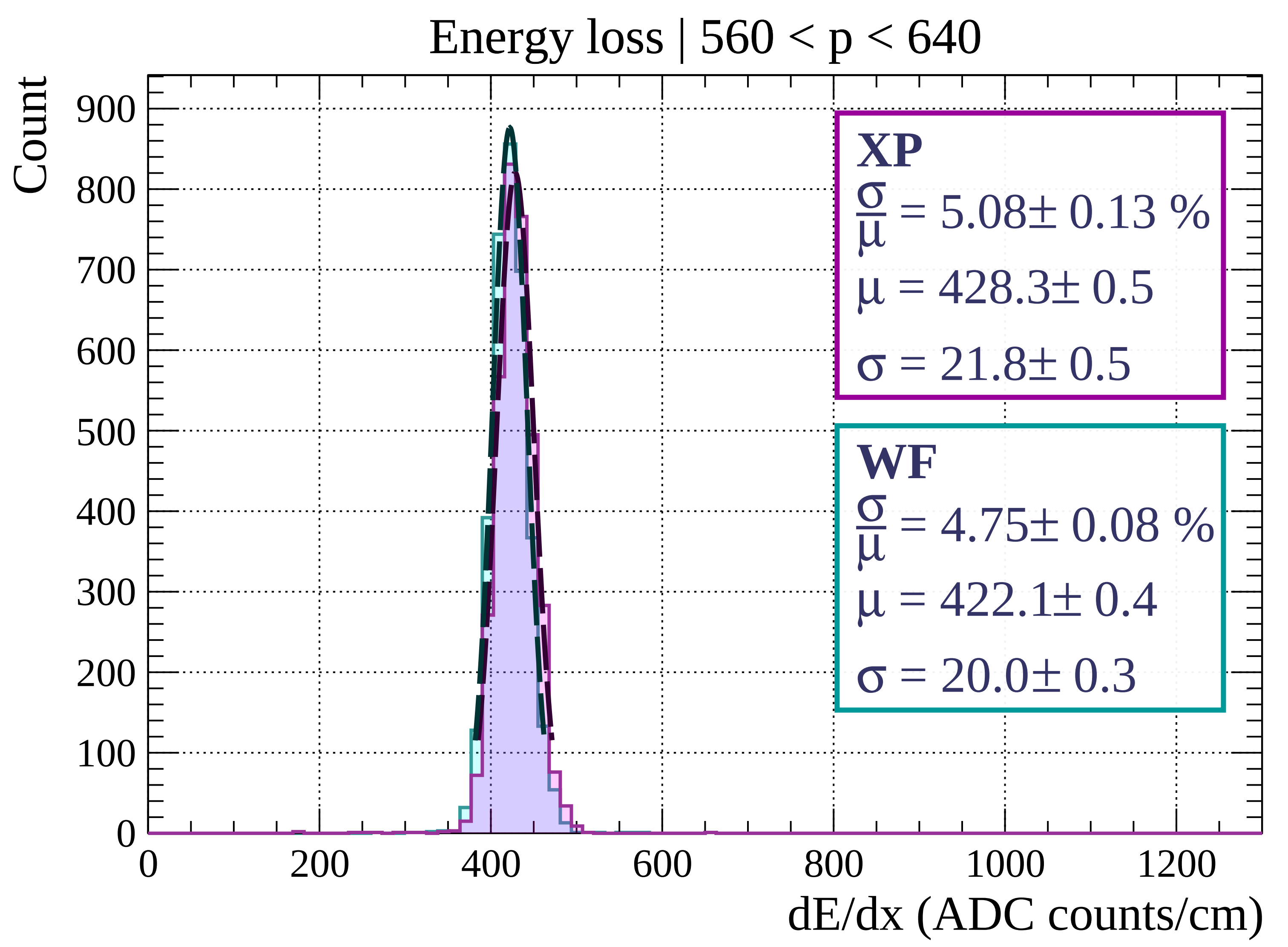
Energy loss | 320 < p < 400
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
σ/μ = 5.46± 0.14 %
μ = 418.6± 0.6
σ = 22.8± 0.6
WF
σ/μ = 5.21± 0.13 %
μ = 412.8± 0.5
σ = 21.5± 0.5

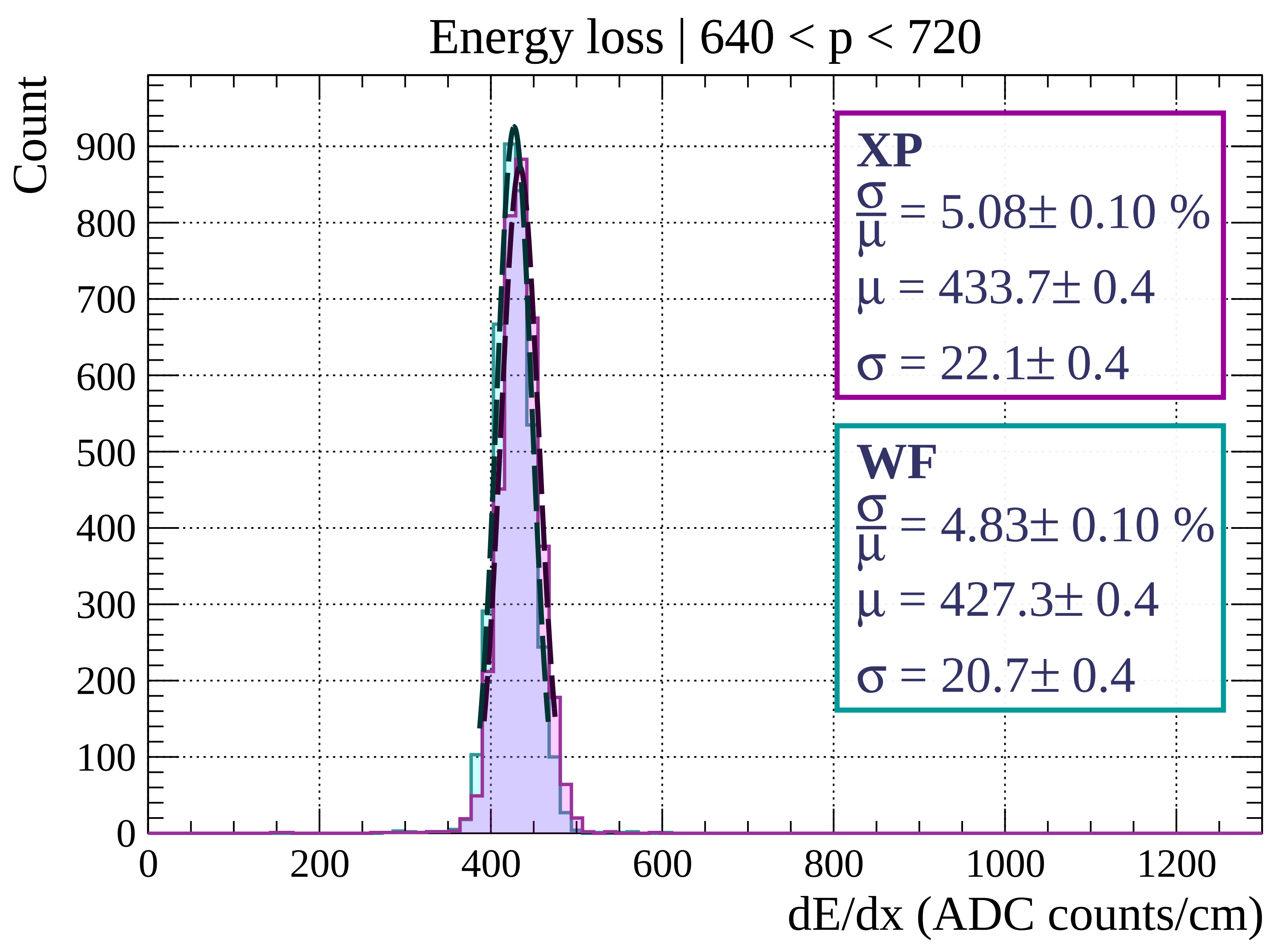


Energy loss | 400 < p < 480
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 5.23± 0.12 %
μ = 421.3± 0.5
σ = 22.0± 0.5
WF
$\frac{\sigma}{\mu}$ = 4.96± 0.13 %
μ = 415.2± 0.5
σ = 20.6± 0.5

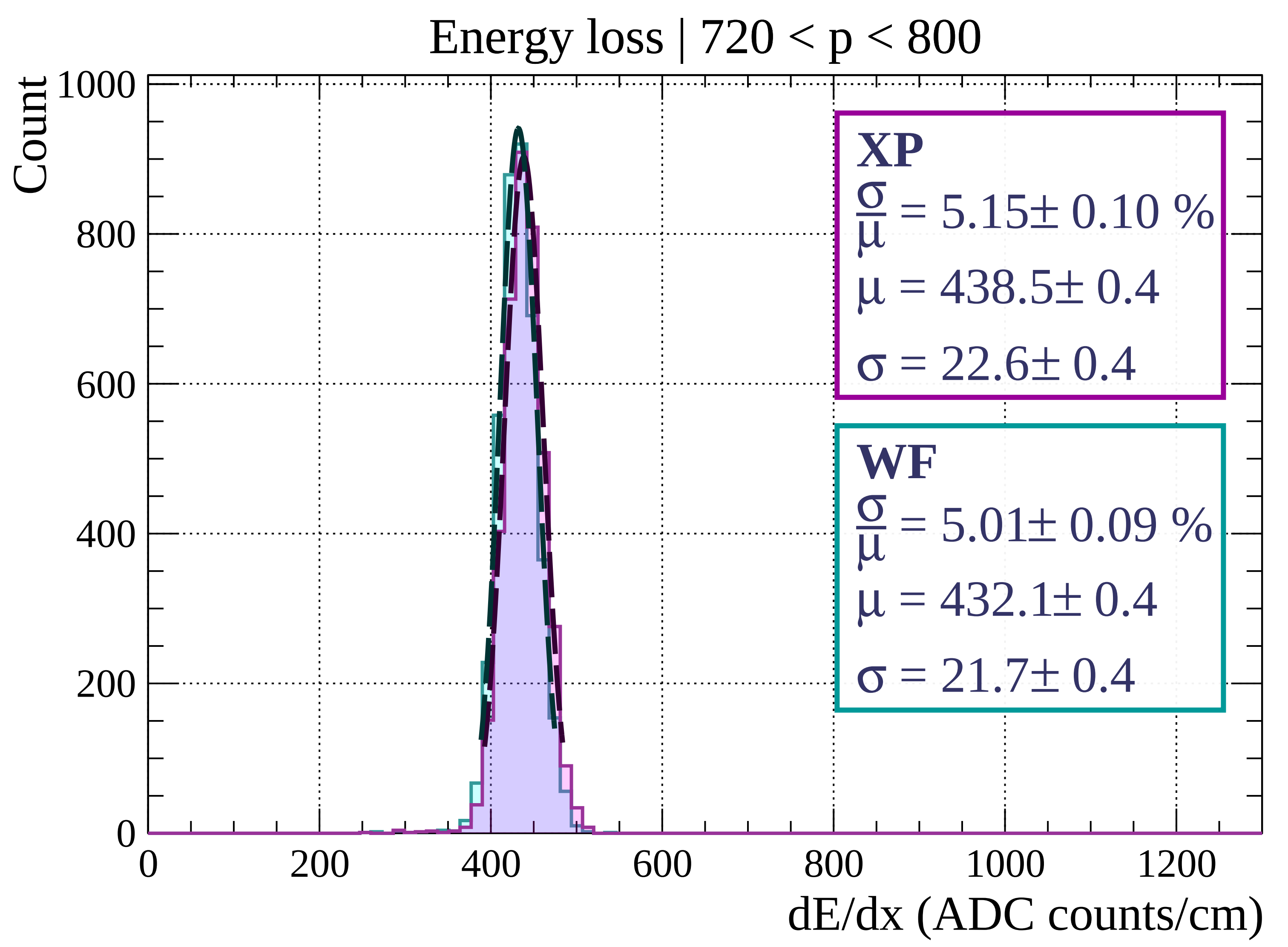


Energy loss | 480 < p < 560
Count
800
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 5.10± 0.11 %
μ = 424.0± 0.4
σ = 21.6± 0.4
WF
σ/μ = 4.89± 0.12 %
μ = 417.9± 0.4
σ = 20.4± 0.5

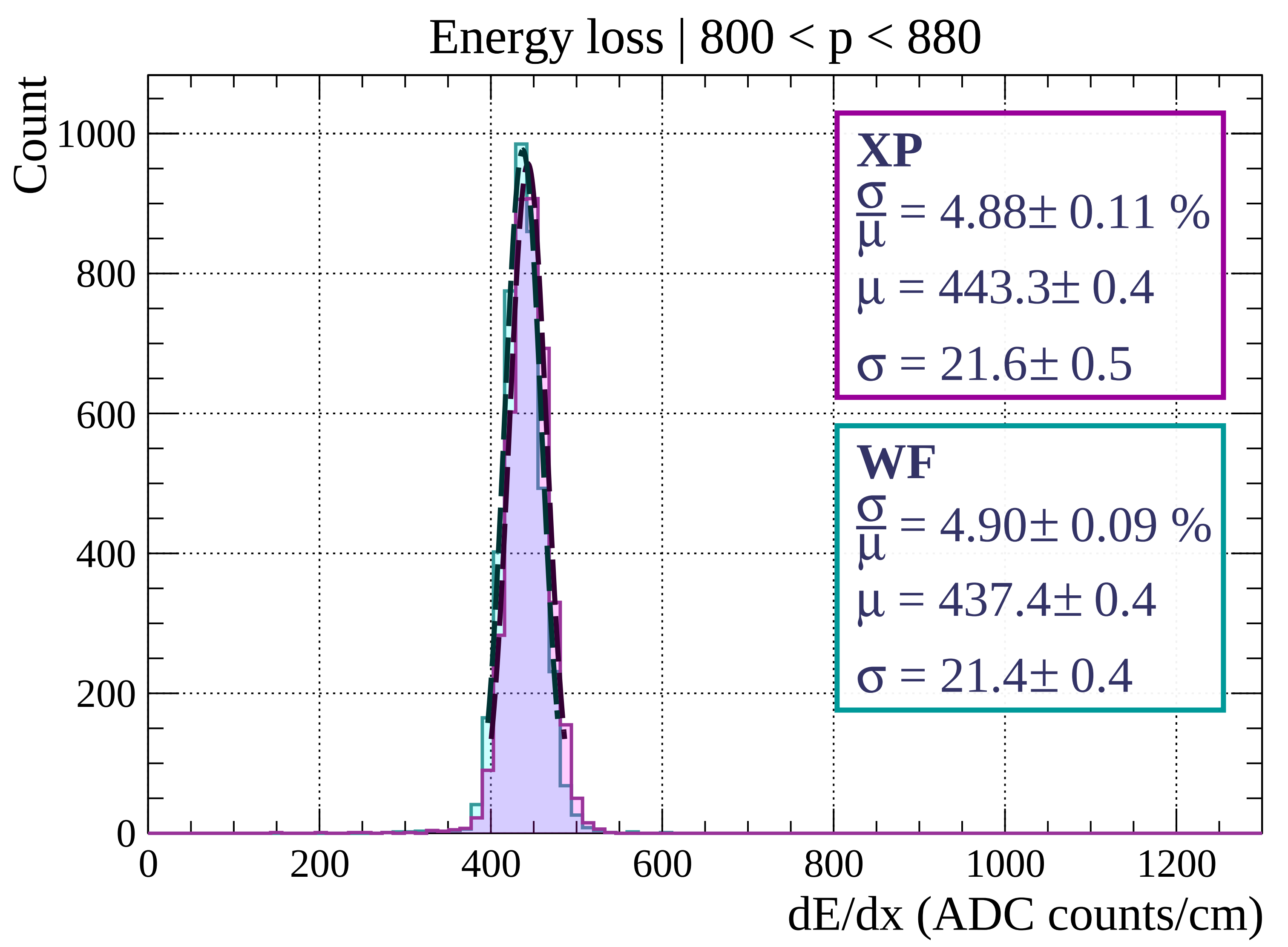

Energy loss | 560 < p < 640
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200
XP
σ/μ = 5.08± 0.13 %
μ = 428.3± 0.5
σ = 21.8± 0.5
WF
σ/μ = 4.75± 0.08 %
μ = 422.1± 0.4
σ = 20.0± 0.3



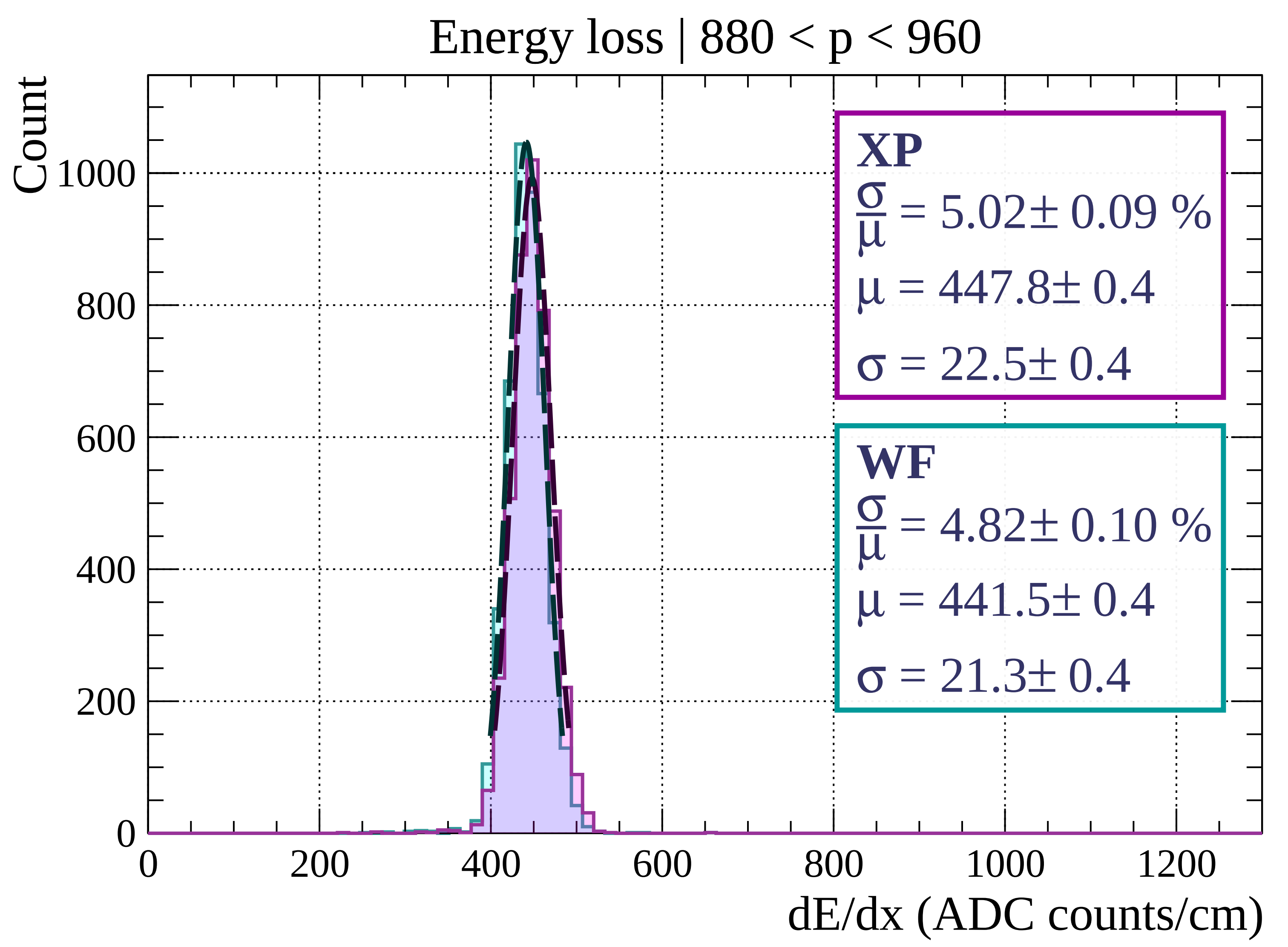
Energy loss | 640 < p < 720
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 5.08± 0.10 %
μ = 433.7± 0.4
σ = 22.1± 0.4
WF
$\frac{\sigma}{\mu}$ = 4.83± 0.10 %
μ = 427.3± 0.4
σ = 20.7± 0.4



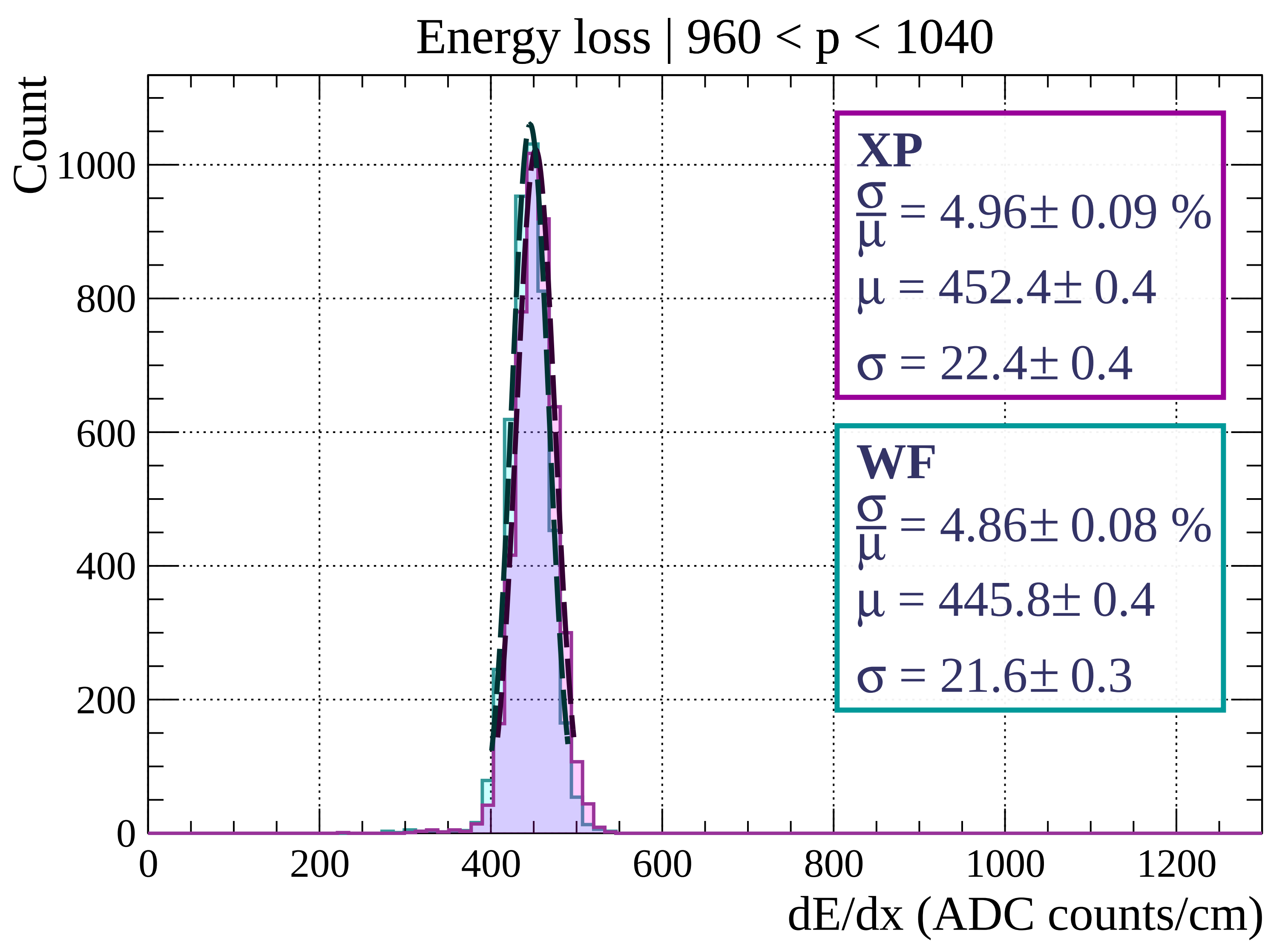
Energy loss | 720 < p < 800
Count
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 5.15± 0.10 %
μ = 438.5± 0.4
σ = 22.6± 0.4
WF
σ/μ = 5.01± 0.09 %
μ = 432.1± 0.4
σ = 21.7± 0.4



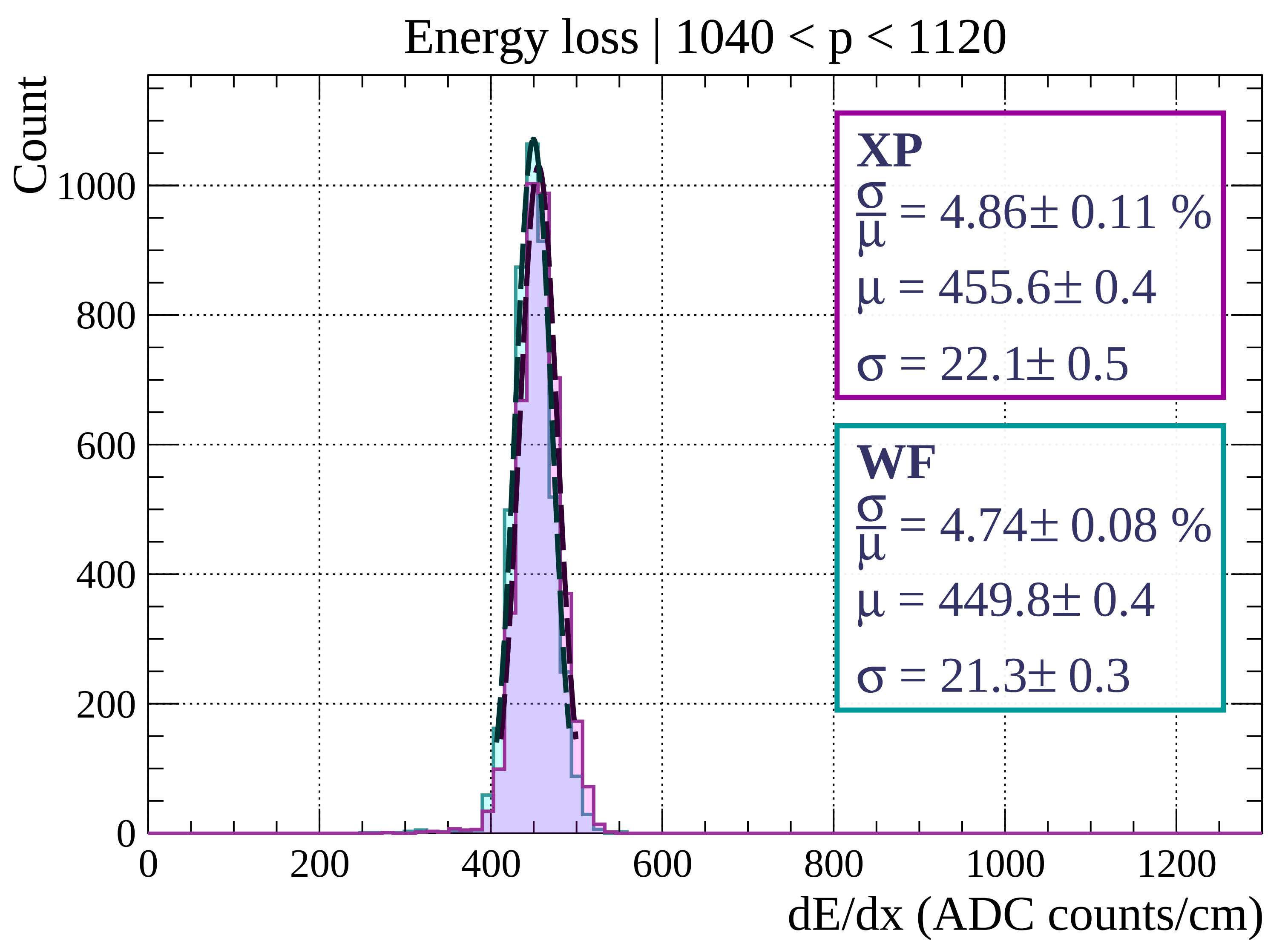
Energy loss | 800 < p < 880
Count
XP
σ/μ = 4.88± 0.11 %
μ = 443.3± 0.4
σ = 21.6± 0.5
WF
σ/μ = 4.90± 0.09 %
μ = 437.4± 0.4
σ = 21.4± 0.4
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)



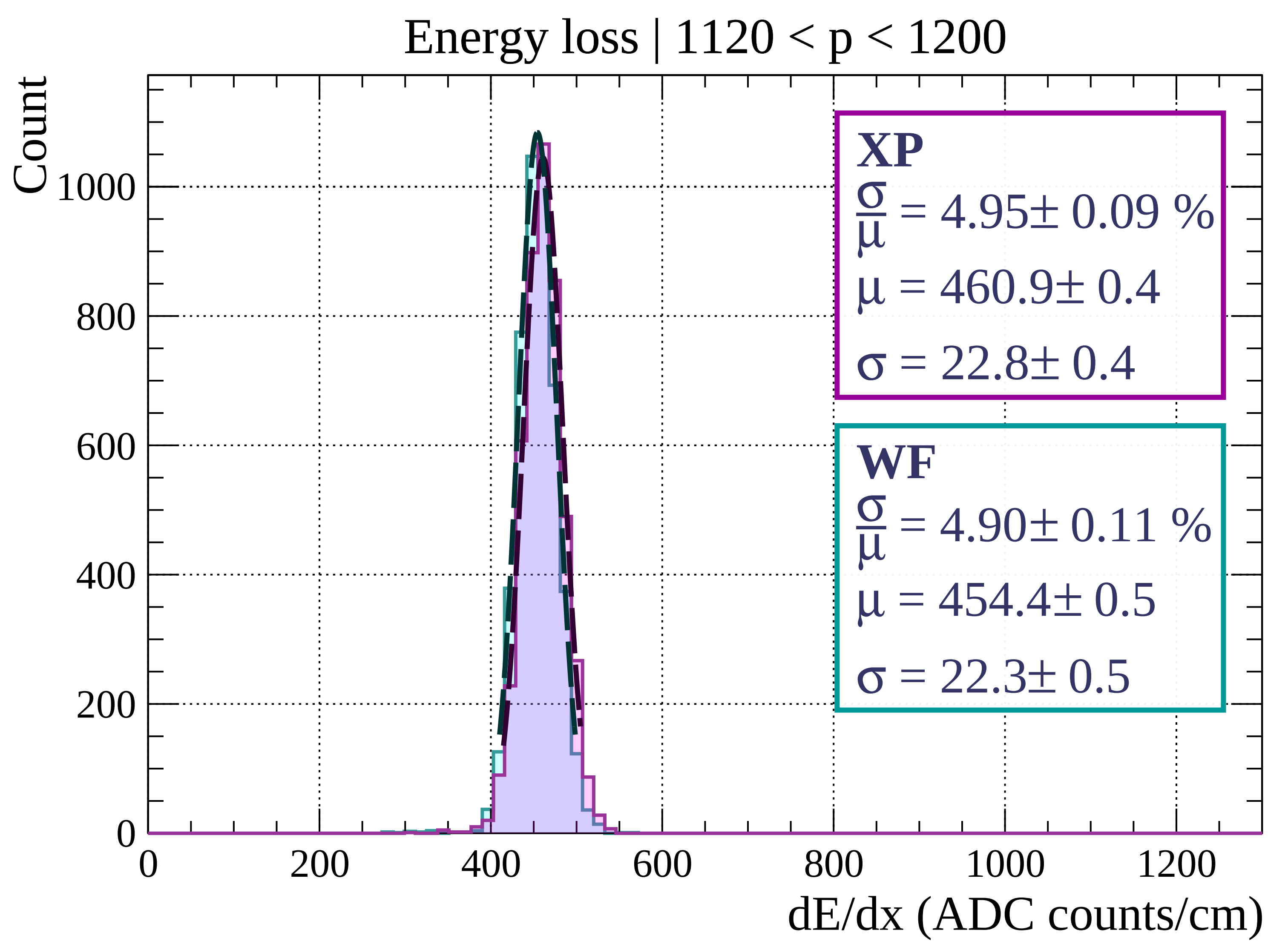
Energy loss | 880 < p < 960
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 5.02± 0.09 %
μ = 447.8± 0.4
σ = 22.5± 0.4
WF
$\frac{\sigma}{\mu}$ = 4.82± 0.10 %
μ = 441.5± 0.4
σ = 21.3± 0.4



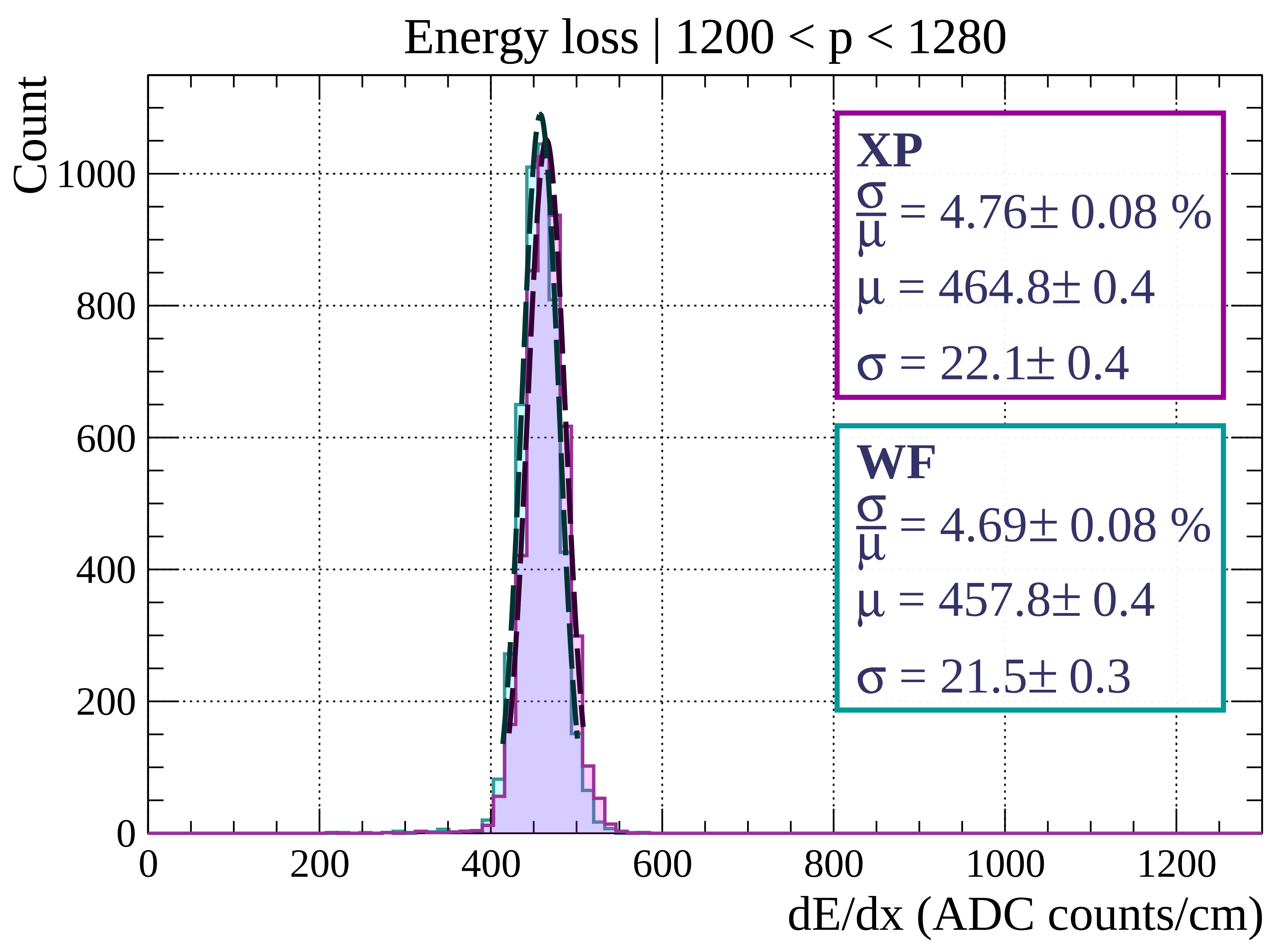
Energy loss | 960 < p < 1040
Count
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 4.96± 0.09 %
μ = 452.4± 0.4
σ = 22.4± 0.4
WF
σ/μ = 4.86± 0.08 %
μ = 445.8± 0.4
σ = 21.6± 0.3



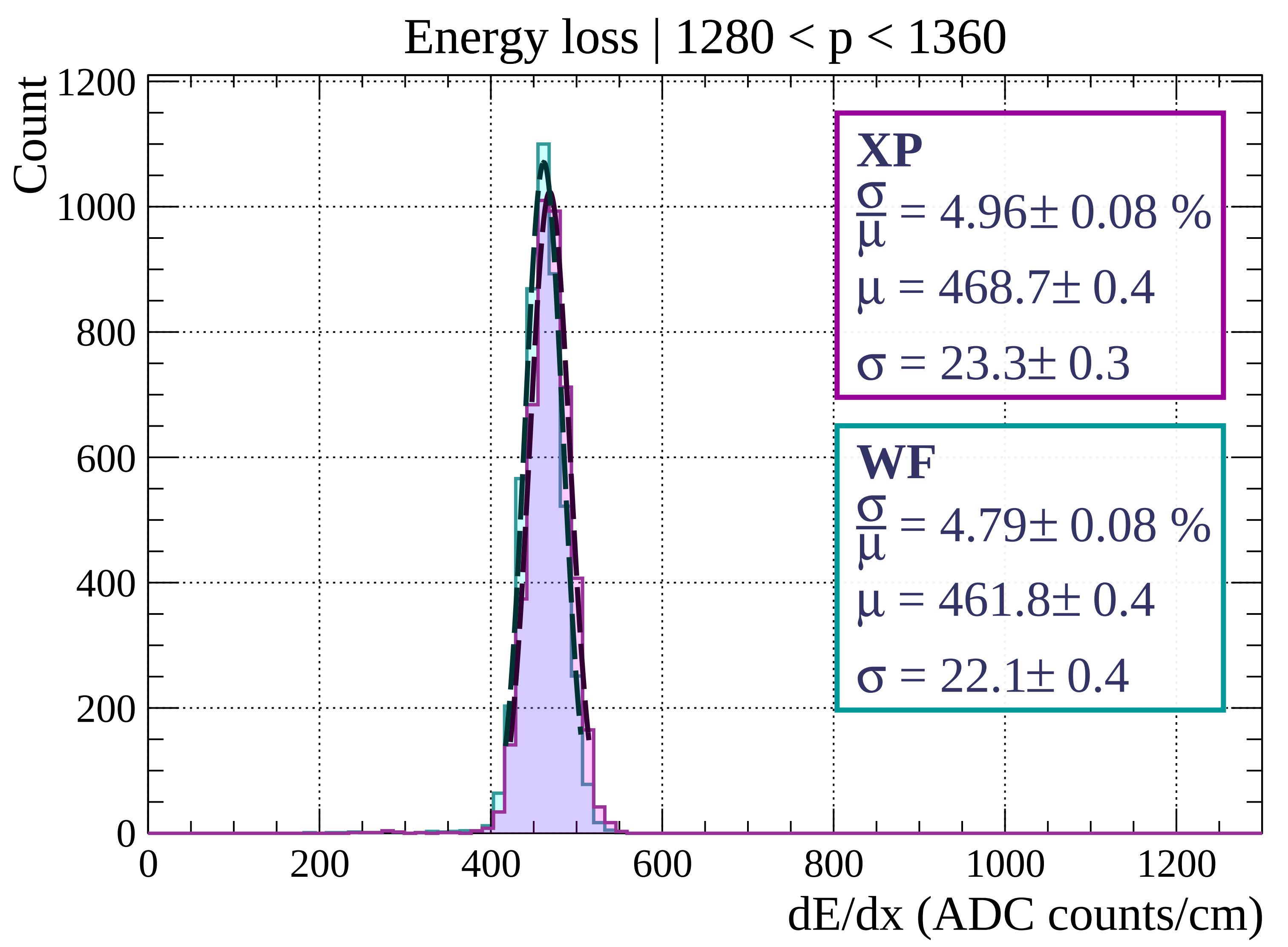
Energy loss | 1040 < p < 1120
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 4.86± 0.11 %
μ = 455.6± 0.4
σ = 22.1± 0.5
WF
σ/μ = 4.74± 0.08 %
μ = 449.8± 0.4
σ = 21.3± 0.3



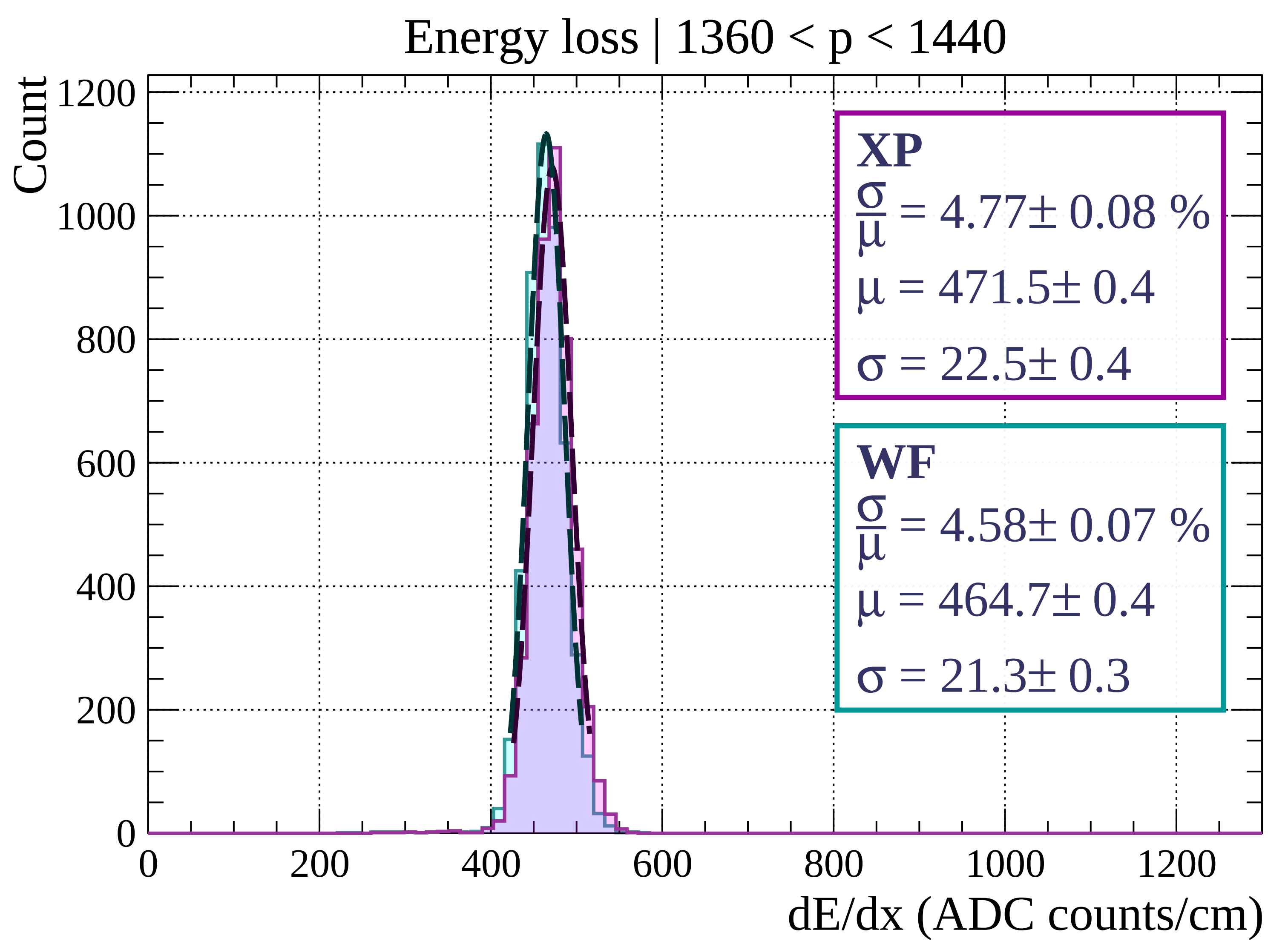
Energy loss | 1120 < p < 1200
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 4.95± 0.09 %
μ = 460.9± 0.4
σ = 22.8± 0.4
WF
$\frac{\sigma}{\mu}$ = 4.90± 0.11 %
μ = 454.4± 0.5
σ = 22.3± 0.5



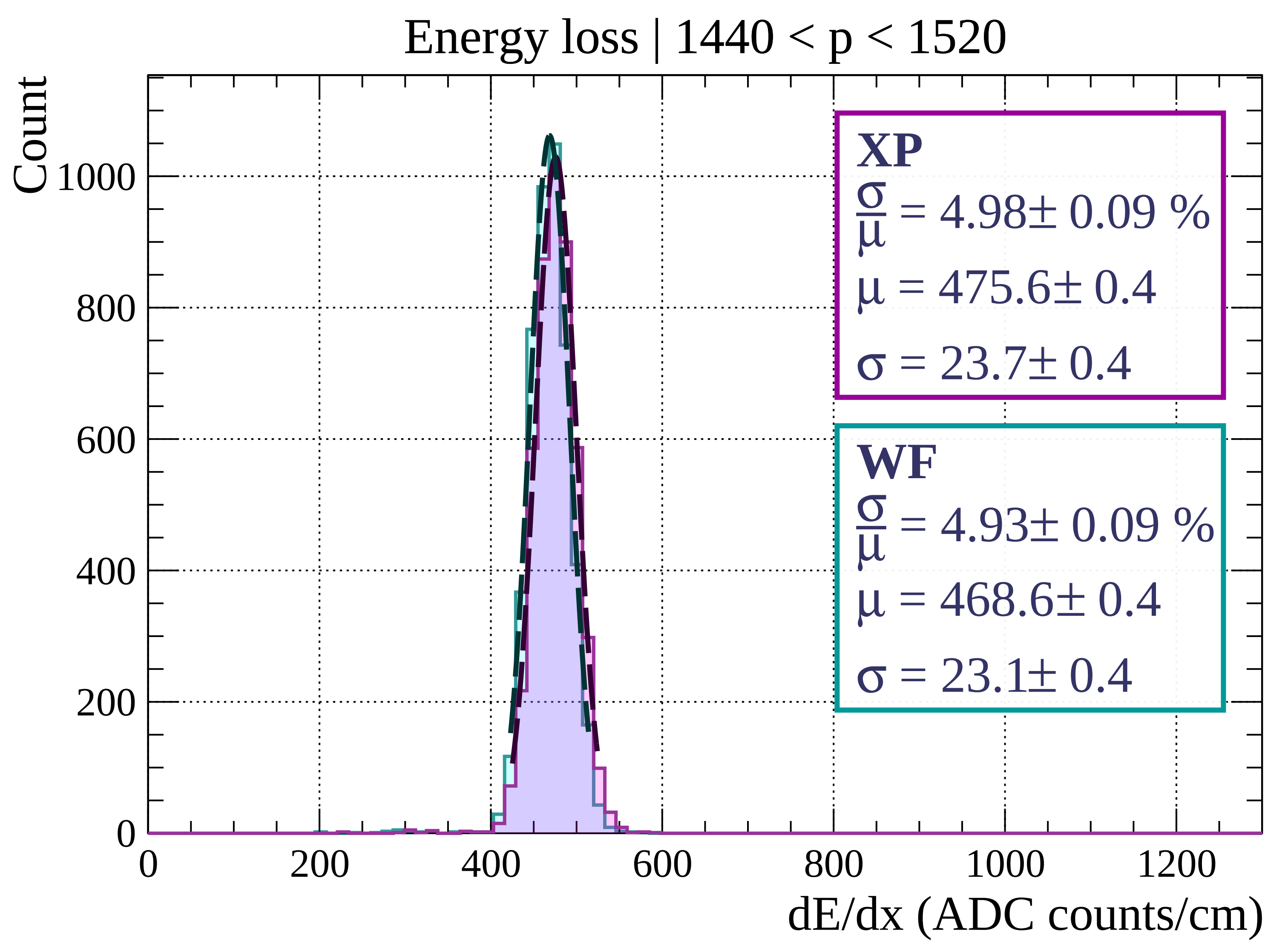
Energy loss | 1200 < p < 1280
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 4.76± 0.08 %
μ = 464.8± 0.4
σ = 22.1± 0.4
WF
$\frac{\sigma}{\mu}$ = 4.69± 0.08 %
μ = 457.8± 0.4
σ = 21.5± 0.3



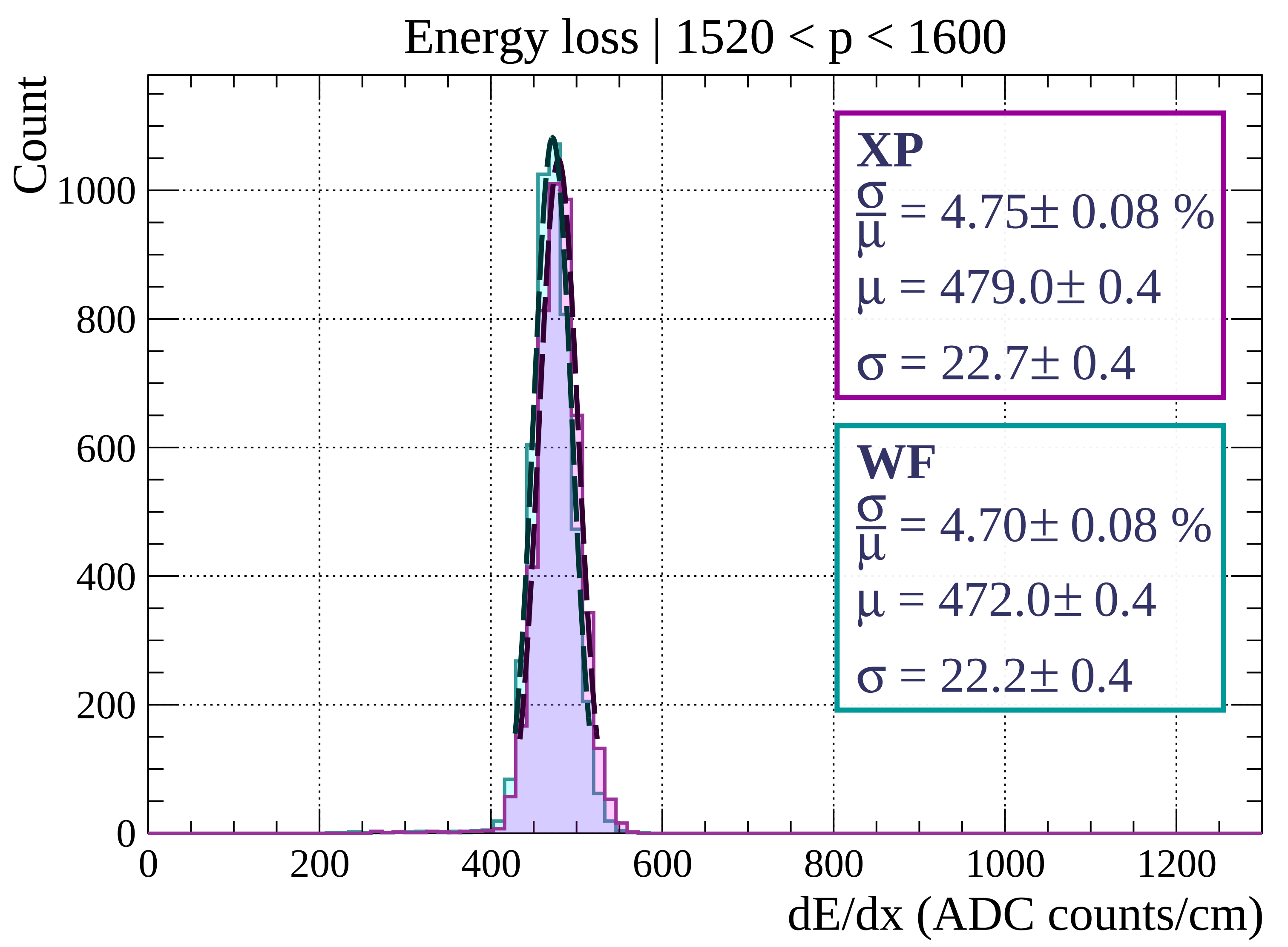
Energy loss | 1280 < p < 1360
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP
σ/μ = 4.96± 0.08 %
μ = 468.7± 0.4
σ = 23.3± 0.3
WF
σ/μ = 4.79± 0.08 %
μ = 461.8± 0.4
σ = 22.1± 0.4



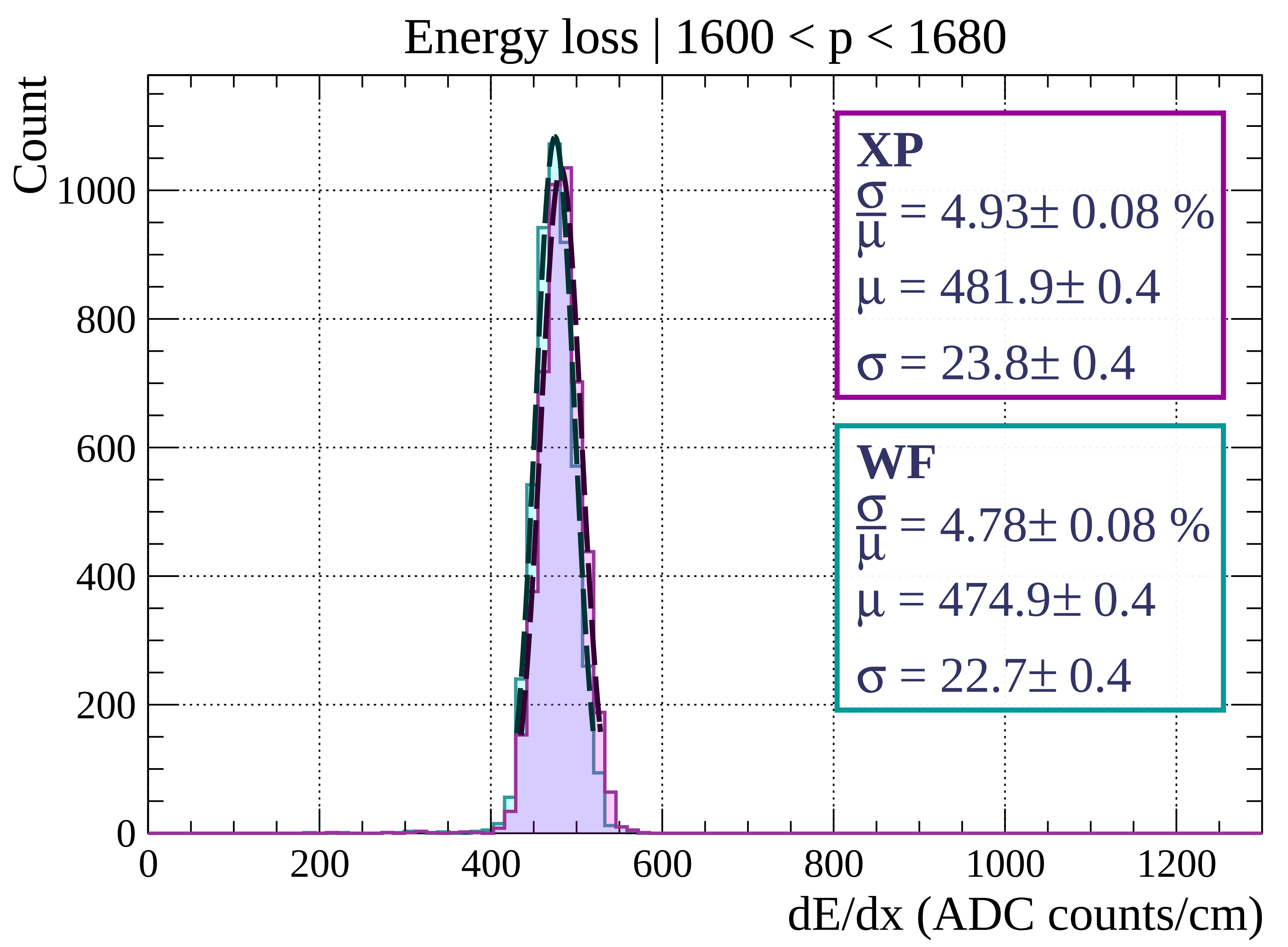
Energy loss | 1360 < p < 1440
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 4.77± 0.08 %
μ = 471.5± 0.4
σ = 22.5± 0.4
WF
σ/μ = 4.58± 0.07 %
μ = 464.7± 0.4
σ = 21.3± 0.3

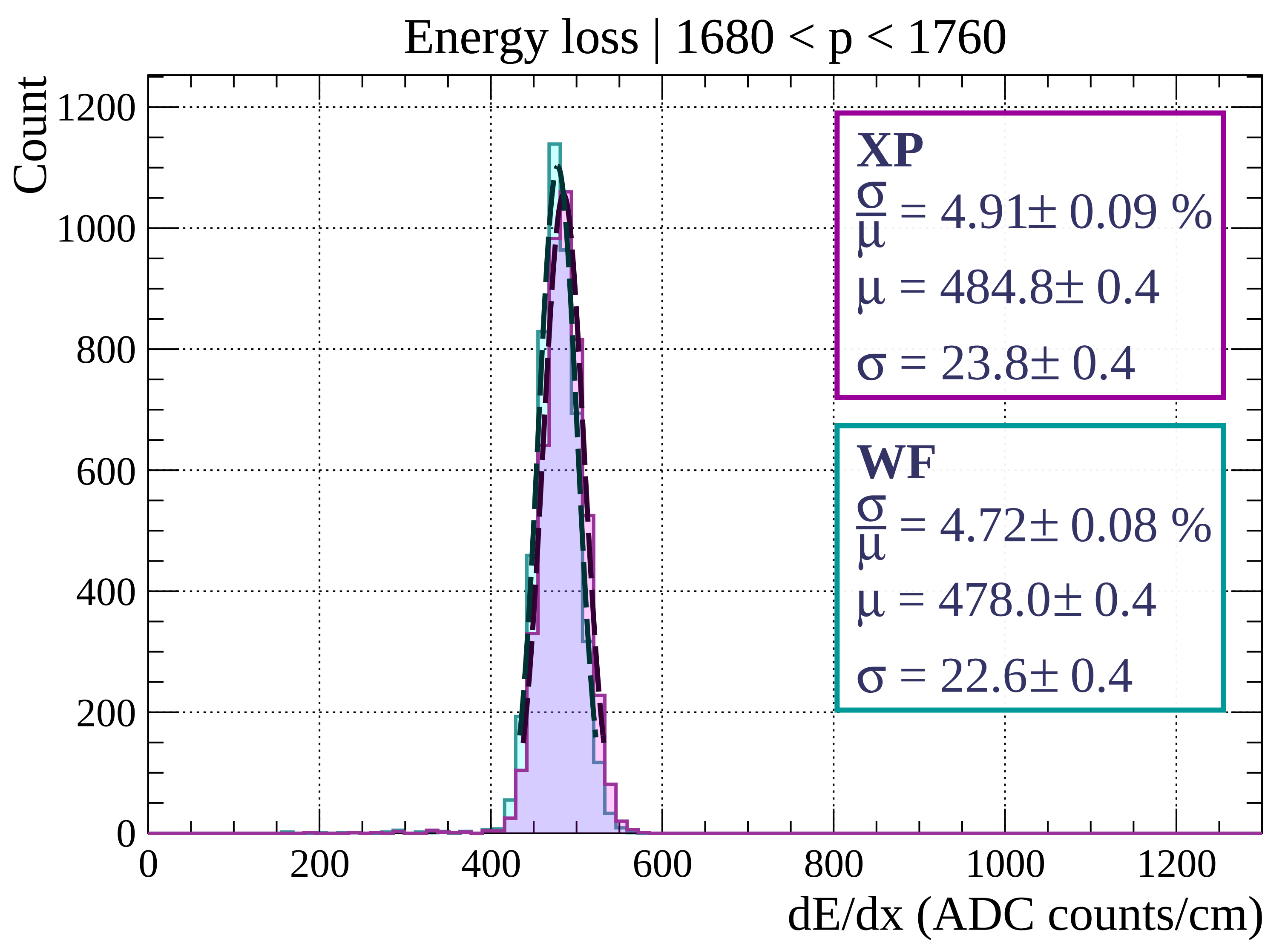


Energy loss | 1440 < p < 1520
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 4.98± 0.09 %
μ = 475.6± 0.4
σ = 23.7± 0.4
WF
$\frac{\sigma}{\mu}$ = 4.93± 0.09 %
μ = 468.6± 0.4
σ = 23.1± 0.4

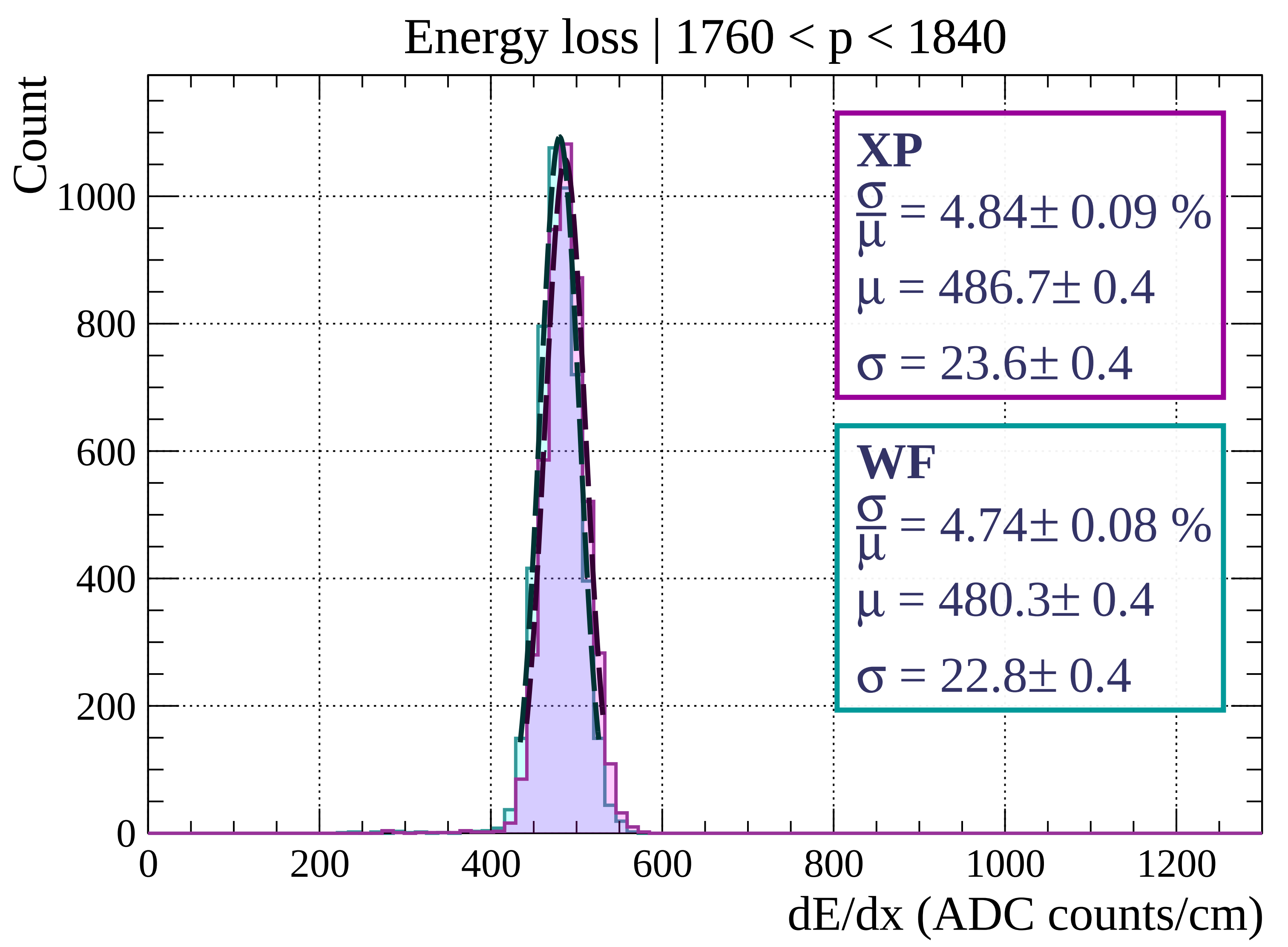


Energy loss | 1520 < p < 1600
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP
σ/μ = 4.75± 0.08 %
μ = 479.0± 0.4
σ = 22.7± 0.4
WF
σ/μ = 4.70± 0.08 %
μ = 472.0± 0.4
σ = 22.2± 0.4

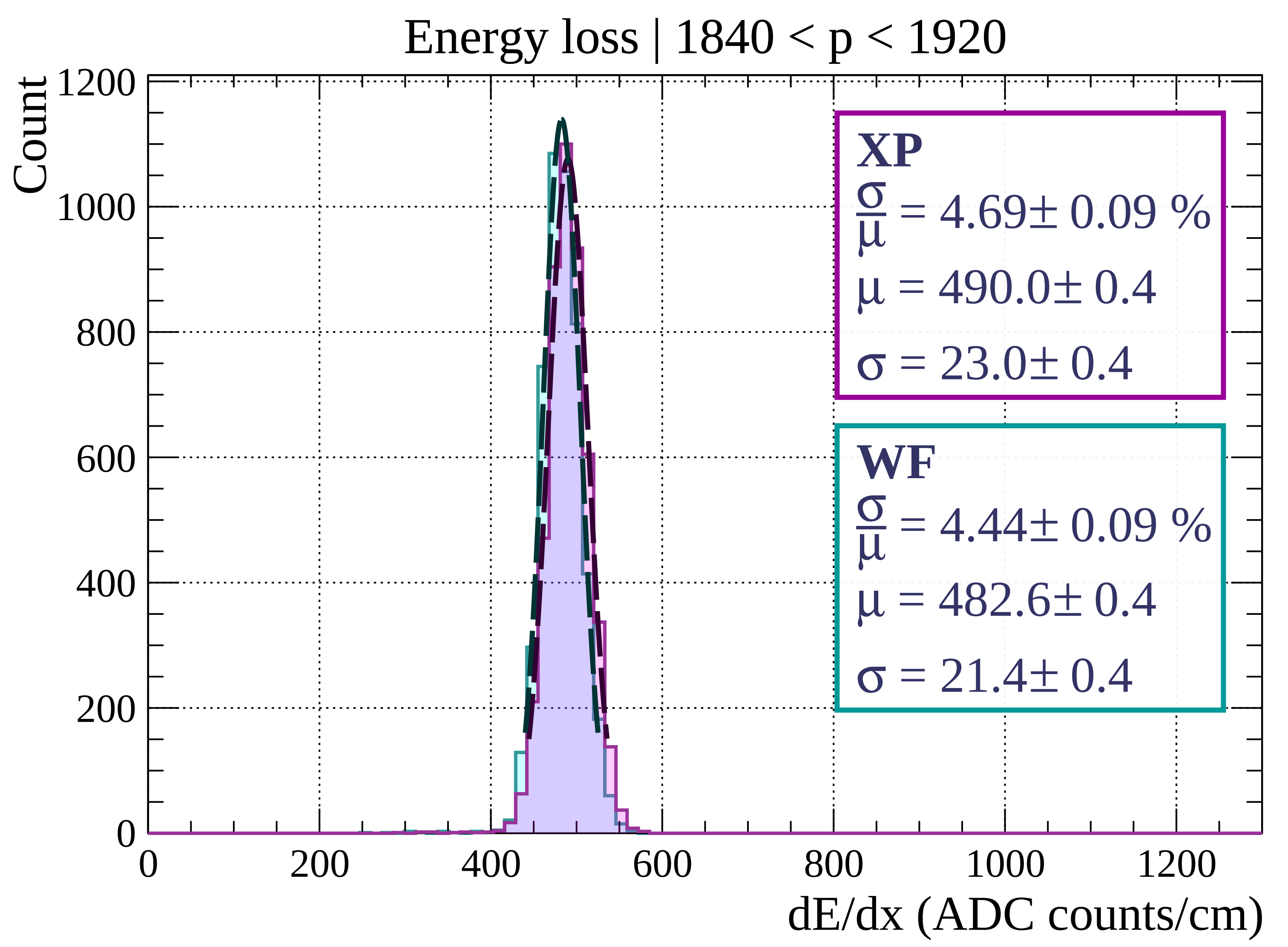

Energy loss | 1600 < p < 1680
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 4.93± 0.08 %
μ = 481.9± 0.4
σ = 23.8± 0.4
WF
$\frac{\sigma}{\mu}$ = 4.78± 0.08 %
μ = 474.9± 0.4
σ = 22.7± 0.4

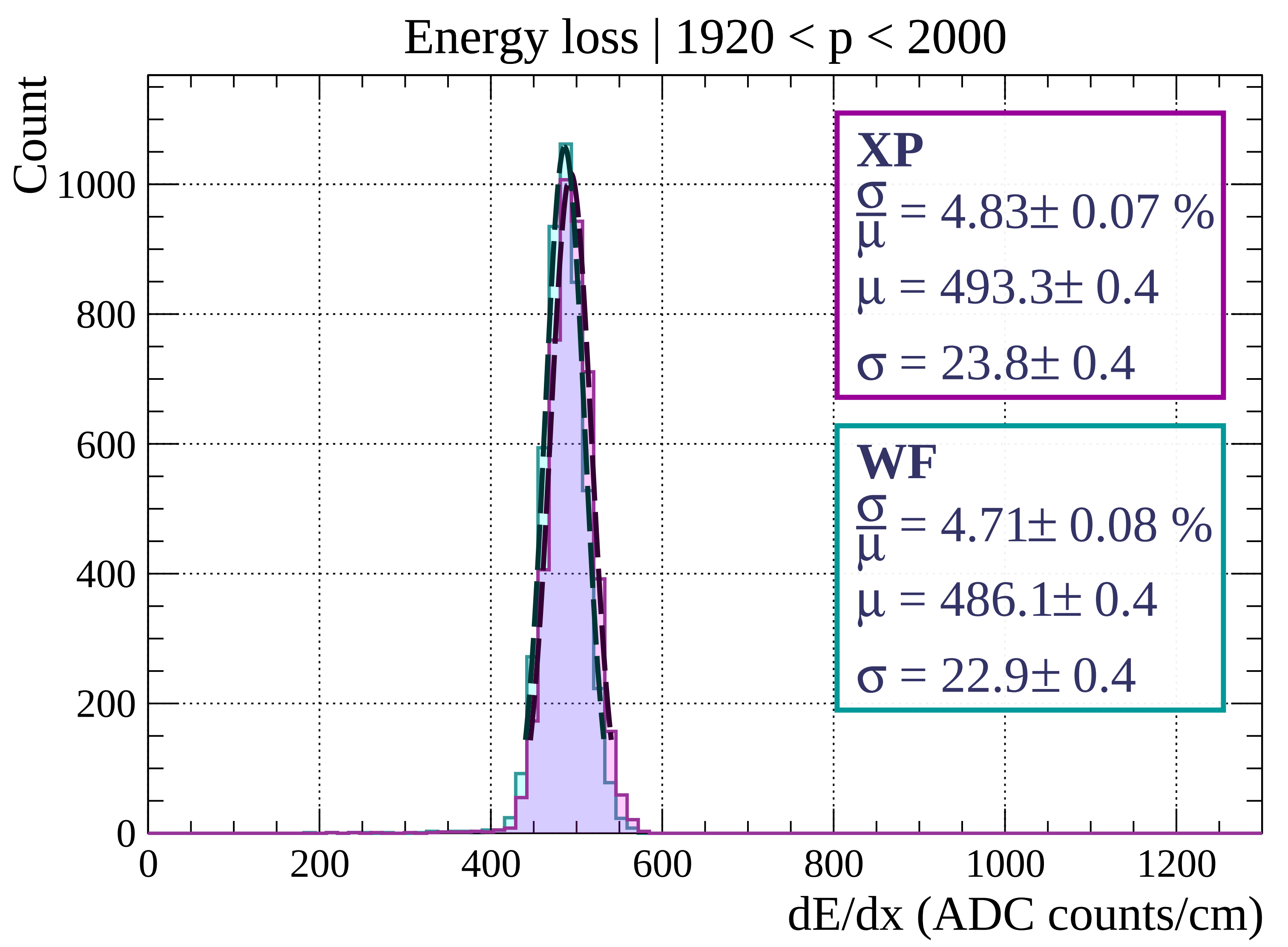


Energy loss | 1680 < p < 1760
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 4.91± 0.09 %
μ = 484.8± 0.4
σ = 23.8± 0.4
WF
$\frac{\sigma}{\mu}$ = 4.72± 0.08 %
μ = 478.0± 0.4
σ = 22.6± 0.4

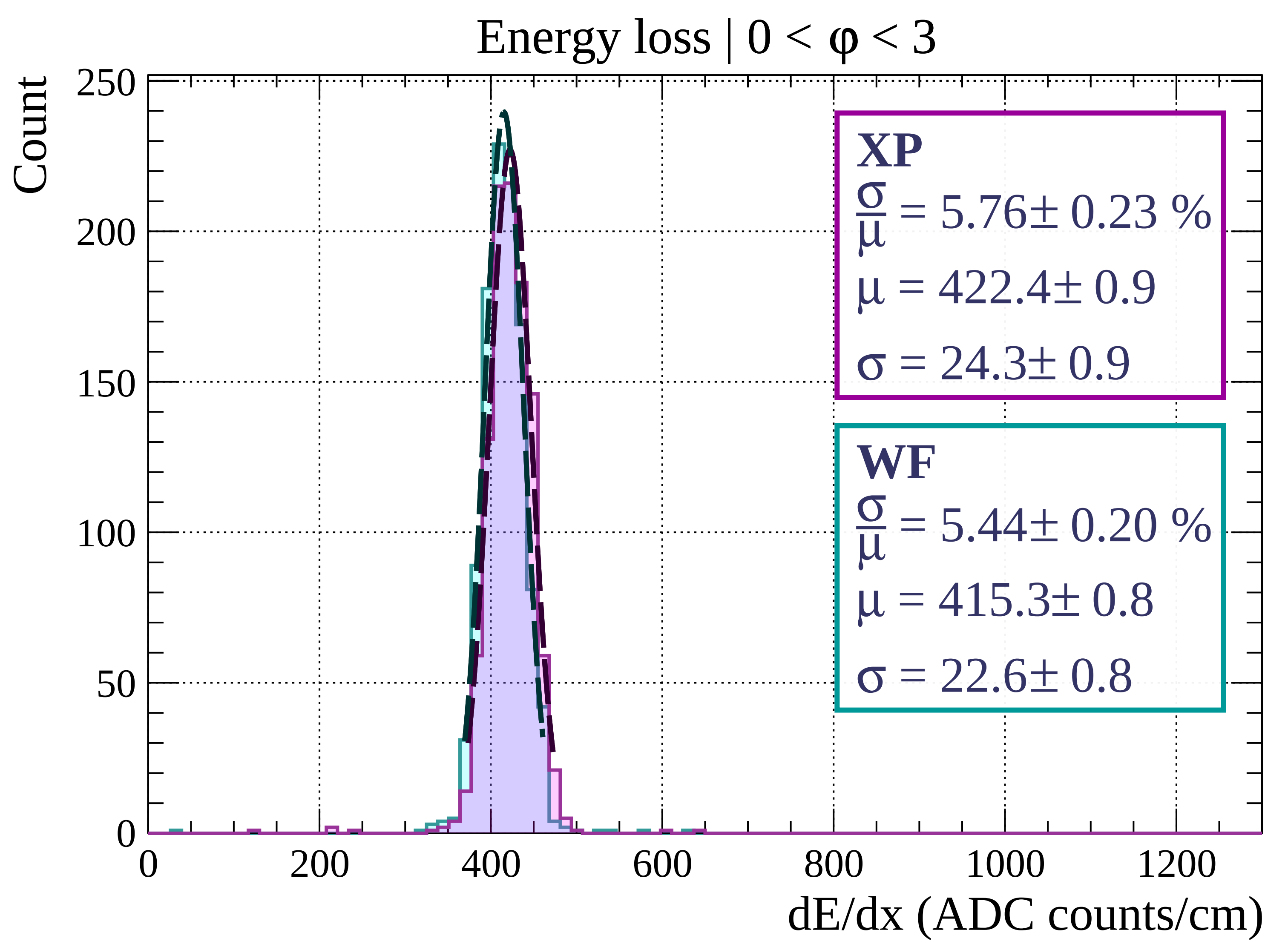


Energy loss | 1760 < p < 1840
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 4.84± 0.09 %
μ = 486.7± 0.4
σ = 23.6± 0.4
WF
$\frac{\sigma}{\mu}$ = 4.74± 0.08 %
μ = 480.3± 0.4
σ = 22.8± 0.4

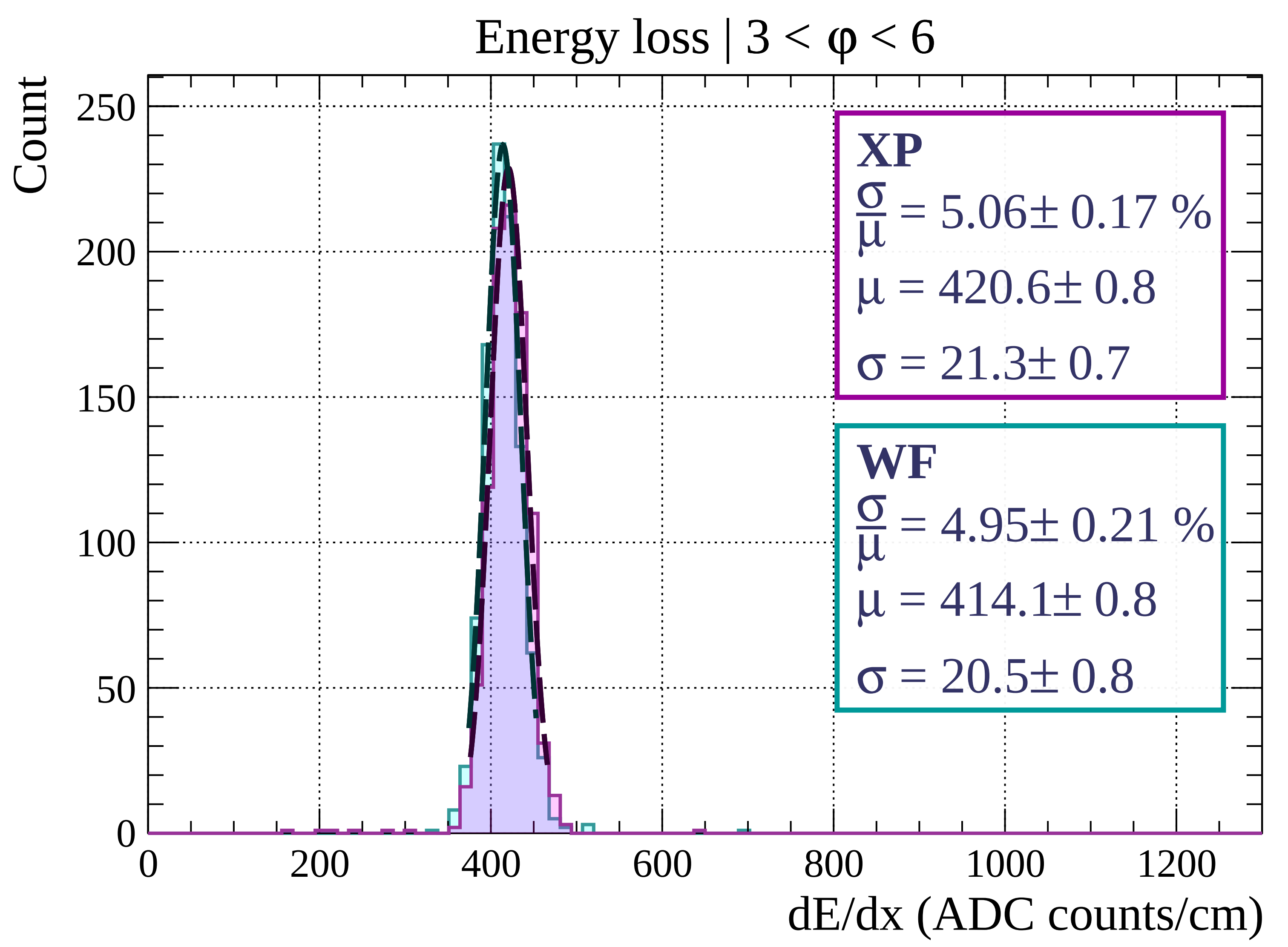

Energy loss | 1840 < p < 1920
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP
σ/μ = 4.69± 0.09 %
μ = 490.0± 0.4
σ = 23.0± 0.4
WF
σ/μ = 4.44± 0.09 %
μ = 482.6± 0.4
σ = 21.4± 0.4



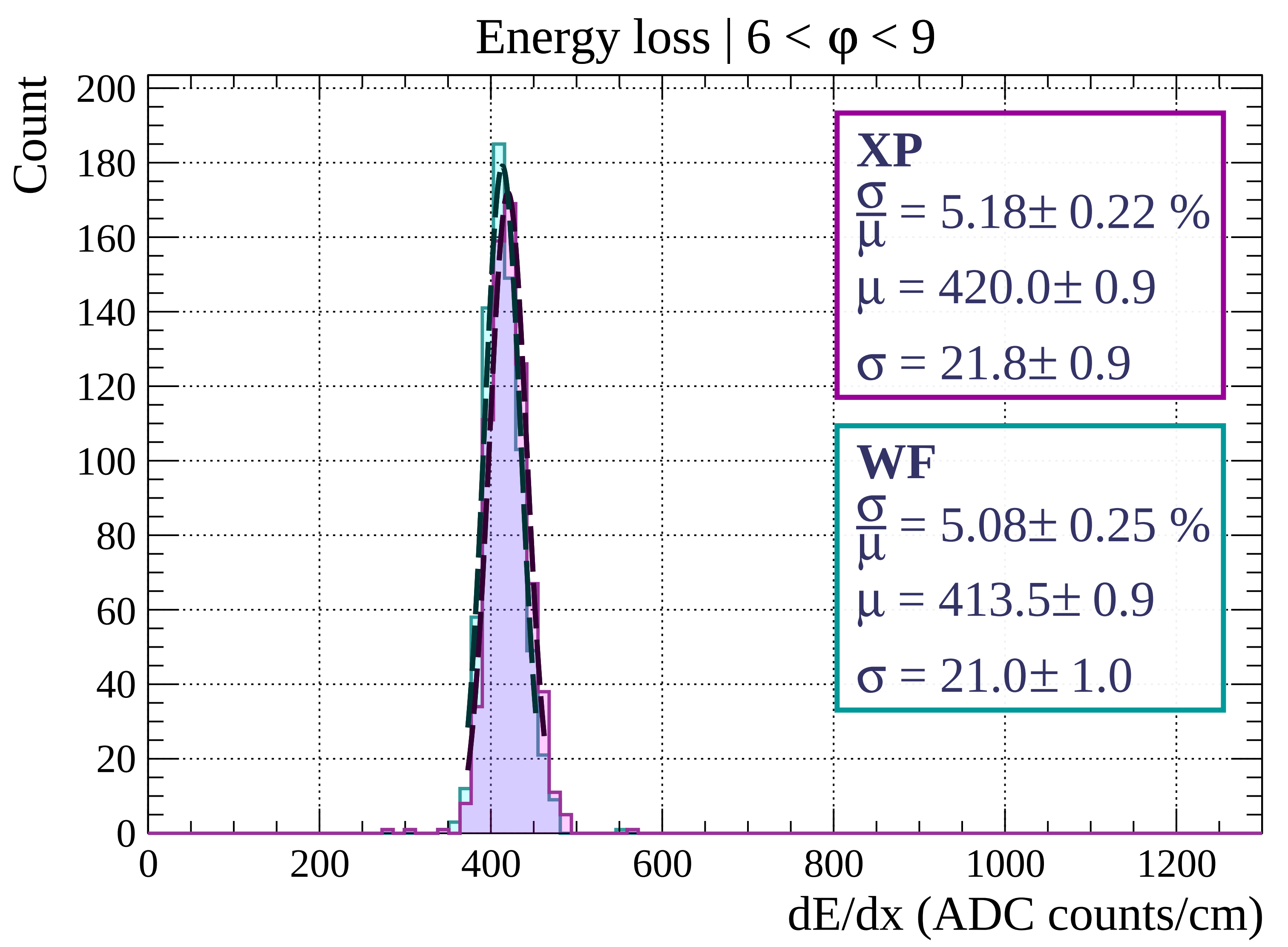
Energy loss | 1920 < p < 2000
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 4.83± 0.07 %
μ = 493.3± 0.4
σ = 23.8± 0.4
WF
$\frac{\sigma}{\mu}$ = 4.71± 0.08 %
μ = 486.1± 0.4
σ = 22.9± 0.4



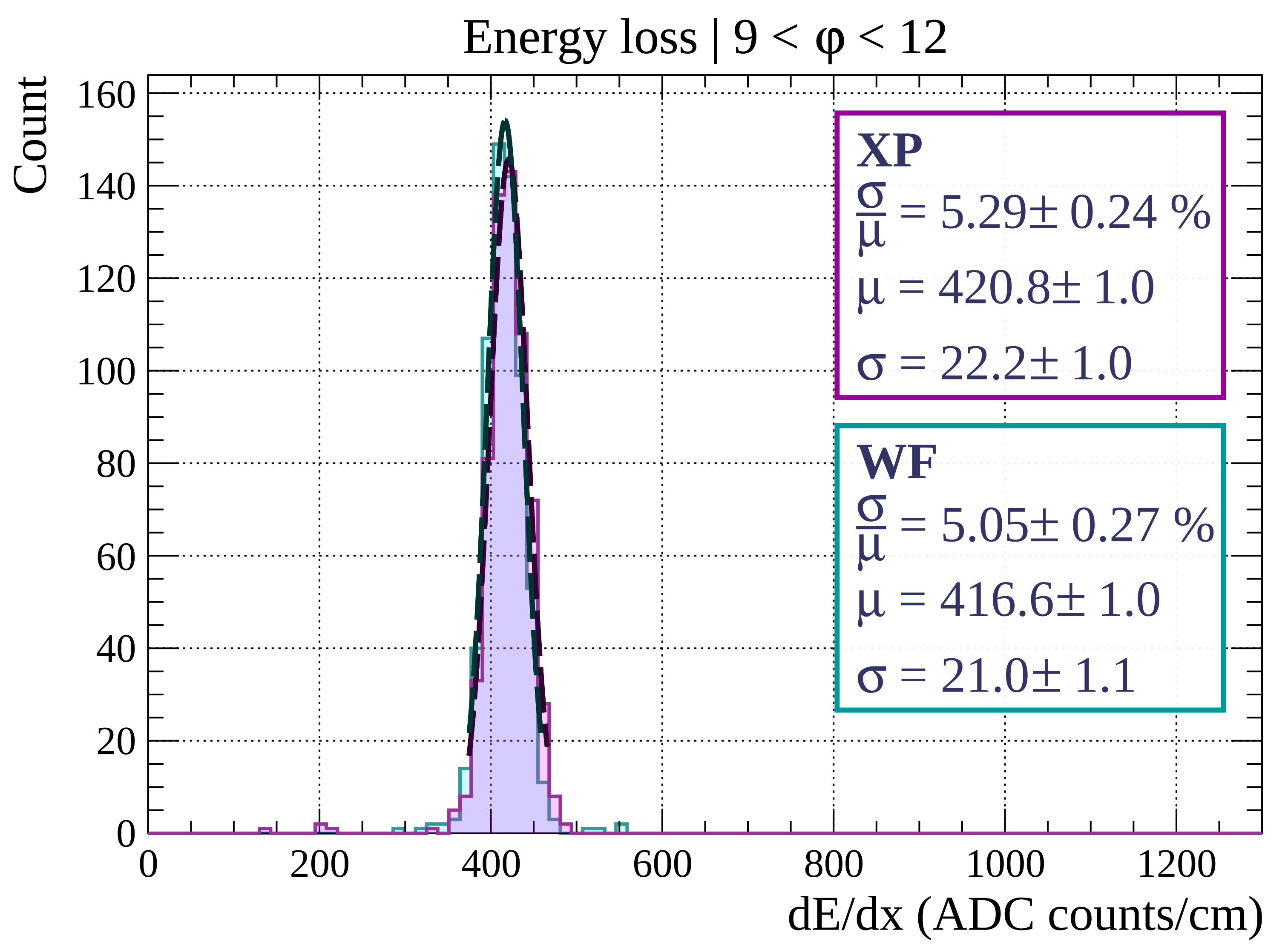
Energy loss | 0 < φ < 3
Count
0
50
100
150
200
250
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 5.76± 0.23 %
μ = 422.4± 0.9
σ = 24.3± 0.9
WF
σ/μ = 5.44± 0.20 %
μ = 415.3± 0.8
σ = 22.6± 0.8



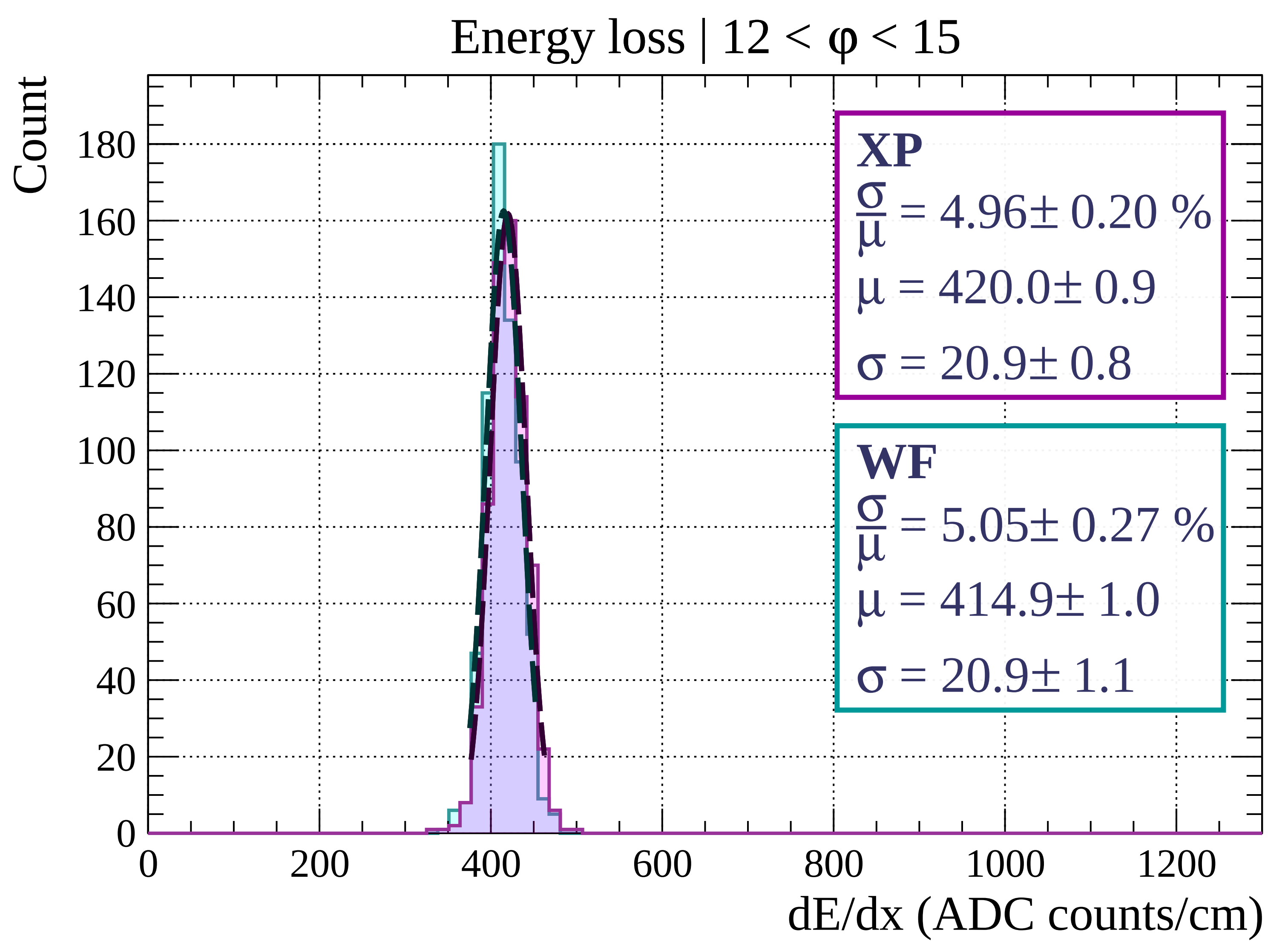
Energy loss | 3 < φ < 6
Count
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 5.06± 0.17 %
μ = 420.6± 0.8
σ = 21.3± 0.7
WF
σ/μ = 4.95± 0.21 %
μ = 414.1± 0.8
σ = 20.5± 0.8



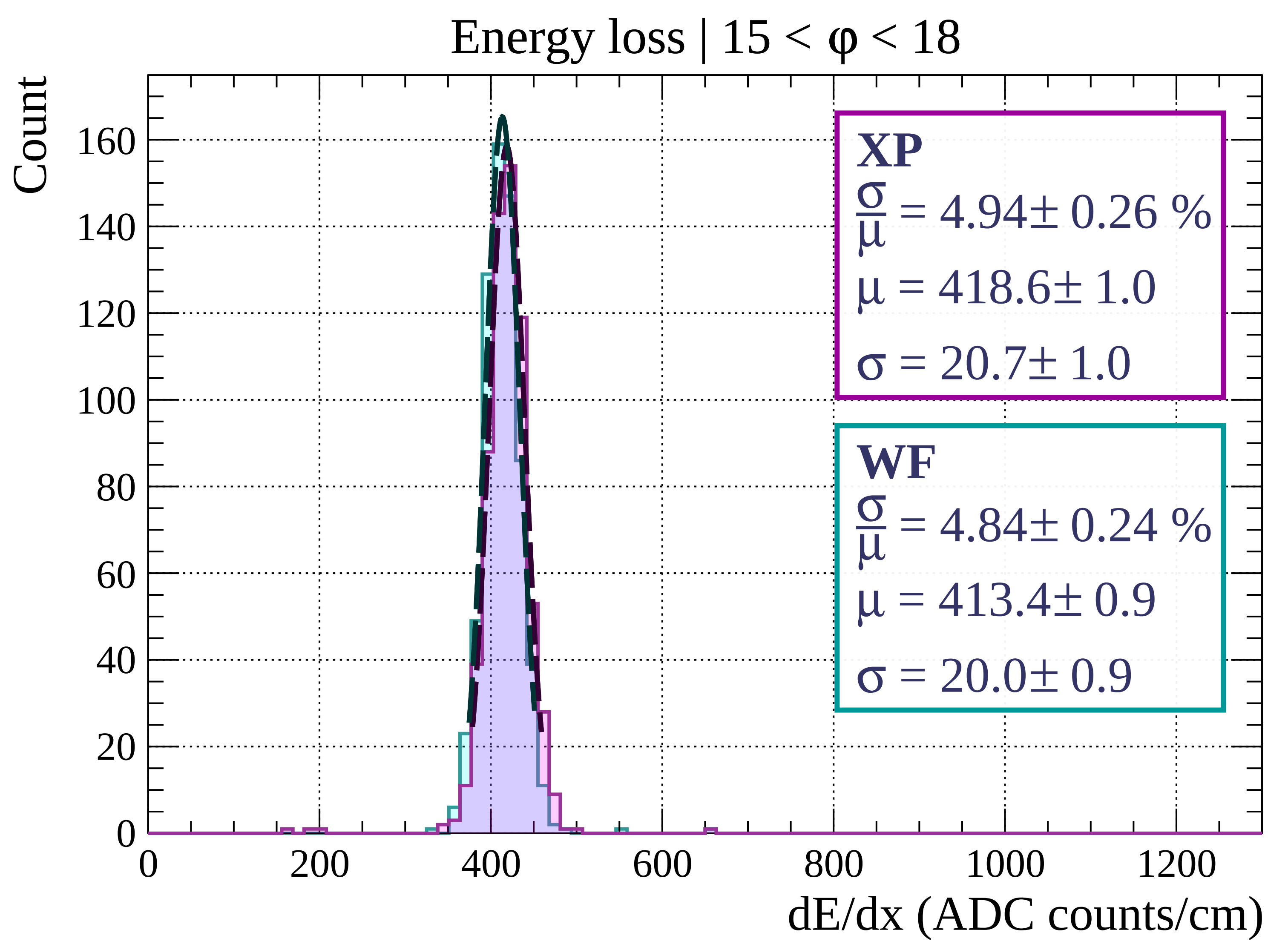
Energy loss | 6 < φ < 9
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 5.18± 0.22 %
μ = 420.0± 0.9
σ = 21.8± 0.9
WF
$\frac{\sigma}{\mu}$ = 5.08± 0.25 %
μ = 413.5± 0.9
σ = 21.0± 1.0



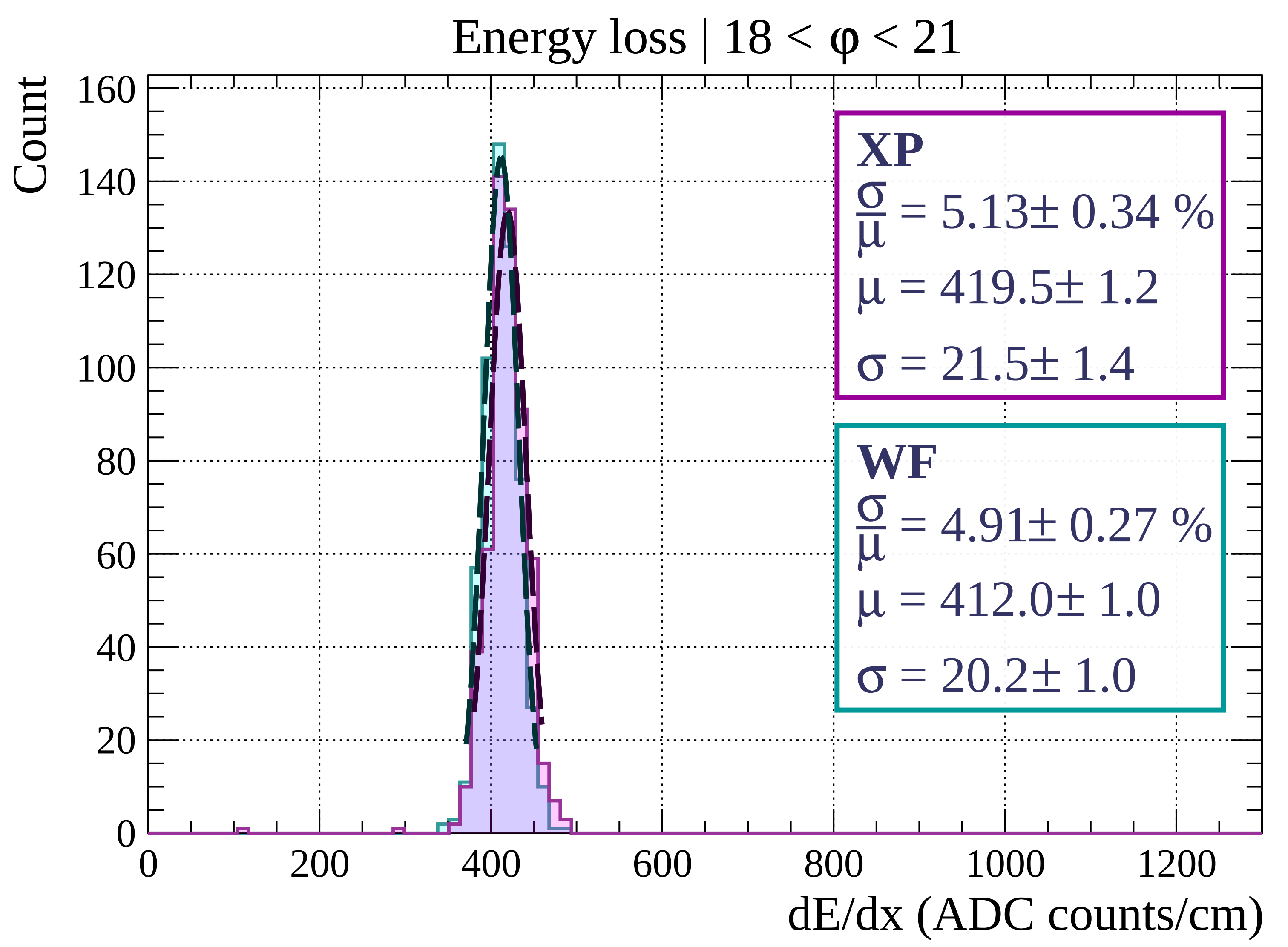
Energy loss | 9 < φ < 12
Count
160
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 5.29± 0.24 %
μ = 420.8± 1.0
σ = 22.2± 1.0
WF
σ/μ = 5.05± 0.27 %
μ = 416.6± 1.0
σ = 21.0± 1.1

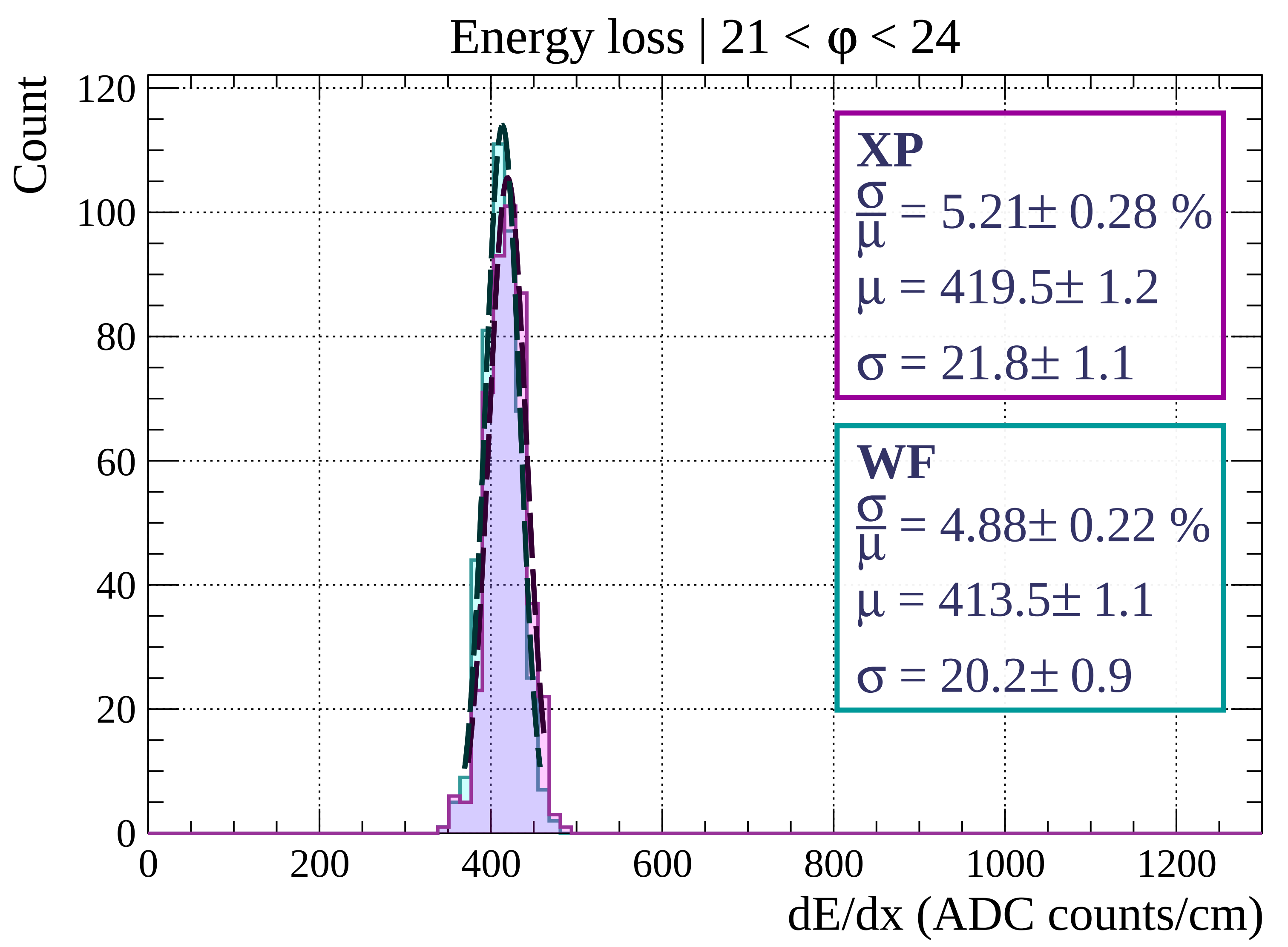


Energy loss | 12 < φ < 15
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 4.96± 0.20 %
μ = 420.0± 0.9
σ = 20.9± 0.8
WF
$\frac{\sigma}{\mu}$ = 5.05± 0.27 %
μ = 414.9± 1.0
σ = 20.9± 1.1

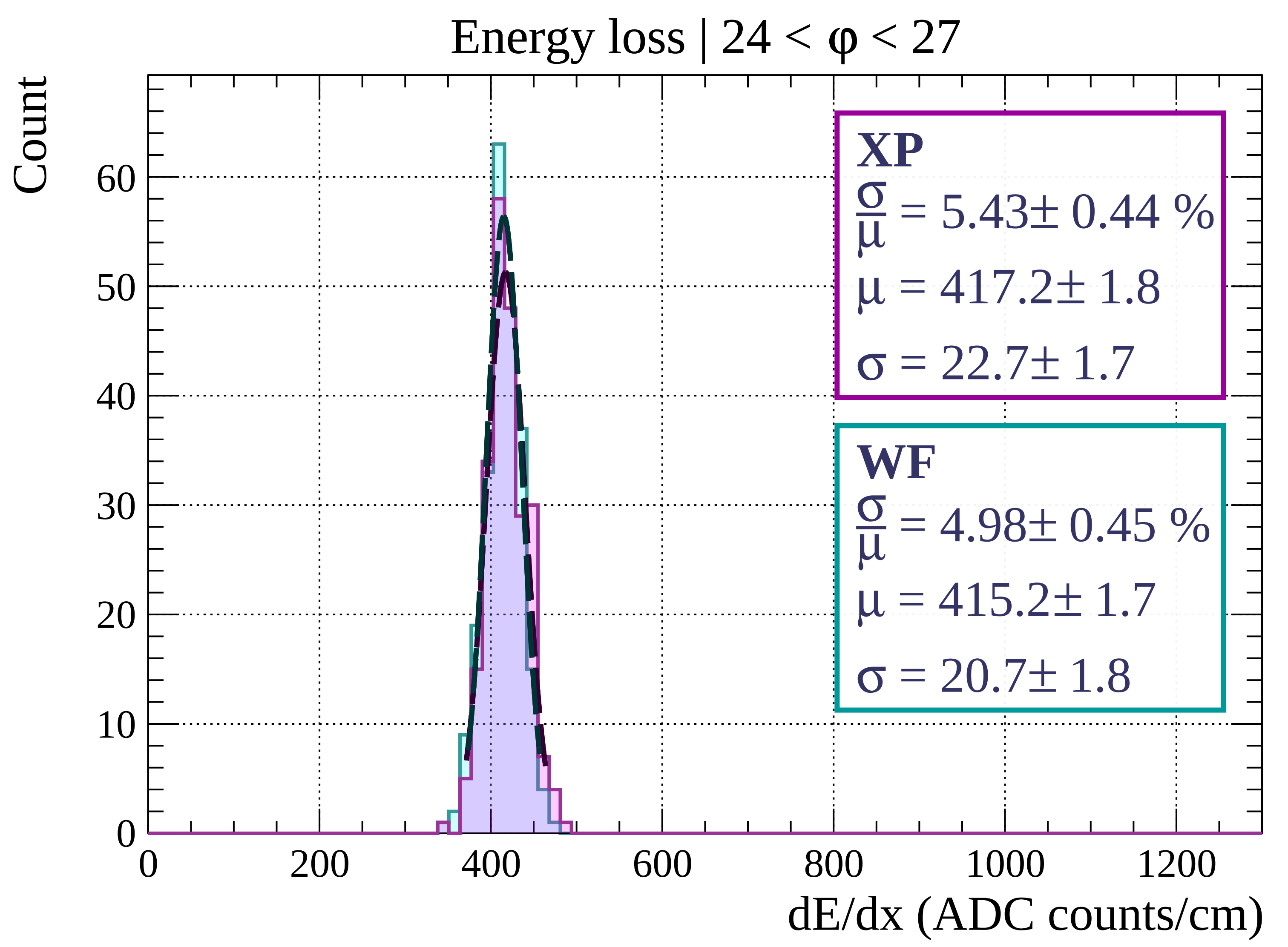


Energy loss | 15 < φ < 18
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
XP
σ/μ = 4.94± 0.26 %
μ = 418.6± 1.0
σ = 20.7± 1.0
WF
σ/μ = 4.84± 0.24 %
μ = 413.4± 0.9
σ = 20.0± 0.9

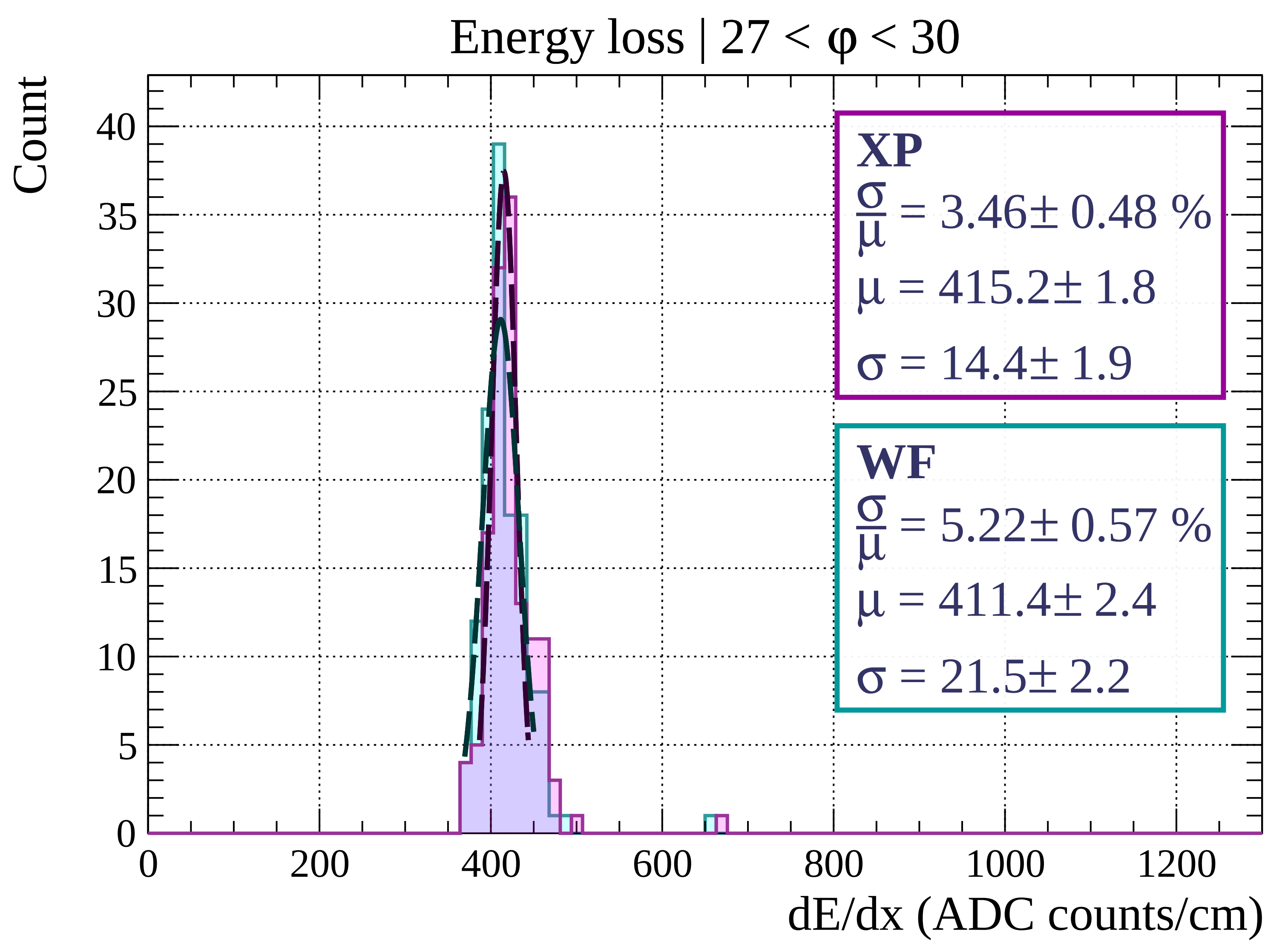

Energy loss | 18 < φ < 21
Count
160
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 5.13± 0.34 %
μ = 419.5± 1.2
σ = 21.5± 1.4
WF
σ/μ = 4.91± 0.27 %
μ = 412.0± 1.0
σ = 20.2± 1.0

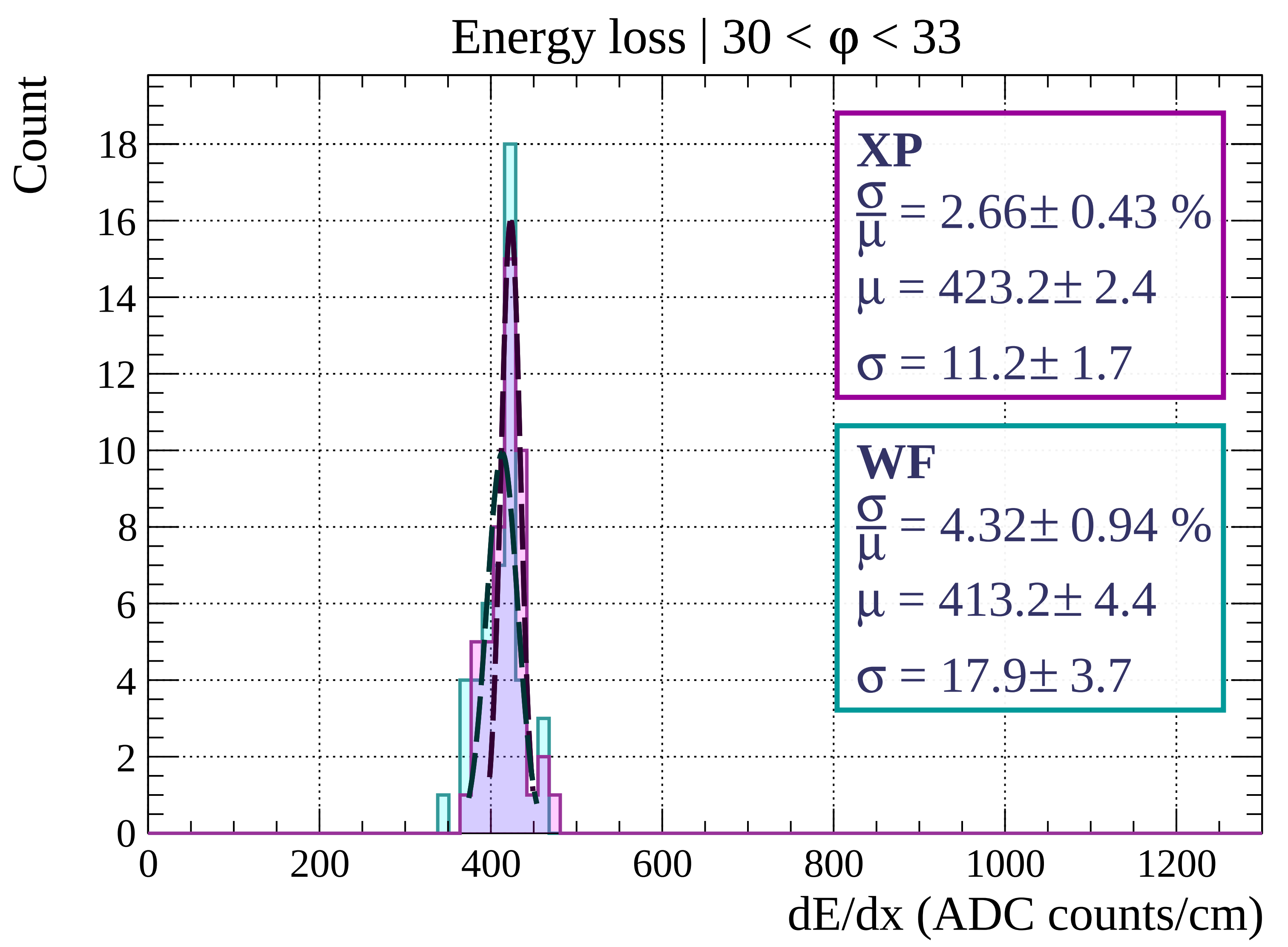


Energy loss | 21 < φ < 24
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 5.21± 0.28 %
μ = 419.5± 1.2
σ = 21.8± 1.1
WF
$\frac{\sigma}{\mu}$ = 4.88± 0.22 %
μ = 413.5± 1.1
σ = 20.2± 0.9

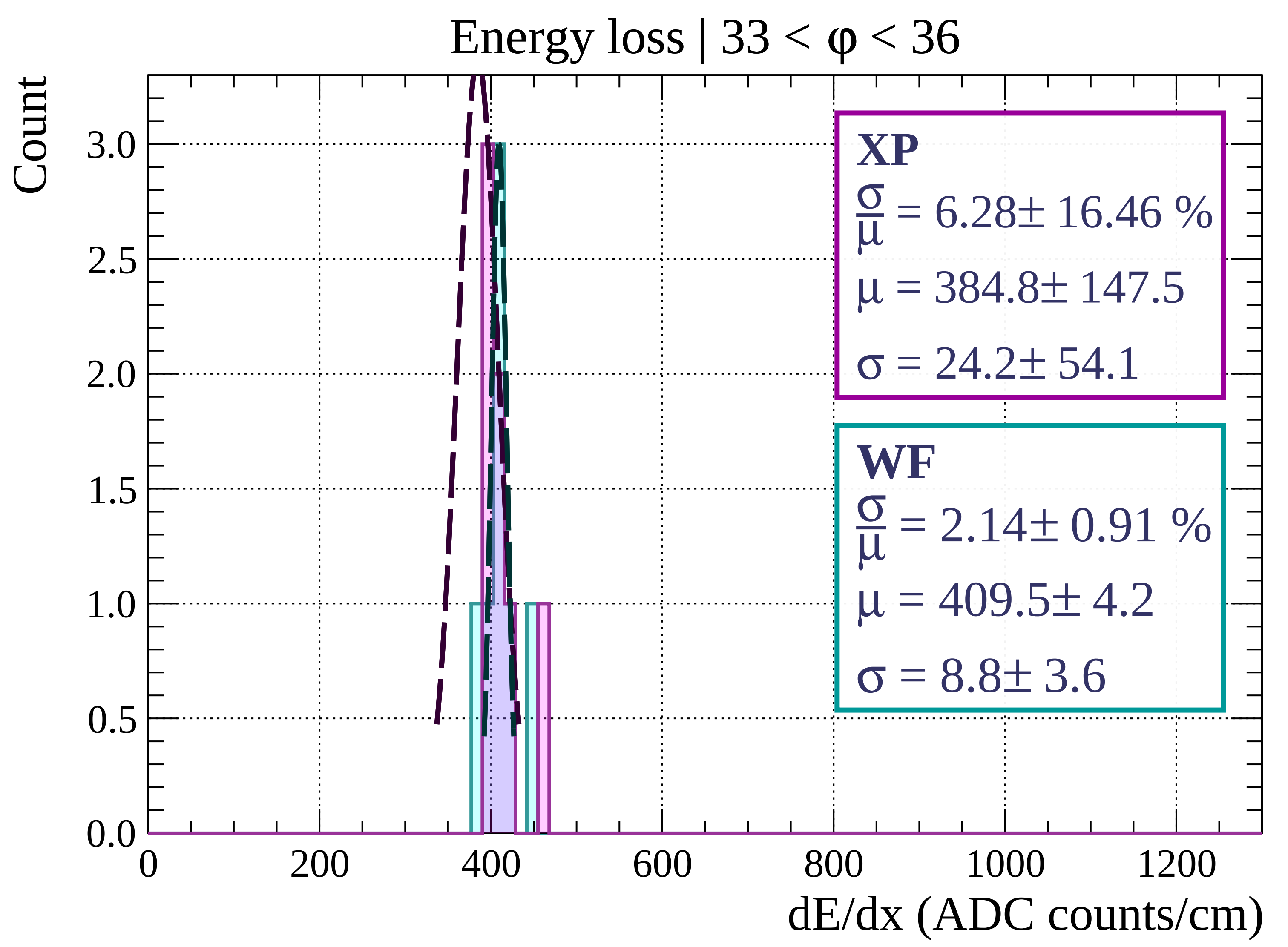


Energy loss | 24 < φ < 27
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
10
20
30
40
50
60
XP
$\frac{\sigma}{\mu}$ = 5.43± 0.44 %
μ = 417.2± 1.8
σ = 22.7± 1.7
WF
$\frac{\sigma}{\mu}$ = 4.98± 0.45 %
μ = 415.2± 1.7
σ = 20.7± 1.8

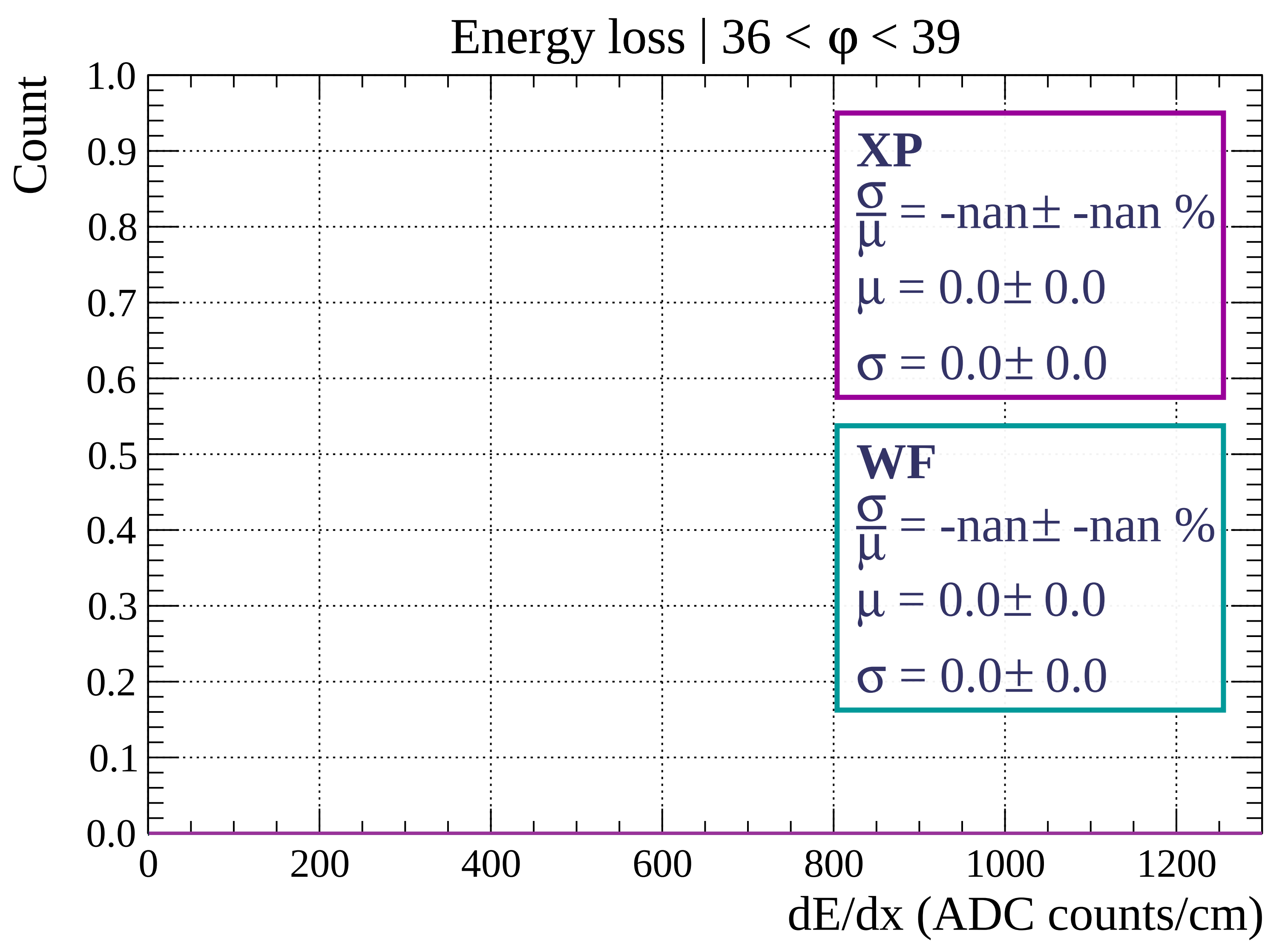

Energy loss | 27 < φ < 30
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
5
10
15
20
25
30
35
40
XP
σ/μ = 3.46± 0.48 %
μ = 415.2± 1.8
σ = 14.4± 1.9
WF
σ/μ = 5.22± 0.57 %
μ = 411.4± 2.4
σ = 21.5± 2.2



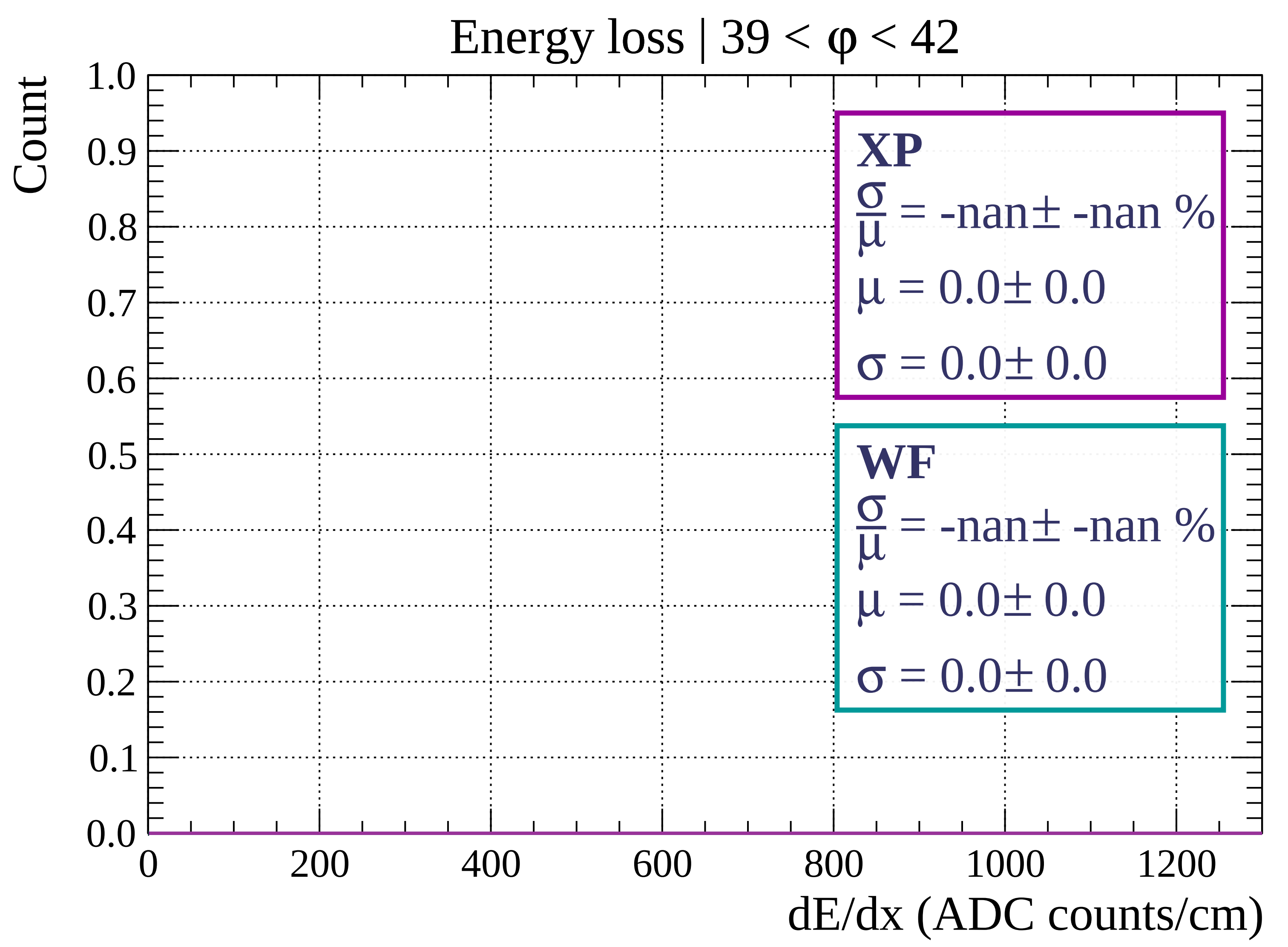
Energy loss | 30 < φ < 33
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
2
4
6
8
10
12
14
16
18
XP
σ/μ = 2.66± 0.43 %
μ = 423.2± 2.4
σ = 11.2± 1.7
WF
σ/μ = 4.32± 0.94 %
μ = 413.2± 4.4
σ = 17.9± 3.7



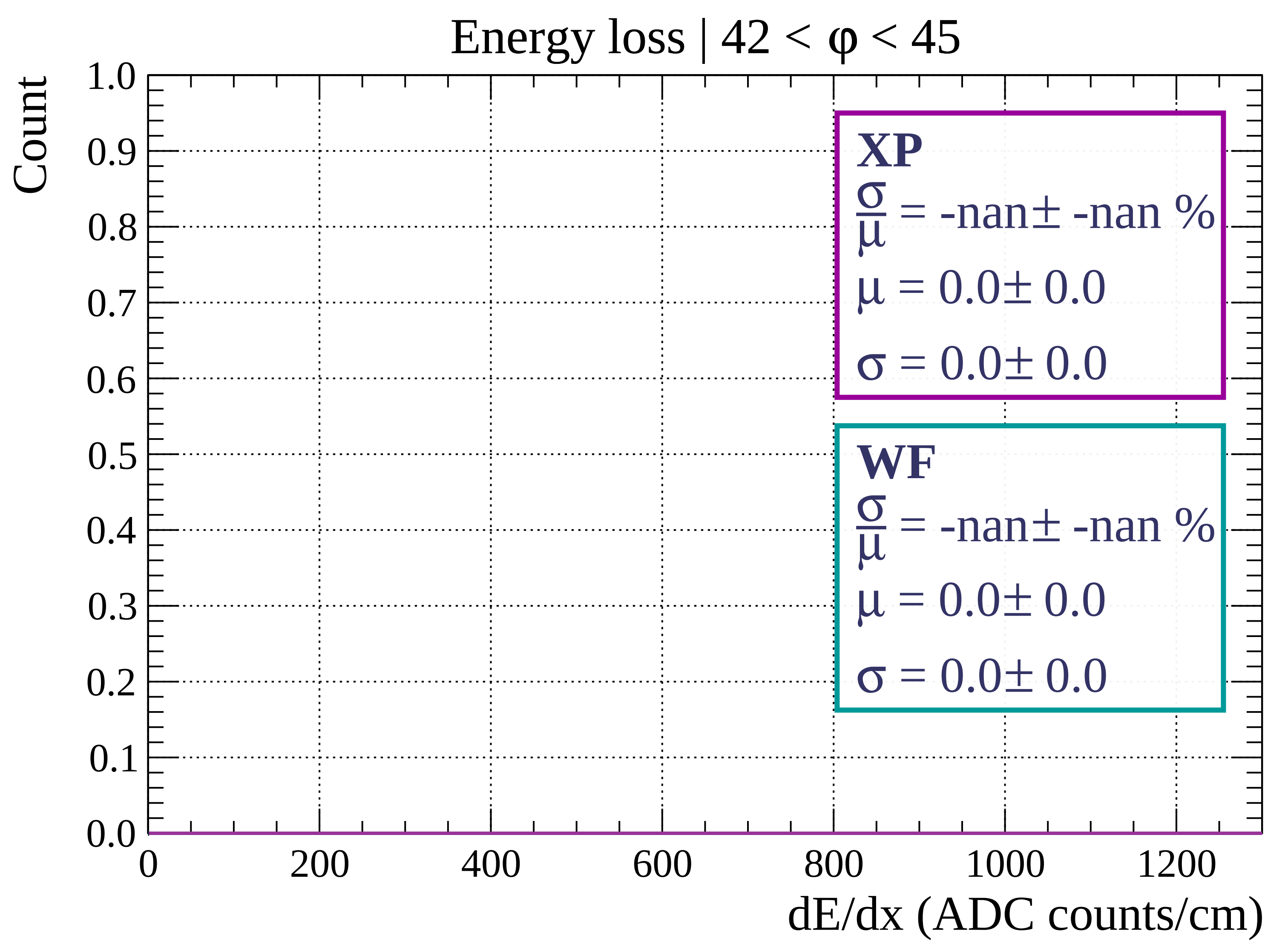
Energy loss | 33 < φ < 36
Count
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 6.28± 16.46 %
μ = 384.8± 147.5
σ = 24.2± 54.1
WF
σ/μ = 2.14± 0.91 %
μ = 409.5± 4.2
σ = 8.8± 3.6



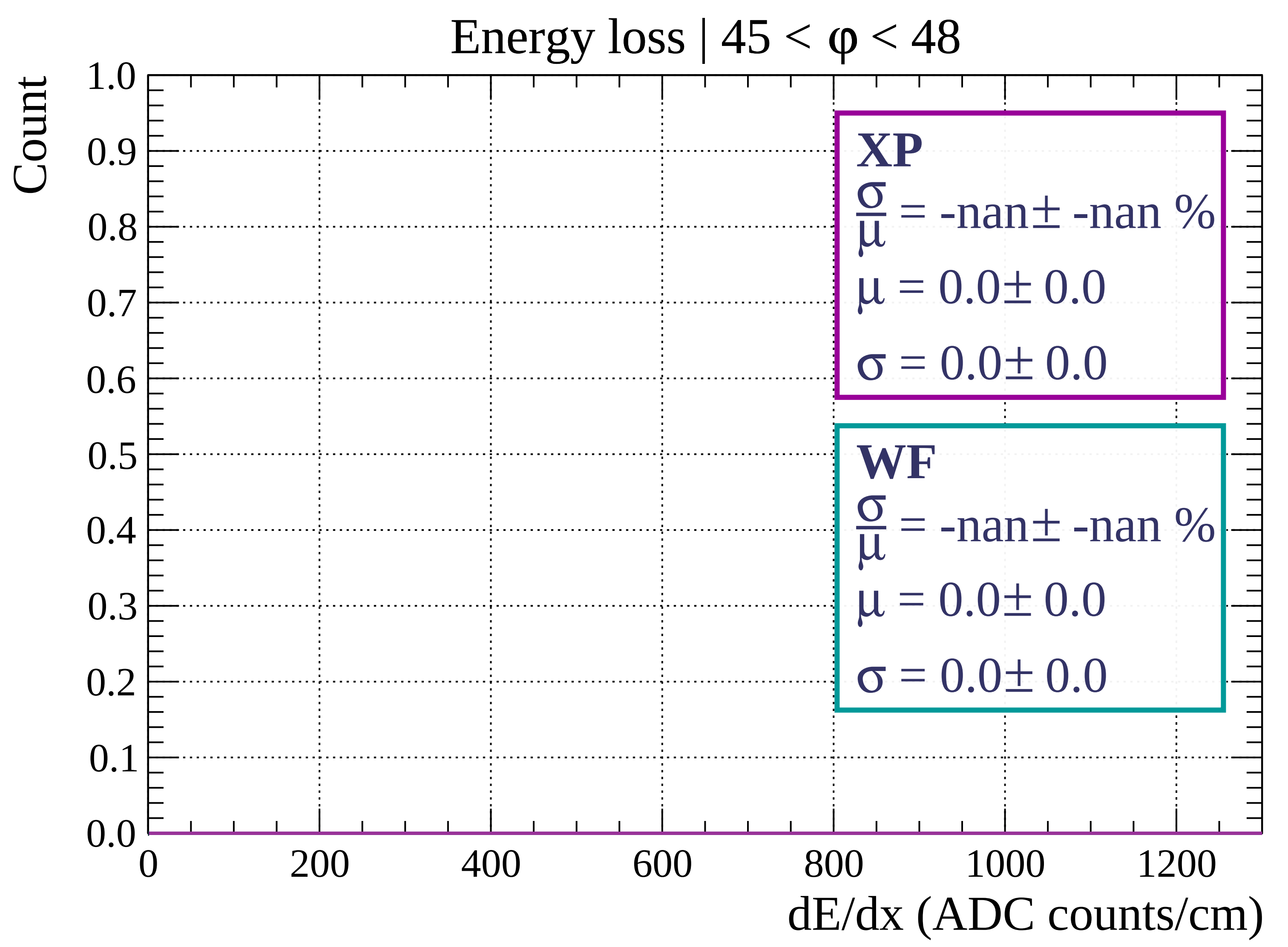
Energy loss | 36 < φ < 39
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



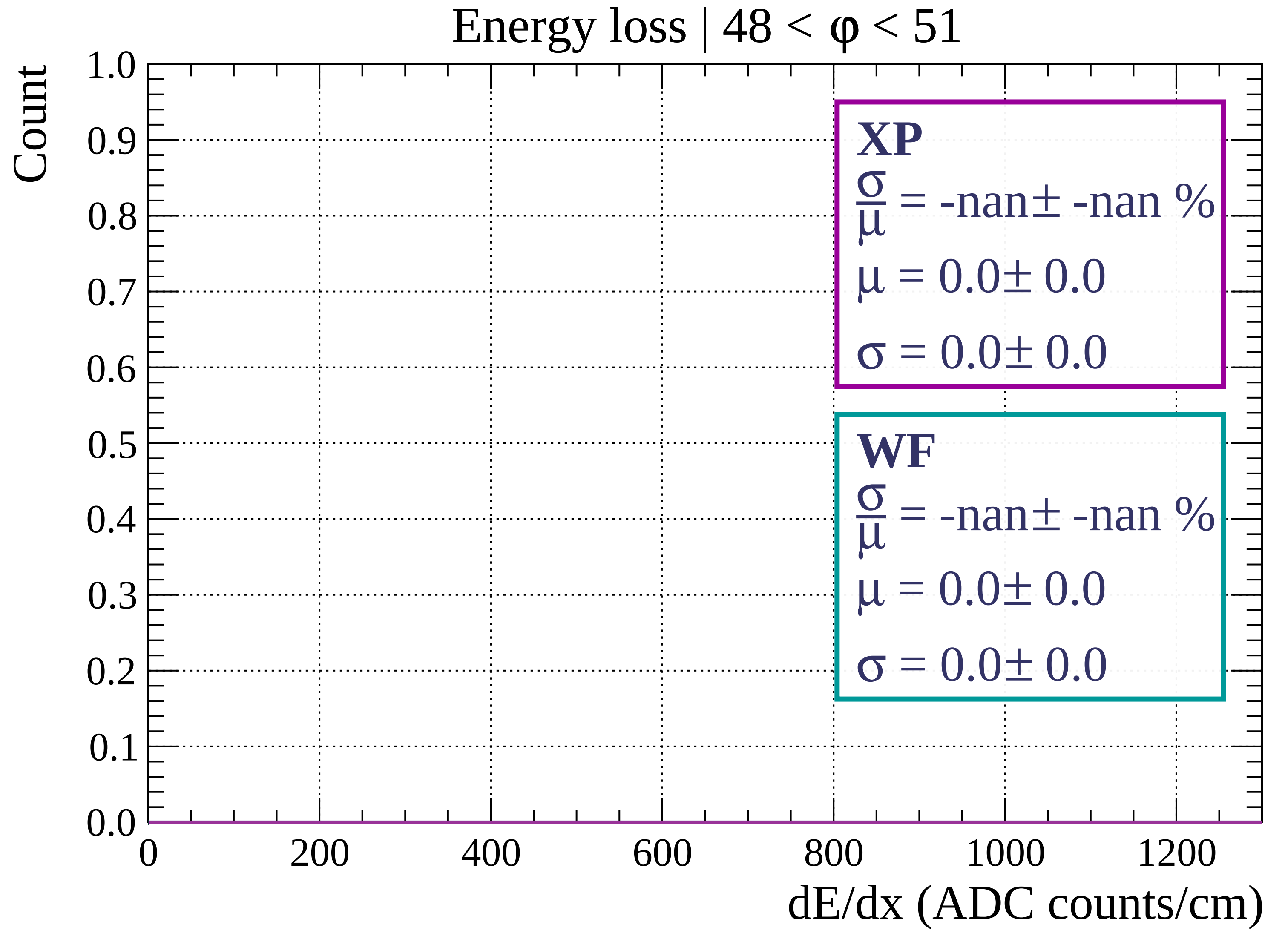
Energy loss | 39 < φ < 42
Count
dE/dx (ADC counts/cm)
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



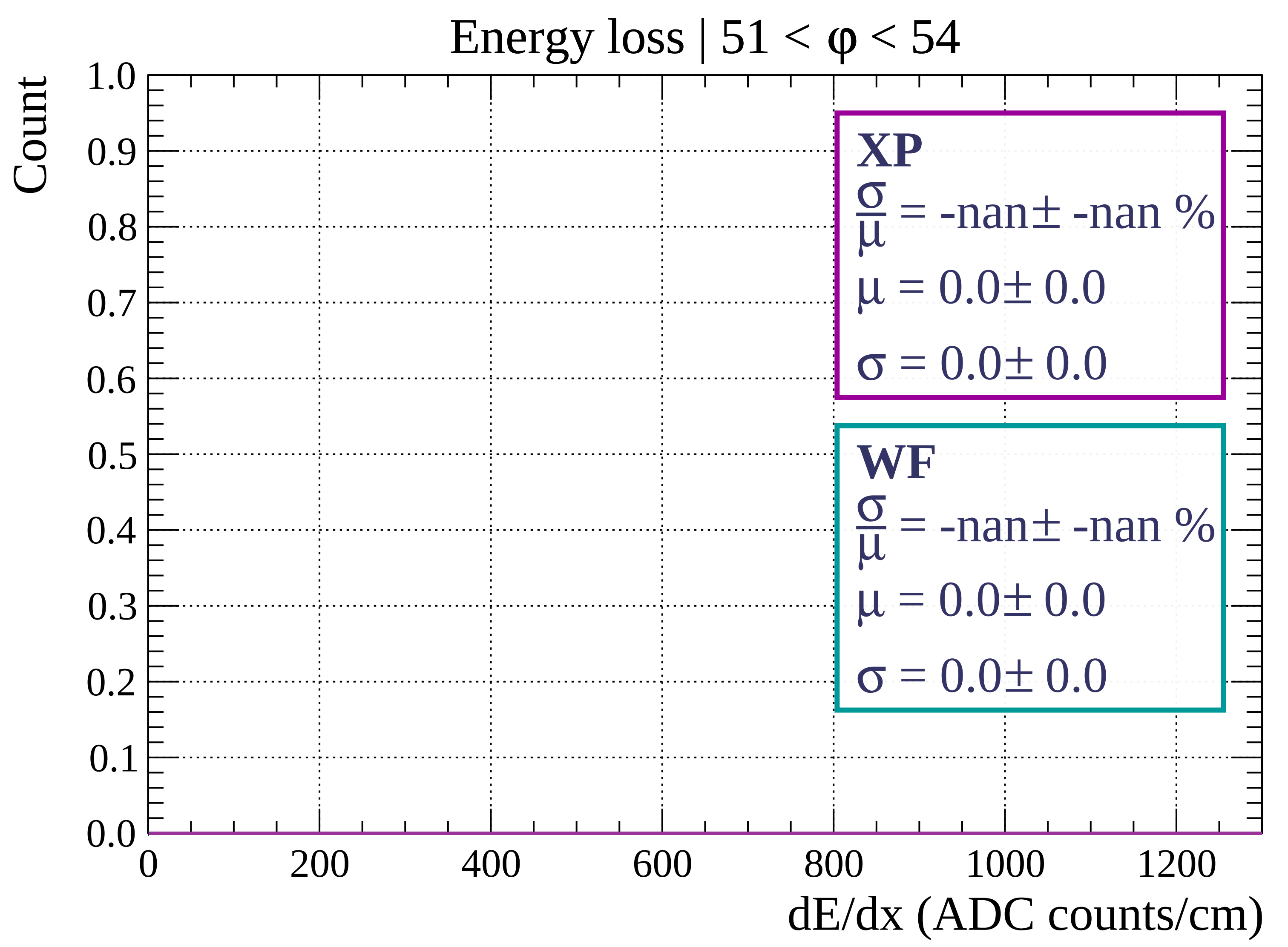
Energy loss | 42 < φ < 45
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



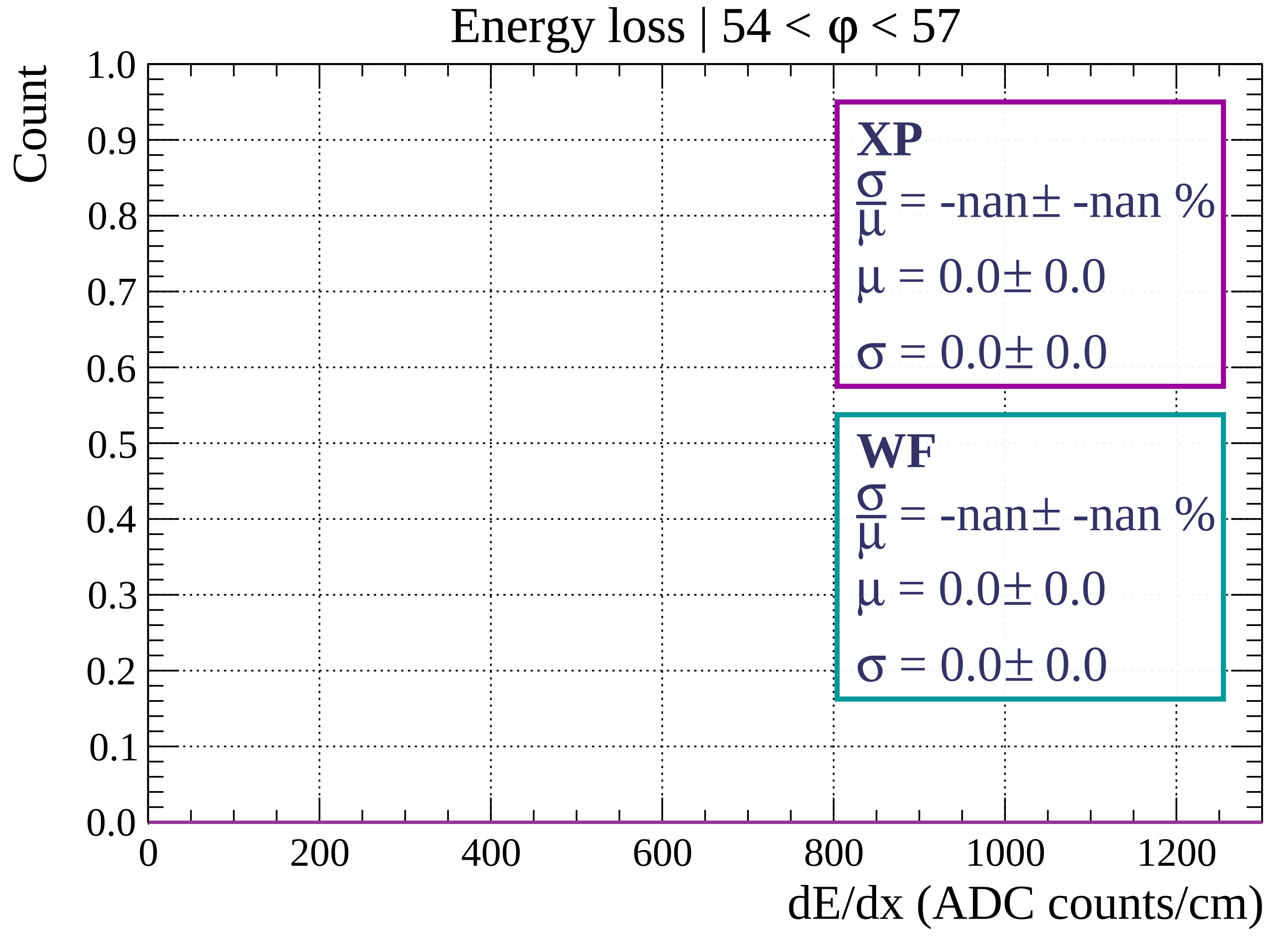
Energy loss | 45 < φ < 48
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



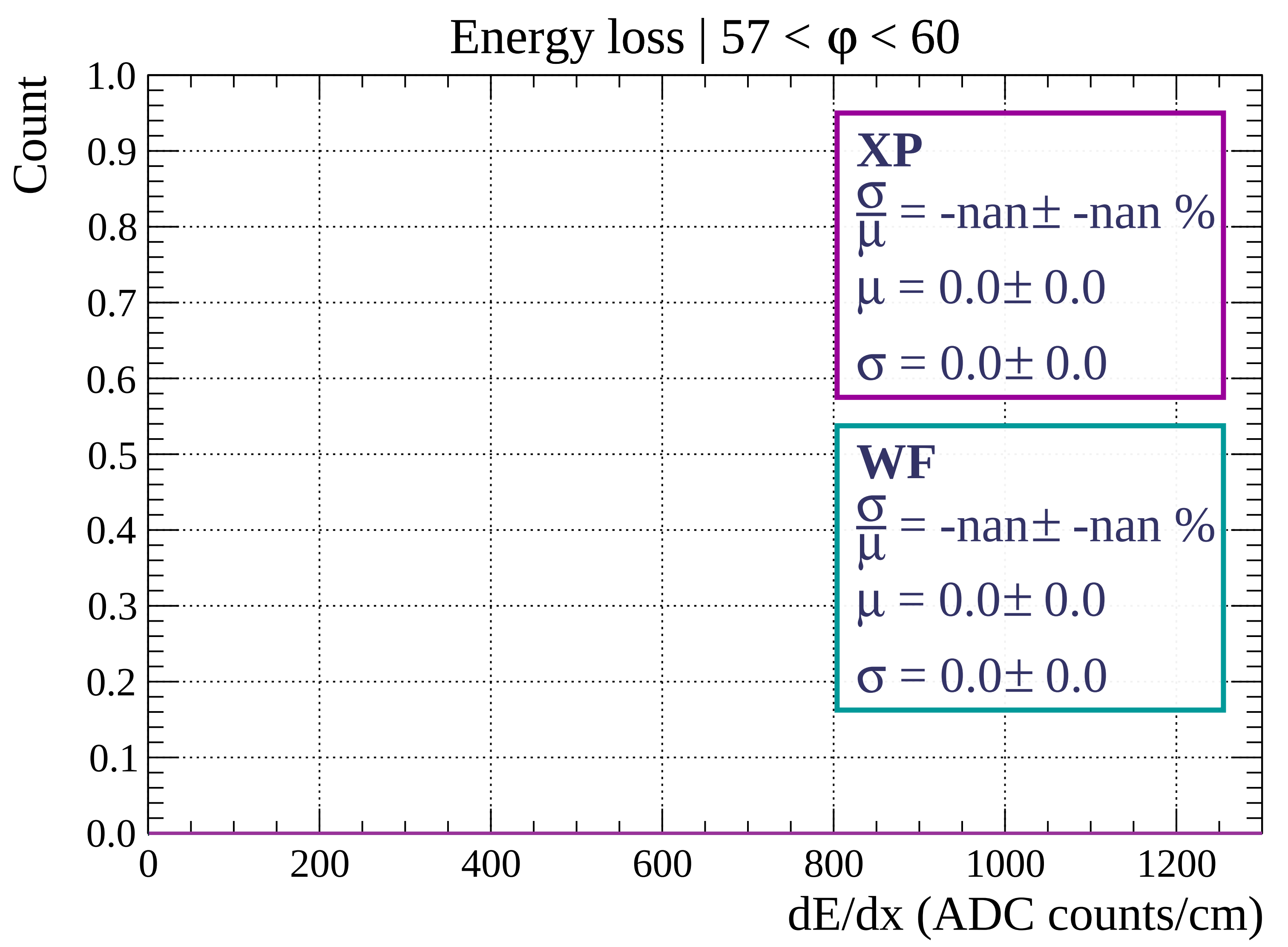
Energy loss | 48 < φ < 51
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



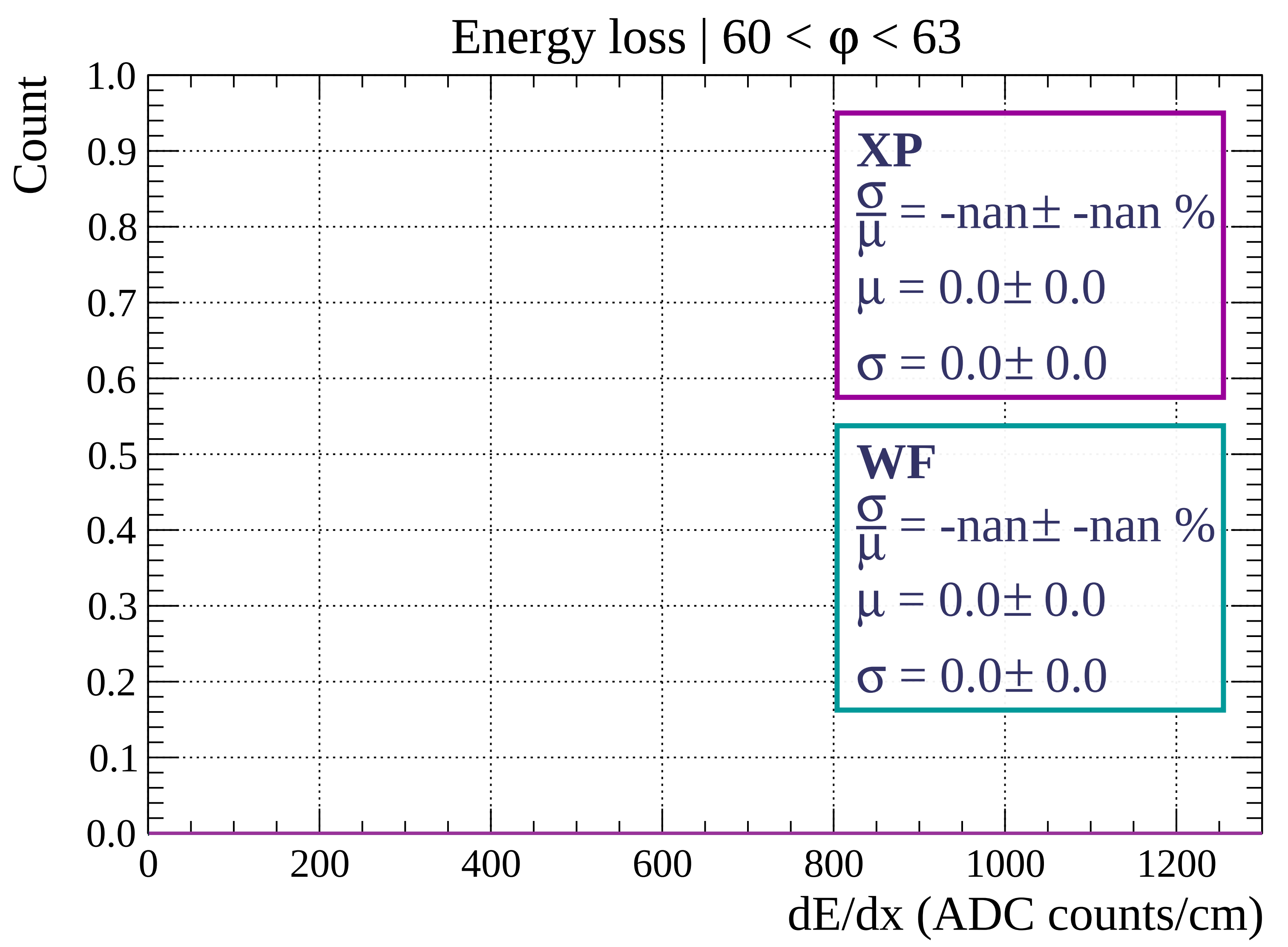
Energy loss | 51 < φ < 54
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



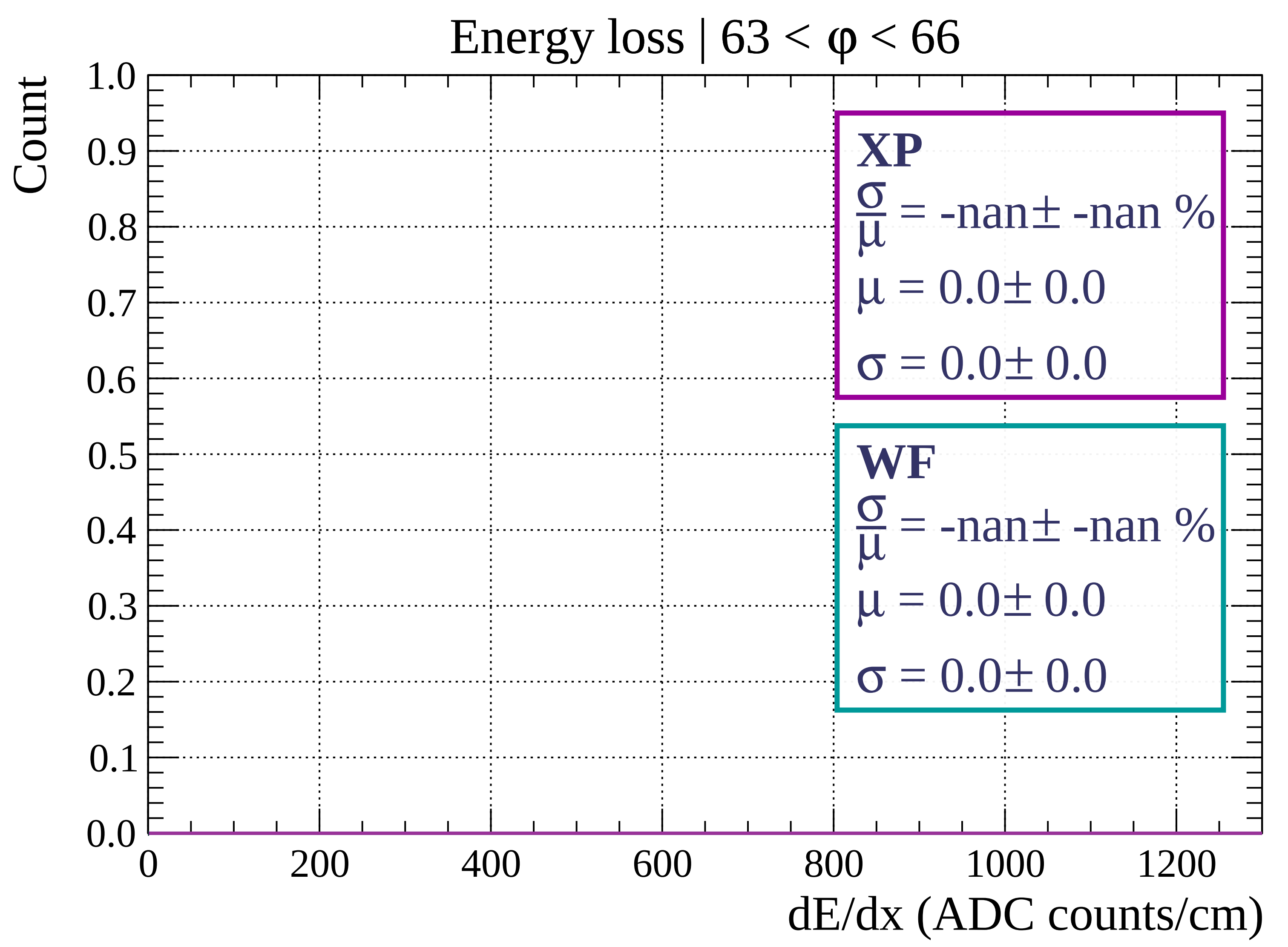
Energy loss | 54 < φ < 57
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



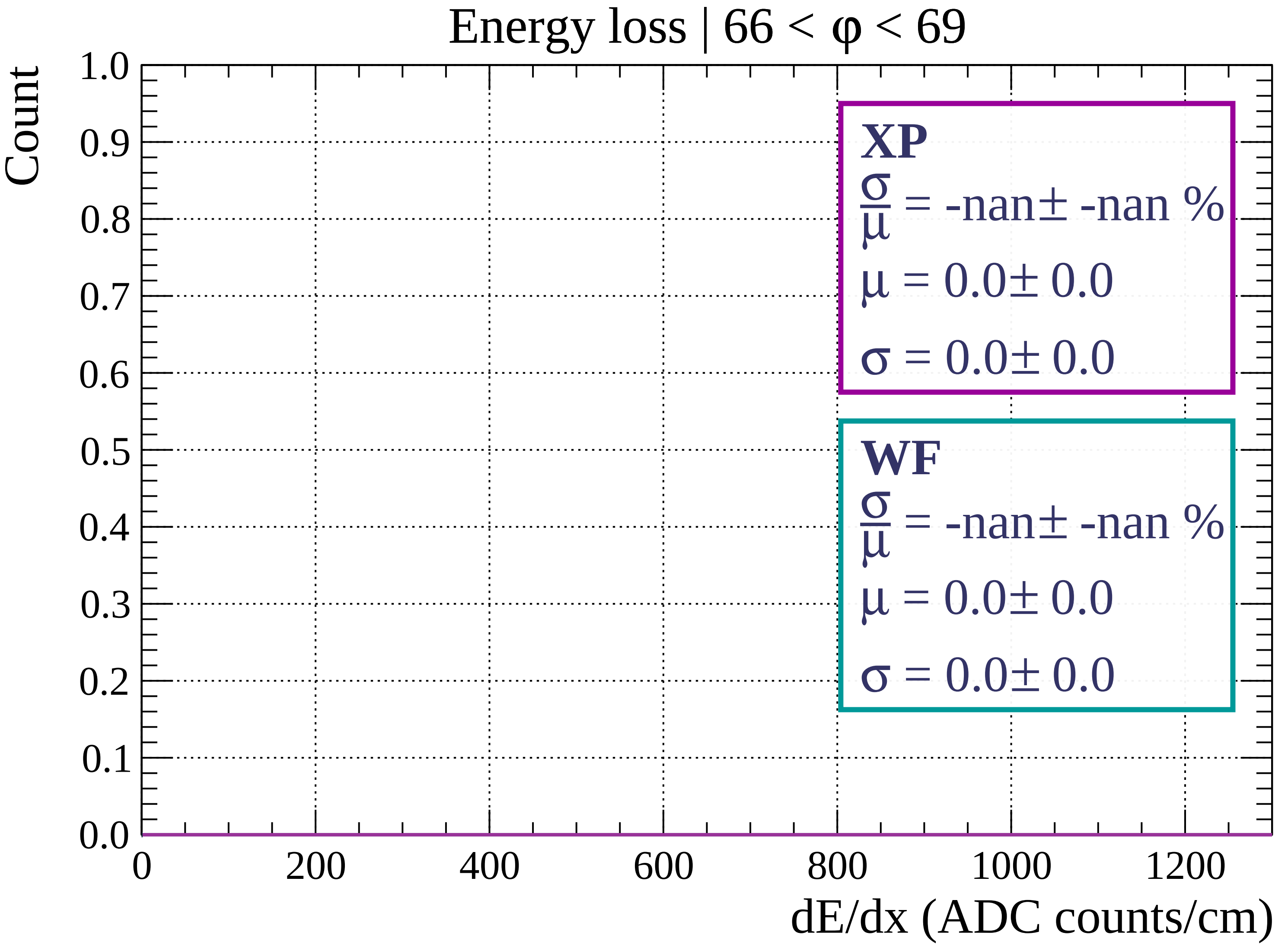
Energy loss | 57 < φ < 60
Count
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



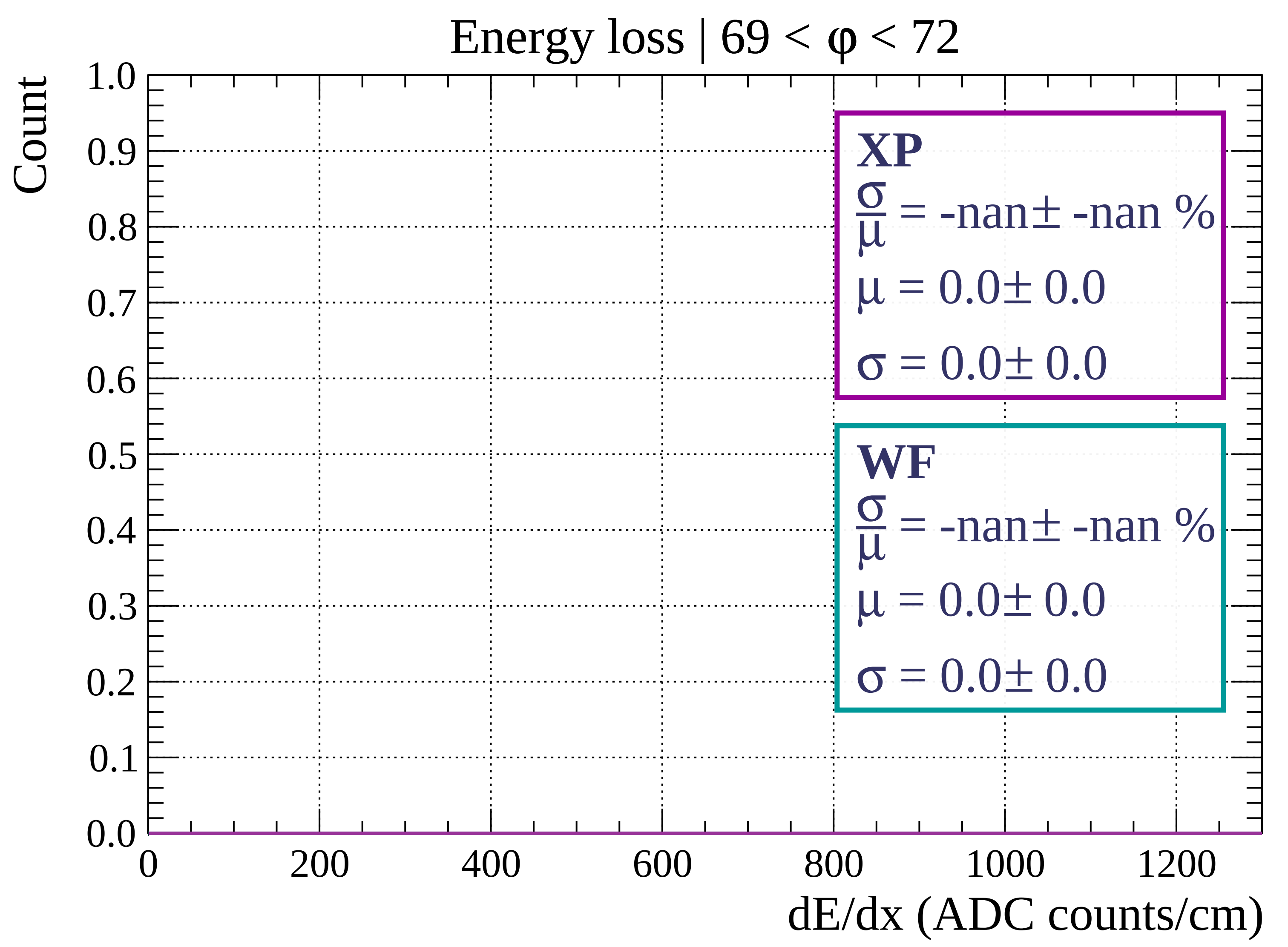
Energy loss | 60 < φ < 63
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



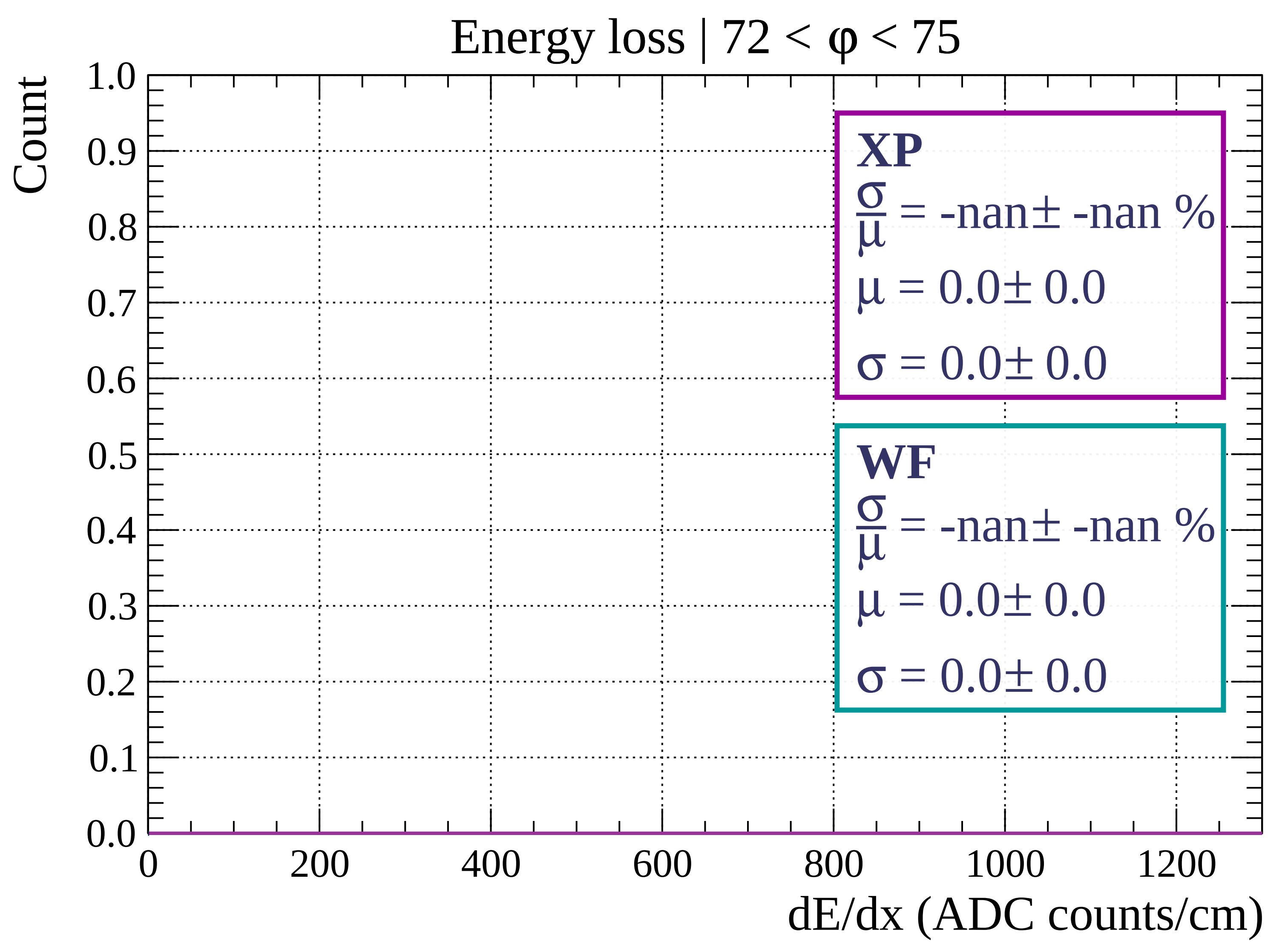
Energy loss | 63 < φ < 66
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



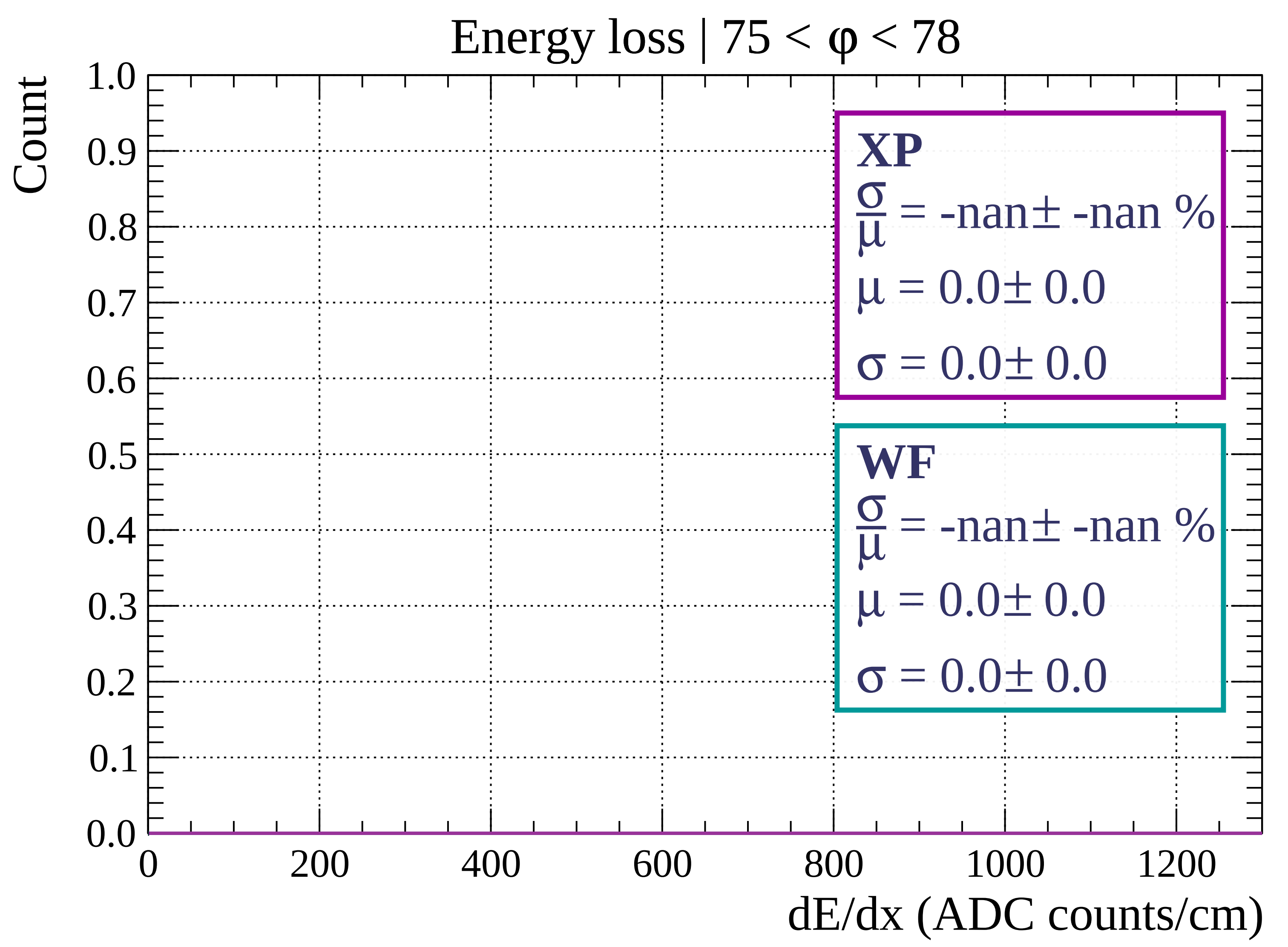
Energy loss | 66 < φ < 69
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

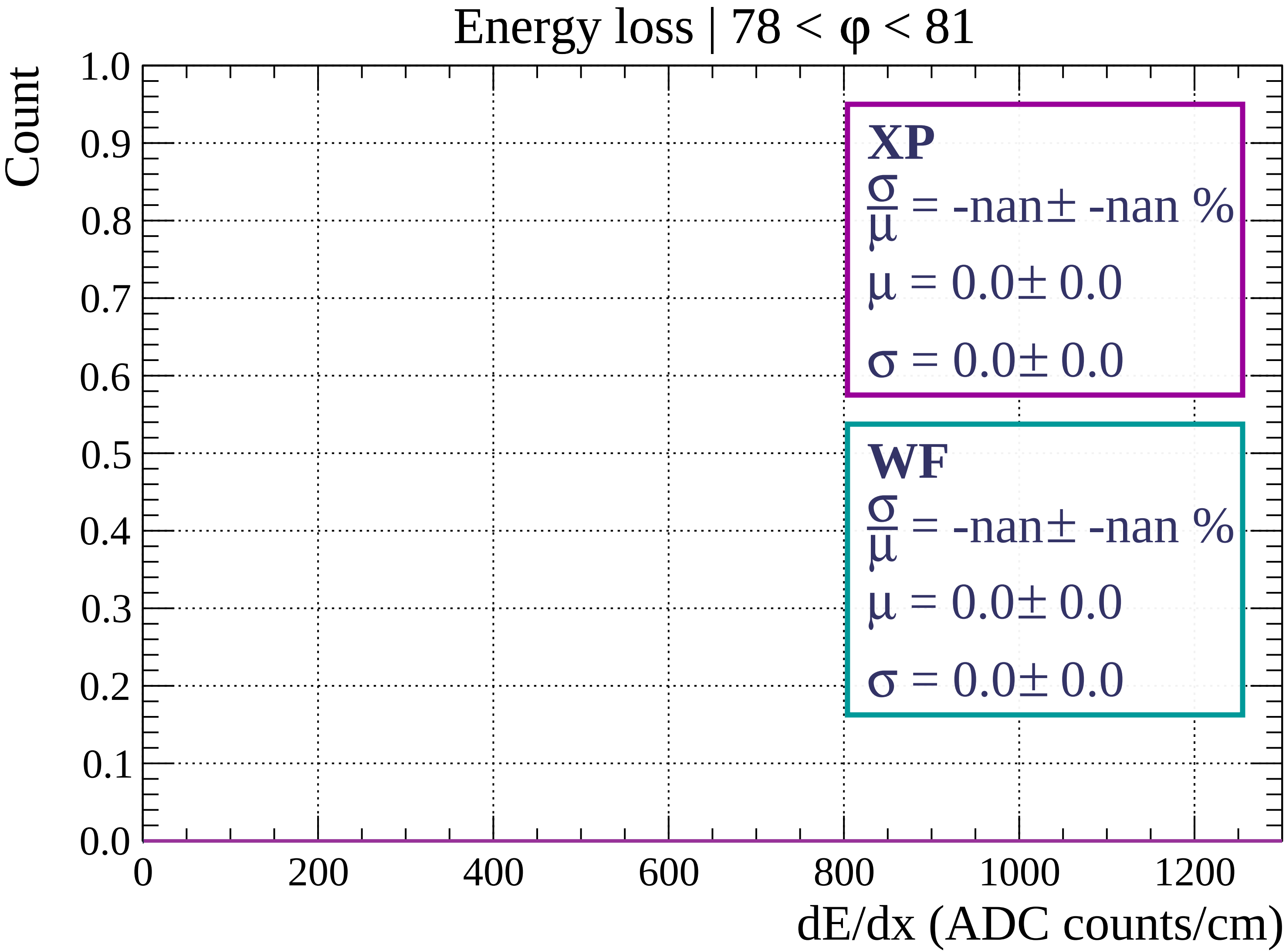


Energy loss | 69 < φ < 72
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

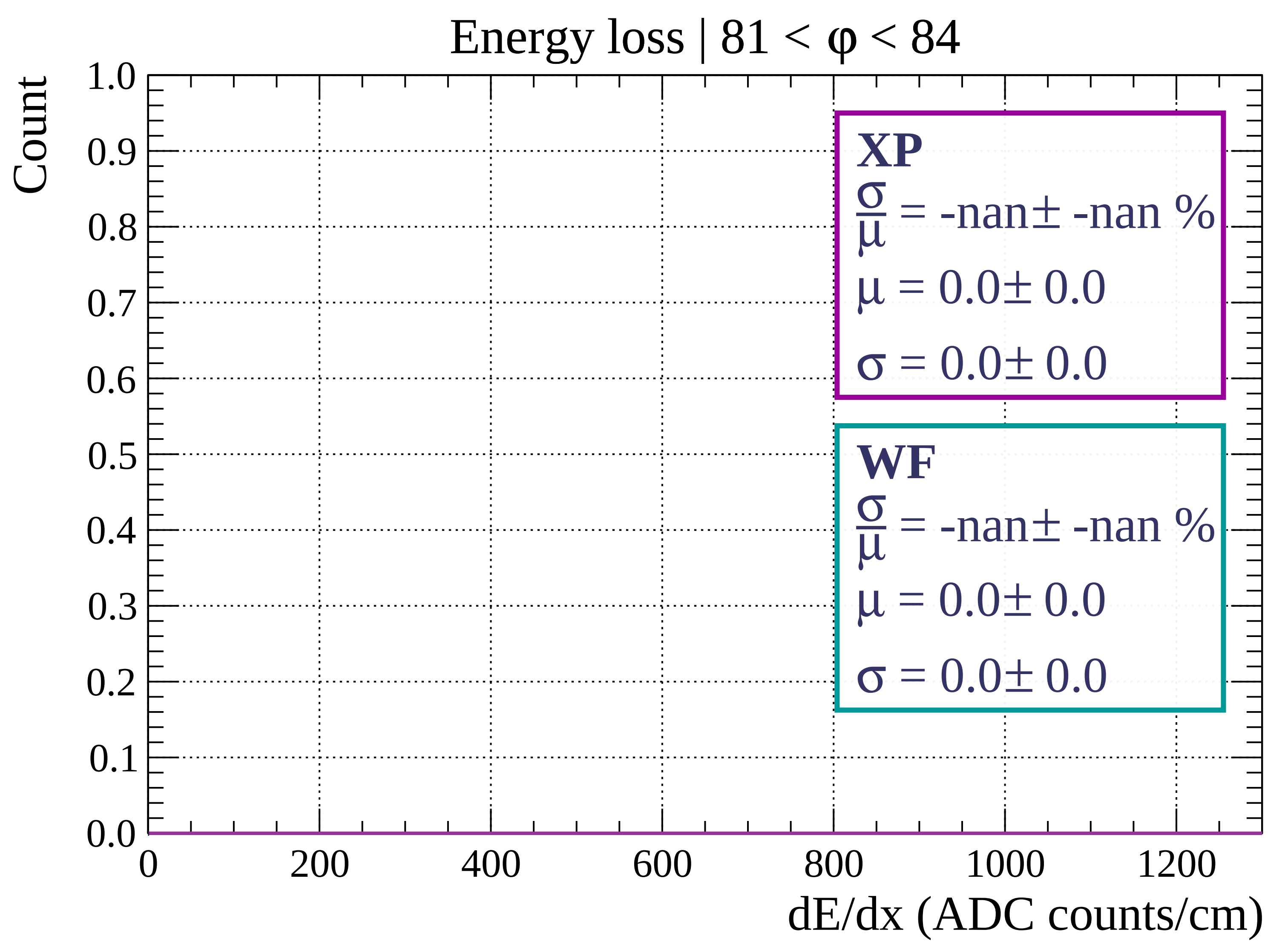


Energy loss | 72 < φ < 75
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

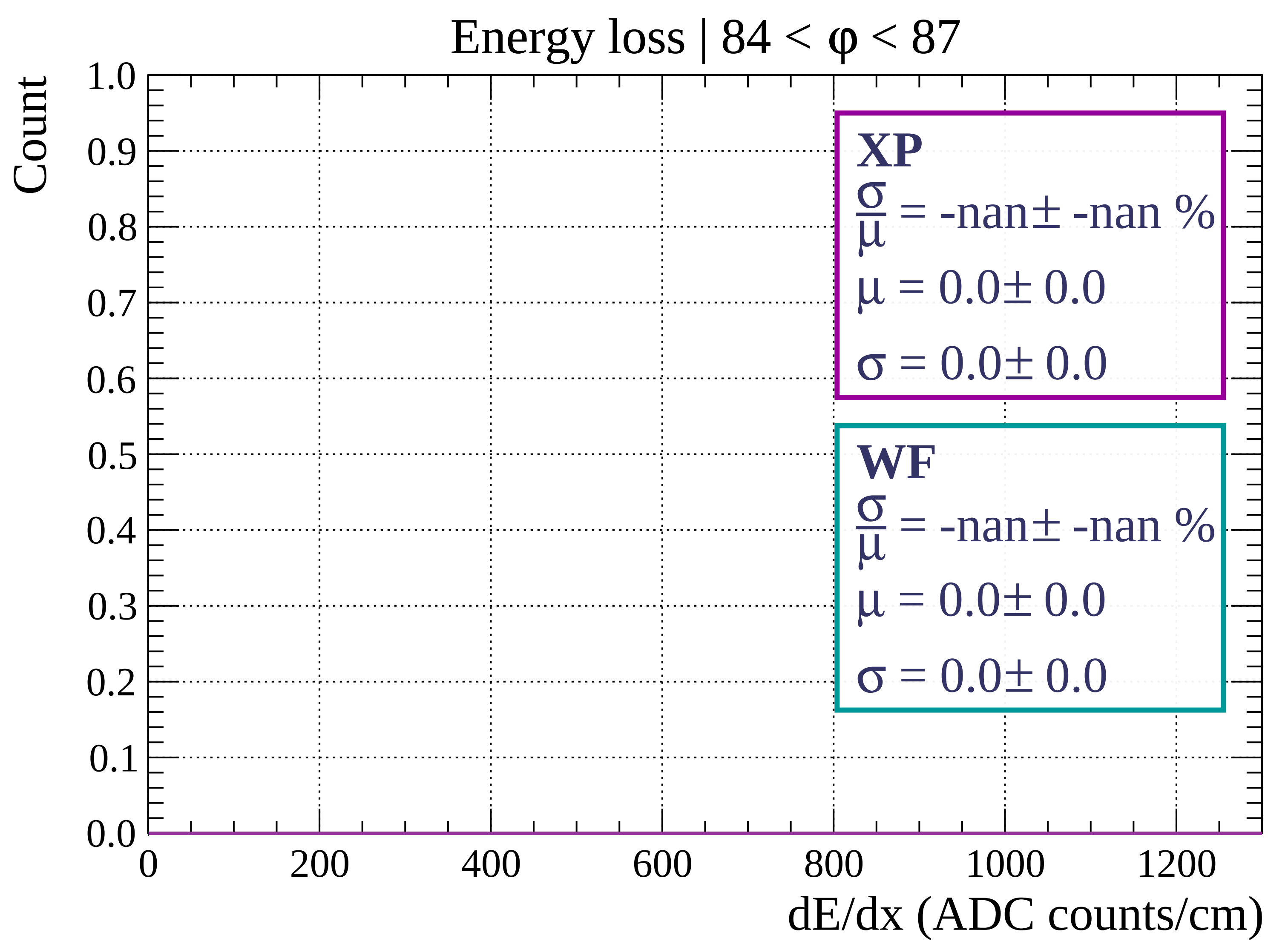

Energy loss | 75 < φ < 78
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



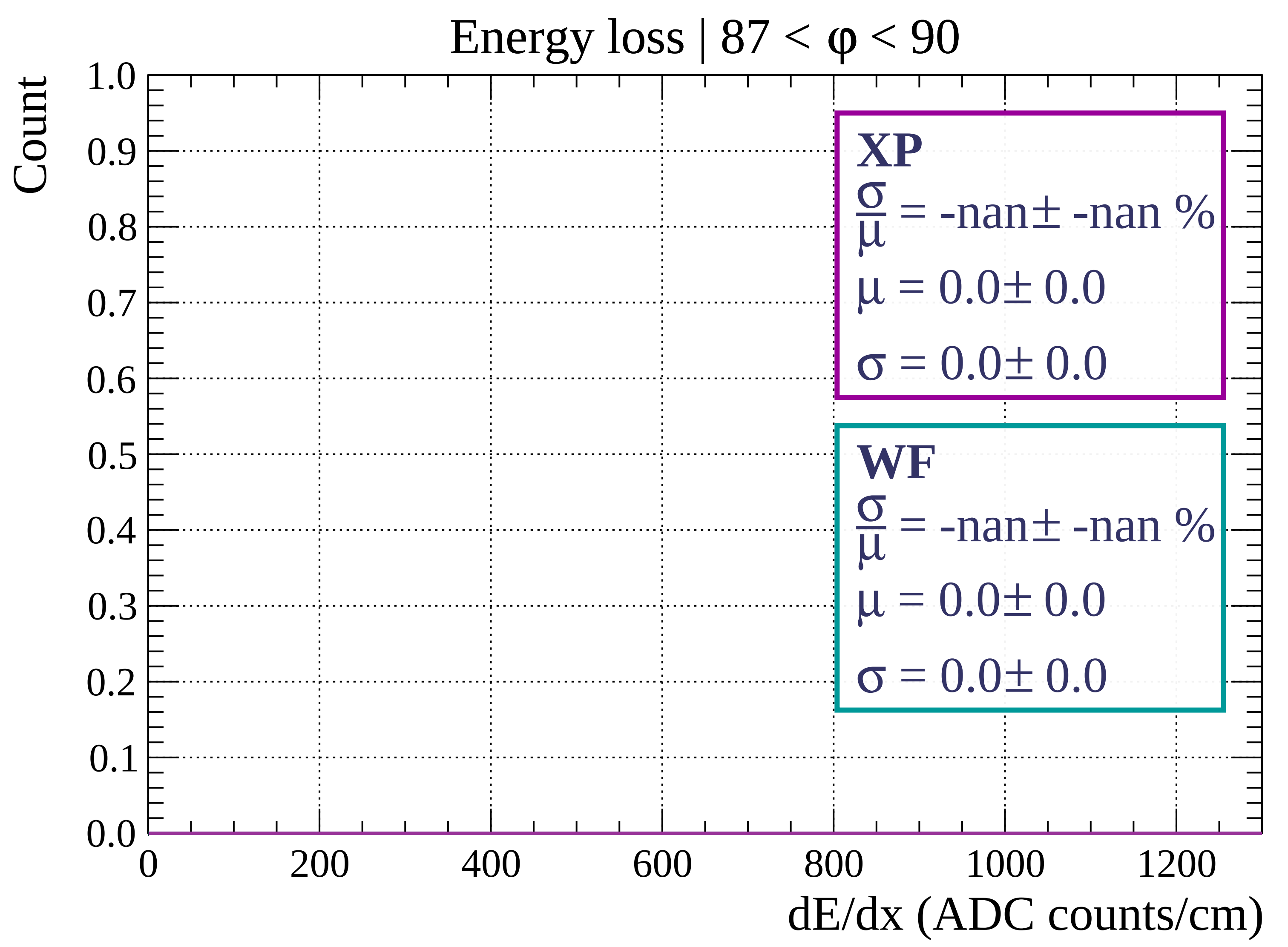
Energy loss | 78 < φ < 81
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



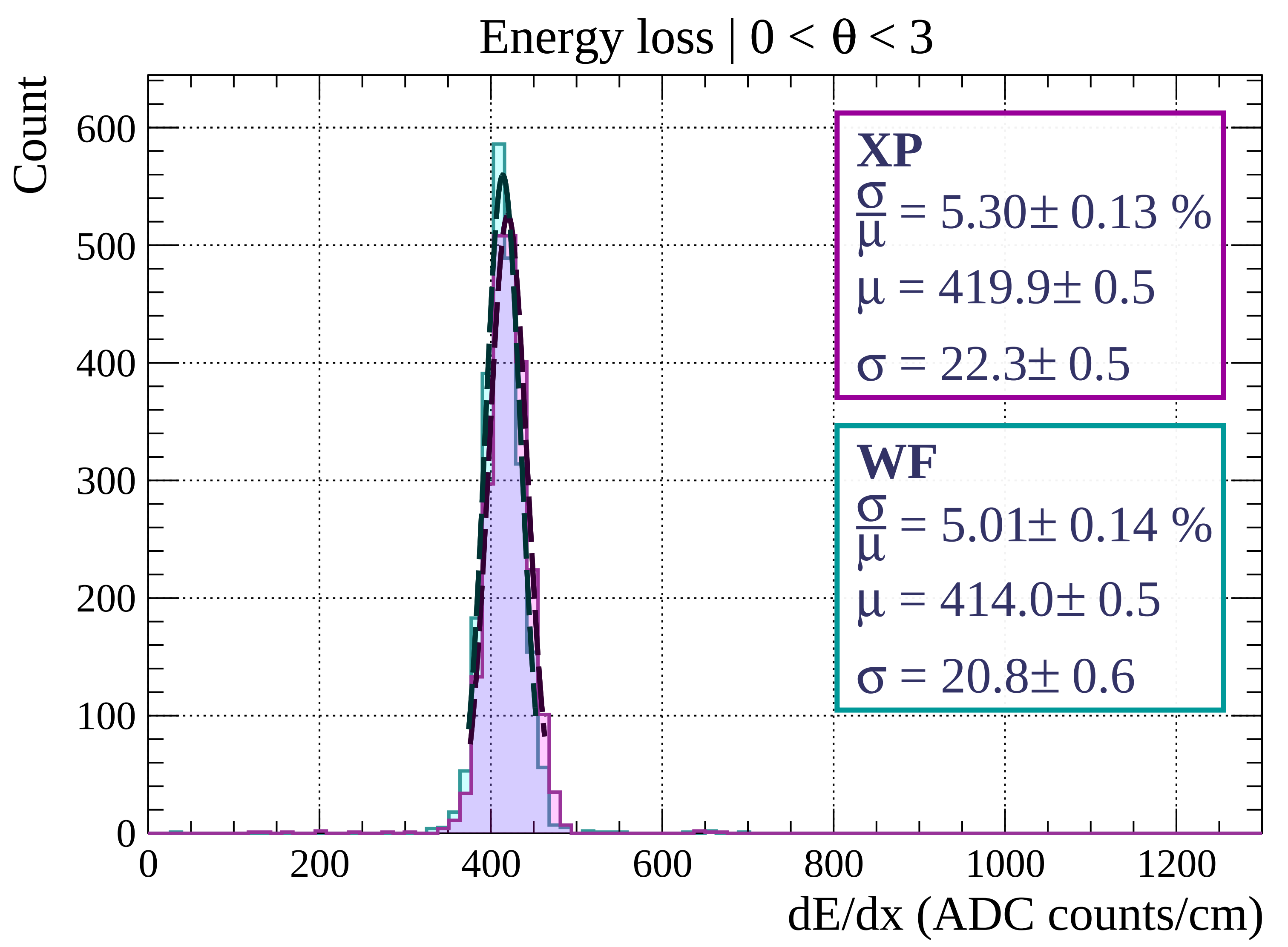
Energy loss | 81 < φ < 84
Count
dE/dx (ADC counts/cm)
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



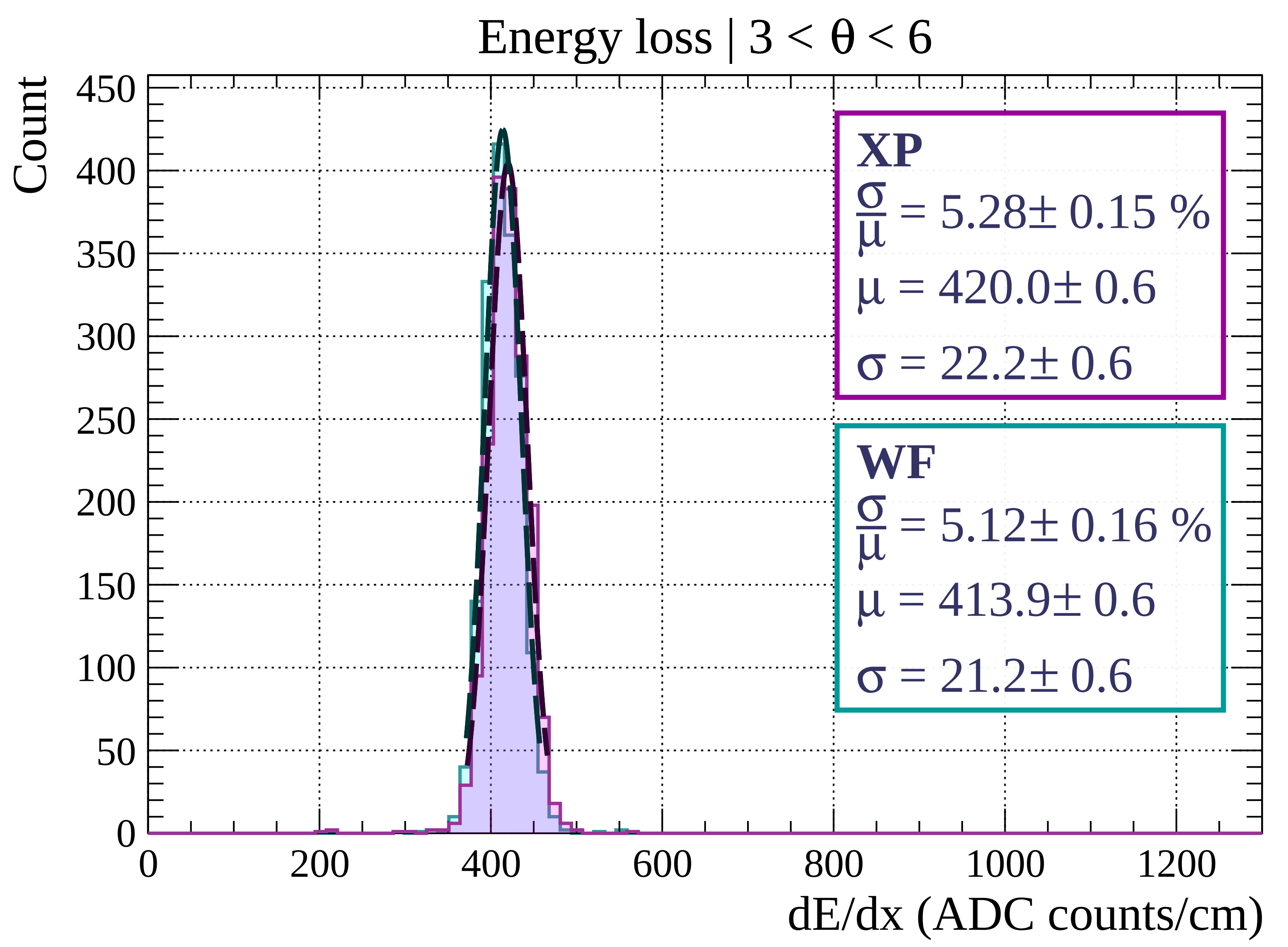
Energy loss | 84 < φ < 87
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



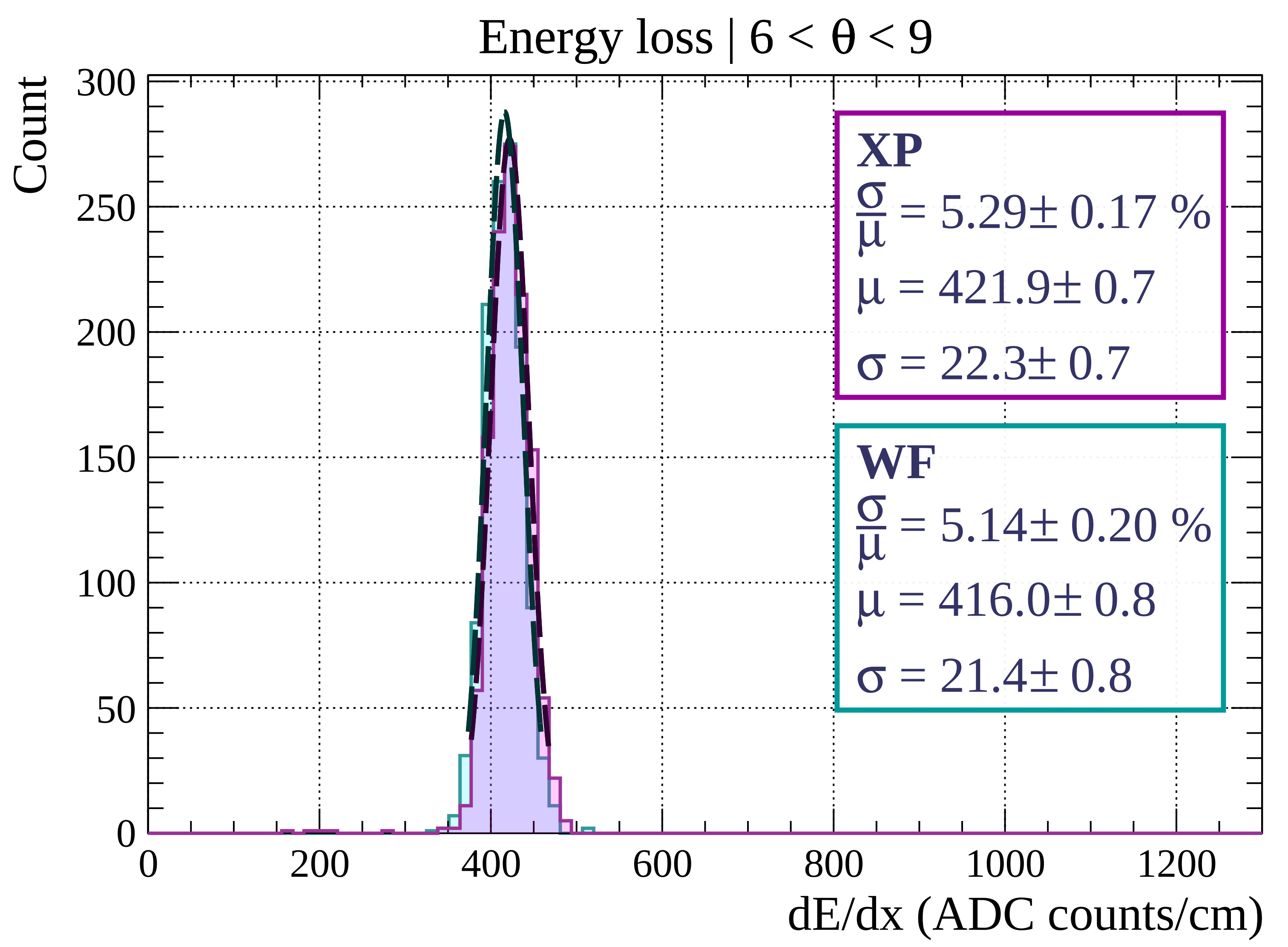
Energy loss | 87 < φ < 90
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

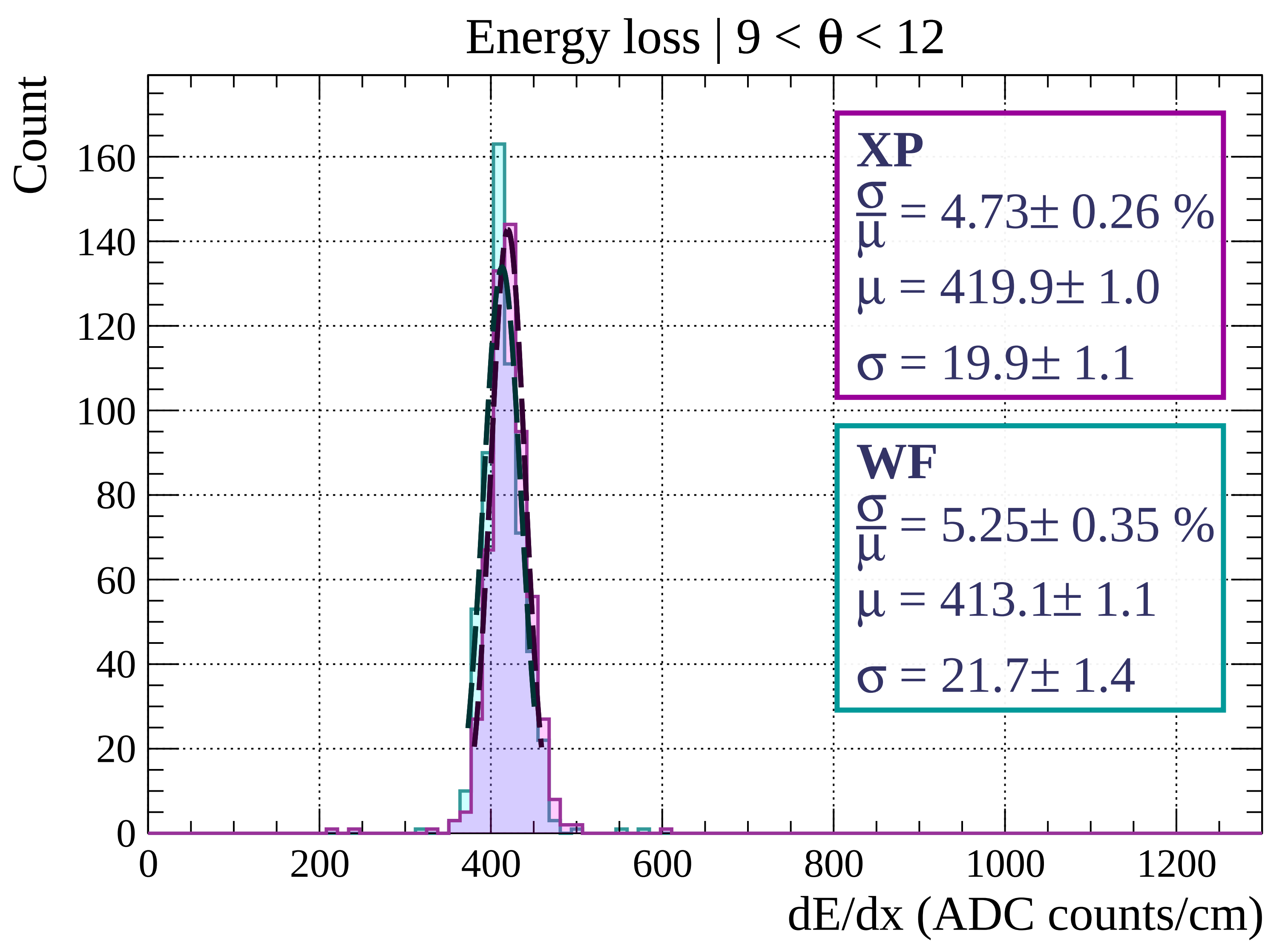


Energy loss | 0 < θ < 3
Count
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 5.30± 0.13 %
μ = 419.9± 0.5
σ = 22.3± 0.5
WF
σ/μ = 5.01± 0.14 %
μ = 414.0± 0.5
σ = 20.8± 0.6

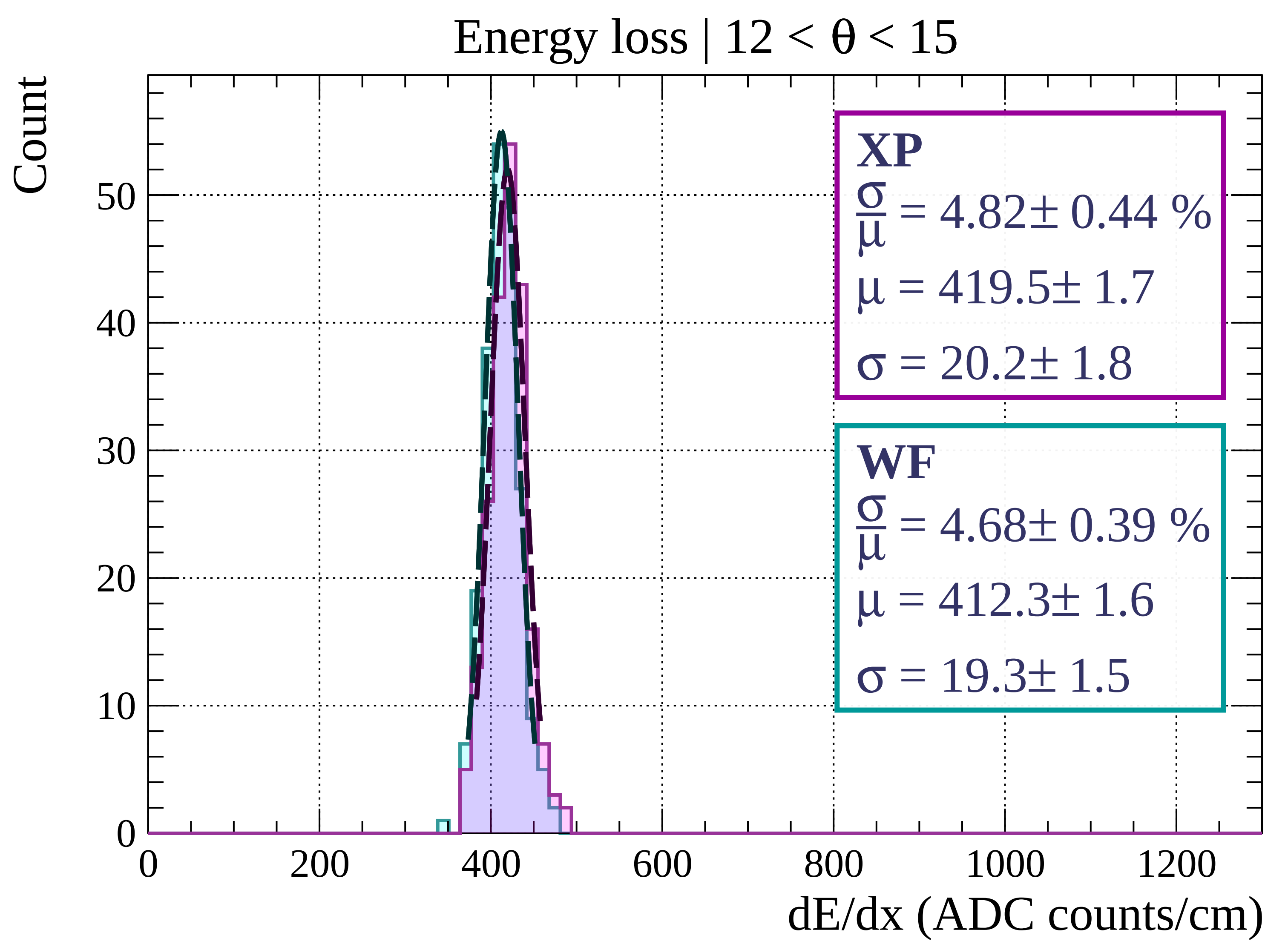


Energy loss | 3 < θ < 6
Count
450
400
350
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 5.28± 0.15 %
μ = 420.0± 0.6
σ = 22.2± 0.6
WF
$\frac{\sigma}{\mu}$ = 5.12± 0.16 %
μ = 413.9± 0.6
σ = 21.2± 0.6

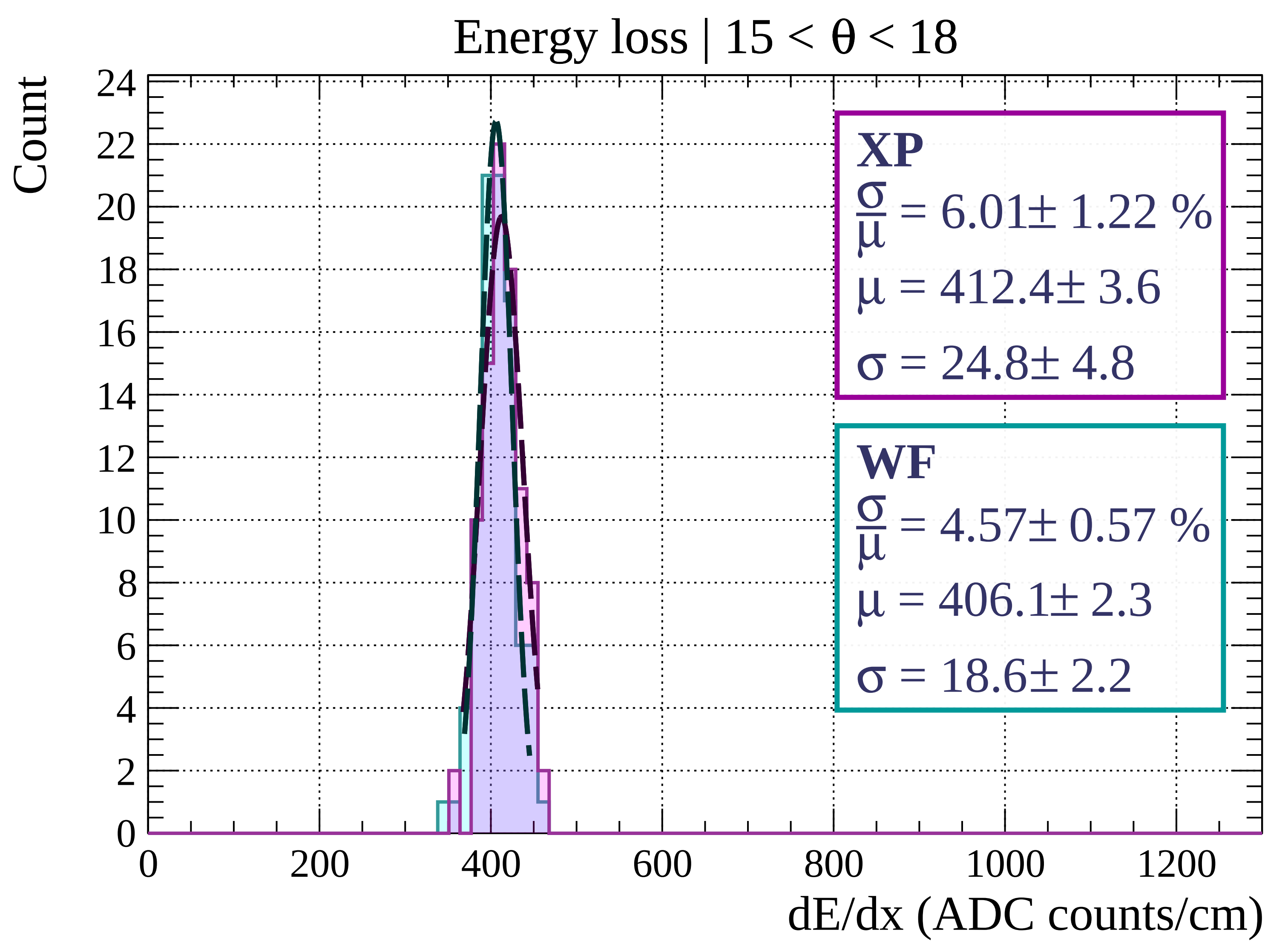

Energy loss | 6 < θ < 9
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP
σ/μ = 5.29± 0.17 %
μ = 421.9± 0.7
σ = 22.3± 0.7
WF
σ/μ = 5.14± 0.20 %
μ = 416.0± 0.8
σ = 21.4± 0.8



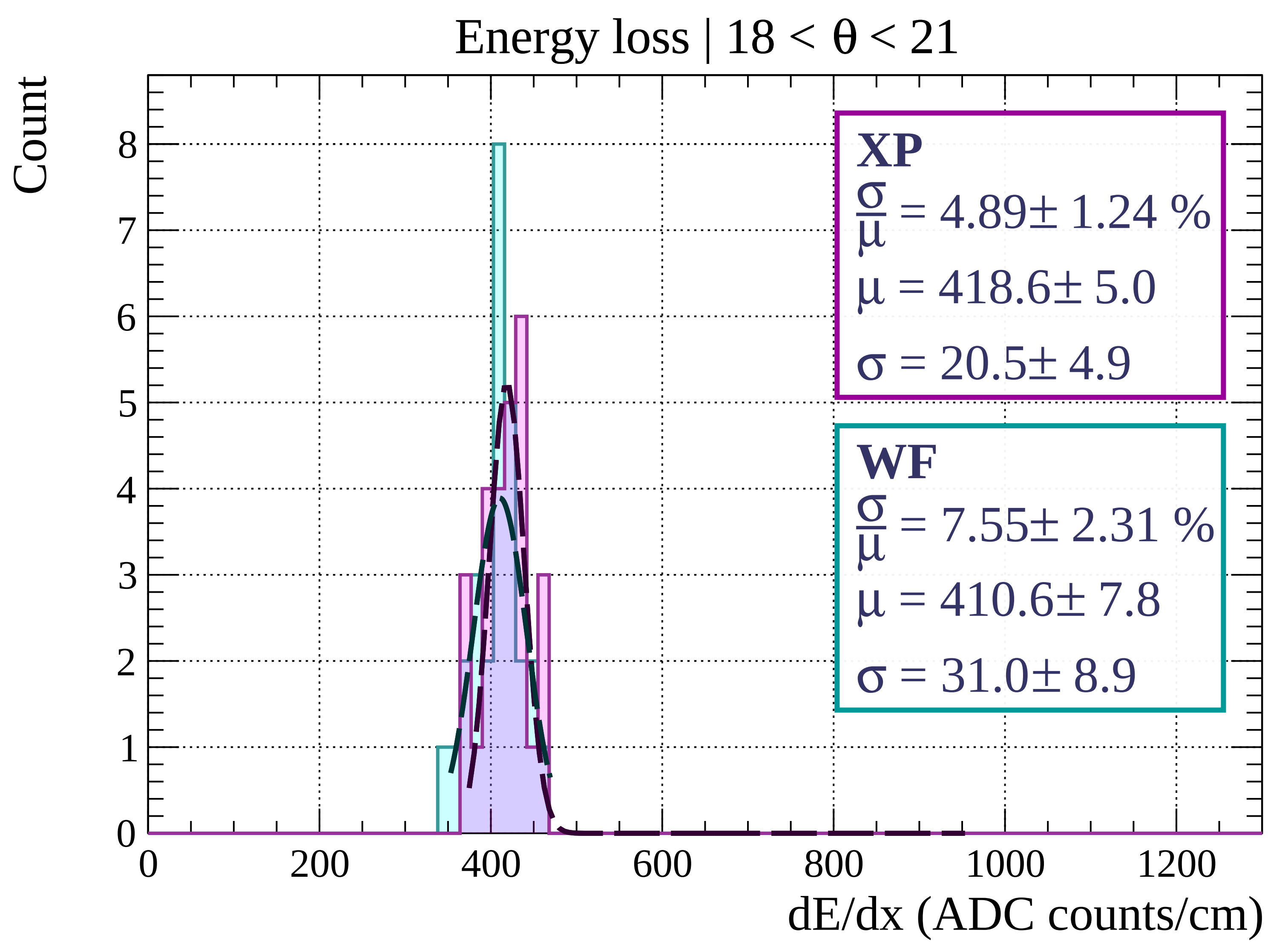
Energy loss | 9 < θ < 12
Count
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 4.73± 0.26 %
μ = 419.9± 1.0
σ = 19.9± 1.1
WF
$\frac{\sigma}{\mu}$ = 5.25± 0.35 %
μ = 413.1± 1.1
σ = 21.7± 1.4



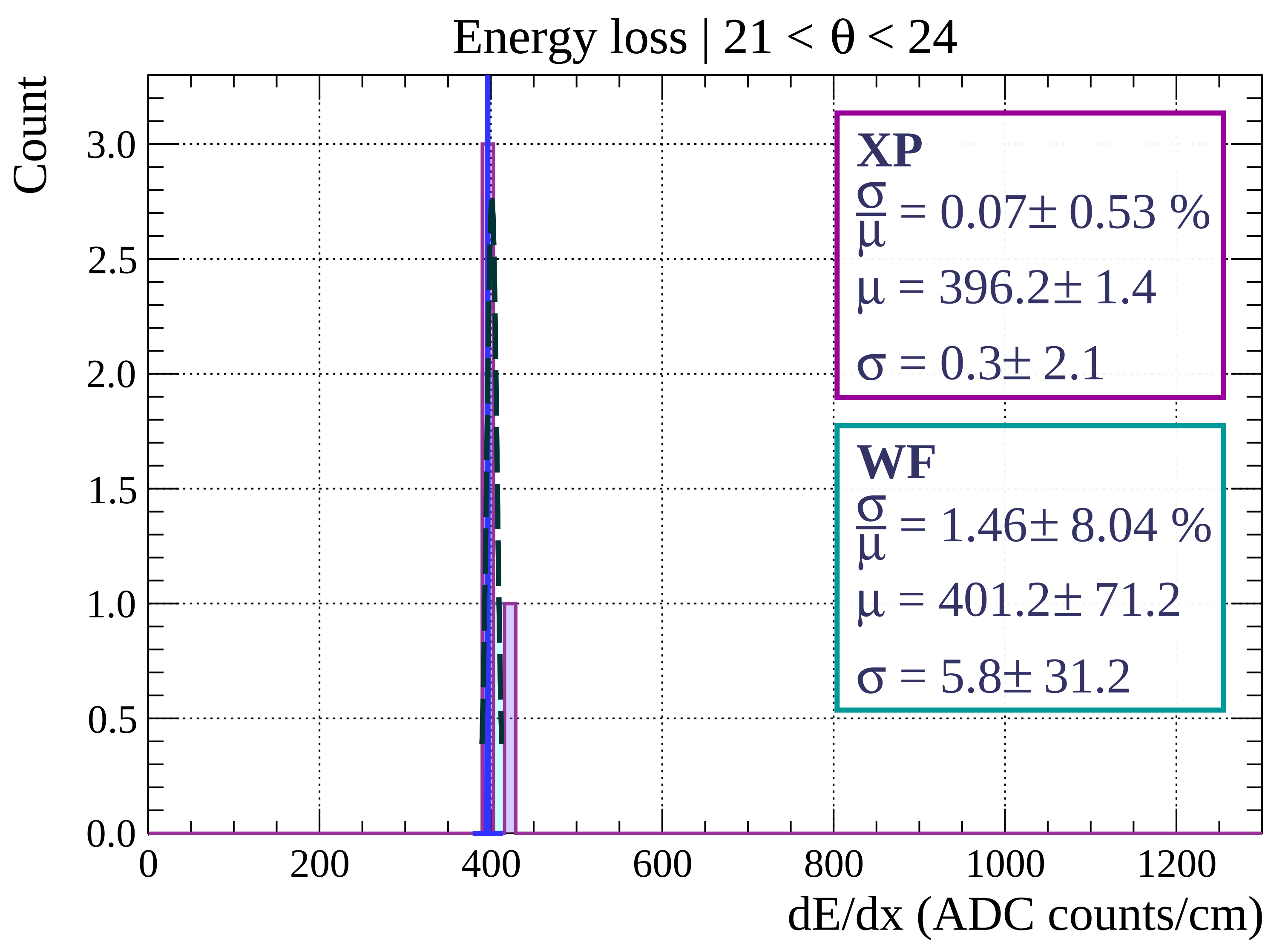
Energy loss | 12 < θ < 15
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
0
200
400
600
800
1000
1200
XP
σ/μ = 4.82± 0.44 %
μ = 419.5± 1.7
σ = 20.2± 1.8
WF
σ/μ = 4.68± 0.39 %
μ = 412.3± 1.6
σ = 19.3± 1.5



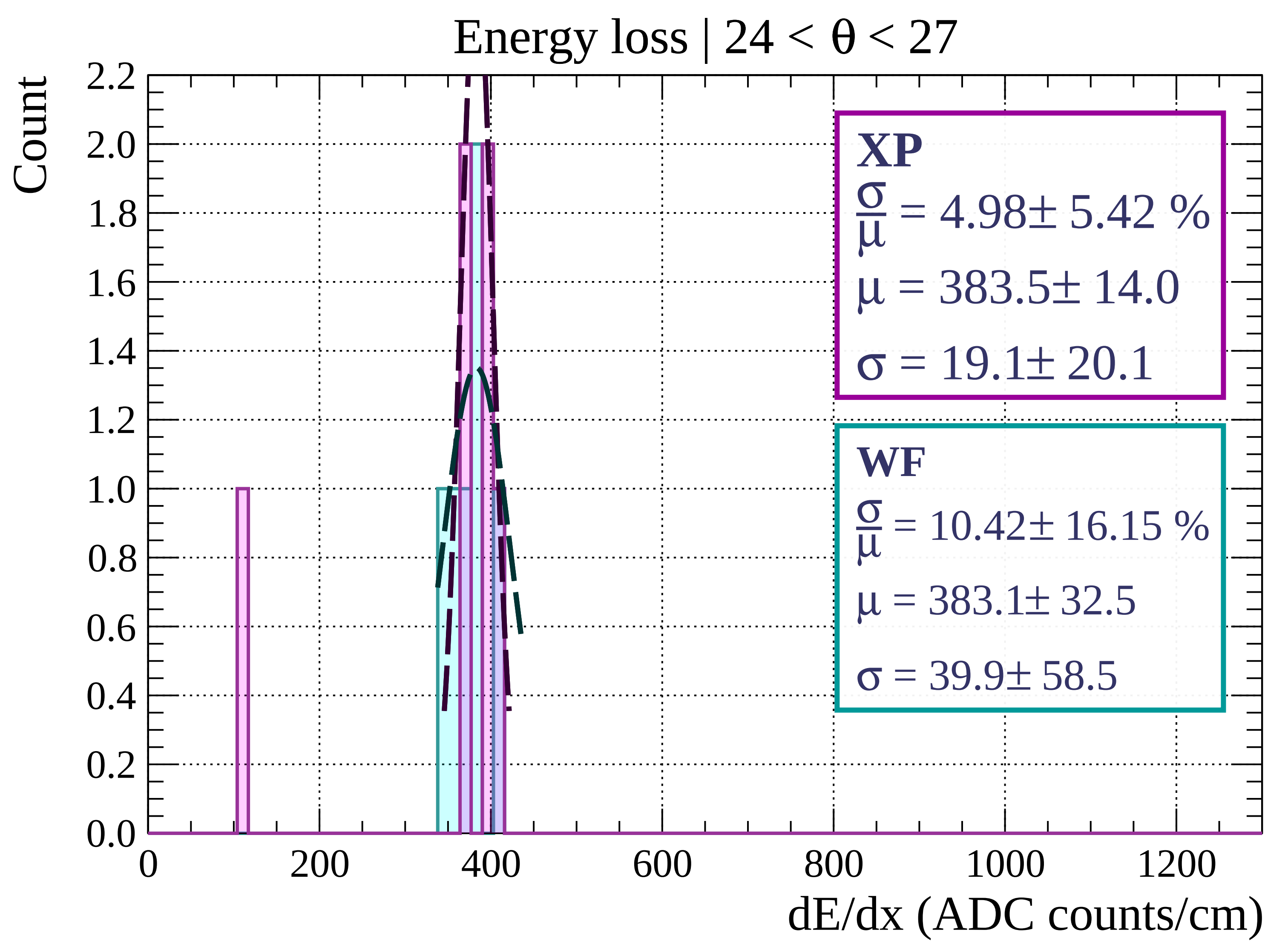
Energy loss | 15 < θ < 18
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
14
16
18
20
22
24
0
200
400
600
800
1000
1200
XP
σ/μ = 6.01± 1.22 %
μ = 412.4± 3.6
σ = 24.8± 4.8
WF
σ/μ = 4.57± 0.57 %
μ = 406.1± 2.3
σ = 18.6± 2.2



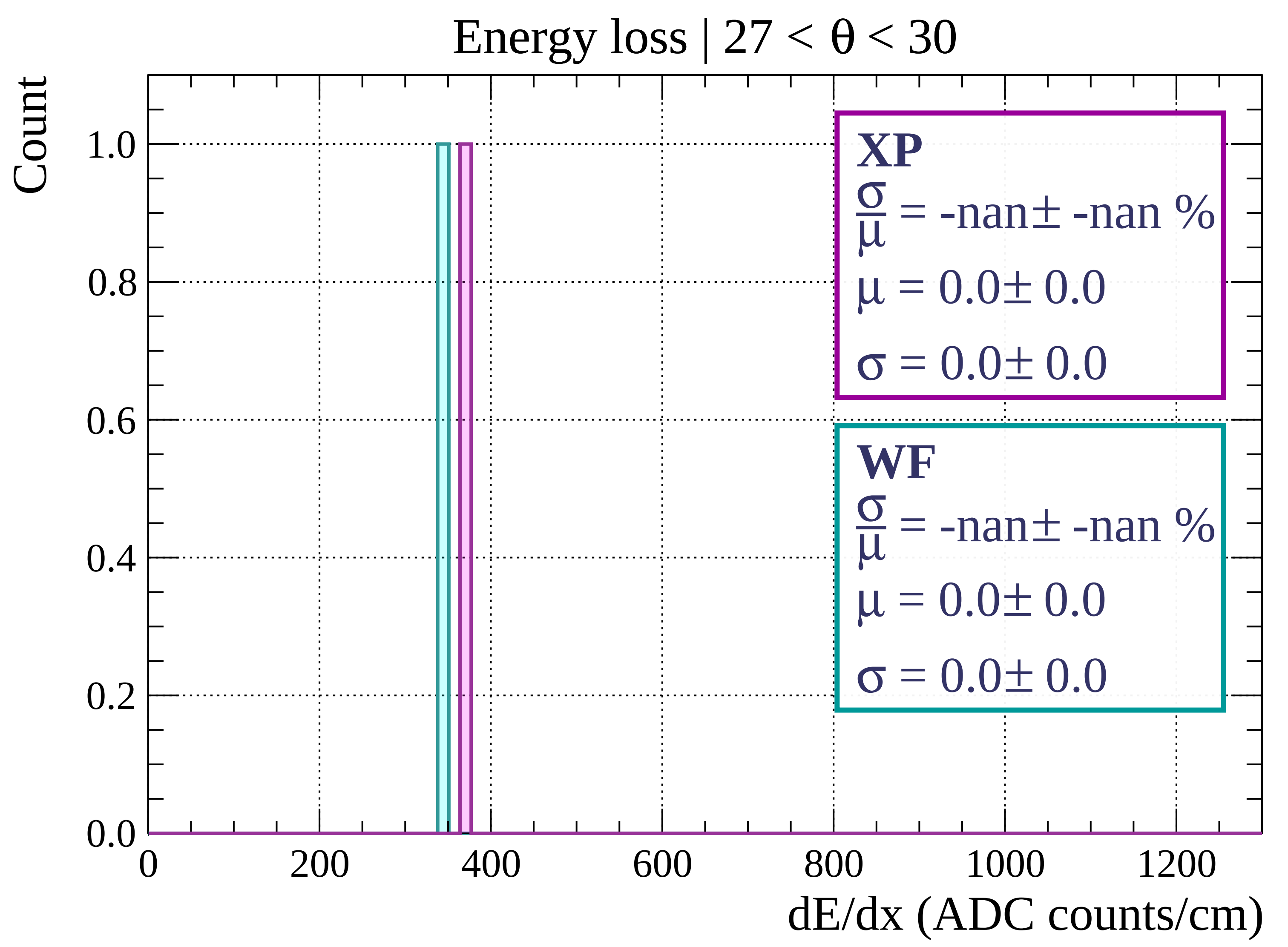
Energy loss | 18 < θ < 21
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1
2
3
4
5
6
7
8
XP
σ/μ = 4.89± 1.24 %
μ = 418.6± 5.0
σ = 20.5± 4.9
WF
σ/μ = 7.55± 2.31 %
μ = 410.6± 7.8
σ = 31.0± 8.9



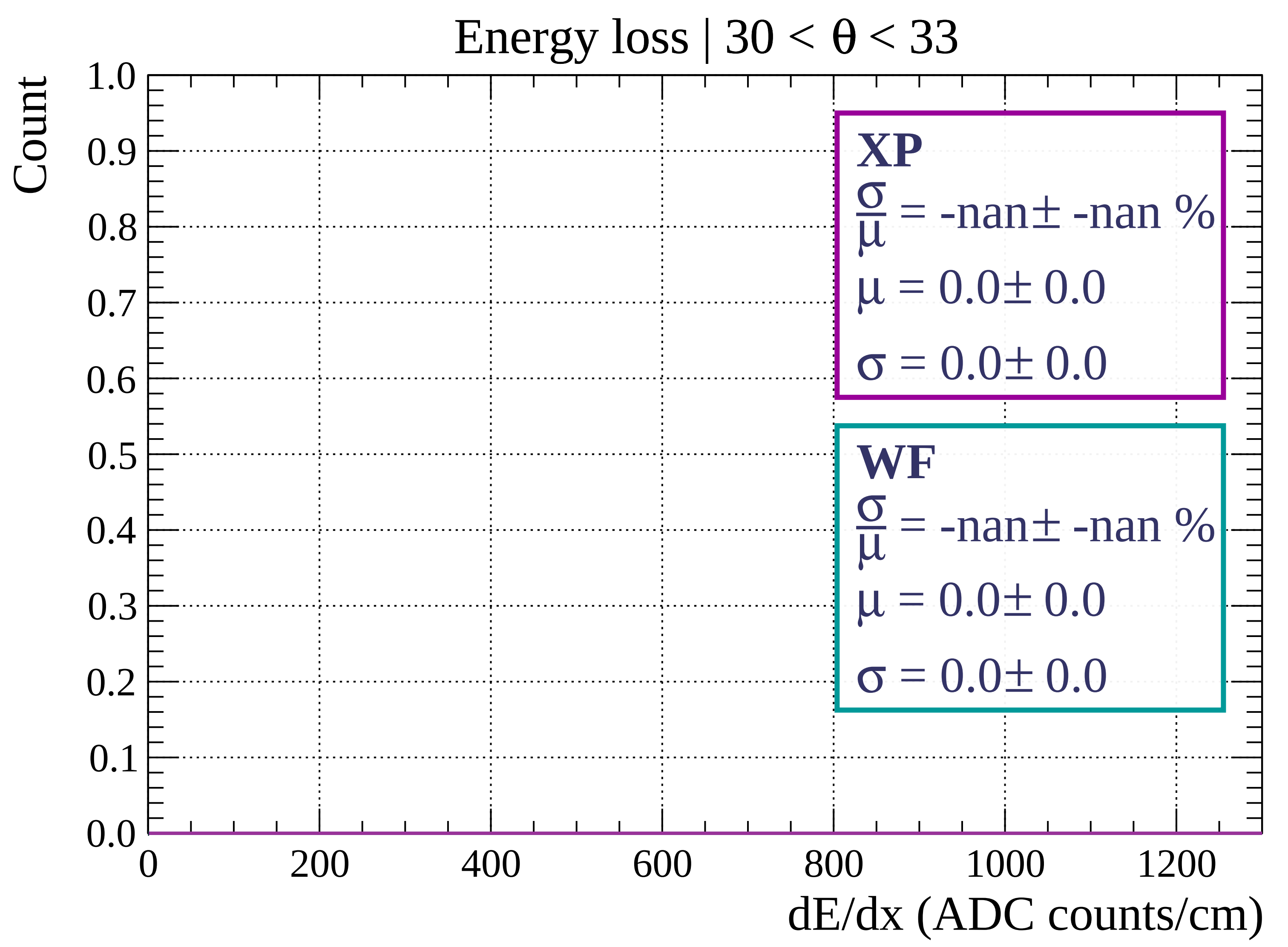
Energy loss | 21 < θ < 24
Count
dE/dx (ADC counts/cm)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
0
200
400
600
800
1000
1200
XP
σ/μ = 0.07± 0.53 %
μ = 396.2± 1.4
σ = 0.3± 2.1
WF
σ/μ = 1.46± 8.04 %
μ = 401.2± 71.2
σ = 5.8± 31.2

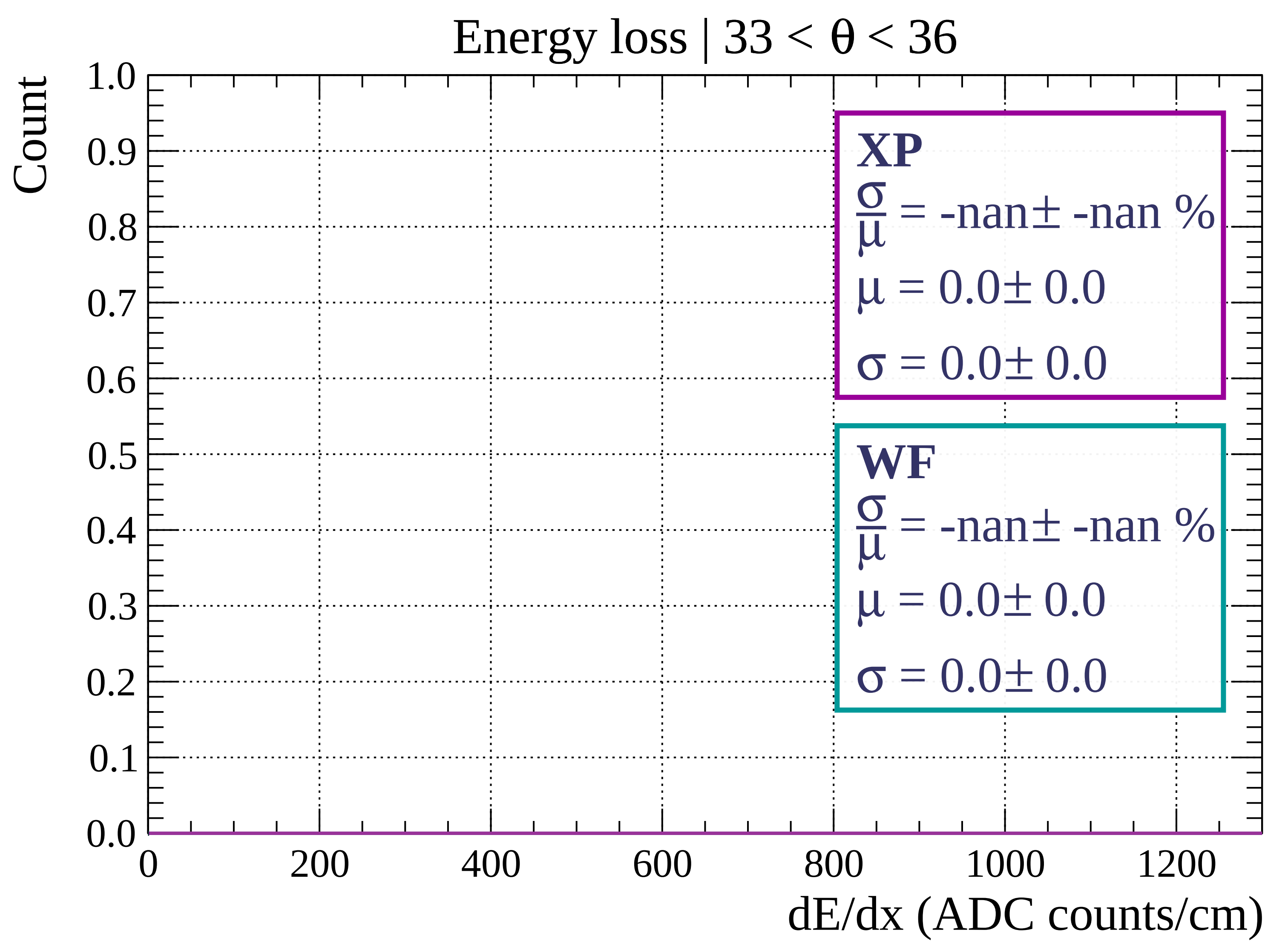


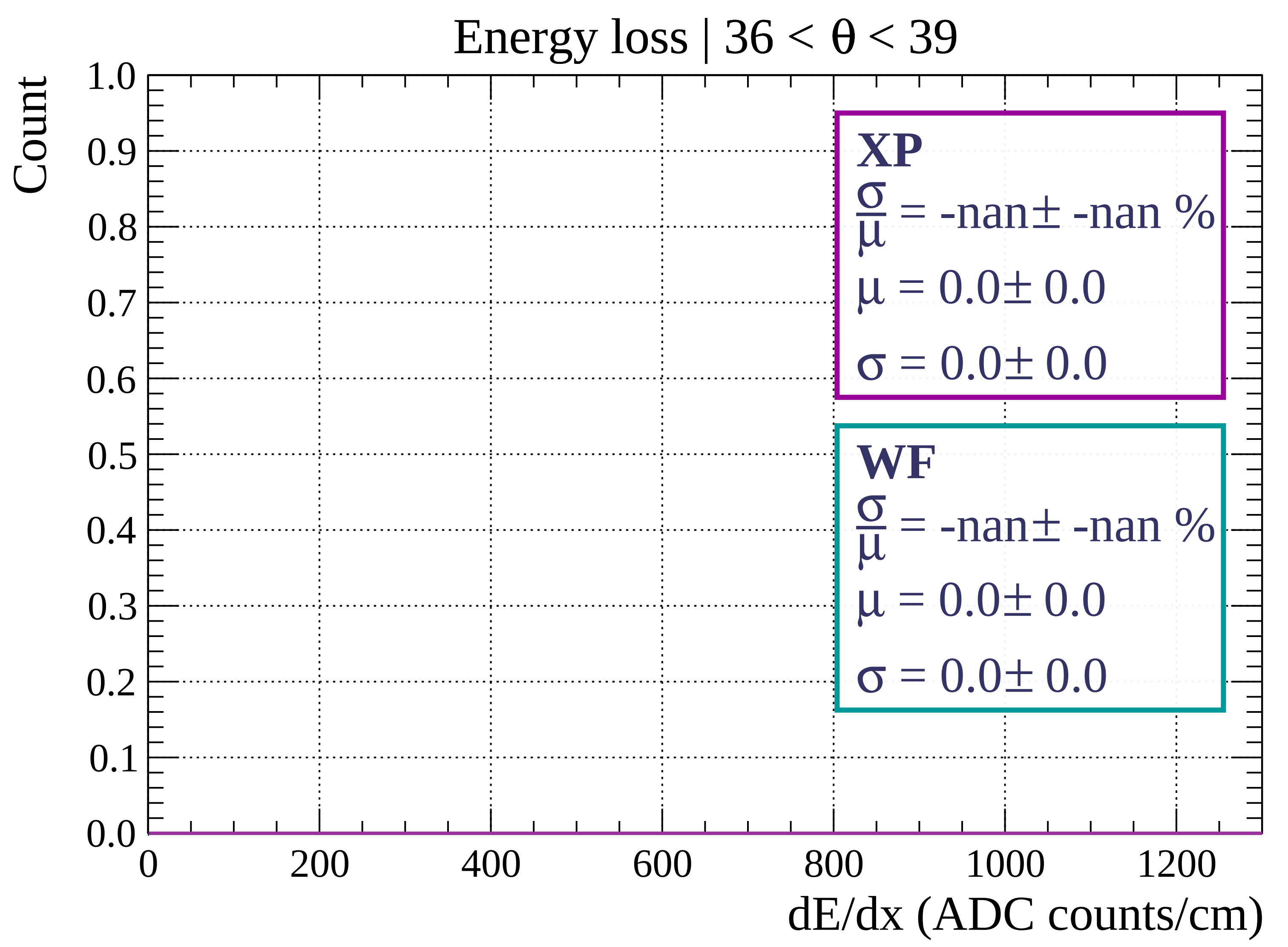
Energy loss | 24 < θ < 27
Count
2.2
2.0
1.8
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 4.98± 5.42 %
μ = 383.5± 14.0
σ = 19.1± 20.1
WF
σ/μ = 10.42± 16.15 %
μ = 383.1± 32.5
σ = 39.9± 58.5

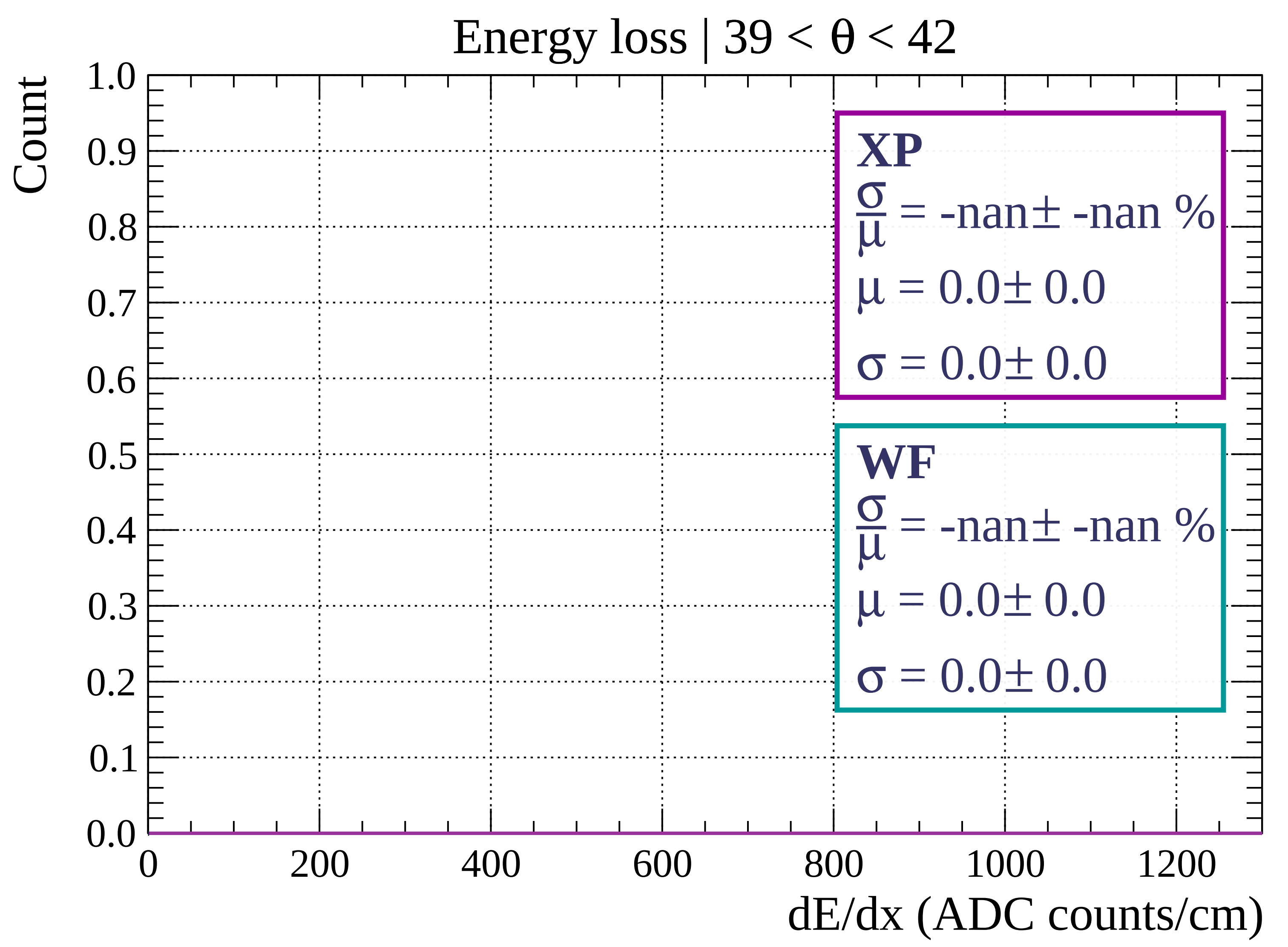


Energy loss | 27 < θ < 30
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

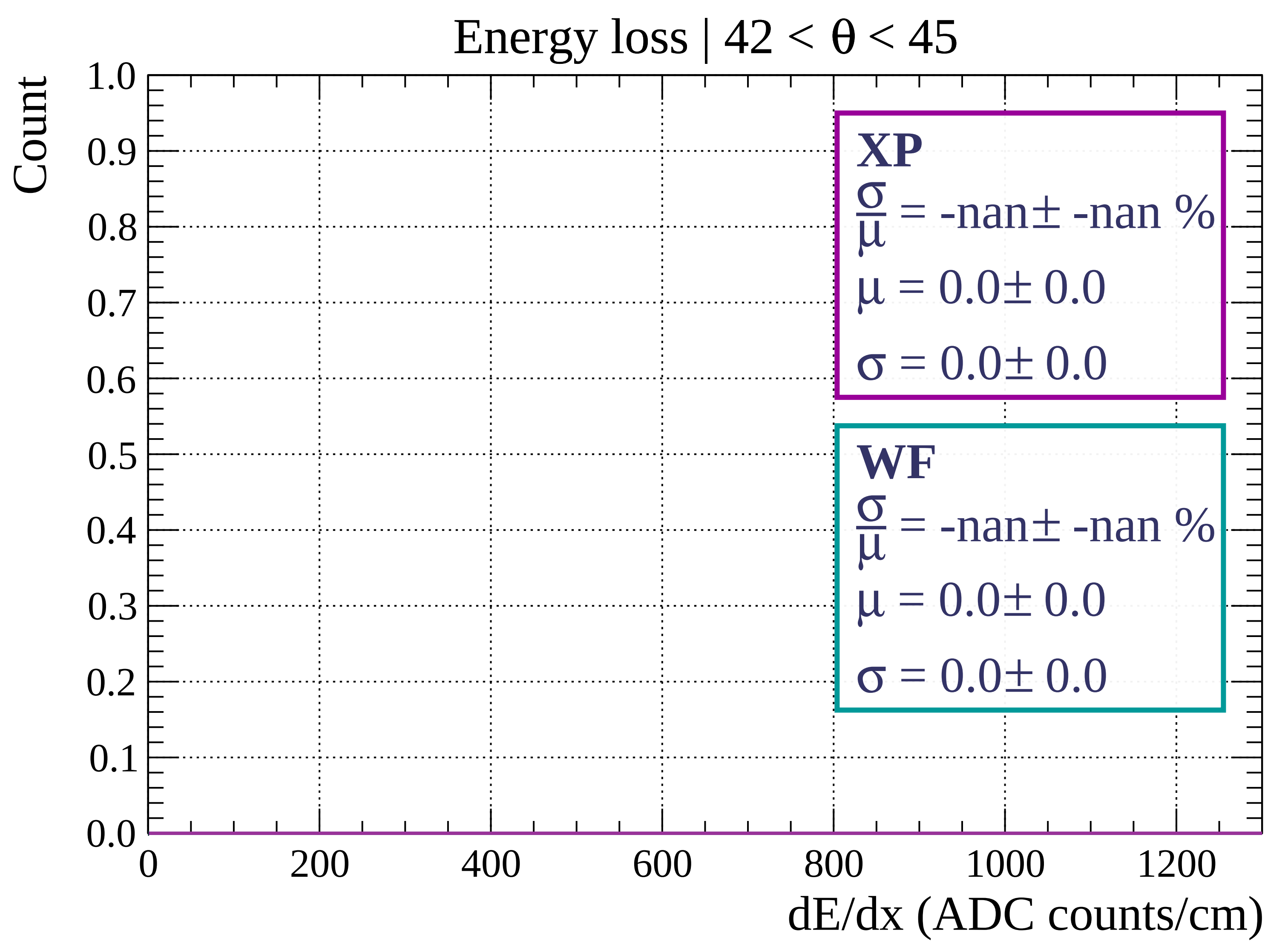

Energy loss | 30 < θ < 33
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | 33 < θ < 36
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

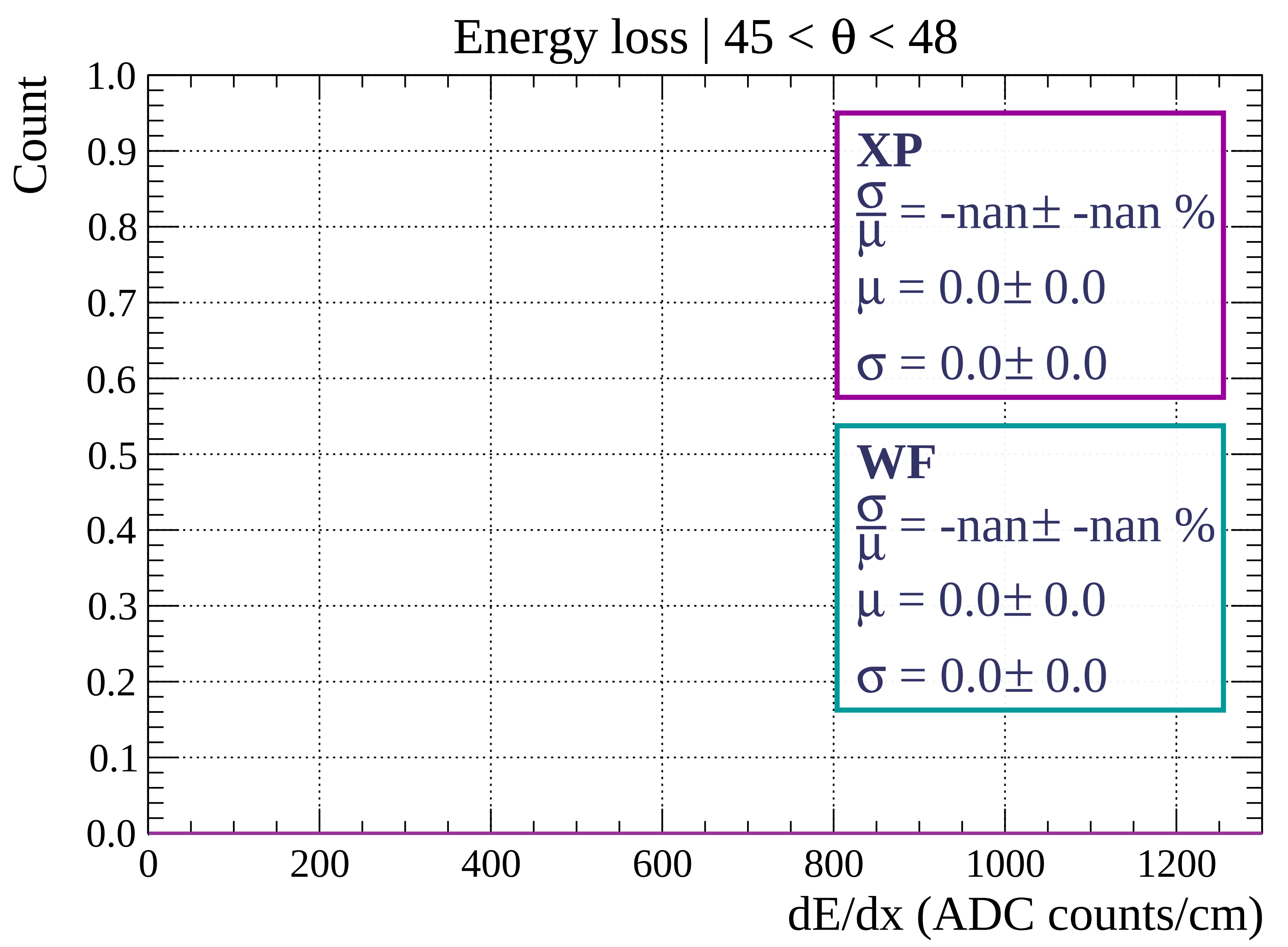

Energy loss | 36 < θ < 39
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



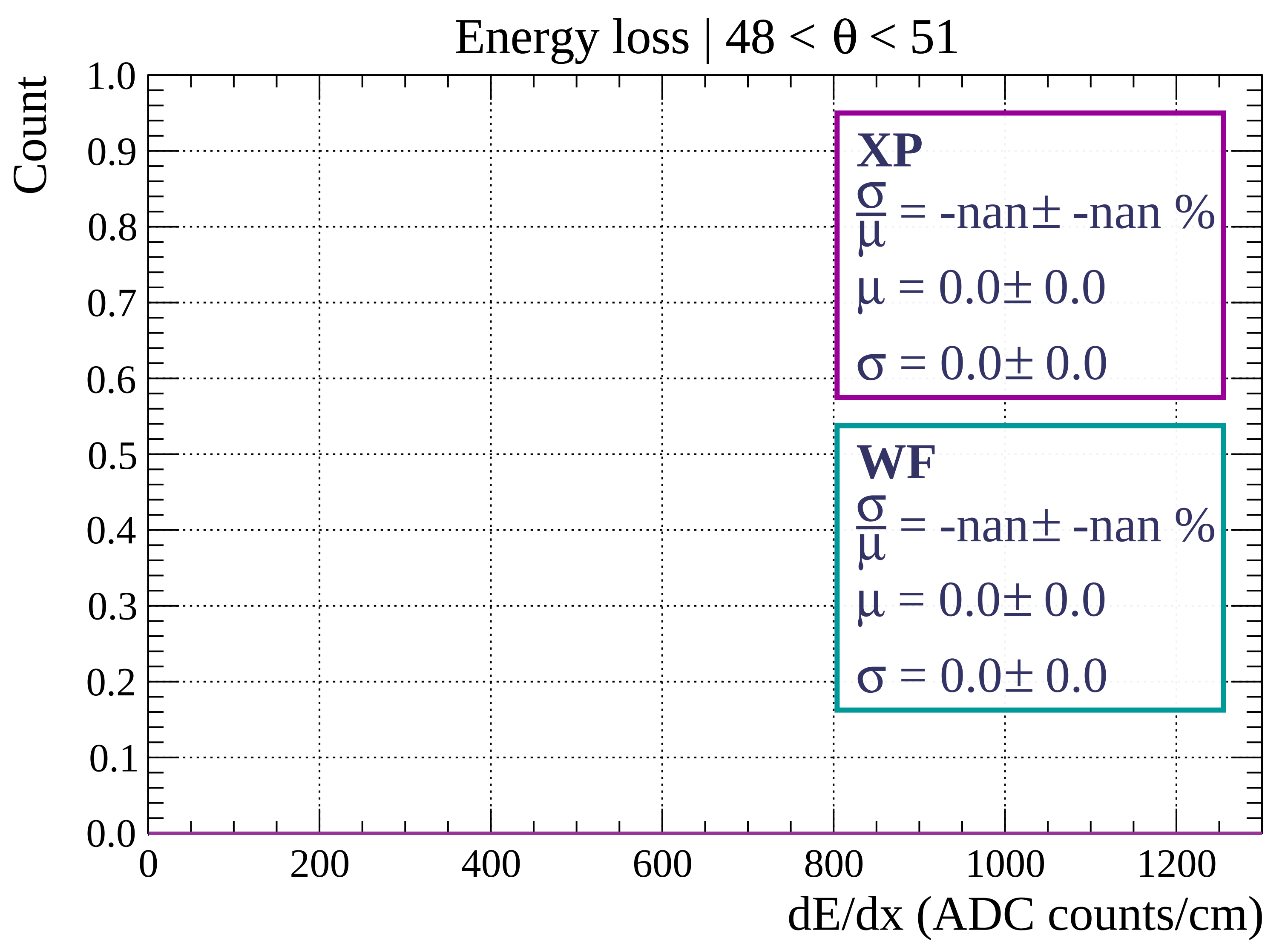
Energy loss | 39 < θ < 42
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



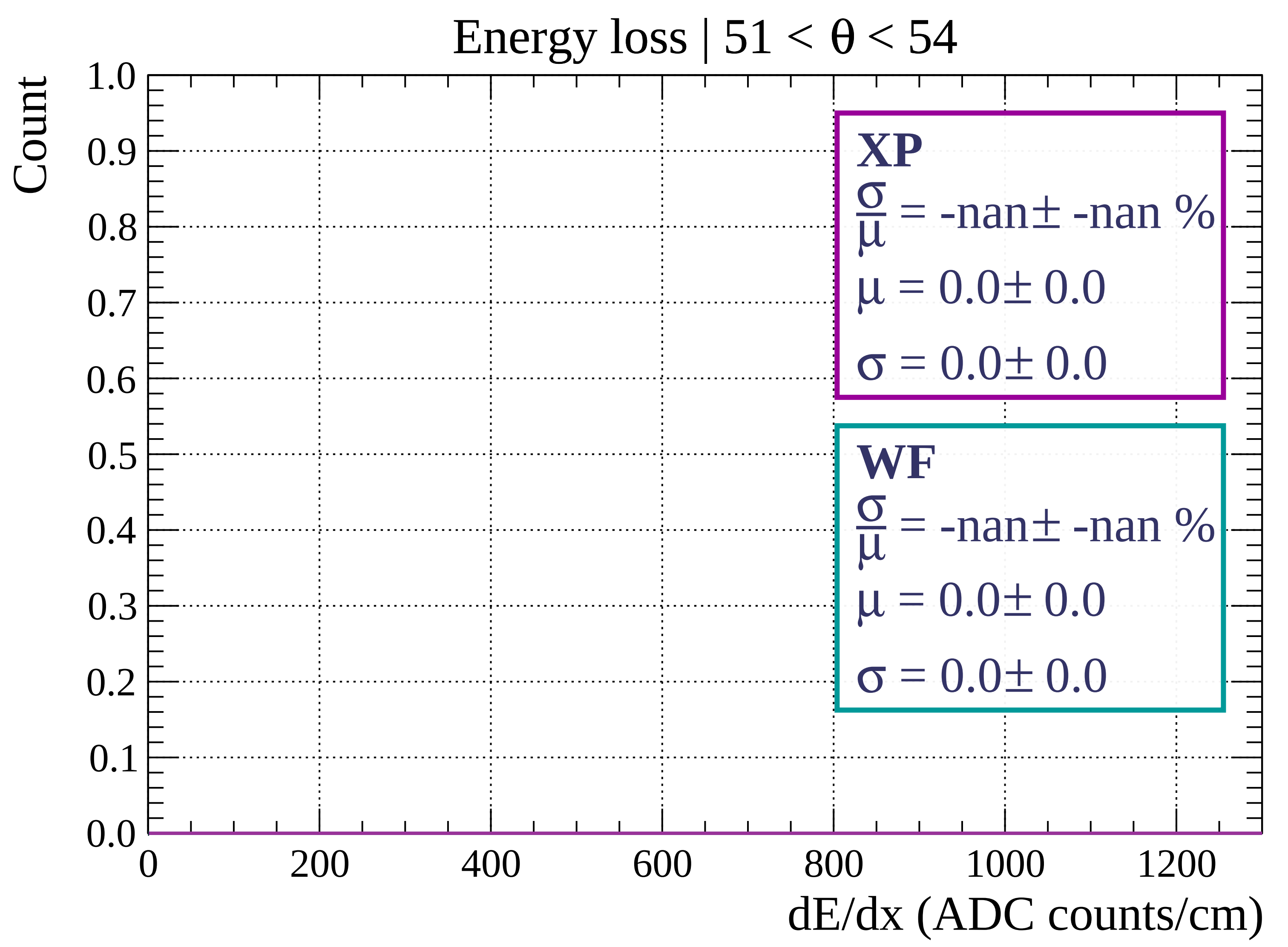
Energy loss | 42 < θ < 45
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

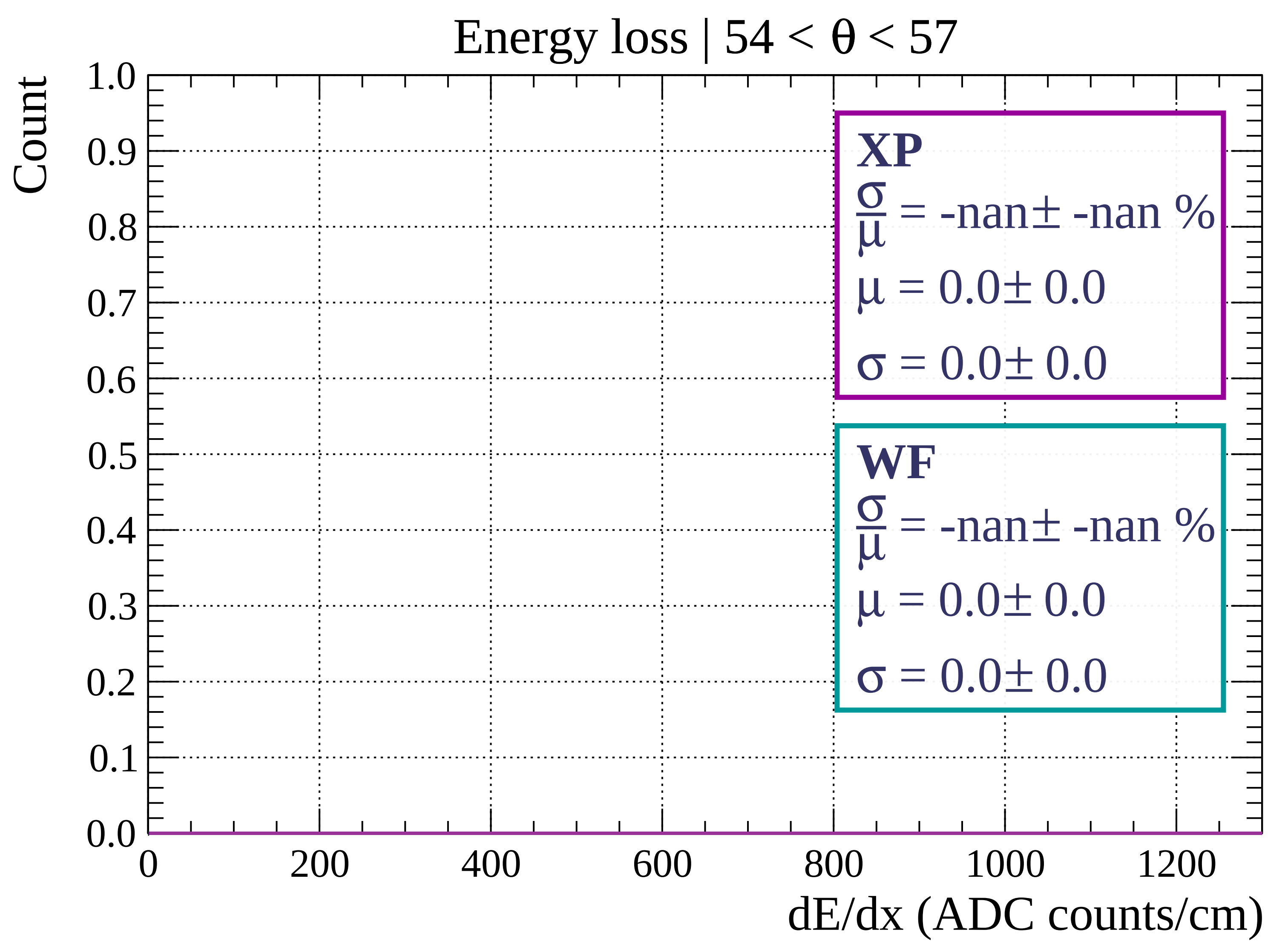


Energy loss | 45 < θ < 48
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

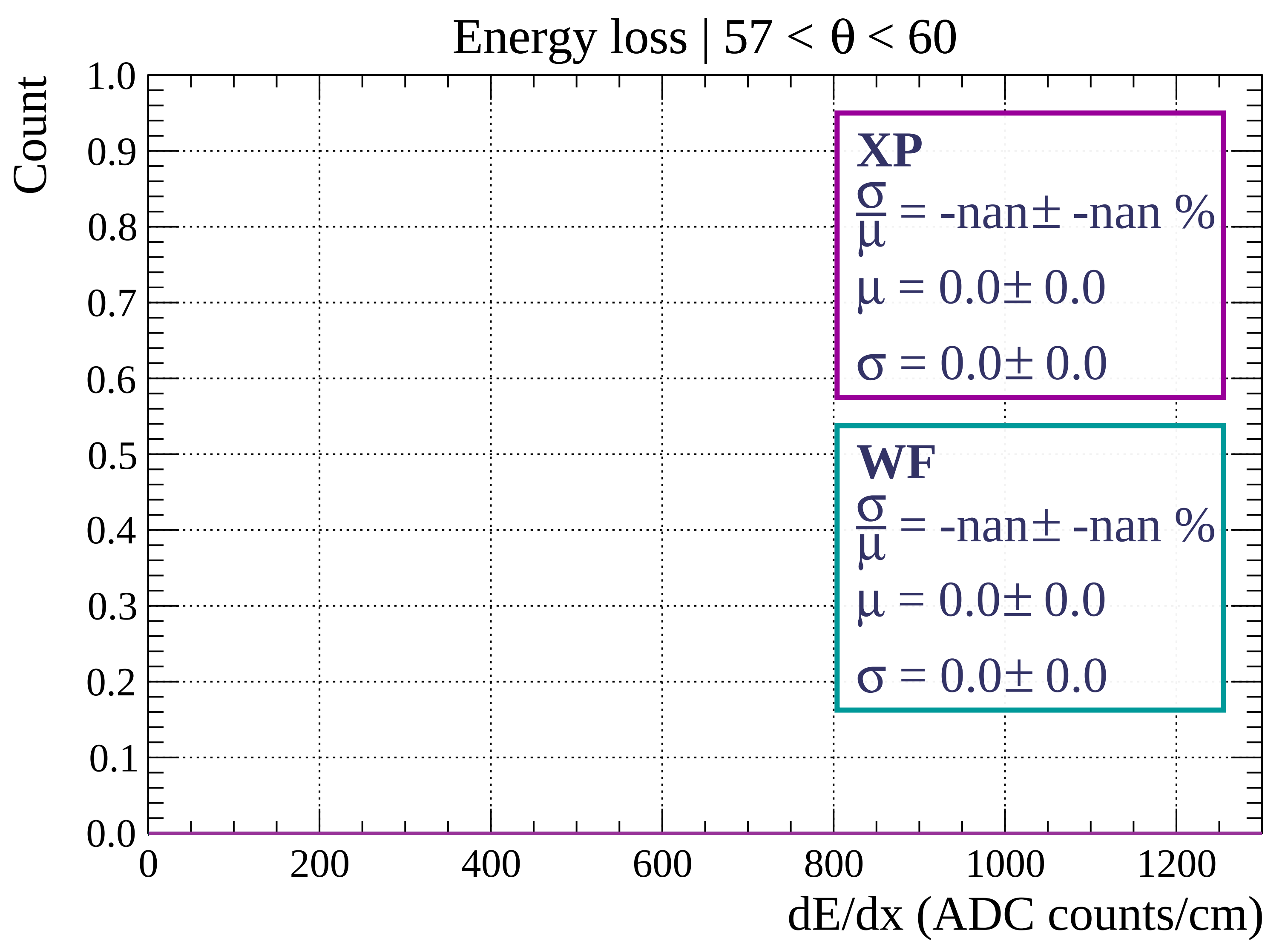


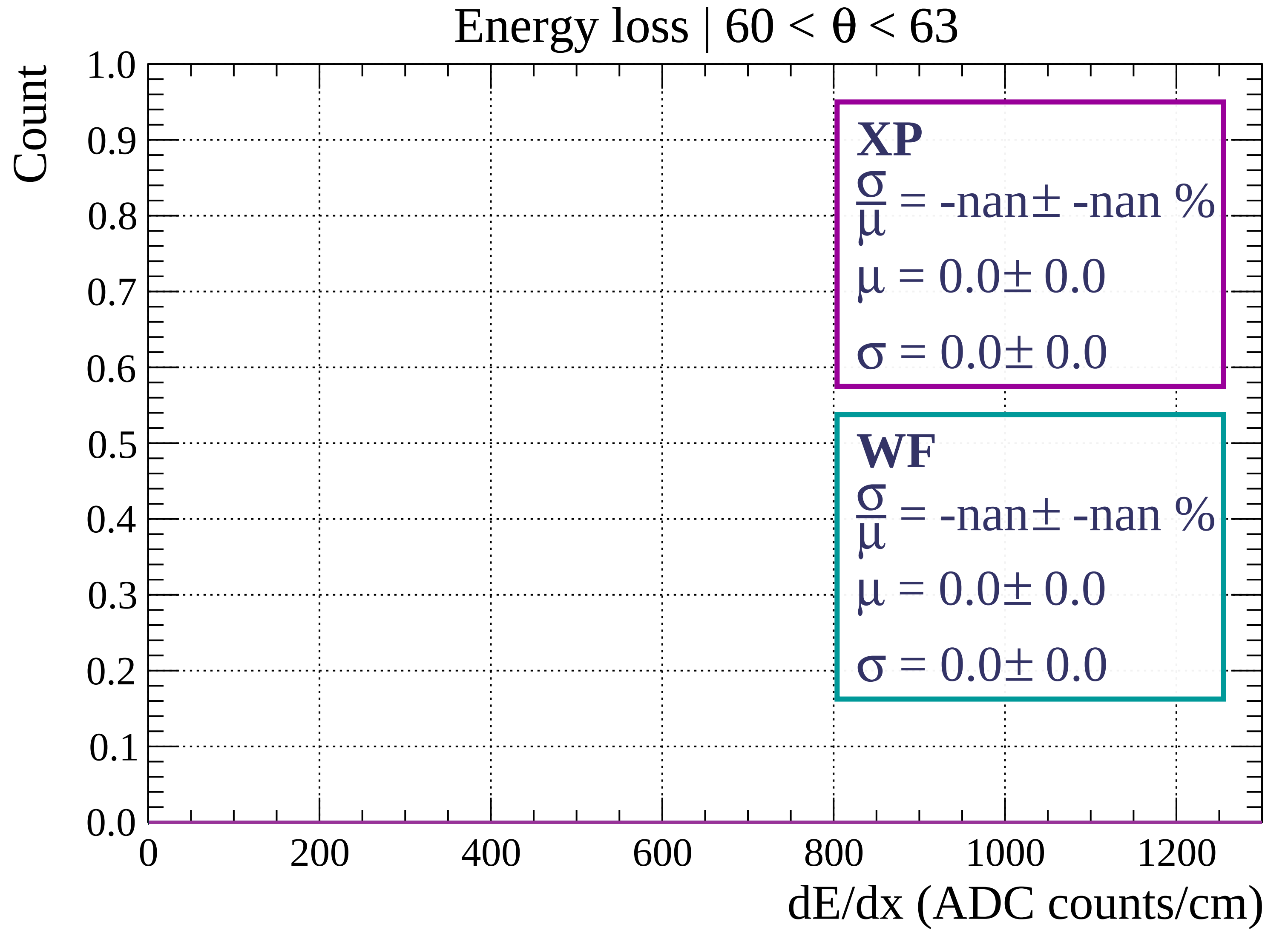
Energy loss | 48 < θ < 51
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | 51 < θ < 54
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

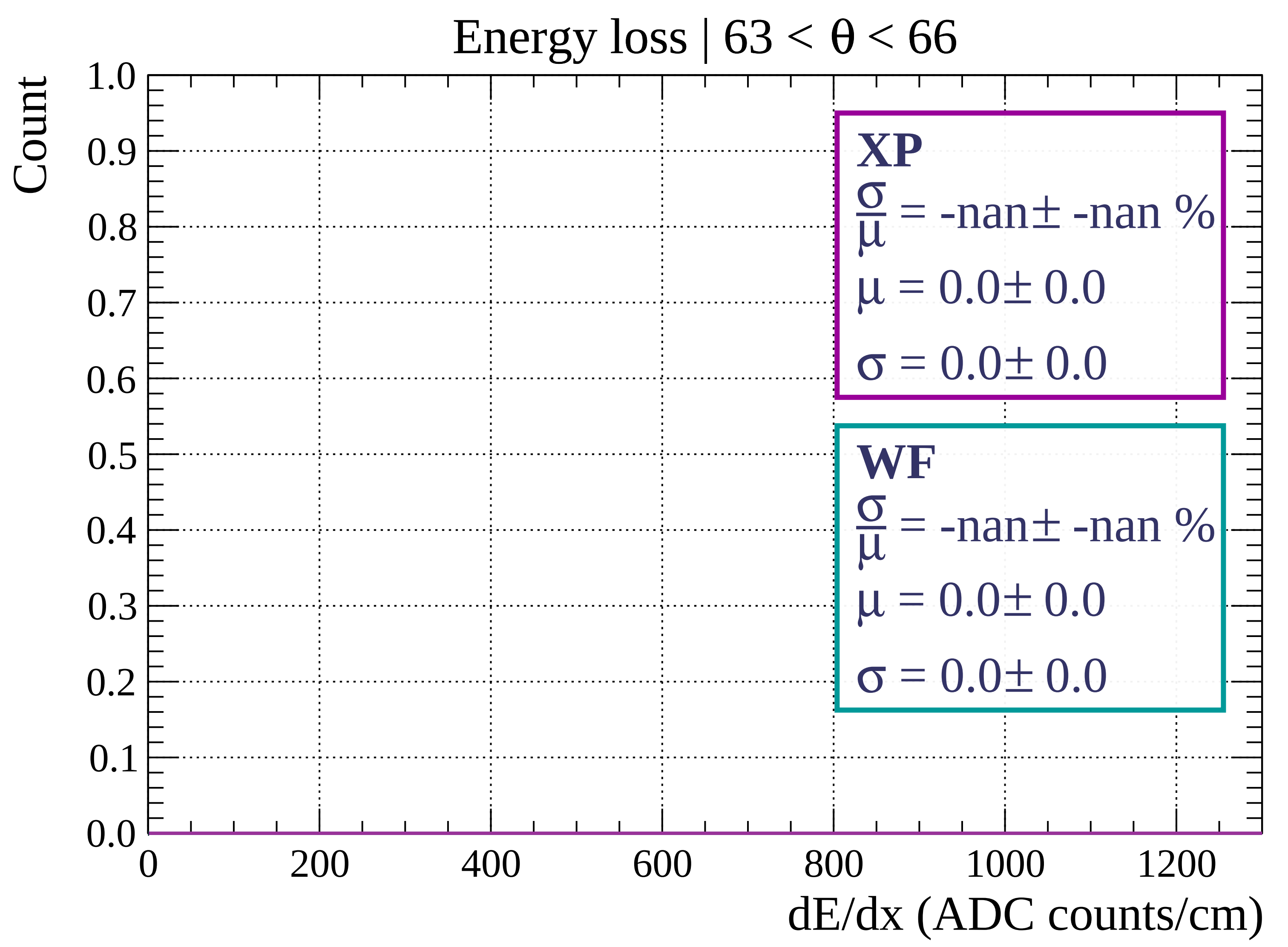


Energy loss | 54 < θ < 57
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

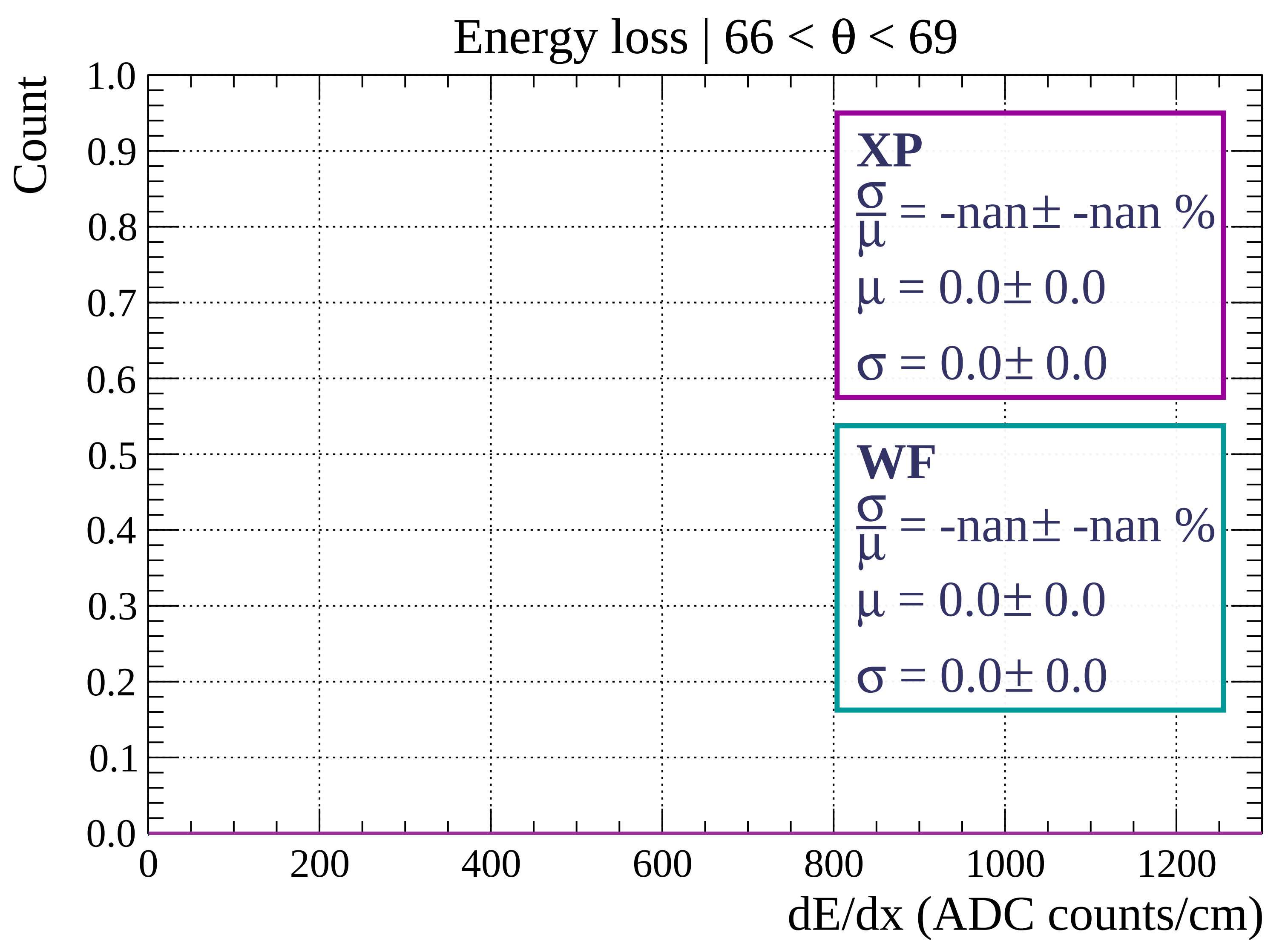

Energy loss | 57 < θ < 60
Count
dE/dx (ADC counts/cm)
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



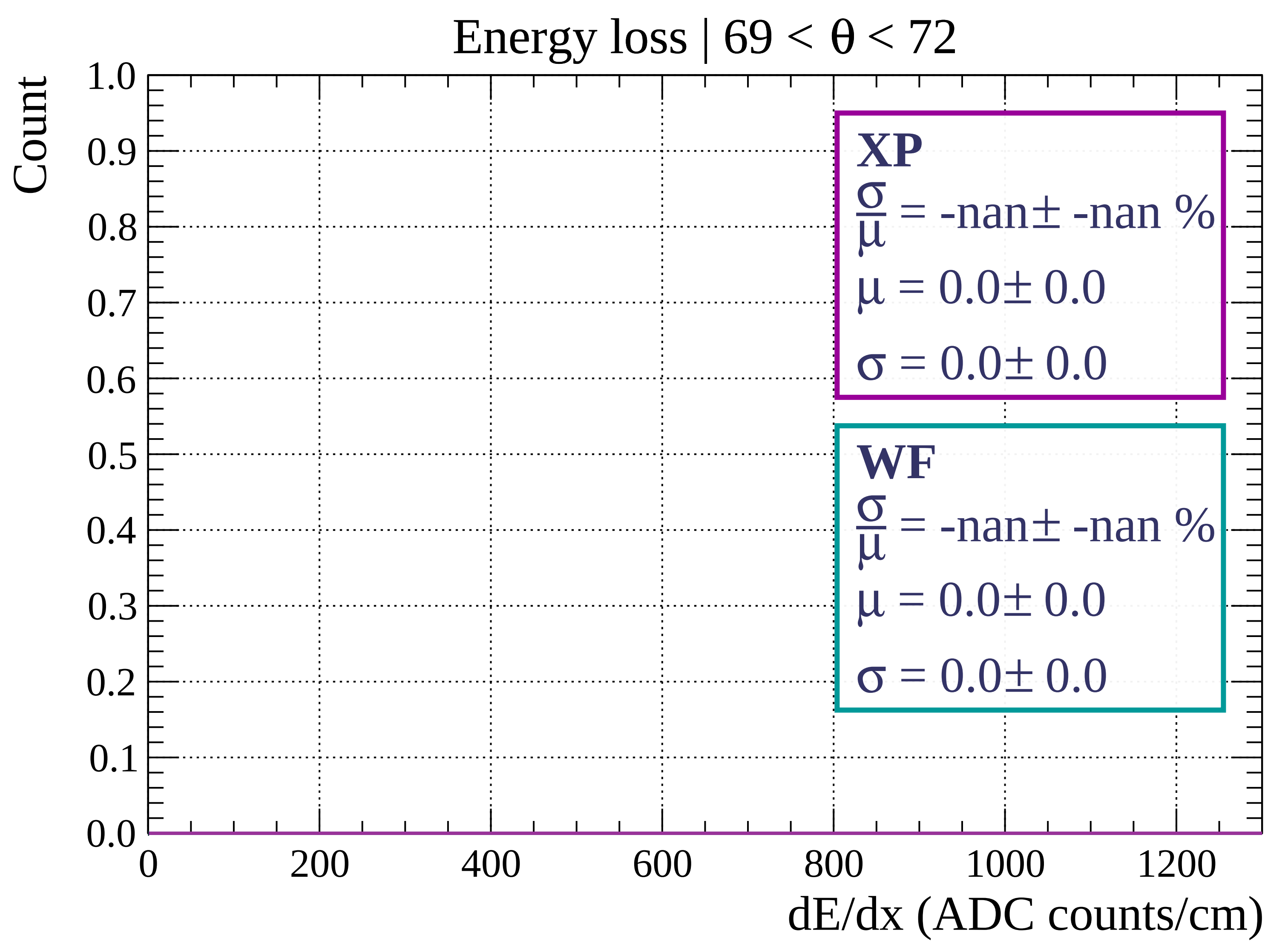
Energy loss | 60 < θ < 63
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



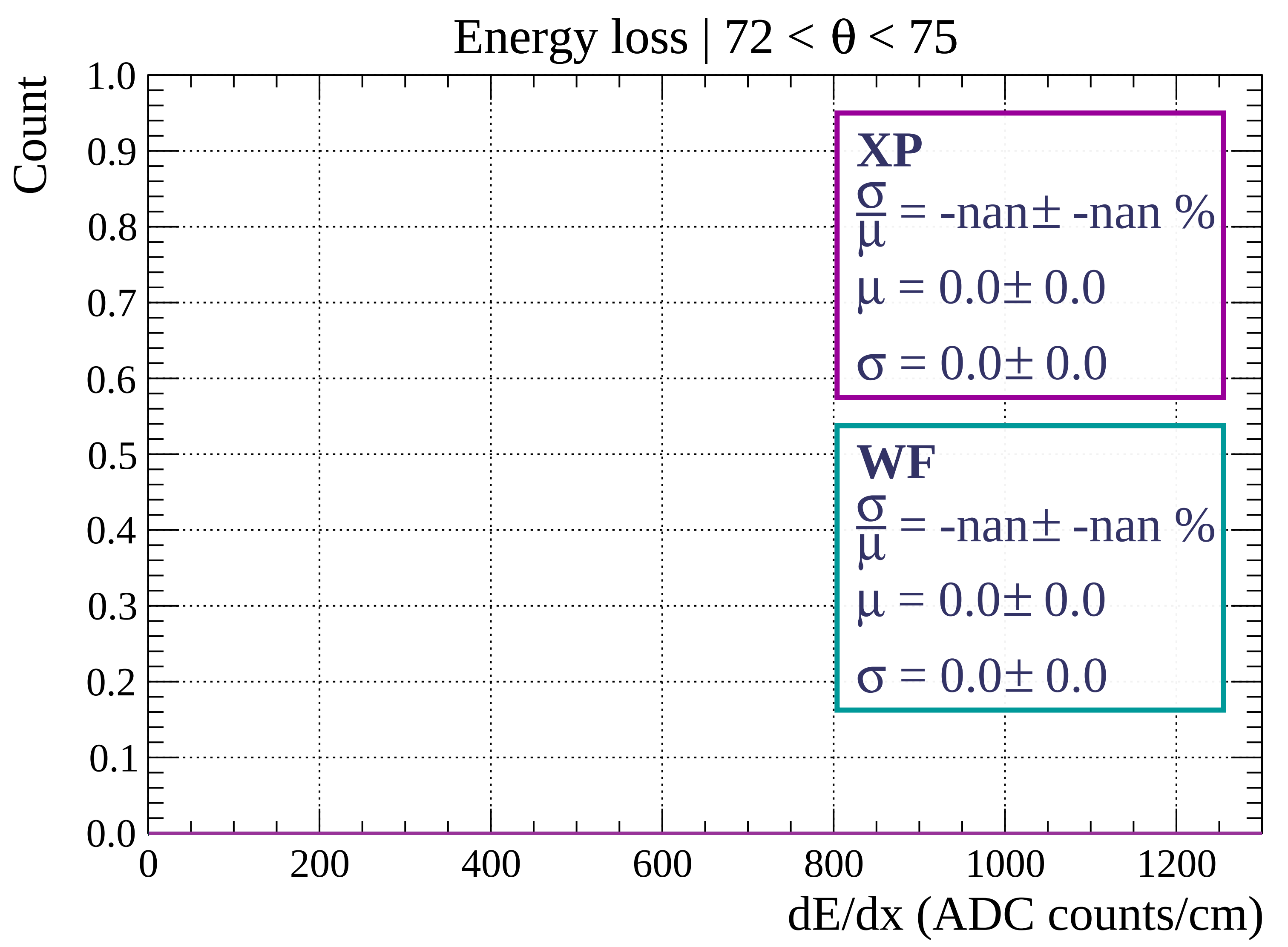
Energy loss | 63 < θ < 66
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



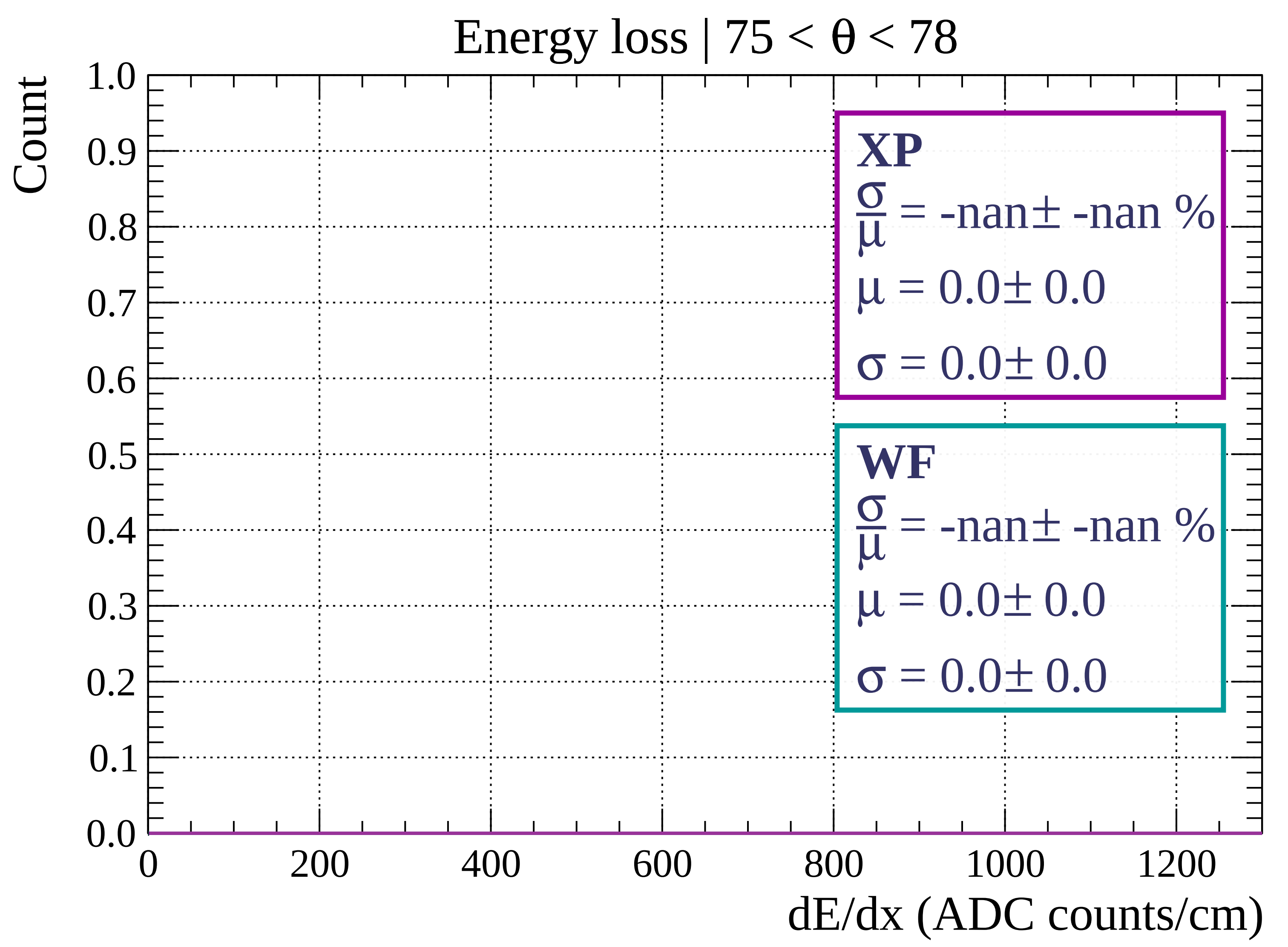
Energy loss | 66 < θ < 69
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

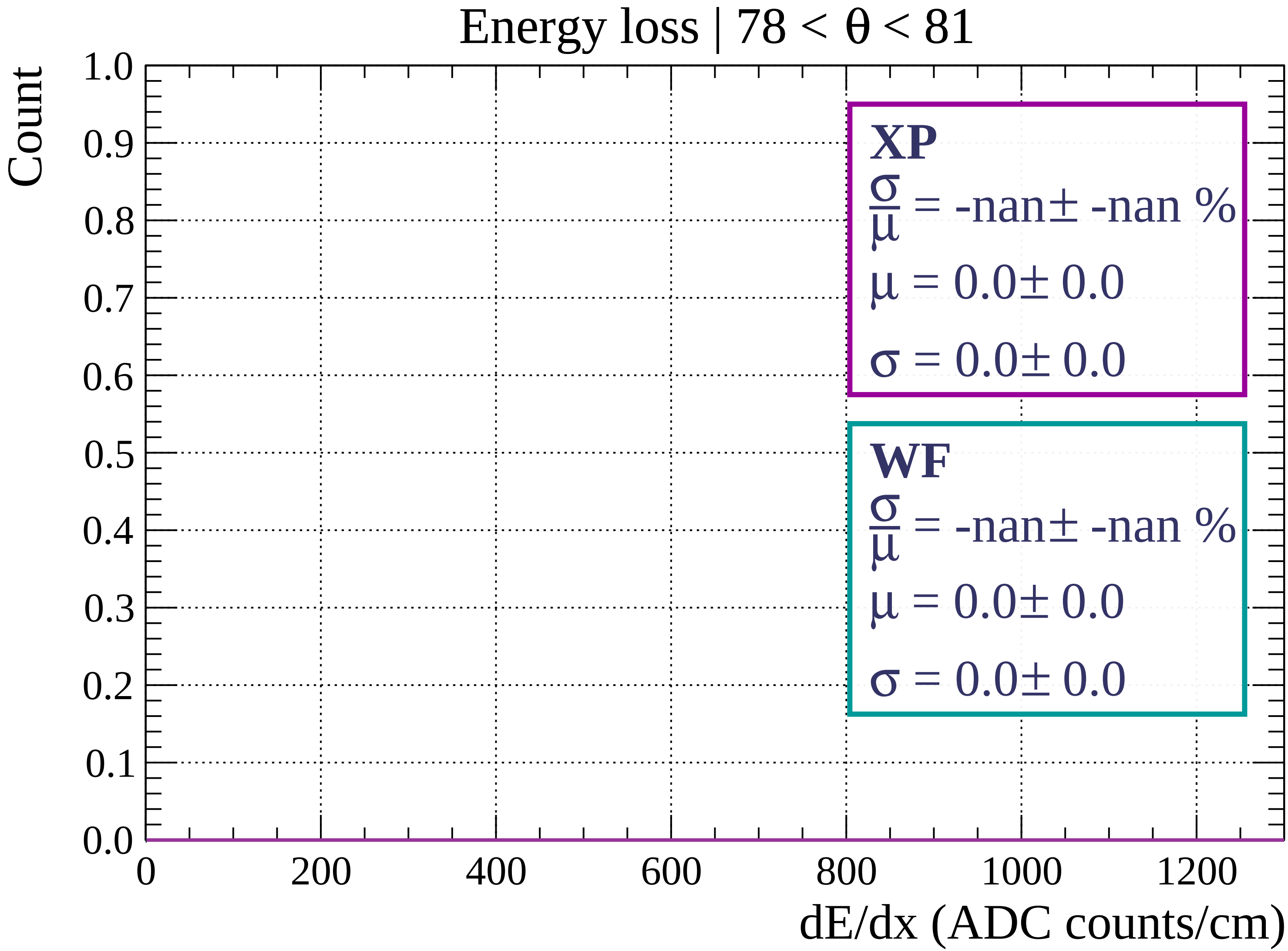


Energy loss | 69 < θ < 72
Count
dE/dx (ADC counts/cm)
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

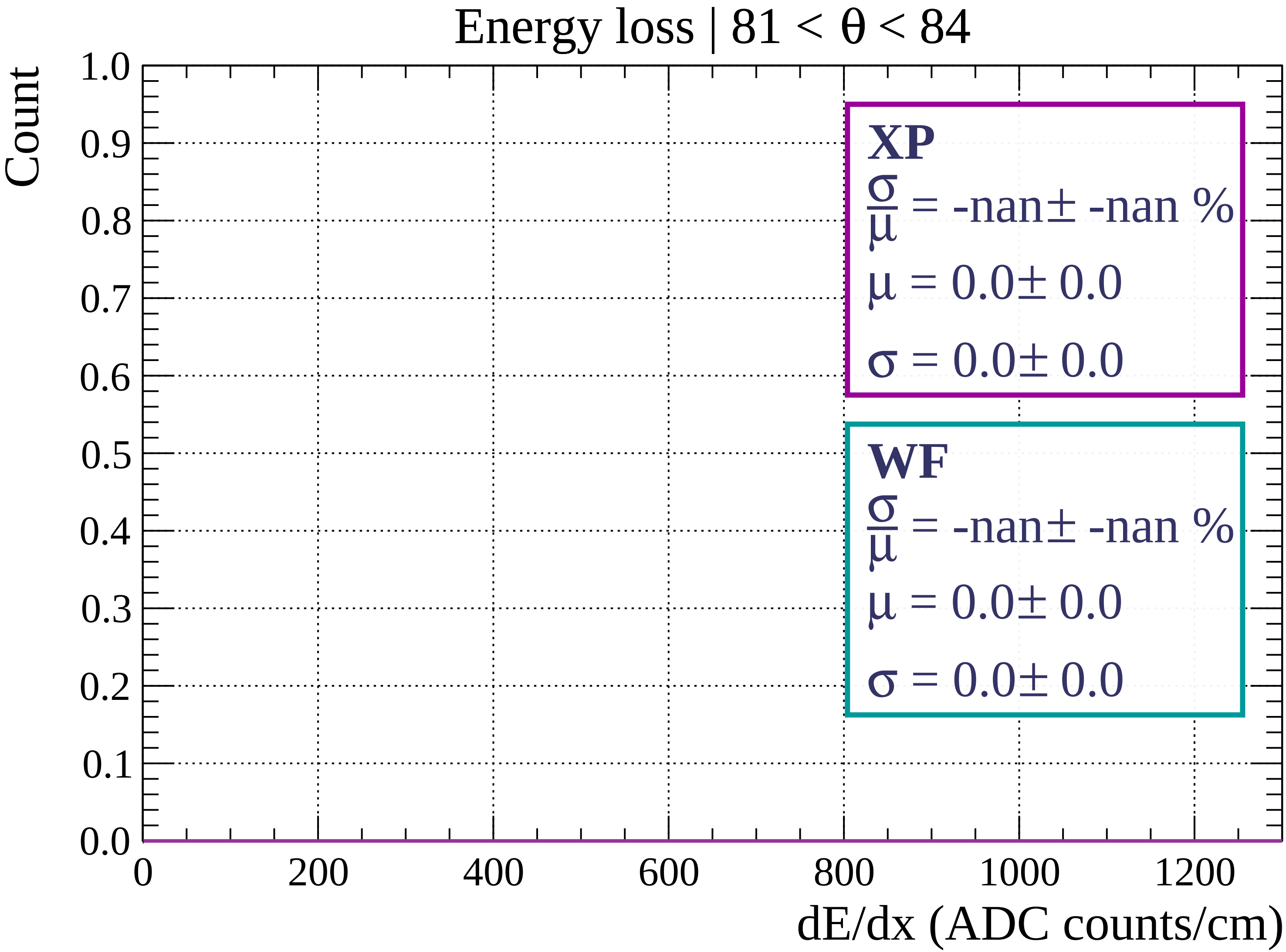


Energy loss | 72 < θ < 75
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

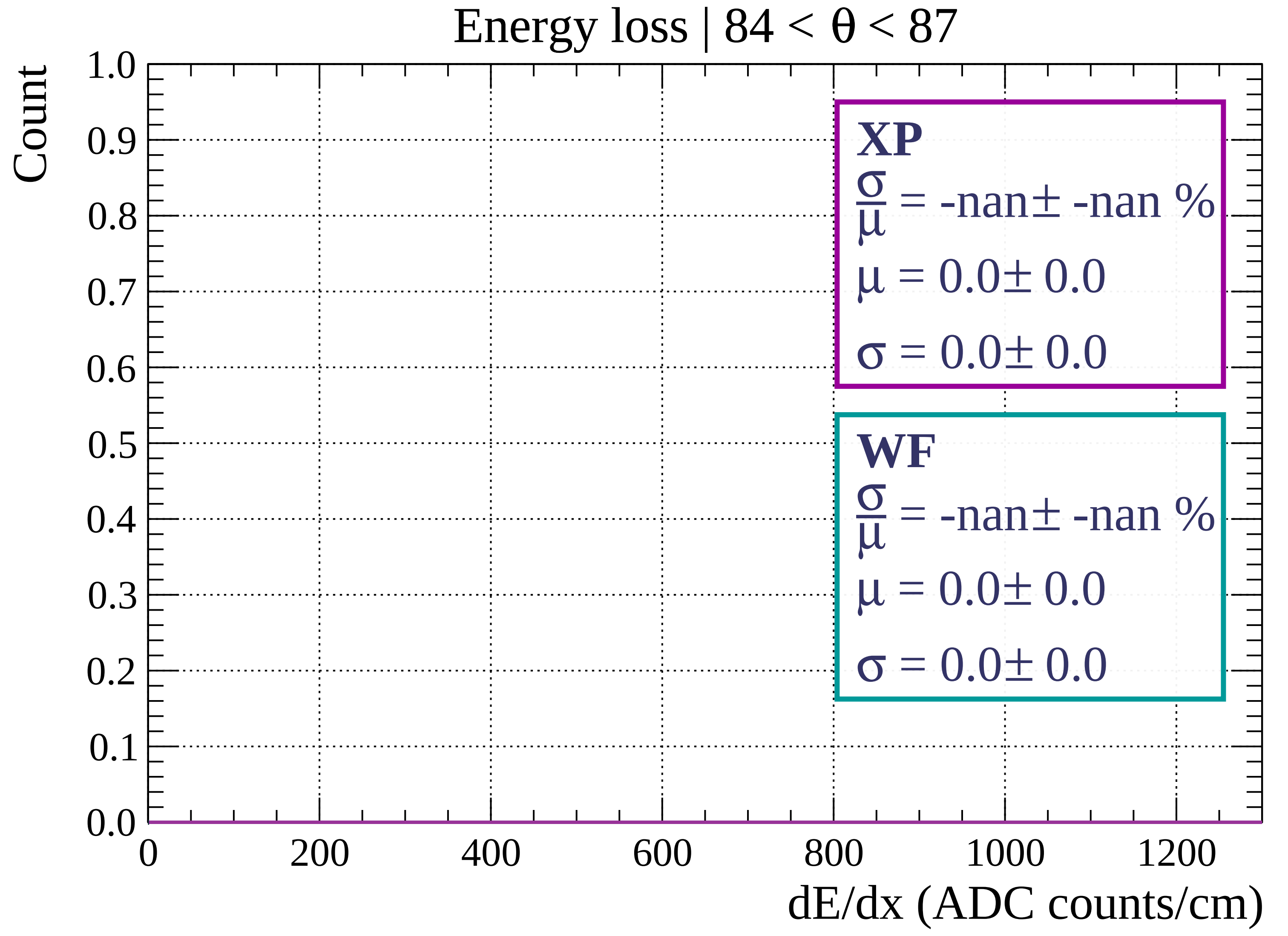

Energy loss | 75 < θ < 78
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

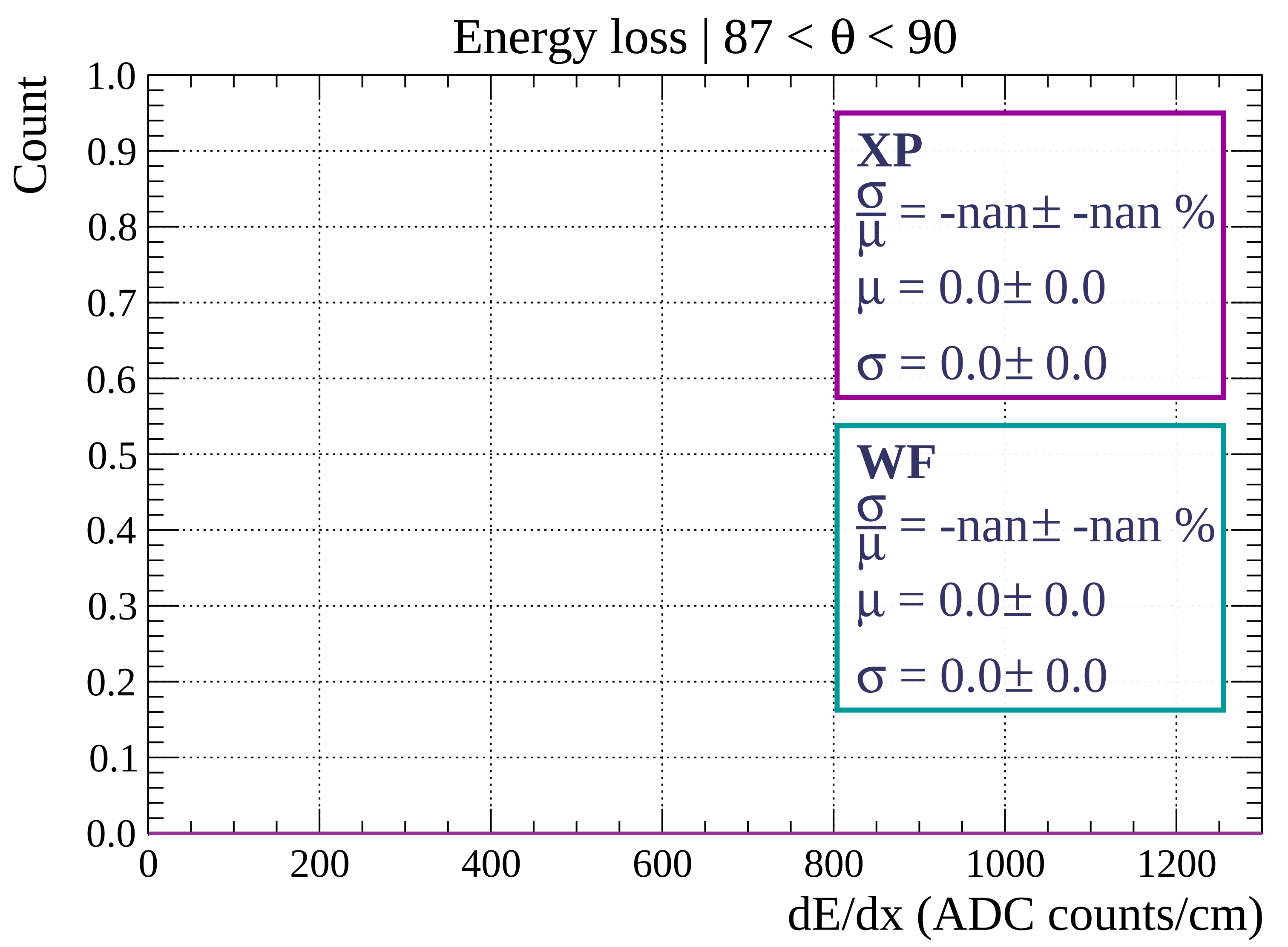


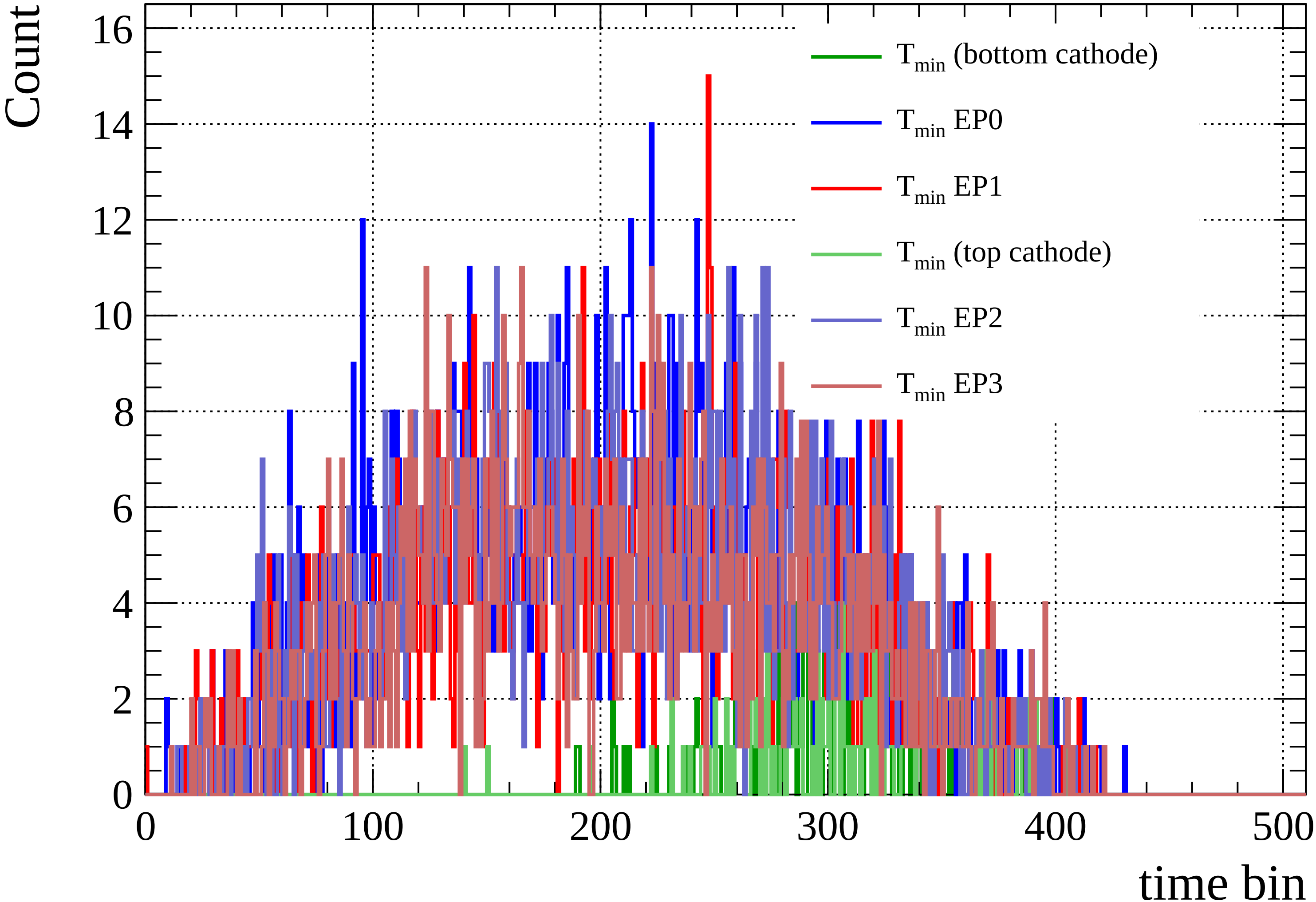
Energy loss | 78 < θ < 81
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



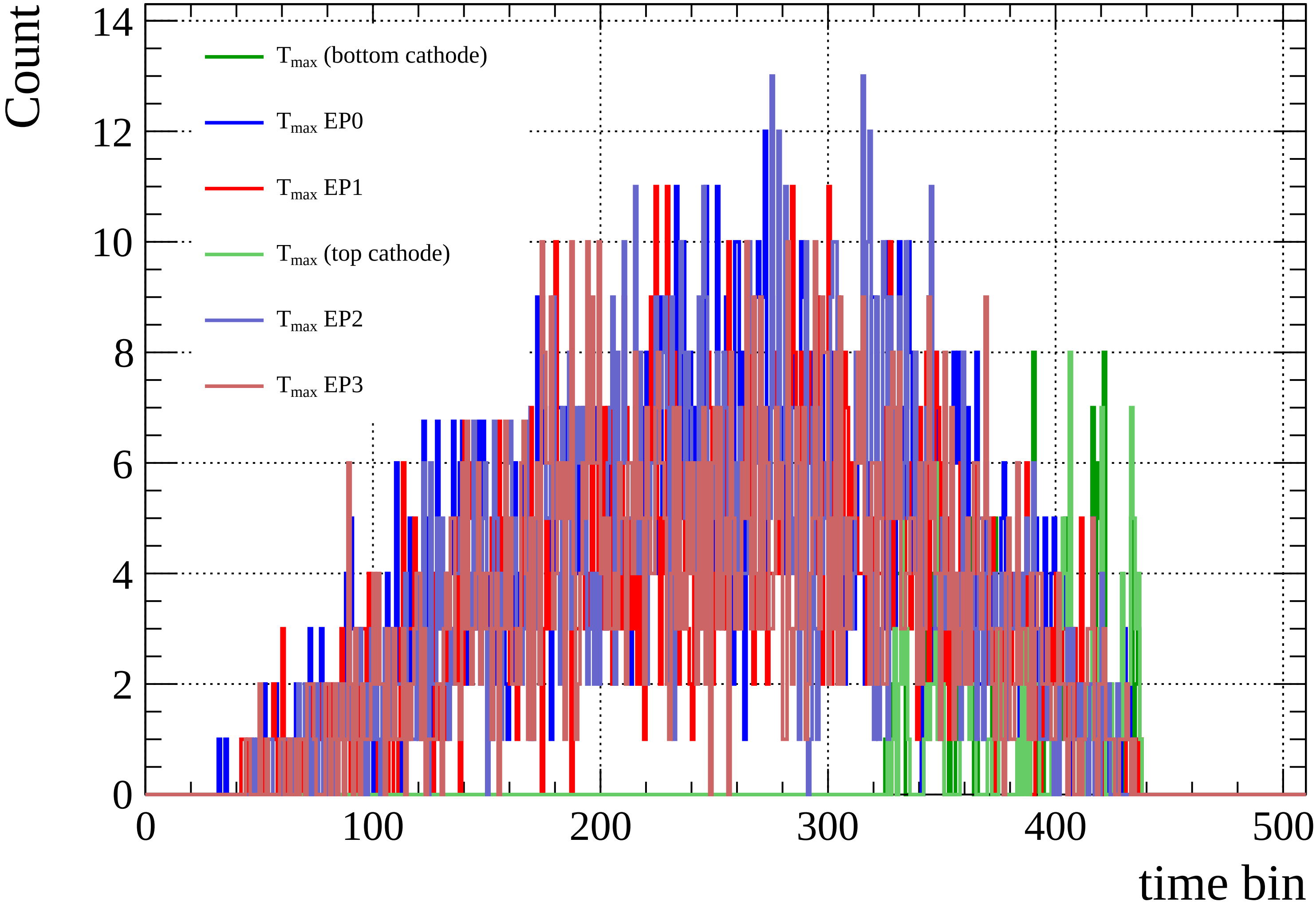
Energy loss | 81 < θ < 84
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | 84 < θ < 87
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | 87 < θ < 90
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



Count
time bin
T_{min} (bottom cathode)
T_{min} EP0
T_{min} EP1
T_{min} (top cathode)
T_{min} EP2
T_{min} EP3
0
2
4
6
8
10
12
14
16
0
100
200
300
400
500



Count
14
12
10
8
6
4
2
0
0
100
200
300
400
500
time bin
T_{max} (bottom cathode)
T_{max} EP0
T_{max} EP1
T_{max} (top cathode)
T_{max} EP2
T_{max} EP3